Mã nguồn và giao diện ứng dụng Ảnh GalleryApp

Báo cáo logic xử lý giao diện và kết quả chạy ứng dụng Android

Ứng dụng Ảnh quản lý ảnh và video, tạo album, tìm kiếm, ẩn hoặc bỏ ẩn, chỉnh sửa ảnh và quản lý hồ sơ tài khoản. Báo cáo trình bày ba phần: mã Java xử lý, mã XML giao diện và ảnh chụp kết quả chạy trên máy ảo Gallery Light Android 11 API 30.

Mã nguồn được lấy từ project GalleryApp tại thời điểm lập báo cáo ngày 18 tháng 10 năm 2026. Phần Java gồm toàn bộ 36 file trong app/src/main/java. Phần XML gồm AndroidManifest.xml và toàn bộ 154 file XML trong app/src/main/res; không lấy file tự sinh trong build hoặc mã kiểm thử.

Bản hiện tại lưu ảnh và video trong bộ nhớ riêng của ứng dụng. Hồ sơ và metadata album sử dụng Firestore. Màn Sao lưu và đồng bộ giữ giao diện minh họa, chưa thực hiện sao lưu cloud. Chức năng ẩn là bộ lọc trong ứng dụng, chưa có khóa mật khẩu.

Nội dung

1 Logic xử lý

2 Giao diện ứng dụng

3 Kết quả sau khi chạy

1 Logic xử lý

Thư mục gốc chứa mã Java: app/src/main/java/com/example/appanh. Các Activity xử lý vòng đời màn hình và sự kiện người dùng; GalleryRepository quản lý dữ liệu; các thư mục editor, models và helpers tách riêng công cụ chỉnh sửa, mô hình và tiện ích.

Luồng thư viện: Activity nhận thao tác → GalleryActivity cập nhật trạng thái tìm kiếm hoặc chọn → MediaAdapter hiển thị dữ liệu → GalleryRepository lưu cache và metadata khi phù hợp. Chọn nhiều dùng LinkedHashSet để giữ thứ tự; adapter hiển thị số 1, 2, 3 theo thứ tự đó.

Luồng chỉnh sửa: EditPhotoActivity nhận photoId → EditorViewModel đọc ảnh trên luồng nền → EditorCanvasView nhận thao tác cắt hoặc vẽ → EditorDocument tạo trạng thái mới → xuất JPEG thành tệp khác và thêm vào thư viện, giữ nguyên ảnh gốc.

1 1 Thư mục appanh

Đường dẫn: app/src/main/java/com/example/appanh

Các Activity, điều hướng, xác thực, repository, adapter và cấu hình dùng chung. Số file Java: 26.

Danh sách: AccountActivity.java, AlbumActivity.java, AlbumDetailActivity.java, AuthActivity.java, BackupActivity.java, BaseActivity.java, CreateAlbumActivity.java, EditPhotoActivity.java, GalleryActivity.java, GalleryApplication.java, GalleryRepository.java, GoogleSignInActivity.java, HiddenMediaActivity.java, LoginActivity.java, MainActivity.java, MediaAdapter.java, NotificationsActivity.java, OtpActivity.java, PhotoDetailActivity.java, RegisterActivity.java, SearchActivity.java, SettingsActivity.java, ShareActivity.java, SplashActivity.java, ThemeManager.java, TrashActivity.java.

1 1 1 AccountActivity

Nguồn: app/src/main/java/com/example/appanh/AccountActivity.java

Vai trò: Đọc hồ sơ users/{uid} từ Firestore, cho phép sửa họ tên và giới thiệu; lưu bằng merge và đồng bộ tên Firebase Auth.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.View;
import android.widget.EditText;
import android.widget.ImageView;
import android.widget.TextView;
```

```
import com.bumptech.glide.Glide;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;
import com.google.firebase.auth.UserProfileChangeRequest;
import com.google.firebase.firestore.DocumentReference;
import com.google.firebase.firestore.DocumentSnapshot;
import com.google.firebase.firestore.FieldValue;
import com.google.firebase.firestore.FirebaseFirestore;
import com.google.firebase.firestore.ListenerRegistration;
import com.google.firebase.firestore.MetadataChanges;
import com.google.firebase.firestore.SetOptions;
import java.util.HashMap;
import java.util.Map;

/** The profile document is always users/{current Firebase Auth uid}. */
public final class AccountActivity extends BaseActivity {
    private FirebaseUser user;
    private DocumentReference profile;
    private ListenerRegistration subscription;
    private EditText name, bio;
    private boolean filling, dirty, saving;
    private final Handler handler = new Handler(Looper.getMainLooper());
    private final Runnable slowSave = () -> {
        if (saving && !isDestroyed()) status("Chưa nhận được xác nhận lưu từ máy chủ. Kiểm
tra kết nối mạng; thay đổi đang chờ đồng bộ.");
    };

    @Override public void onCreate(Bundle state) {
        super.onCreate(state);
        screen(R.layout.activity_account);
        name = findViewById(R.id.profileName);
        bio = findViewById(R.id.profileBio);
        user = FirebaseAuth.getInstance().getCurrentUser();
        click(R.id.saveProfile, this::save);
        click(R.id.accountLogin, () -> go(LoginActivity.class));
        if (user == null) {
            findViewById(R.id.profileForm).setVisibility(View.GONE);
            findViewById(R.id.accountLogin).setVisibility(View.VISIBLE);
            text(R.id.profileHeading, "Bạn đang dùng chế độ khách");
            text(R.id.profileSubtitle, "Đăng nhập để lưu hồ sơ và sao lưu những khoảnh khắc củ
a bạn.");
            return;
        }
        profile = FirebaseFirestore.getInstance().collection("users").document(user.getUid());
        text(R.id.profileEmail, value(user.getEmail(), "Chưa liên kết email"));
        text(R.id.profilePhone, value(user.getPhoneNumber(), "Chưa liên kết số điện thoại"));
        text(R.id.profileVerification, user.getEmail() == null
                ? "Đăng nhập bằng số điện thoại" : user.isEmailVerified() ? "Email đã xác minh" :
 "Email chưa xác minh");
```

```
        text(R.id.profileSubtitle, value(user.getEmail(), user.getPhoneNumber()));
        fill(value(user.getDisplayName(), ""), "", user.getPhotoUrl() == null ? null : user.getPh
otoUrl().toString());
        TextWatcher changes = new TextWatcher() {
            public void beforeTextChanged(CharSequence s,int start,int count,int after) {}
            public void onTextChanged(CharSequence s,int start,int before,int count) {
                if (!filling) { dirty = true; findViewById(R.id.saveProfile).setEnabled(!saving); }
            }
            public void afterTextChanged(Editable text) {}
        };
        name.addTextChangedListener(changes);
        bio.addTextChangedListener(changes);
        findViewById(R.id.saveProfile).setEnabled(false);
        status("Đang tải hồ sơ…");
        if (state != null && state.getBoolean("dirty")) {
            filling = true;
            name.setText(state.getString("name", ""));
            bio.setText(state.getString("bio", ""));
            filling = false;
            dirty = true;
            findViewById(R.id.saveProfile).setEnabled(true);
        }
        getOnBackPressedDispatcher().addCallback(this,new androidx.activity.OnBackPresse
dCallback(true) {
            @Override public void handleOnBackPressed() {
                if (saving) {
                    new androidx.appcompat.app.AlertDialog.Builder(AccountActivity.this)
                            .setTitle("Hồ sơ đang đồng bộ")
                            .setMessage("Máy chủ chưa xác nhận lưu. Bạn có thể ở lại để xem kết quả
 hoặc rời màn hình và kiểm tra lại sau.")
                            .setNegativeButton("Ở lại",null).setPositiveButton("Rời màn hình",(dialog
,which)->finish()).show();
                    return;
                }
                if (!dirty) { finish(); return; }
                new androidx.appcompat.app.AlertDialog.Builder(AccountActivity.this)
                        .setTitle("Thay đổi chưa được lưu")
                        .setMessage("Bạn muốn tiếp tục chỉnh sửa hay bỏ các thay đổi?")
                        .setNegativeButton("Tiếp tục sửa", null)
                        .setPositiveButton("Bỏ thay đổi",(dialog,which)->finish()).show();
            }
        });
    }

    @Override protected void onStart() {
        super.onStart();
        if (profile == null) return;
        subscription = profile.addSnapshotListener(MetadataChanges.INCLUDE, (snapshot,er
ror) -> {
```

```
            if (error != null) {
                status("Không tải được hồ sơ: " + error.getLocalizedMessage());
                return;
            }
            if (snapshot == null || saving) return;
            if (!dirty && snapshot.exists()) show(snapshot);
            if (!dirty) status(snapshot.getMetadata().isFromCache()
                    ? "Đang hiển thị dữ liệu trên thiết bị; chờ kết nối máy chủ."
                    : snapshot.exists() ? "Hồ sơ đã đồng bộ." : "Hoàn thiện thông tin rồi nhấn Lưu t
hay đổi để tạo hồ sơ.");
        });
    }

    private void show(DocumentSnapshot snapshot) {
        fill(value(snapshot.getString("displayName"), value(user.getDisplayName(), "")),
                value(snapshot.getString("bio"), ""),
                value(snapshot.getString("photoUrl"),user.getPhotoUrl()==null?null:user.getPhot
oUrl().toString()));
    }

    private void fill(String displayName,String about,String photoUrl) {
        filling = true;
        name.setText(displayName);
        bio.setText(about);
        filling = false;
        text(R.id.profileHeading, value(displayName,"Hồ sơ của bạn"));
        text(R.id.profileInitial, displayName.isEmpty()?"Ả":displayName.substring(0,displayN
ame.offsetByCodePoints(0,1)).toUpperCase(java.util.Locale.ROOT));
        if (photoUrl != null && !photoUrl.isEmpty()) {
            ImageView avatar = findViewById(R.id.profileAvatar);
            avatar.setVisibility(View.VISIBLE);
            Glide.with(this).load(photoUrl).circleCrop().into(avatar);
        }
    }

    private void save() {
        if (user == null || saving || !dirty) return;
        String displayName = name.getText().toString().trim();
        String about = bio.getText().toString().trim();
        if (displayName.isEmpty() || displayName.length() > 80) { name.setError("Họ tên cần t
ừ 1 đến 80 ký tự."); name.requestFocus(); return; }
        if (about.length() > 240) { bio.setError("Giới thiệu tối đa 240 ký tự."); return; }
        // Prevent saving into a different account after an authentication change.
        FirebaseUser current = FirebaseAuth.getInstance().getCurrentUser();
        if (current == null || !current.getUid().equals(user.getUid())) { status("Tài khoản đã tha
y đổi. Mở lại màn hồ sơ."); return; }
        saving = true;
        name.setEnabled(false); bio.setEnabled(false);
        findViewById(R.id.saveProfile).setEnabled(false);
```

```
        findViewById(R.id.profileProgress).setVisibility(View.VISIBLE);
        status("Đang lưu hồ sơ…");
        handler.postDelayed(slowSave,15000);
        Map<String,Object> data = new HashMap<>();
        data.put("displayName",displayName); data.put("bio",about);
        data.put("email",user.getEmail()); data.put("phoneNumber",user.getPhoneNumber());
        data.put("emailVerified",user.isEmailVerified()); data.put("updatedAt",FieldValue.ser
verTimestamp());
        // Merge preserves photos/albums and any other user preferences.
        profile.set(data, SetOptions.merge()).addOnCompleteListener(task -> {
            handler.removeCallbacks(slowSave);
            if (isDestroyed()) return;
            saving = false;
            name.setEnabled(true); bio.setEnabled(true);
            findViewById(R.id.profileProgress).setVisibility(View.GONE);
            if (!task.isSuccessful()) {
                findViewById(R.id.saveProfile).setEnabled(true);
                status("Chưa lưu được hồ sơ: " + (task.getException()==null?"Vui lòng thử lại.":t
ask.getException().getLocalizedMessage()));
                return;
            }
            dirty = false;
            text(R.id.profileHeading,displayName);
            text(R.id.profileInitial,displayName.substring(0,displayName.offsetByCodePoints(0,
1)).toUpperCase(java.util.Locale.ROOT));
            status("Đã lưu hồ sơ trên Firestore.");
            user.updateProfile(new UserProfileChangeRequest.Builder().setDisplayName(displa
yName).build())
                    .addOnFailureListener(error -> { if(!isDestroyed())status("Hồ sơ đã lưu; tên tro
ng phiên đăng nhập chưa đồng bộ. Mở lại ứng dụng để thử lại."); });
        });
    }

    private static String value(String text,String fallback) { return text==null||text.trim().isEm
pty()?fallback:text; }
    private void text(int id,String value) { ((TextView)findViewById(id)).setText(value); }
    @Override protected void onSaveInstanceState(Bundle state) {
        state.putBoolean("dirty",dirty); state.putString("name",name.getText().toString()); state
.putString("bio",bio.getText().toString());
        super.onSaveInstanceState(state);
    }
    @Override protected void onStop() { if(subscription!=null){subscription.remove();subscr
iption=null;}super.onStop(); }
    @Override protected void onDestroy() { handler.removeCallbacks(slowSave);super.onD
estroy(); }
}
```

1 1 2 AlbumActivity

Nguồn: app/src/main/java/com/example/appanh/AlbumActivity.java

Vai trò: Màn danh sách album; kế thừa xử lý chung từ GalleryActivity.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class AlbumActivity extends GalleryActivity {}
```

1 1 3 AlbumDetailActivity

Nguồn: app/src/main/java/com/example/appanh/AlbumDetailActivity.java

Vai trò: Màn nội dung album, nhận albumId và dùng xử lý thêm ảnh, đổi tên, chọn nhiều từ GalleryActivity.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class AlbumDetailActivity extends GalleryActivity {}
```

1 1 4 AuthActivity

Nguồn: app/src/main/java/com/example/appanh/AuthActivity.java

Vai trò: Xử lý đăng nhập và đăng ký email hoặc số điện thoại, xác minh email, gửi SMS, đặt lại mật khẩu và ghi hồ sơ Firestore.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.content.*;
import android.os.Bundle;
import android.util.Patterns;
import android.view.View;
import com.google.firebase.auth.*;
import com.google.firebase.firestore.*;
import java.util.*;
import java.util.concurrent.TimeUnit;

public abstract class AuthActivity extends BaseActivity {
 protected boolean register;
 private boolean busy;
 @Override public void onCreate(Bundle state){super.onCreate(state);register=this instance of RegisterActivity;screen(register?R.layout.activity_register:R.layout.activity_login);click(R.id.submit,this::submit);click(R.id.switchAuth,()->{go(register?LoginActivity.class:RegisterActivity.class);finish();});click(R.id.google,()->go(GoogleSignInActivity.class));click(R.id.forgot,this::reset);}
```

```
 private void setBusy(boolean value){busy=value;findViewById(R.id.submit).setEnabled(!v
alue);android.view.View progress=findViewById(R.id.authProgress);if(progress!=null)prog
ress.setVisibility(value?View.VISIBLE:View.GONE);}
 private void submit(){
  if(busy)return;String email=input(R.id.email),password=((android.widget.EditText)findVi
ewById(R.id.password)).getText().toString();
  if(email.startsWith("+")){phone(email);return;}
  if(!Patterns.EMAIL_ADDRESS.matcher(email).matches()){toast("Nhập email hợp lệ hoặc
 số điện thoại dạng +84…");return;}
  if(password.length()<6){toast("Mật khẩu cần ít nhất 6 ký tự.");return;}
  if(register&&input(R.id.name).isEmpty()){toast("Vui lòng nhập họ tên.");return;}
  setBusy(true);status("Đang kết nối…");
  com.google.android.gms.tasks.Task<AuthResult> task=register?FirebaseAuth.getInstance(
).createUserWithEmailAndPassword(email,password):FirebaseAuth.getInstance().signInWi
thEmailAndPassword(email,password);
  task.addOnCompleteListener(this,t->{setBusy(false);if(!t.isSuccessful()){status(t.getExcep
tion()==null?"Đăng nhập thất bại":t.getException().getLocalizedMessage());return;}
   FirebaseUser user=FirebaseAuth.getInstance().getCurrentUser();if(user==null)return;
   if(register){user.updateProfile(new UserProfileChangeRequest.Builder().setDisplayName(
input(R.id.name)).build());saveProfile(user,input(R.id.name));user.sendEmailVerification().
addOnCompleteListener(this,sent->{if(!sent.isSuccessful())toast("Chưa gửi được email. Bạ
n có thể gửi lại ở màn hình tiếp theo.");go(OtpActivity.class);finish();});}
   else {saveProfile(user,user.getDisplayName());go(user.isEmailVerified()?MainActivity.cl
ass:OtpActivity.class);finish();}
  });
 }
 private void reset(){String email=input(R.id.email);if(!Patterns.EMAIL_ADDRESS.match
er(email).matches()){toast("Nhập email trước khi đặt lại mật khẩu.");return;}FirebaseAuth.g
etInstance().sendPasswordResetEmail(email).addOnCompleteListener(this,t->status(t.isSuc
cessful()?"Đã gửi email đặt lại mật khẩu.":String.valueOf(t.getException().getLocalizedMes
sage())));}
 private void phone(String number){
  if(!number.matches("\\+[1-9][0-9]{7,14}")){toast("Số điện thoại cần có mã quốc gia, ví dụ
 +84901234567.");return;}
  setBusy(true);status("Đang gửi SMS…");
  PhoneAuthProvider.verifyPhoneNumber(PhoneAuthOptions.newBuilder(FirebaseAuth.get
Instance()).setPhoneNumber(number).setTimeout(60L,TimeUnit.SECONDS).setActivity(th
is).setCallbacks(new PhoneAuthProvider.OnVerificationStateChangedCallbacks(){
   @Override public void onVerificationCompleted(PhoneAuthCredential credential){Fireba
seAuth.getInstance().signInWithCredential(credential).addOnCompleteListener(AuthActivit
y.this,t->{setBusy(false);if(t.isSuccessful()){saveProfile(FirebaseAuth.getInstance().getCurr
entUser(),null);go(MainActivity.class);finish();}else status(t.getException().getLocalizedMe
ssage());});}
   @Override public void onVerificationFailed(com.google.firebase.FirebaseException e){se
tBusy(false);status(e.getLocalizedMessage());}
   @Override public void onCodeSent(String id,PhoneAuthProvider.ForceResendingToken t
oken){setBusy(false);startActivity(new Intent(AuthActivity.this,OtpActivity.class).putExtra
("verificationId",id).putExtra("phone",number));}
  }).build());
```

```
 }
 public static void saveProfile(FirebaseUser user,String name){
  if(user==null)return;
  FirebaseFirestore db=FirebaseFirestore.getInstance();
  DocumentReference ref=db.collection("users").document(user.getUid());
  db.runTransaction(transaction->{
   DocumentSnapshot old=transaction.get(ref);
   Map<String,Object> data=new HashMap<>();
   data.put("email",user.getEmail());data.put("phoneNumber",user.getPhoneNumber());
   data.put("emailVerified",user.isEmailVerified());data.put("lastLoginAt",FieldValue.server
Timestamp());
   if(!old.exists())data.put("createdAt",FieldValue.serverTimestamp());
   // Keep the name edited in the user's Firestore profile across logins.
   String display=name==null?user.getDisplayName():name;
   if(display!=null&&(!old.contains("displayName")||old.getString("displayName")==null||o
ld.getString("displayName").trim().isEmpty()))data.put("displayName",display);
   transaction.set(ref,data,SetOptions.merge());return null;
  });
 }
}
```

1 1 5 BackupActivity

Nguồn: app/src/main/java/com/example/appanh/BackupActivity.java

Vai trò: Giữ màn Sao lưu và đồng bộ để minh họa. Nút sao lưu giải thích trạng thái demo; không tải file lên Storage.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.os.Bundle;
import com.google.android.material.switchmaterial.SwitchMaterial;

/** Presentation screen only: cloud backup is deliberately not activated in this demo. */
public class BackupActivity extends BaseActivity {
    @Override public void onCreate(Bundle state){
        super.onCreate(state);screen(R.layout.activity_backup);
        android.content.SharedPreferences prefs=getSharedPreferences("backup-"+repo.uid(),0
);
        SwitchMaterial auto=findViewById(R.id.autoBackup),wifi=findViewById(R.id.wifiO
nly);
        auto.setChecked(prefs.getBoolean("auto",false));wifi.setChecked(prefs.getBoolean("wi
fi",true));
        auto.setOnCheckedChangeListener((view,checked)->prefs.edit().putBoolean("auto",ch
ecked).apply());
        wifi.setOnCheckedChangeListener((view,checked)->prefs.edit().putBoolean("wifi",che
cked).apply());
```

```
        status("Giao diện minh họa • Chưa kích hoạt sao lưu cloud.");
        click(R.id.backupNow,()->new androidx.appcompat.app.AlertDialog.Builder(this)
                .setTitle("Sao lưu và đồng bộ")
                .setMessage("Màn hình này dùng để minh họa giao diện trong bản demo. Ảnh và
video hiện được lưu trên thiết bị; chưa tải lên Firebase Storage.")
                .setPositiveButton("Đã hiểu",null).show());
    }
    public static void backup(BaseActivity activity,GalleryRepository repo,boolean manual,R
unnable done){
        if(manual)activity.status("Giao diện minh họa • Chưa kích hoạt sao lưu cloud.");
        done.run();
    }
}
```

1 1 6 BaseActivity

Nguồn: app/src/main/java/com/example/appanh/BaseActivity.java

Vai trò: Thiết lập giao diện chung, xử lý khoảng cách thanh hệ thống và bàn phím, điều hướng bốn tab, hộp thoại và repository.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.content.*;
import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.view.*;
import com.google.android.material.appbar.MaterialToolbar;
import com.google.android.material.bottomnavigation.BottomNavigationView;

public abstract class BaseActivity extends AppCompatActivity {
    protected GalleryRepository repo;
    @Override public void onCreate(Bundle state) {
        super.onCreate(state);
        // Authentication screens do not read the media cache. In particular, the
        // welcome screen must draw before Firebase authentication is warmed up.
        if (!(this instanceof SplashActivity) && !(this instanceof AuthActivity)
                && !(this instanceof OtpActivity) && !(this instanceof GoogleSignInActivity)
                && !(this instanceof AccountActivity)) {
            repo = new GalleryRepository(this);
        }
    }
    protected void screen(int layout) {
        // Use one inset policy on Android 7–16. Otherwise pre-Android-15 windows
        // already inset the content and our listener would add the bars twice.
```

```
        WindowCompat.setDecorFitsSystemWindows(getWindow(), false);
        boolean dark = (getResources().getConfiguration().uiMode
                & android.content.res.Configuration.UI_MODE_NIGHT_MASK)
                == android.content.res.Configuration.UI_MODE_NIGHT_YES;
        getWindow().setStatusBarColor(android.graphics.Color.TRANSPARENT);
        getWindow().setNavigationBarColor(android.graphics.Color.TRANSPARENT);
        WindowInsetsControllerCompat barsController = WindowCompat.getInsetsController(
getWindow(),getWindow().getDecorView());
        barsController.setAppearanceLightStatusBars(!dark);
        barsController.setAppearanceLightNavigationBars(!dark);
        setContentView(layout);
        View root=findViewById(R.id.screenRoot);
        root.setFocusableInTouchMode(true);
        root.requestFocus();
        ViewCompat.setOnApplyWindowInsetsListener(root,(v,insets)->{
            androidx.core.graphics.Insets bars=insets.getInsets(WindowInsetsCompat.Type.syst
emBars()|WindowInsetsCompat.Type.ime());
            // This root owns system/keyboard spacing. Material navigation must
            // not add the same bottom inset again inside its fixed-height bar.
            v.setPadding(bars.left,bars.top,bars.right,bars.bottom);
            return WindowInsetsCompat.CONSUMED;
        });
        MaterialToolbar toolbar=findViewById(R.id.toolbar);
        if(toolbar!=null) toolbar.setNavigationOnClickListener(v->getOnBackPressedDispatch
er().onBackPressed());
        BottomNavigationView nav=findViewById(R.id.navigation);
        if(nav!=null) {
            nav.setItemActiveIndicatorEnabled(false);
            if(toolbar!=null && !(this instanceof SettingsActivity))toolbar.setNavigationIcon(nu
ll);
            int selected=this instanceof AlbumActivity ? R.id.nav_albums : this instanceof Setti
ngsActivity ? R.id.nav_profile : this instanceof SearchActivity ? R.id.nav_search : R.id.nav_
photos;
            nav.setSelectedItemId(selected);
            nav.setOnItemSelectedListener(item->{
                if(item.getItemId()==selected)return true;
                Class<?> target=item.getItemId()==R.id.nav_albums?AlbumActivity.class:item.g
etItemId()==R.id.nav_profile?SettingsActivity.class:item.getItemId()==R.id.nav_search?Se
archActivity.class:MainActivity.class;
                Intent intent=new Intent(this,target).addFlags(Intent.FLAG_ACTIVITY_REORD
ER_TO_FRONT); startActivity(intent); return true;
            });
        }
    }
    protected void click(int id,Runnable action){View view=findViewById(id);if(view!=null)
view.setOnClickListener(v->action.run());}
    @Override protected void onResume() {
        super.onResume();
        BottomNavigationView nav = findViewById(R.id.navigation);
```

```
        if (nav != null) {
            // REORDER_TO_FRONT reuses Activities; restore their own tab when
            // returning instead of keeping the tab tapped before leaving.
            nav.setSelectedItemId(this instanceof AlbumActivity ? R.id.nav_albums
                    : this instanceof SettingsActivity ? R.id.nav_profile
                    : this instanceof SearchActivity ? R.id.nav_search : R.id.nav_photos);
        }
    }
    protected String input(int id){return ((EditText)findViewById(id)).getText().toString().trim();}
    protected void go(Class<?> target){Intent intent=new Intent(this,target);if(target==MainActivity.class&&(this instanceof AuthActivity||this instanceof OtpActivity||this instanceof GoogleSignInActivity))intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK|Intent.FLAG_ACTIVITY_CLEAR_TASK);startActivity(intent);}
    protected void status(String value){TextView text=findViewById(R.id.status);if(text!=null){text.setText(value);text.setVisibility(value==null||value.isEmpty()?View.GONE:View.VISIBLE);}}
    protected void toast(String value){Toast.makeText(this,value,Toast.LENGTH_LONG).show();}
    protected void result(String error){if(error!=null)toast(error);}
    protected void confirm(String title,Runnable yes){new androidx.appcompat.app.AlertDialog.Builder(this).setTitle(title).setMessage("Bạn có chắc muốn tiếp tục?").setNegativeButton("Hủy",null).setPositiveButton("Đồng ý",(d,w)->yes.run()).show();}
    protected void ask(String title,String initial,java.util.function.Consumer<String> done){EditText text=new EditText(this);text.setText(initial);text.setSingleLine();new androidx.appcompat.app.AlertDialog.Builder(this).setTitle(title).setView(text).setNegativeButton("Hủy",null).setPositiveButton("Lưu",(d,w)->{String value=text.getText().toString().trim();if(value.isEmpty()||value.length()>80){toast("Nhập từ 1 đến 80 ký tự.");return;}done.accept(value);}).show();}
    @Override protected void onDestroy(){if(repo!=null)repo.close();super.onDestroy();}
}
```

1 1 7 CreateAlbumActivity

Nguồn: app/src/main/java/com/example/appanh/CreateAlbumActivity.java

Vai trò: Màn tạo album; dùng GalleryActivity để chọn ảnh, nhập tên và ghi album cùng quan hệ thành viên.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class CreateAlbumActivity extends GalleryActivity {}
```

1 1 8 EditPhotoActivity

Nguồn: app/src/main/java/com/example/appanh/EditPhotoActivity.java

Vai trò: Điều phối công cụ cắt, xoay, bộ lọc, sáng và tương phản, vẽ, hoàn tác và lưu bản sao; xác nhận khi bỏ thay đổi.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.content.res.ColorStateList;
import android.graphics.Color;
import android.graphics.Rect;
import android.os.Bundle;
import android.view.View;
import android.widget.SeekBar;
import androidx.appcompat.app.AlertDialog;
import androidx.lifecycle.ViewModelProvider;
import com.example.appanh.editor.EditorCanvasView;
import com.example.appanh.editor.EditorDocument;
import com.example.appanh.editor.EditorViewModel;
import com.example.appanh.models.Photo;
import com.google.android.material.button.MaterialButton;

public class EditPhotoActivity extends BaseActivity {
   EditorViewModel model;
   EditorCanvasView canvas;
   private String shownTool;
   private Rect restoredCrop;
   private static final int[] COLORS={Color.WHITE,Color.BLACK,0xFFEF4444,0xFF256
3EB,0xFF16A34A,0xFFFACC15};
   private static final int[] COLOR_IDS={R.id.brushWhite,R.id.brushBlack,R.id.brushRed,
R.id.brushBlue,R.id.brushGreen,R.id.brushYellow};
   @Override public void onCreate(Bundle state){
      super.onCreate(state);screen(R.layout.activity_edit_photo);
      model=new ViewModelProvider(this).get(EditorViewModel.class);canvas=findViewB
yId(R.id.photo);
      if(state!=null)restoredCrop=state.getParcelable("cropRect");
      canvas.setOnChanged(model::changed);
      click(R.id.crop,()->tool("crop"));click(R.id.draw,()->tool("draw"));click(R.id.filter,()->
tool("filter"));click(R.id.adjust,()->tool("adjust"));
      click(R.id.rotate,()->model.rotate());
      click(R.id.applyCrop,()->{if(ready())model.crop(canvas.cropPixels(),false);});
      click(R.id.undo,()->{if(ready()){model.document.undo();model.changed();}});
      click(R.id.redo,()->{if(ready()){model.document.redo();model.changed();}});
      click(R.id.resetEditor,()->{if(ready()&&model.document.dirty())new AlertDialog.Buil
der(this).setTitle("Khôi phục ảnh gốc?").setMessage("Bỏ các thay đổi hiện tại. Bạn vẫn có t
hể nhấn Hoàn tác sau thao tác này.").setNegativeButton("Hủy",null).setPositiveButton("Kh
ôi phục",(dialog,which)->{model.document.reset();model.tool="view";model.changed();}).s
how();});
      click(R.id.clearDrawing,()->{if(ready()){model.document.clearStrokes();model.change
d();}});
      click(R.id.save,this::save);click(R.id.cancelEdit,this::leave);
```

```
        int[] ratios={R.id.ratioFree,R.id.ratioSquare,R.id.ratioPortrait,R.id.ratioWide};float[] v
alues={0,1,4f/3,16f/9};
        for(int i=0;i<ratios.length;i++){final float value=values[i];click(ratios[i],()->{model.as
pect=value;canvas.setAspect(value);ratios();});}
        int[] filters={R.id.filterOriginal,R.id.filterMono,R.id.filterSepia,R.id.filterWarm,R.id.fi
lterCool};
        for(int i=0;i<filters.length;i++){final int value=i;click(filters[i],()->{if(ready()){model.
document.filter(value);model.changed();}});}
        for(int i=0;i<COLOR_IDS.length;i++){final int color=COLORS[i];click(COLOR_IDS
[i],()->{model.brushColor=color;brush();});}
        ((SeekBar)findViewById(R.id.brushWidth)).setOnSeekBarChangeListener(new SeekB
ar.OnSeekBarChangeListener(){public void onProgressChanged(SeekBar bar,int value,bool
ean user){if(user){model.brushWidth=value+1;brush();}}public void onStartTrackingTouch
(SeekBar bar){}public void onStopTrackingTouch(SeekBar bar){}});
        SeekBar.OnSeekBarChangeListener adjust=new SeekBar.OnSeekBarChangeListener()
{
            public void onStartTrackingTouch(SeekBar bar){if(ready())model.document.beginA
djustment();}
            public void onProgressChanged(SeekBar bar,int value,boolean user){if(user&&read
y()){model.document.adjust(((SeekBar)findViewById(R.id.brightness)).getProgress()-100,(
(SeekBar)findViewById(R.id.contrast)).getProgress()-50);model.changed();}}
            public void onStopTrackingTouch(SeekBar bar){}
        };
        ((SeekBar)findViewById(R.id.brightness)).setOnSeekBarChangeListener(adjust);((See
kBar)findViewById(R.id.contrast)).setOnSeekBarChangeListener(adjust);
        model.updates.observe(this,ignored->refresh());
        model.message.observe(this,this::status);
        model.saved.observe(this,photo->{if(photo!=null){NotificationsActivity.record(this,"Đ
ã lưu bản chỉnh sửa "+photo.getFileName());toast("Đã lưu bản sao. Ảnh gốc được giữ nguy
ên.");finish();}});
        getOnBackPressedDispatcher().addCallback(this,new androidx.activity.OnBackPresse
dCallback(true){public void handleOnBackPressed(){leave();}});
        if(model.document==null&&!model.busy)repo.load((photos,albums,error)->loadPhoto
(true),()->loadPhoto(false));
    }
    private void loadPhoto(boolean finalResult){if(isDestroyed()||model.document!=null||mo
del.busy)return;Photo photo=repo.find(getIntent().getStringExtra("photoId"));if(photo!=null
&&!photo.isVideo())model.load(repo,photo);else if(finalResult)status(photo==null?"Không
 tìm thấy ảnh. Quay lại thư viện để chọn lại.":"Công cụ chỉnh sửa chỉ hỗ trợ ảnh.");}
    private boolean ready(){return model.document!=null&&!model.busy;}
    private void tool(String value){if(!ready())return;model.tool=value.equals(model.tool)?"v
iew":value;status(model.tool.equals("crop")?"Kéo bốn góc để đổi khung, kéo bên trong để d
i chuyển.":model.tool.equals("draw")?"Chọn màu và nét bút, sau đó vẽ trực tiếp lên ảnh.":"
Các thay đổi chỉ áp dụng lên bản sao khi lưu.");refresh();}
    private void refresh(){
        boolean enabled=ready();findViewById(R.id.editorBusy).setVisibility(model.busy?Vie
w.VISIBLE:View.GONE);
        for(int id:new int[]{R.id.crop,R.id.draw,R.id.filter,R.id.adjust,R.id.rotate,R.id.applyCro
p,R.id.save,R.id.resetEditor,R.id.clearDrawing,R.id.brightness,R.id.contrast,R.id.brushWidt
```

```
h})findViewById(id).setEnabled(enabled);
        canvas.setEnabled(enabled);findViewById(R.id.undo).setEnabled(enabled&&model.document.canUndo());findViewById(R.id.redo).setEnabled(enabled&&model.document.canRedo());
        if(model.document==null)return;
        canvas.bind(model.document);
        if(!model.tool.equals(shownTool)){shownTool=model.tool;canvas.setMode(model.tool.equals("crop")?EditorCanvasView.Mode.CROP:model.tool.equals("draw")?EditorCanvasView.Mode.DRAW:EditorCanvasView.Mode.VIEW);if(model.tool.equals("crop"))canvas.setAspect(model.aspect);}
        if(restoredCrop!=null){canvas.restoreCrop(restoredCrop);restoredCrop=null;}
        findViewById(R.id.cropPanel).setVisibility(model.tool.equals("crop")?View.VISIBLE:View.GONE);findViewById(R.id.drawPanel).setVisibility(model.tool.equals("draw")?View.VISIBLE:View.GONE);findViewById(R.id.adjustPanel).setVisibility(model.tool.equals("adjust")?View.VISIBLE:View.GONE);findViewById(R.id.filterPanel).setVisibility(model.tool.equals("filter")?View.VISIBLE:View.GONE);
        ((SeekBar)findViewById(R.id.brightness)).setProgress(model.document.state().brightness+100);((SeekBar)findViewById(R.id.contrast)).setProgress(model.document.state().contrast+50);
        ((SeekBar)findViewById(R.id.brushWidth)).setProgress((int)model.brushWidth-1);brush();ratios();
    }
    private void ratios(){int[] ids={R.id.ratioFree,R.id.ratioSquare,R.id.ratioPortrait,R.id.ratioWide};float[] values={0,1,4f/3,16f/9};for(int i=0;i<ids.length;i++){MaterialButton button=findViewById(ids[i]);button.setBackgroundTintList(ColorStateList.valueOf(getColor(Math.abs(model.aspect-values[i])<.01?R.color.primary_light:R.color.viewer_controls)));}}
    private void brush(){canvas.setBrush(model.brushColor,model.brushWidth);for(int i=0;i<COLOR_IDS.length;i++){MaterialButton button=findViewById(COLOR_IDS[i]);button.setStrokeWidth((int)((model.brushColor==COLORS[i]?3:1)*getResources().getDisplayMetrics().density));button.setStrokeColor(ColorStateList.valueOf(getColor(model.brushColor==COLORS[i]?R.color.primary:R.color.field_border)));}}
    private boolean pendingCrop(){return ready()&&canvas.mode()==EditorCanvasView.Mode.CROP&&!canvas.cropPixels().equals(new Rect(0,0,model.document.state().image.getWidth(),model.document.state().image.getHeight()));}
    private void save(){if(!ready())return;if(pendingCrop()){model.crop(canvas.cropPixels(),true);return;}if(!model.document.dirty()){toast("Chưa có thay đổi để lưu.");return;}model.save();}
    private void leave(){if(model.busy){toast("Đang xử lý ảnh, vui lòng chờ một chút.");return;}if(model.document==null||!model.document.dirty()&&!pendingCrop()){finish();return;}
new AlertDialog.Builder(this).setTitle("Bỏ các thay đổi?").setMessage("Ảnh gốc vẫn được giữ nguyên. Bản chỉnh sửa chưa được lưu.").setNegativeButton("Tiếp tục sửa",null).setPositiveButton("Bỏ thay đổi",(dialog,which)->finish()).show();}
    @Override protected void onSaveInstanceState(Bundle state){if(canvas!=null&&model.document!=null)state.putParcelable("cropRect",canvas.cropPixels());super.onSaveInstanceState(state);}
}
```

1 1 9 GalleryActivity

Nguồn: app/src/main/java/com/example/appanh/GalleryActivity.java

Vai trò: Xử lý thư viện, tìm kiếm, lọc, sắp xếp, nhập tệp, chọn nhiều theo thứ tự, ẩn và bỏ ẩn, album và thùng rác.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.content.*;
import android.database.Cursor;
import android.net.Uri;
import android.os.Bundle;
import android.provider.OpenableColumns;
import android.text.*;
import android.view.*;
import android.widget.*;
import androidx.activity.result.*;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.recyclerview.widget.*;
import com.example.appanh.models.*;
import com.google.android.material.appbar.MaterialToolbar;
import com.google.android.material.bottomsheet.BottomSheetDialog;
import java.io.*;
import java.util.*;

/** Gallery behavior is Java; screen structure is defined in res/layout. */
public class GalleryActivity extends BaseActivity {
 protected final Set<String> selected=new LinkedHashSet<>();
 protected String type="all",query="",sort="newest",albumId;
 protected boolean importing,selecting,showHidden;
 private final java.util.concurrent.ExecutorService io=java.util.concurrent.Executors.newSin
gleThreadExecutor();
 private final ActivityResultLauncher<String[]> picker=registerForActivityResult(new Acti
vityResultContracts.OpenMultipleDocuments(),uris->{if(!uris.isEmpty())importFiles(uris);}
);
 @Override public void onCreate(Bundle state){super.onCreate(state);albumId=getIntent().
getStringExtra("albumId");
  int layout=this instanceof AlbumActivity?R.layout.activity_album:this instanceof CreateAl
bumActivity?R.layout.activity_create_album:this instanceof AlbumDetailActivity?R.layout.
activity_album_detail:this instanceof TrashActivity?R.layout.activity_trash:this instanceof S
earchActivity?R.layout.activity_search:R.layout.activity_main;
  screen(layout);MaterialToolbar toolbar=findViewById(R.id.toolbar);
  if(this instanceof MainActivity||this instanceof SearchActivity||this instanceof HiddenMedi
aActivity){toolbar.inflateMenu(R.menu.gallery_actions);for(int i=0;i<toolbar.getMenu().siz
e();i++){android.graphics.drawable.Drawable icon=toolbar.getMenu().getItem(i).getIcon();i
f(icon!=null)icon.mutate().setTint(androidx.core.content.ContextCompat.getColor(this,R.co
lor.text_primary));}toolbar.setOnMenuItemClickListener(item->{if(item.getItemId()==R.id.
galleryFilter)filter();else options();return true;});}
```

```
  if(this instanceof AlbumDetailActivity){toolbar.inflateMenu(R.menu.album_actions);for(i
nt i=0;i<toolbar.getMenu().size();i++){android.graphics.drawable.Drawable icon=toolbar.ge
tMenu().getItem(i).getIcon();if(icon!=null)icon.mutate().setTint(getColor(R.color.text_prim
ary));}toolbar.setOnMenuItemClickListener(item->{if(item.getItemId()==R.id.addPhotos)a
ddPhotos();else albumOptions();return true;});selecting=getIntent().getBooleanExtra("select
ing",false);}
  EditText search=findViewById(R.id.search);if(search!=null){search.addTextChangedListe
ner(new TextWatcher(){public void beforeTextChanged(CharSequence s,int a,int c,int f){}p
ublic void onTextChanged(CharSequence s,int a,int b,int c){query=s.toString().toLowerCas
e(Locale.ROOT);render();}public void afterTextChanged(Editable e){}});search.setOnEdit
orActionListener((view,action,event)->{if(action!=android.view.inputmethod.EditorInfo.IM
E_ACTION_SEARCH)return false;((android.view.inputmethod.InputMethodManager)getS
ystemService(INPUT_METHOD_SERVICE)).hideSoftInputFromWindow(view.getWindo
wToken(),0);view.clearFocus();return true;});}
  click(R.id.all,()->{type="all";render();});click(R.id.images,()->{type="images";render();})
;click(R.id.videos,()->{type="videos";render();});click(R.id.favorites,()->{type="favorites";
render();});
  com.google.android.material.floatingactionbutton.FloatingActionButton importButton=fin
dViewById(R.id.importMedia);if(importButton!=null){importButton.setImageDrawable(an
droidx.appcompat.content.res.AppCompatResources.getDrawable(this,R.drawable.ic_add).
mutate());importButton.setImageTintList(android.content.res.ColorStateList.valueOf(getCol
or(R.color.white)));}
  click(R.id.importMedia,()->picker.launch(new String[]{"image/*","video/*"}));click(R.id.
sort,this::filter);
  click(R.id.createAlbum,()->go(CreateAlbumActivity.class));click(R.id.create,this::createA
lbum);
  click(R.id.addPhotos,this::addPhotos);click(R.id.albumOptions,this::albumOptions);click(
R.id.emptyTrash,this::emptyTrash);
  if(state!=null){selected.addAll(state.getStringArrayList("selected")==null?Collections.emp
tyList():state.getStringArrayList("selected"));sort=state.getString("sort","newest");type=stat
e.getString("type","all");query=state.getString("query","");showHidden=state.getBoolean("s
howHidden");selecting=state.getBoolean("selecting");if(search!=null)search.setText(query);
int chip=type.equals("images")?R.id.images:type.equals("videos")?R.id.videos:type.equals(
"favorites")?R.id.favorites:R.id.all;android.widget.CompoundButton selectedChip=findVie
wById(chip);if(selectedChip!=null)selectedChip.setChecked(true);}
 }
 @Override protected void onResume(){super.onResume();if(!importing)reload();}
 @Override protected void onSaveInstanceState(Bundle out){out.putStringArrayList("select
ed",new ArrayList<>(selected));out.putString("sort",sort);out.putString("type",type);out.put
String("query",query);out.putBoolean("showHidden",showHidden);out.putBoolean("selecti
ng",selecting);super.onSaveInstanceState(out);}
 protected void reload(){repo.load((p,a,error)->{if(isFinishing()||isDestroyed())return;render
();if(error!=null)status(error);else if(this instanceof TrashActivity){List<Photo> expired=ne
w ArrayList<>();long cutoff=System.currentTimeMillis()-30L*24*60*60*1000;for(Photo p
hoto:p)if(photo.isDeleted()&&photo.getDeletedAt()!=null&&photo.getDeletedAt().getTime
()<cutoff)expired.add(photo);if(!expired.isEmpty())removeNext(expired,0);}else if(this inst
anceof MainActivity&&getSharedPreferences("backup-"+repo.uid(),0).getBoolean("auto",f
alse))BackupActivity.backup(this,repo,false,()->{});},()->{if(!isFinishing()&&!isDestroyed
())render();});}
```

```
 protected List<Photo> visible(){ArrayList<Photo> list=new ArrayList<>();for(Photo p:repo.photos){if(p.isDeleted()!=(this instanceof TrashActivity))continue;if(!(this instanceof TrashActivity)&&p.isHidden()!=showHidden)continue;if(this instanceof AlbumDetailActivity&&!p.getAlbumIds().contains(albumId))continue;if(type.equals("images")&&p.isVideo()||type.equals("videos")&&!p.isVideo()||type.equals("favorites")&&!p.isFavorite())continue;if(!String.valueOf(p.getFileName()).toLowerCase(Locale.ROOT).contains(query))continue;list.add(p);}Comparator<Photo> comparator=Comparator.comparingLong(p->p.getDateTaken()==null?0:p.getDateTaken().getTime());if(sort.equals("newest"))comparator=comparator.reversed();if(sort.equals("size"))comparator=Comparator.comparingLong(Photo::getFileSize).reversed();if(sort.equals("type"))comparator=Comparator.comparing(p->String.valueOf(p.getFileName()),String.CASE_INSENSITIVE_ORDER);list.sort(comparator);return list;}
 protected void render(){
  RecyclerView grid=findViewById(R.id.grid);if(grid==null)return;boolean albumScreen=this instanceof AlbumActivity;
  grid.setLayoutManager(new GridLayoutManager(this,this instanceof TrashActivity?1:albumScreen?2:3));List<Album> orderedAlbums=new ArrayList<>(repo.albums);orderedAlbums.sort(Comparator.comparingLong((Album album)->album.getCreatedAt()==null?0:album.getCreatedAt().getTime()).reversed());List<?> items=albumScreen?orderedAlbums:visible();View empty=findViewById(R.id.emptyState);if(empty!=null)empty.setVisibility(items.isEmpty()?View.VISIBLE:View.GONE);
  grid.setAdapter(new MediaAdapter(items,repo,selected,new MediaAdapter.Events(){public void tap(Object item){if(item instanceof Album){startActivity(new Intent(GalleryActivity.this,AlbumDetailActivity.class).putExtra("albumId",((Album)item).getId()));return;}Photo p=(Photo)item;if(GalleryActivity.this instanceof CreateAlbumActivity||selecting){if(!selected.add(p.getId()))selected.remove(p.getId());render();}else if(GalleryActivity.this instanceof TrashActivity)trashActions(p);else startActivity(new Intent(GalleryActivity.this,PhotoDetailActivity.class).putExtra("photoId",p.getId()));}public void hold(Object item){if(item instanceof Photo)photoMenu((Photo)item);else{albumId=((Album)item).getId();albumOptions();}}},this instanceof TrashActivity));
  status(items.isEmpty()?(empty!=null?"":albumScreen?"Chưa có album. Nhấn Tạo album để bắt đầu.":this instanceof TrashActivity?"Thùng rác đang trống.":"Thêm ảnh từ tab Ảnh để bắt đầu."):(showHidden?"Đã ẩn • ":"")+items.size()+" "+(albumScreen?"album":"mục")+(selected.isEmpty()?"":" • Đã chọn "+selected.size()));
  if(this instanceof AlbumDetailActivity){Album album=repo.album(albumId);if(album!=null)((MaterialToolbar)findViewById(R.id.toolbar)).setTitle(album.getName());}
 }
 private void filter(){BottomSheetDialog dialog=sheet(R.layout.sheet_filter);for(int id:new int[]{R.id.newest,R.id.oldest,R.id.size,R.id.type})dialog.findViewById(id).setOnClickListener(v->{sort=id==R.id.newest?"newest":id==R.id.oldest?"oldest":id==R.id.size?"size":"type";dialog.dismiss();render();});dialog.show();}
 private BottomSheetDialog sheet(int layout){BottomSheetDialog d=new BottomSheetDialog(this);d.setContentView(layout);d.setOnShowListener(dialog -> d.getBehavior().setState(com.google.android.material.bottomsheet.BottomSheetBehavior.STATE_EXPANDED));MaterialToolbar bar=d.findViewById(R.id.toolbar);if(bar!=null)bar.setNavigationOnClickListener(v->d.dismiss());return d;}
 private android.app.Dialog optionsPopup(){android.app.Dialog dialog=new android.app.Dialog(this);dialog.setContentView(R.layout.sheet_options);android.view.Window window=dialog.getWindow();if(window!=null){window.setBackgroundDrawableResource(android.R.color.transparent);window.setGravity(android.view.Gravity.TOP|android.view.Gravity.EN
```

```
D);window.setLayout((int)(264*getResources().getDisplayMetrics().density),android.view.
WindowManager.LayoutParams.WRAP_CONTENT);android.view.WindowManager.Layo
utParams params=window.getAttributes();params.y=(int)(64*getResources().getDisplayMet
rics().density);params.x=(int)(12*getResources().getDisplayMetrics().density);params.dim
Amount=.28f;window.setAttributes(params);}com.google.android.material.appbar.Material
Toolbar bar=dialog.findViewById(R.id.toolbar);if(bar!=null)bar.setVisibility(View.GONE);
return dialog;}
 private void options(){android.app.Dialog d=optionsPopup();com.google.android.material.
button.MaterialButton hide=d.findViewById(R.id.hide);hide.setText(showHidden?"Bỏ ẩn c
ác mục đã chọn":"Ẩn các mục đã chọn");hide.setEnabled(!selected.isEmpty());d.findViewB
yId(R.id.hiddenMedia).setVisibility(showHidden?View.GONE:View.VISIBLE);d.findView
ById(R.id.hiddenMedia).setOnClickListener(v->{d.dismiss();go(HiddenMediaActivity.clas
s);});d.findViewById(R.id.select).setOnClickListener(v->{selecting=!selecting;selected.cle
ar();render();status(selecting?"Chạm vào từng ảnh để chọn, sau đó mở menu → Thêm vào a
lbum":"Đã thoát chọn nhiều");d.dismiss();});d.findViewById(R.id.addAlbum).setOnClickLi
stener(v->{d.dismiss();chooseAlbum(new ArrayList<>(selected));});d.findViewById(R.id.h
ide).setOnClickListener(v->{List<String> ids=new ArrayList<>(selected);repo.setHidden(i
ds,!showHidden,error->{result(error);selected.clear();selecting=false;render();});d.dismiss()
;});d.findViewById(R.id.settings).setOnClickListener(v->{go(SettingsActivity.class);d.dis
miss();});d.findViewById(R.id.help).setOnClickListener(v->{toast("Thêm ảnh bằng nút +.
Nhấn giữ ảnh để mở menu. Nhấn ảnh để xem và chỉnh sửa.");d.dismiss();});d.show();}
 private void photoMenu(Photo p){if(this instanceof TrashActivity){trashActions(p);return;
}BottomSheetDialog d=sheet(R.layout.sheet_photo_context);((com.google.android.material
.button.MaterialButton)d.findViewById(R.id.hide)).setText(p.isHidden()?"Bỏ ẩn":"Ẩn ảnh/
video");d.findViewById(R.id.edit).setVisibility(p.isVideo()?View.GONE:View.VISIBLE);i
nt[] ids={R.id.info,R.id.addAlbum,R.id.favorite,R.id.hide,R.id.edit,R.id.share,R.id.trash};fo
r(int id:ids)d.findViewById(id).setOnClickListener(v->{d.dismiss();if(id==R.id.info)Photo
DetailActivity.info(this,repo,p);else if(id==R.id.addAlbum)chooseAlbum(Collections.single
tonList(p.getId()));else if(id==R.id.favorite){p.setFavorite(!p.isFavorite());repo.save(p,error
->{result(error);render();});}else if(id==R.id.hide){repo.setHidden(Collections.singletonLis
t(p.getId()),!p.isHidden(),error->{result(error);render();});}else if(id==R.id.edit)startActivit
y(new Intent(this,EditPhotoActivity.class).putExtra("photoId",p.getId()));else if(id==R.id.sh
are)startActivity(new Intent(this,ShareActivity.class).putExtra("photoId",p.getId()));else co
nfirm("Chuyển ảnh vào thùng rác?",()->{p.setDeleted(true);p.setDeletedAt(new Date());rep
o.save(p,error->{result(error);render();});});});d.show();}
 private void chooseAlbum(List<String> ids){if(ids.isEmpty()){toast("Nhấn Chọn nhiều và
chọn ảnh trước.");return;}if(repo.albums.isEmpty()){toast("Bạn cần tạo album trước.");retur
n;}String[] names=new String[repo.albums.size()];for(int i=0;i<names.length;i++)names[i]
=repo.albums.get(i).getName();new androidx.appcompat.app.AlertDialog.Builder(this).setT
itle("Thêm vào album").setItems(names,(d,index)->{String target=repo.albums.get(index).g
etId();saveMembership(ids,target,0,()->{selected.clear();selecting=false;render();});}).show
();}
 private void createAlbum(){String name=input(R.id.albumName);if(name.isEmpty()||name.
length()>80){toast("Tên album cần từ 1 đến 80 ký tự.");return;}findViewById(R.id.create).s
etEnabled(false);Album a=new Album(repo.uid(),name,"");a.setCreatedAt(new Date());repo
.saveAlbum(a,error->{if(error!=null){result(error);findViewById(R.id.create).setEnabled(tr
ue);return;}saveMembership(new ArrayList<>(selected),a.getId(),0,()->{NotificationsActiv
ity.record(this,"Đã tạo album "+name);finish();});});}
 private void saveMembership(List<String> ids,String target,int index,Runnable done){if(in
```

```
dex>=ids.size()){done.run();return;}Photo p=repo.find(ids.get(index));if(p==null){saveMembership(ids,target,index+1,done);return;}if(!p.getAlbumIds().contains(target))p.getAlbumIds().add(target);repo.save(p,error->{if(error!=null){result(error);View create=findViewById(R.id.create);if(create!=null)create.setEnabled(true);return;}saveMembership(ids,target,index+1,done);});}
 private void addPhotos(){List<Photo> candidates=new ArrayList<>();for(Photo p:repo.photos)if(!p.isDeleted()&&!p.isHidden()&&!p.getAlbumIds().contains(albumId))candidates.add(p);if(candidates.isEmpty()){toast("Không còn ảnh để thêm. Hãy nhập ảnh ở tab Ảnh.");return;}String[] labels=new String[candidates.size()];boolean[] checked=new boolean[candidates.size()];for(int i=0;i<labels.length;i++)labels[i]=candidates.get(i).getFileName();new androidx.appcompat.app.AlertDialog.Builder(this).setTitle("Chọn ảnh").setMultiChoiceItems(labels,checked,(d,i,c)->checked[i]=c).setNegativeButton("Hủy",null).setPositiveButton("Thêm",(d,w)->{List<String> ids=new ArrayList<>();for(int i=0;i<checked.length;i++)if(checked[i])ids.add(candidates.get(i).getId());saveMembership(ids,albumId,0,this::render);}).show();}
 private void albumOptions(){
  Album album=repo.album(albumId);if(album==null)return;
  BottomSheetDialog dialog=sheet(R.layout.sheet_album_options);
  dialog.findViewById(R.id.removeSelected).setVisibility(this instanceof AlbumDetailActivity&&!selected.isEmpty()?View.VISIBLE:View.GONE);
  dialog.findViewById(R.id.select).setOnClickListener(view->{dialog.dismiss();if(this instanceof AlbumDetailActivity){selecting=!selecting;selected.clear();render();if(selecting)status("Chạm để chọn ảnh → menu → Bỏ ảnh đã chọn khỏi album");}else startActivity(new Intent(this,AlbumDetailActivity.class).putExtra("albumId",albumId).putExtra("selecting",true));});
  dialog.findViewById(R.id.addPhotos).setOnClickListener(view->{dialog.dismiss();addPhotos();});
  dialog.findViewById(R.id.renameAlbum).setOnClickListener(view->{dialog.dismiss();ask("Tên album",album.getName(),name->{album.setName(name);repo.saveAlbum(album,error->{result(error);render();});});});
  dialog.findViewById(R.id.removeSelected).setOnClickListener(view->{dialog.dismiss();removeMembership(new ArrayList<>(selected),0);});
  dialog.findViewById(R.id.deleteAlbum).setOnClickListener(view->{dialog.dismiss();confirm("Xóa album? Ảnh vẫn được giữ trong thư viện.",()->repo.deleteAlbum(album,error->{if(error!=null){result(error);return;}if(this instanceof AlbumDetailActivity)finish();else render();}));});
  dialog.show();
 }
 private void removeMembership(List<String> ids,int index){if(index>=ids.size()){selected.clear();selecting=false;render();return;}Photo photo=repo.find(ids.get(index));if(photo==null){removeMembership(ids,index+1);return;}photo.getAlbumIds().remove(albumId);repo.save(photo,error->{if(error!=null){result(error);return;}removeMembership(ids,index+1);});}
 private void trashActions(Photo p){new androidx.appcompat.app.AlertDialog.Builder(this).setTitle(p.getFileName()).setItems(new String[]{"Khôi phục","Xóa vĩnh viễn"},(d,i)->{if(i==0){p.setDeleted(false);p.setDeletedAt(null);repo.save(p,error->{result(error);render();});}else confirm("Xóa vĩnh viễn ảnh này?",()->repo.remove(p,error->{result(error);render();}));}).show();}
 private void emptyTrash(){List<Photo> trash=visible();if(trash.isEmpty())return;confirm("Xóa vĩnh viễn "+trash.size()+" mục?",()->removeNext(trash,0));}
```

```
 private void removeNext(List<Photo> list,int i){if(i>=list.size()){render();return;}repo.rem
ove(list.get(i),error->{if(error!=null){result(error);render();return;}removeNext(list,i+1);});
}
 private void importFiles(List<Uri> uris){importing=true;status("Đang nhập "+uris.size()+"
mục…");io.execute(()->{ArrayList<Photo> imported=new ArrayList<>();String failure=nul
l;for(Uri uri:uris){File target=null;try{String mime=getContentResolver().getType(uri);if(m
ime==null||!(mime.startsWith("image/")||mime.startsWith("video/")))continue;String name=
"Media";try(Cursor c=getContentResolver().query(uri,new String[]{OpenableColumns.DIS
PLAY_NAME},null,null,null)){if(c!=null&&c.moveToFirst())name=c.getString(0);}File di
r=new File(getFilesDir(),"media");if(!dir.exists()&&!dir.mkdirs())throw new IOException("
Không tạo được thư mục ảnh");String id=UUID.randomUUID().toString();target=new File(
dir,id);try(InputStream in=getContentResolver().openInputStream(uri);OutputStream out=n
ew FileOutputStream(target)){if(in==null)throw new IOException("Không mở được ảnh");
byte[] buffer=new byte[65536];int n;while((n=in.read(buffer))!=-1)out.write(buffer,0,n);}Ph
oto p=new Photo(repo.uid(),name,target.getPath(),target.length(),0,0,mime);p.setId(id);p.set
CreatedAt(new Date());p.setDateTaken(new Date());p.setDeviceName(android.os.Build.M
ODEL);if(!p.isVideo()){android.graphics.BitmapFactory.Options o=new android.graphics.
BitmapFactory.Options();o.inJustDecodeBounds=true;android.graphics.BitmapFactory.deco
deFile(target.getPath(),o);p.setWidth(o.outWidth);p.setHeight(o.outHeight);}imported.add(
p);}catch(Exception e){if(target!=null)target.delete();failure=e.getLocalizedMessage();}}Str
ing error=failure;runOnUiThread(()->{if(isDestroyed())return;repo.photos.addAll(imported)
;repo.persist(e->{importing=false;render();if(error!=null)toast("Một số file chưa nhập được:
 "+error);result(e);if(repo.uid()!=null&&getSharedPreferences("backup-"+repo.uid(),0).get
Boolean("auto",false))BackupActivity.backup(this,repo,false,()->{});});});});}
 @Override protected void onDestroy(){io.shutdown();super.onDestroy();}
}
```

1 1 10 GalleryApplication

Nguồn: app/src/main/java/com/example/appanh/GalleryApplication.java

Vai trò: Khởi tạo ứng dụng, áp dụng chế độ sáng tối và chuẩn bị Firebase để màn chào có thể hiển thị sớm.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.app.Application;
import android.os.Handler;
import android.os.Looper;
import com.google.firebase.auth.FirebaseAuth;
import java.util.ArrayList;
import java.util.List;

/** Warm up the cached authentication state without blocking the first frame. */
public final class GalleryApplication extends Application {
    private final Handler main = new Handler(Looper.getMainLooper());
    private final List<Runnable> waiting = new ArrayList<>();
```

```
    private boolean ready;
    private RuntimeException initializationError;

    @Override public void onCreate() {
        super.onCreate();
        ThemeManager.initialize(this);
        new Thread(() -> {
            try { FirebaseAuth.getInstance(); }
            catch (RuntimeException error) { initializationError = error; }
            main.post(() -> {
                ready = true;
                for (Runnable callback : new ArrayList<>(waiting)) callback.run();
                waiting.clear();
            });
        }, "authentication-startup").start();
    }

    public void whenReady(Runnable callback) {
        if (ready) callback.run(); else waiting.add(callback);
    }

    public RuntimeException getInitializationError() { return initializationError; }
}
```

1 1 11 GalleryRepository

Nguồn: app/src/main/java/com/example/appanh/GalleryRepository.java

Vai trò: Quản lý cache riêng của khách hoặc tài khoản, file nội bộ và metadata Firestore. Có mã Storage dự phòng, nhưng màn sao lưu của bản demo không gọi upload.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.content.Context;
import android.net.Uri;
import android.os.Handler;
import android.os.Looper;
import com.example.appanh.models.Photo;
import com.example.appanh.models.Album;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.*;
import com.google.firebase.storage.FirebaseStorage;
import java.io.*;
import java.util.*;
import java.util.concurrent.Executors;

/** Account-scoped local cache; Firestore metadata and Storage originals are separate. */
```

```
public final class GalleryRepository {
   public interface Result { void done(String error); }
   public interface Loaded { void done(List<Photo> photos, List<Album> albums, String er
ror); }
   private final Context context;
   public final ArrayList<Photo> photos = new ArrayList<>();
   public final ArrayList<Album> albums = new ArrayList<>();
   private final String owner;
   private static final java.util.concurrent.ExecutorService io = Executors.newSingleThread
Executor();
   private final Handler main = new Handler(Looper.getMainLooper());
   public GalleryRepository(Context context) {
      this.context = context.getApplicationContext();
      owner = FirebaseAuth.getInstance().getUid();
   }
   public String uid() { return owner; }
   private CollectionReference collection(String name) {
      return FirebaseFirestore.getInstance().collection("users").document(owner).collection(
name);
   }
   private File cache() { return new File(context.getFilesDir(), "gallery-" + (owner == null ?
"guest" : owner) + ".bin"); }
   public void load(Loaded result) {
      load(result, null);
   }
   public void load(Loaded result, Runnable localReady) {
      io.execute(() -> {
         ArrayList<Photo> localPhotos = new ArrayList<>(); ArrayList<Album> localAlbu
ms = new ArrayList<>();
         String failure = null;
         if (cache().exists()) try (ObjectInputStream in = new ObjectInputStream(new FileInp
utStream(cache()))) {
            for (Object item : (List<?>) in.readObject()) if (item instanceof Photo) localPhoto
s.add((Photo)item);
            for (Object item : (List<?>) in.readObject()) if (item instanceof Album) localAlbu
ms.add((Album)item);
         } catch (Exception e) { failure = "Không đọc được bộ nhớ cục bộ: " + e.getMessage(
); }
         String localError = failure;
         main.post(() -> {
            for(Photo photo:localPhotos)applyPendingHidden(photo);
            photos.clear(); photos.addAll(localPhotos); albums.clear(); albums.addAll(localAl
bums);
            // Gallery screens can draw their local content while cloud reads
            // are pending; detail screens still wait for the final result.
            if (localReady != null) localReady.run();
            if (owner == null) { result.done(photos, albums, localError); return; }
            collection("photos").get().addOnSuccessListener(snapshot -> {
               Map<String, Photo> merged = new LinkedHashMap<>();
```

```
                // Retain imports not yet backed up. Synced records removed on another device
                // must disappear instead of being resurrected from this device's cache.
                for (Photo p : photos) if(p.getFirebaseUrl()==null||p.getFirebaseUrl().isEmpty()
) merged.put(p.getId(), p);
                for (DocumentSnapshot doc : snapshot) {
                    Photo p = doc.toObject(Photo.class); if (p == null) continue; p.setId(doc.getI
d());
                    Photo local = find(p.getId());
                    if (local != null && local.getFilePath() != null && new File(local.getFilePat
h()).exists()) p.setFilePath(local.getFilePath());
                    applyPendingHidden(p);merged.put(p.getId(), p);
                }
                photos.clear(); photos.addAll(merged.values());
                collection("albums").get().addOnSuccessListener(a -> {
                    Set<String> pending=context.getSharedPreferences("pending-albums-"+own
er,0).getStringSet("ids",Collections.emptySet());
                    Map<String,Album> mergedAlbums=new LinkedHashMap<>();
                    for(Album local:albums)if(pending.contains(local.getId()))mergedAlbums.pu
t(local.getId(),local);
                    for (DocumentSnapshot doc : a) { Album album = doc.toObject(Album.class
); if (album != null) { album.setId(doc.getId());Album local=album(album.getId());if(local!
=null&&local.getCoverPhotoPath()!=null&&new File(local.getCoverPhotoPath()).exists())a
lbum.setCoverPhotoPath(local.getCoverPhotoPath());mergedAlbums.put(album.getId(),albu
m); } }
                    albums.clear();albums.addAll(mergedAlbums.values());
                    persist(error -> result.done(photos, albums, error));
                }).addOnFailureListener(e -> result.done(photos, albums, e.getMessage()));
            }).addOnFailureListener(e -> result.done(photos, albums, "Dùng dữ liệu trên máy.
 Firebase: " + e.getMessage()));
        });
    });
  }
  public void persist(Result result) {
    ArrayList<Photo> p = new ArrayList<>(photos); ArrayList<Album> a = new ArrayLis
t<>(albums);
    io.execute(() -> {
      String error = null; android.util.AtomicFile atomic = new android.util.AtomicFile(ca
che()); FileOutputStream stream = null;
      try { stream=atomic.startWrite(); ObjectOutputStream out=new ObjectOutputStream
(stream); out.writeObject(p); out.writeObject(a); out.flush(); atomic.finishWrite(stream); }
      catch (Exception e) { if(stream!=null)atomic.failWrite(stream);error=e.getMessage()
; }
      String finalError = error; main.post(() -> result.done(finalError));
    });
  }
  /** Append an editor export against the latest disk snapshot, preserving concurrent backu
p updates. */
  public void appendLocal(Photo photo,Result result){
    io.execute(()->{
```

```
        ArrayList<Photo> latestPhotos=new ArrayList<>();ArrayList<Album> latestAlbum
s=new ArrayList<>();
        try{
          if(cache().exists())try(ObjectInputStream in=new ObjectInputStream(new FileInp
utStream(cache()))){for(Object item:(List<?>)in.readObject())if(item instanceof Photo)lates
tPhotos.add((Photo)item);for(Object item:(List<?>)in.readObject())if(item instanceof Albu
m)latestAlbums.add((Album)item);}
          latestPhotos.removeIf(existing->Objects.equals(existing.getId(),photo.getId()));lat
estPhotos.add(photo);
          android.util.AtomicFile atomic=new android.util.AtomicFile(cache());FileOutput
Stream stream=atomic.startWrite();
          try{ObjectOutputStream out=new ObjectOutputStream(stream);out.writeObject(la
testPhotos);out.writeObject(latestAlbums);out.flush();atomic.finishWrite(stream);}catch(Ex
ception error){atomic.failWrite(stream);throw error;}
          main.post(()->{photos.clear();photos.addAll(latestPhotos);albums.clear();albums.
addAll(latestAlbums);result.done(null);});
        }catch(Exception error){main.post(()->result.done(error.getLocalizedMessage()));}
      });
    }
    public Photo find(String id) { for (Photo p : photos) if (Objects.equals(p.getId(),id)) retur
n p; return null; }
    private android.content.SharedPreferences hiddenChanges(){return context.getSharedPref
erences("pending-hidden-"+(owner==null?"guest":owner),0);}
    private void applyPendingHidden(Photo photo){android.content.SharedPreferences prefs
=hiddenChanges();if(prefs.contains(photo.getId()))photo.setHidden(prefs.getBoolean(photo.
getId(),false));}
    /** Persist visibility locally first; a pending cloud write must never block hiding. */
    public void setHidden(List<String> ids,boolean hidden,Result result){
      android.content.SharedPreferences.Editor prefs=hiddenChanges().edit();List<Photo> c
hanged=new ArrayList<>();
      for(String id:ids){Photo photo=find(id);if(photo==null||photo.isDeleted())continue;pho
to.setHidden(hidden);prefs.putBoolean(id,hidden);changed.add(photo);}
      prefs.apply();persist(error->{result.done(error);if(error!=null||owner==null)return;for(P
hoto photo:changed)save(photo,syncError->{if(syncError==null&&hiddenChanges().contai
ns(photo.getId())&&hiddenChanges().getBoolean(photo.getId(),!hidden)==hidden)hiddenC
hanges().edit().remove(photo.getId()).apply();});});
    }
    public Album album(String id) { for (Album a : albums) if (Objects.equals(a.getId(),id)) r
eturn a; return null; }
    public void save(Photo p, Result result) {
      p.setUserId(owner); p.setUpdatedAt(new Date());
      if (find(p.getId()) == null) photos.add(p);
      if (owner == null) { persist(result); return; }
      // Device-local paths must not be copied to another device through Firestore.
      Map<String,Object> data = new HashMap<>();
      data.put("fileName",p.getFileName()); data.put("firebaseUrl",p.getFirebaseUrl()); data.
put("userId",owner);
      data.put("fileSize",p.getFileSize()); data.put("width",p.getWidth()); data.put("height",p
.getHeight());
```

```
        data.put("mimeType",p.getMimeType()); data.put("video",p.isVideo()); data.put("favo
rite",p.isFavorite());
        data.put("deleted",p.isDeleted()); data.put("deletedAt",p.getDeletedAt()); data.put("alb
umIds",p.getAlbumIds());
        data.put("hidden",p.isHidden());
        data.put("dateTaken",p.getDateTaken()); data.put("createdAt",p.getCreatedAt()); data.
put("updatedAt",FieldValue.serverTimestamp());
        persist(error -> {
            if (error != null) { result.done(error); return; }
            collection("photos").document(p.getId()).set(data).addOnSuccessListener(v -> result
.done(null)).addOnFailureListener(e -> result.done(e.getMessage()));
        });
    }
    public void saveAlbum(Album a, Result result) {
        a.setUserId(owner); if (a.getId() == null) a.setId(UUID.randomUUID().toString());
        if (album(a.getId()) == null) albums.add(a);
        if (owner == null) { persist(result); return; }
        collection("albums").document(a.getId()).set(a).addOnSuccessListener(v -> {clearPen
dingAlbum(a.getId());persist(result);}).addOnFailureListener(e -> result.done(e.getMessage
()));
    }
    public void deleteAlbum(Album a, Result result) {
        // Membership lives on the photo; an orphan ID is harmless and excluded by the UI.
        if (owner == null) { albums.remove(a); persist(result); return; }
        collection("albums").document(a.getId()).delete().addOnSuccessListener(v -> { clearP
endingAlbum(a.getId());albums.remove(a); persist(result); }).addOnFailureListener(e -> res
ult.done(e.getMessage()));
    }
    private void clearPendingAlbum(String id){android.content.SharedPreferences prefs=con
text.getSharedPreferences("pending-albums-"+owner,0);Set<String> ids=new HashSet<>(p
refs.getStringSet("ids",Collections.emptySet()));ids.remove(id);prefs.edit().putStringSet("id
s",ids).apply();}
    public void upload(Photo p, Result result) {
        if (owner == null) { result.done("Hãy đăng nhập để sao lưu."); return; }
        if (p.getFirebaseUrl() != null && !p.getFirebaseUrl().isEmpty()) { result.done(null); ret
urn; }
        if (p.getFilePath() == null || !new File(p.getFilePath()).exists()) { result.done("Không tì
m thấy file gốc: " + p.getFileName()); return; }
        com.google.firebase.storage.StorageReference ref = FirebaseStorage.getInstance().get
Reference().child("users/"+owner+"/media/"+p.getId());
        com.google.firebase.storage.StorageMetadata metadata = new com.google.firebase.stor
age.StorageMetadata.Builder().setContentType(p.getMimeType()).build();
        ref.putFile(Uri.fromFile(new File(p.getFilePath())),metadata).continueWithTask(task -
> { if (!task.isSuccessful()) throw Objects.requireNonNull(task.getException()); return ref.g
etDownloadUrl(); })
            .addOnSuccessListener(uri -> { p.setFirebaseUrl(uri.toString()); save(p,error -> {
                if(error==null)result.done(null);
                else {p.setFirebaseUrl(null);persist(cacheError->result.done(error));}
            }); }).addOnFailureListener(e -> result.done(e.getMessage()));
```

```
    }
    public void remove(Photo p, Result result) {
        Runnable removeMetadata = () -> {
            Runnable removeLocal = () -> { photos.remove(p); if (p.getFilePath()!=null) { File f
=new File(p.getFilePath()); if (f.getAbsolutePath().startsWith(new File(context.getFilesDir(
),"media").getAbsolutePath()+File.separator)) f.delete(); } persist(result); };
            if (owner == null) removeLocal.run();
            else collection("photos").document(p.getId()).delete().addOnSuccessListener(v -> re
moveLocal.run()).addOnFailureListener(e -> result.done(e.getMessage()));
        };
        if (owner != null && p.getFirebaseUrl()!=null) FirebaseStorage.getInstance().getRefer
ence().child("users/"+owner+"/media/"+p.getId()).delete()
            .addOnSuccessListener(v -> removeMetadata.run()).addOnFailureListener(e -> {
                if (e instanceof com.google.firebase.storage.StorageException && ((com.google.f
irebase.storage.StorageException)e).getErrorCode()==com.google.firebase.storage.Storage
Exception.ERROR_OBJECT_NOT_FOUND) removeMetadata.run(); else result.done(e.get
Message());
            });
        else removeMetadata.run();
    }
    public Object source(Photo p) { return p.getFilePath()!=null && new File(p.getFilePath()
).exists() ? new File(p.getFilePath()) : p.getFirebaseUrl(); }
    public void close() { /* Shared I/O executor also finishes in-flight uploads after Activity c
hanges. */ }
}
```

1 1 12 GoogleSignInActivity

Nguồn: app/src/main/java/com/example/appanh/GoogleSignInActivity.java

Vai trò: Mở luồng chọn tài khoản Google thật, lấy thông tin xác thực và đăng nhập Firebase.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.os.Bundle;
import androidx.activity.result.*;
import androidx.activity.result.contract.ActivityResultContracts;
import com.google.android.gms.auth.api.signin.*;
import com.google.android.gms.common.api.ApiException;
import com.google.firebase.auth.*;
public class GoogleSignInActivity extends BaseActivity {
 private final ActivityResultLauncher<android.content.Intent> launcher=registerForActivity
Result(new ActivityResultContracts.StartActivityForResult(),r->{try{GoogleSignInAccount
 a=GoogleSignIn.getSignedInAccountFromIntent(r.getData()).getResult(ApiException.class
);if(a==null||a.getIdToken()==null){status("Không nhận được token Google.");return;}Fireb
aseAuth.getInstance().signInWithCredential(GoogleAuthProvider.getCredential(a.getIdTok
en(),null)).addOnCompleteListener(this,t->{if(t.isSuccessful()){AuthActivity.saveProfile(Fi
rebaseAuth.getInstance().getCurrentUser(),a.getDisplayName());go(MainActivity.class);fini
```

sh();}else status(t.getException().getLocalizedMessage());});}catch(ApiException e){status("Đăng nhập Google chưa hoàn tất ("+e.getStatusCode()+"). Kiểm tra SHA-1/SHA-256 và google-services.json.");}});
@Override public void onCreate(Bundle state){super.onCreate(state);screen(R.layout.activity_google_sign_in);click(R.id.google,()->{int id=getResources().getIdentifier("default_web_client_id","string",getPackageName());if(id==0){status("Bật Google provider trong Firebase, thêm SHA và tải lại google-services.json.");return;}GoogleSignInOptions options=new GoogleSignInOptions.Builder(GoogleSignInOptions.DEFAULT_SIGN_IN).requestIdToken(getString(id)).requestEmail().build();launcher.launch(GoogleSignIn.getClient(this,options).getSignInIntent());});}
}

1 1 13 HiddenMediaActivity

Nguồn: app/src/main/java/com/example/appanh/HiddenMediaActivity.java

Vai trò: Màn riêng để xem, tìm và lọc ảnh hoặc video đã ẩn; kế thừa thao tác bỏ ẩn và chọn nhiều.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.os.Bundle;
import android.view.View;
import com.google.android.material.appbar.MaterialToolbar;

/** A separate, discoverable collection for hidden photos and videos. */
public class HiddenMediaActivity extends GalleryActivity {
    @Override public void onCreate(Bundle state){
        super.onCreate(state);showHidden=true;
        MaterialToolbar toolbar=findViewById(R.id.toolbar);
        toolbar.setTitle("Ảnh và video đã ẩn");toolbar.setNavigationIcon(R.drawable.ic_arrow_back);
        findViewById(R.id.navigation).setVisibility(View.GONE);
        findViewById(R.id.importMedia).setVisibility(View.GONE);
        View empty=findViewById(R.id.emptyState);
        if(empty instanceof android.view.ViewGroup){android.view.ViewGroup group=(android.view.ViewGroup)empty;((android.widget.TextView)group.getChildAt(1)).setText("Chưa có ảnh hoặc video đã ẩn");((android.widget.TextView)group.getChildAt(2)).setText("Nhấn giữ một mục trong thư viện rồi chọn Ẩn. Nhấn giữ mục tại đây để bỏ ẩn và đưa về thư viện.");}
    }
}
```

1 1 14 LoginActivity

Nguồn: app/src/main/java/com/example/appanh/LoginActivity.java

Vai trò: Điểm vào màn đăng nhập; kế thừa AuthActivity.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class LoginActivity extends AuthActivity {}
```

1 1 15 MainActivity

Nguồn: app/src/main/java/com/example/appanh/MainActivity.java

Vai trò: Điểm vào thư viện Ảnh; kế thừa GalleryActivity.

Mã nguồn đầy đủ

```
package com.example.appanh; public class MainActivity extends GalleryActivity {}
```

1 1 16 MediaAdapter

Nguồn: app/src/main/java/com/example/appanh/MediaAdapter.java

Vai trò: Hiển thị ô ảnh, video, album và danh sách. Nạp hình bằng Glide, đếm thành viên album và hiển thị huy hiệu số theo thứ tự chọn.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.view.*;
import android.widget.*;
import androidx.recyclerview.widget.RecyclerView;
import com.bumptech.glide.Glide;
import com.example.appanh.models.*;
import java.util.*;
public final class MediaAdapter extends RecyclerView.Adapter<MediaAdapter.Holder> {
 public interface Events { void tap(Object item); void hold(Object item); }
 private final List<?> items; private final GalleryRepository repo; private final Set<String>
selected; private final Events events; private final boolean rows;
 private final Map<String,Integer> selectionOrder=new LinkedHashMap<>();
 public MediaAdapter(List<?> items,GalleryRepository repo,Set<String> selected,Events ev
ents,boolean rows){this.items=items;this.repo=repo;this.selected=selected;this.events=event
s;this.rows=rows;int number=1;for(String id:selected)selectionOrder.put(id,number++);}
 @Override public Holder onCreateViewHolder(ViewGroup parent,int type){return new Ho
lder(LayoutInflater.from(parent.getContext()).inflate(type==1?R.layout.item_album:type==
2?R.layout.item_media_row:R.layout.item_media,parent,false));}
 @Override public void onBindViewHolder(Holder h,int position){Object item=items.get(p
osition);Object image=null;String caption,id;
  if(item instanceof Photo){Photo p=(Photo)item;image=repo.source(p);id=p.getId();caption
```

```
=(p.isVideo()?"▶ Video":"")+(p.isFavorite()?" ♥":"");if(rows)caption=p.getFileName()+"\
n"+com.example.appanh.helpers.PhotoHelper.formatFileSize(p.getFileSize());h.image.setC
ontentDescription(p.getFileName());}
  else{Album a=(Album)item;id=a.getId();if(a.getCoverPhotoPath()!=null&&new java.io.Fi
le(a.getCoverPhotoPath()).exists())image=new java.io.File(a.getCoverPhotoPath());else if(a.
getCoverPhotoUrl()!=null&&!a.getCoverPhotoUrl().isEmpty())image=a.getCoverPhotoUrl(
);boolean hiddenCover=false;for(Photo p:repo.photos)if(p.isHidden()&&p.getAlbumIds().c
ontains(a.getId()))hiddenCover=true;if(hiddenCover)image=null;int images=0,videos=0;for(
Photo p:repo.photos)if(!p.isDeleted()&&!p.isHidden()&&p.getAlbumIds().contains(a.getId(
))){if(image==null)image=repo.source(p);if(p.isVideo())videos++;else images++;}caption=
a.getName()+"\n"+(images==0&&videos>0?videos+" video":videos==0?images+" ảnh":im
ages+" ảnh · "+videos+" video");View badge=h.itemView.findViewById(R.id.albumVideo
Badge);if(badge!=null)badge.setVisibility(videos>0?View.VISIBLE:View.GONE);View m
ore=h.itemView.findViewById(R.id.albumMore);if(more!=null){more.setContentDescripti
on("Tùy chọn album "+a.getName());more.setOnClickListener(view->events.hold(item));}}
  Glide.with(h.image).load(image).placeholder(R.drawable.art_landscape).error(R.drawable.
ic_photo).centerCrop().into(h.image);h.caption.setText(caption);if(item instanceof Album){
android.text.SpannableString styled=new android.text.SpannableString(caption);int newline
=caption.indexOf('\n');styled.setSpan(new android.text.style.StyleSpan(android.graphics.Ty
peface.BOLD),0,newline,0);styled.setSpan(new android.text.style.ForegroundColorSpan(h.i
temView.getContext().getColor(R.color.text_secondary)),newline+1,caption.length(),0);styl
ed.setSpan(new android.text.style.RelativeSizeSpan(.85f),newline+1,caption.length(),0);h.c
aption.setText(styled);}h.caption.setVisibility(caption.trim().isEmpty()?View.GONE:View.
VISIBLE);boolean chosen=selected.contains(id);h.check.setVisibility(chosen?View.VISIB
LE:View.GONE);if(h.check instanceof TextView){((TextView)h.check).setText(chosen?St
ring.valueOf(selectionOrder.get(id)):"");h.check.setContentDescription(chosen?"Đã chọn th
ứ "+selectionOrder.get(id):null);}if(item instanceof Photo&&h.itemView instanceof com.go
ogle.android.material.card.MaterialCardView){com.google.android.material.card.MaterialC
ardView card=(com.google.android.material.card.MaterialCardView)h.itemView;card.setSt
rokeWidth(chosen?(int)(2*card.getResources().getDisplayMetrics().density):0);card.setStro
keColor(card.getContext().getColor(R.color.button_primary));}h.itemView.setOnClickListe
ner(v->events.tap(item));h.itemView.setOnLongClickListener(v->{events.hold(item);return
true;});
 }
 @Override public int getItemViewType(int position){return items.get(position) instanceof
Album?1:rows?2:0;}
 @Override public int getItemCount(){return items.size();}
 static final class Holder extends RecyclerView.ViewHolder {final ImageView image;final
TextView caption;final View check;Holder(View view){super(view);image=view.findView
ById(R.id.thumbnail);caption=view.findViewById(R.id.caption);check=view.findViewByI
d(R.id.check);}}
}
```

1 1 17 NotificationsActivity

Nguồn: app/src/main/java/com/example/appanh/NotificationsActivity.java

Vai trò: Lưu và đọc lịch sử hoạt động cục bộ và Firestore; hỗ trợ đánh dấu đã đọc. Không phải thông báo đẩy FCM.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.content.*;
import android.os.Bundle;
import android.view.*;
import android.widget.*;
import androidx.recyclerview.widget.*;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.firestore.*;
import org.json.*;
import java.util.*;
public class NotificationsActivity extends BaseActivity {
 private final List<String> messages=new ArrayList<>();
 private static String key(){return "notifications-"+FirebaseAuth.getInstance().getUid();}
 public static void record(Context context,String message){String uid=FirebaseAuth.getInst
ance().getUid();SharedPreferences prefs=context.getSharedPreferences(key(),0);try{JSONA
rray old=new JSONArray(prefs.getString("items","[]")),all=new JSONArray();all.put(messa
ge+"\n"+new java.text.SimpleDateFormat("dd/MM/yyyy HH:mm",new Locale("vi","VN"))
.format(new Date()));for(int i=0;i<Math.min(old.length(),99);i++)all.put(old.get(i));prefs.ed
it().putString("items",all.toString()).apply();}catch(JSONException ignored){}if(uid!=null)
{Map<String,Object> data=new HashMap<>();data.put("message",message);data.put("creat
edAt",FieldValue.serverTimestamp());data.put("read",false);FirebaseFirestore.getInstance().
collection("users").document(uid).collection("notifications").add(data);}}
 @Override public void onCreate(Bundle state){super.onCreate(state);screen(R.layout.activi
ty_notifications);try{JSONArray items=new JSONArray(getSharedPreferences(key(),0).get
String("items","[]"));for(int i=0;i<items.length();i++)messages.add(items.getString(i));}catc
h(JSONException e){status("Không đọc được lịch sử trên máy.");}render();if(repo.uid()!=n
ull)FirebaseFirestore.getInstance().collection("users").document(repo.uid()).collection("noti
fications").orderBy("createdAt",Query.Direction.DESCENDING).limit(100).get().addOnSu
ccessListener(s->{messages.clear();for(DocumentSnapshot d:s)messages.add(String.valueO
f(d.getString("message")));render();}).addOnFailureListener(e->status(e.getLocalizedMessa
ge()));click(R.id.markRead,()->{if(repo.uid()==null){status("Đã đọc tất cả thông báo trên m
áy.");return;}FirebaseFirestore db=FirebaseFirestore.getInstance();db.collection("users").do
cument(repo.uid()).collection("notifications").limit(100).get().addOnSuccessListener(s->{
WriteBatch batch=db.batch();for(DocumentSnapshot d:s)batch.update(d.getReference(),"rea
d",true);batch.commit().addOnSuccessListener(v->status("Đã đọc tất cả thông báo.")).addO
nFailureListener(e->status(e.getLocalizedMessage()));});});}
 private void render(){status(messages.isEmpty()?"Chưa có thông báo.":messages.size()+" h
oạt động gần đây");RecyclerView list=findViewById(R.id.grid);list.setLayoutManager(new
 LinearLayoutManager(this));list.setAdapter(new RecyclerView.Adapter<Row>(){public R
ow onCreateViewHolder(ViewGroup parent,int type){return new Row(LayoutInflater.from(
parent.getContext()).inflate(R.layout.item_notification,parent,false));}public void onBindVi
ewHolder(Row row,int index){((TextView)row.itemView.findViewById(R.id.notification
Message)).setText(messages.get(index));}public int getItemCount(){return messages.size();
}});}
 static class Row extends RecyclerView.ViewHolder {Row(View v){super(v);}}
```

}

1 1 18 OtpActivity

Nguồn: app/src/main/java/com/example/appanh/OtpActivity.java

Vai trò: Xử lý các ô mã xác minh SMS và bước kiểm tra liên kết xác minh email; cho phép gửi lại theo luồng tương ứng.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.os.Bundle;
import android.view.View;
import com.google.firebase.auth.*;
public class OtpActivity extends BaseActivity {
 private String verificationId; private long resendAt;
 @Override public void onCreate(Bundle state){
  super.onCreate(state);screen(R.layout.activity_otp);verificationId=getIntent().getStringExt
ra("verificationId");
  if(verificationId==null){
   findViewById(R.id.otpBoxes).setVisibility(View.GONE);
   ((android.widget.Button)findViewById(R.id.verify)).setText("Tôi đã xác minh email");
   status("Liên kết xác minh đã được gửi đến email của bạn. Mở email, nhấn xác minh rồi qu
ay lại đây.");
  }else{
   status("Nhập mã xác thực đã được gửi đến\n"+getIntent().getStringExtra("phone"));
   connectCodeBoxes();
  }
  click(R.id.verify,this::verify);click(R.id.resend,this::resend);
 }
 private boolean changingCode;
 private void connectCodeBoxes(){
  int[] ids={R.id.otp1,R.id.otp2,R.id.otp3,R.id.otp4,R.id.otp5,R.id.otp6};
  android.widget.EditText[] boxes=new android.widget.EditText[6];
  for(int i=0;i<6;i++)boxes[i]=findViewById(ids[i]);
  for(int i=0;i<6;i++){
   final int index=i;
   boxes[i].addTextChangedListener(new android.text.TextWatcher(){
    public void beforeTextChanged(CharSequence s,int start,int count,int after){}
    public void onTextChanged(CharSequence s,int start,int before,int count){}
    public void afterTextChanged(android.text.Editable value){
     if(changingCode)return;changingCode=true;String digits=value.toString();
     if(digits.length()>1){for(int j=0;j<6;j++)boxes[j].setText(j<digits.length()?String.valueO
f(digits.charAt(j)):"");boxes[Math.min(5,digits.length()-1)].requestFocus();}
     else if(!digits.isEmpty()&&index<5)boxes[index+1].requestFocus();
     StringBuilder code=new StringBuilder();for(android.widget.EditText box:boxes)code.ap
pend(box.getText());
     ((android.widget.EditText)findViewById(R.id.otp)).setText(code);changingCode=false;
```

```
    }
   });
   boxes[i].setOnKeyListener((v,key,event)->{if(key==android.view.KeyEvent.KEYCODE
_DEL&&event.getAction()==android.view.KeyEvent.ACTION_DOWN&&boxes[index].g
etText().length()==0&&index>0){boxes[index-1].setText("");boxes[index-1].requestFocus(
);return true;}return false;});
  }
 }
 private void verify(){if(verificationId!=null){String code=input(R.id.otp);if(!code.matches(
"[0-9]{6}")){toast("Nhập đủ 6 chữ số.");return;}FirebaseAuth.getInstance().signInWithCred
ential(PhoneAuthProvider.getCredential(verificationId,code)).addOnCompleteListener(this,
t->{if(t.isSuccessful())enter();else status(t.getException().getLocalizedMessage());});return;
}FirebaseUser u=FirebaseAuth.getInstance().getCurrentUser();if(u==null){go(LoginActivit
y.class);finish();return;}u.reload().addOnCompleteListener(this,t->{if(t.isSuccessful()&&u.i
sEmailVerified())enter();else status("Email chưa được xác minh. Kiểm tra cả thư mục Spam
.");});}
 private void enter(){AuthActivity.saveProfile(FirebaseAuth.getInstance().getCurrentUser(),
null);go(MainActivity.class);finish();}
 private void resend(){if(System.currentTimeMillis()<resendAt){toast("Vui lòng chờ 60 giâ
y giữa các lần gửi.");return;}if(verificationId!=null){toast("Quay lại đăng nhập và nhập lại s
ố điện thoại để gửi SMS mới.");return;}FirebaseUser u=FirebaseAuth.getInstance().getCurr
entUser();if(u==null)return;resendAt=System.currentTimeMillis()+60000;u.sendEmailVerif
ication().addOnCompleteListener(this,t->status(t.isSuccessful()?"Đã gửi lại email xác minh.
":t.getException().getLocalizedMessage()));}
}
```

1 1 19 PhotoDetailActivity

Nguồn: app/src/main/java/com/example/appanh/PhotoDetailActivity.java

Vai trò: Hiển thị ảnh, vuốt giữa các mục cùng trạng thái ẩn, yêu thích, thông tin, sửa, chia sẻ, ẩn và thùng rác; mở video bằng trình phát hệ thống.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.content.*;
import android.os.Bundle;
import android.view.*;
import android.widget.*;
import com.bumptech.glide.Glide;
import com.example.appanh.models.Photo;
import com.example.appanh.helpers.PhotoHelper;
import java.util.*;
public class PhotoDetailActivity extends BaseActivity {
 private Photo photo; private float downX;
 @Override public void onCreate(Bundle state){super.onCreate(state);screen(R.layout.activi
ty_photo_detail);click(R.id.favorite,()->{if(photo==null)return;photo.setFavorite(!photo.isF
avorite());repo.save(photo,error->{result(error);display();});});click(R.id.edit,()->{if(photo!
```

```
=null)startActivity(new Intent(this,EditPhotoActivity.class).putExtra("photoId",photo.getId(
)));});click(R.id.share,()->{if(photo!=null)startActivity(new Intent(this,ShareActivity.class).
putExtra("photoId",photo.getId()));});click(R.id.more,this::more);click(R.id.deletePhoto,this
::trash);
  findViewById(R.id.photo).setOnTouchListener((v,event)->{if(event.getAction()==Motion
Event.ACTION_DOWN){downX=event.getX();return true;}if(event.getAction()==Motion
Event.ACTION_UP){float dx=event.getX()-downX;if(Math.abs(dx)>70)next(dx<0?1:-1);el
se if(photo!=null&&photo.isVideo())play();v.performClick();return true;}return true;});
 }
 @Override protected void onResume(){super.onResume();repo.load((p,a,error)->{photo=re
po.find(getIntent().getStringExtra("photoId"));if(photo==null){toast("Không tìm thấy ảnh.")
;finish();}else display();if(error!=null)toast(error);});}
 private void display(){if(photo==null)return;Glide.with(this).load(repo.source(photo)).error
(R.drawable.ic_photo).into((ImageView)findViewById(R.id.photo));status(photo.isVideo()
?"Chạm để phát video":"");com.google.android.material.button.MaterialButton favorite=fin
dViewById(R.id.favorite);favorite.setText(photo.isFavorite()?"Đã yêu thích":"Yêu thích");f
avorite.setIconResource(photo.isFavorite()?R.drawable.ic_favorite_filled:R.drawable.ic_fav
orite_outline);}
 private void next(int delta){List<Photo> active=new ArrayList<>();for(Photo p:repo.photo
s)if(!p.isDeleted()&&p.isHidden()==photo.isHidden())active.add(p);for(int i=0;i<active.size
();i++)if(active.get(i).getId().equals(photo.getId())){int index=i+delta;if(index>=0&&index
<active.size()){photo=active.get(index);getIntent().putExtra("photoId",photo.getId());displa
y();}return;}}
 private void play(){ShareActivity.openFile(this,repo,photo,false);}
 private void trash(){if(photo==null)return;confirm("Chuyển vào thùng rác?",()->{photo.set
Deleted(true);photo.setDeletedAt(new Date());repo.save(photo,error->{if(error==null)finish
();else result(error);});});}
 private void more(){if(photo==null)return;new androidx.appcompat.app.AlertDialog.Build
er(this).setTitle(photo.getFileName()).setItems(new String[]{"Thông tin ảnh",photo.isHidde
n()?"Bỏ ẩn":"Ẩn ảnh/video","Chuyển vào thùng rác"},(d,i)->{if(i==0)info(this,repo,photo);
else if(i==1)repo.setHidden(Collections.singletonList(photo.getId()),!photo.isHidden(),error
->{if(error==null)finish();else result(error);});else trash();}).show();}
 public static void info(android.app.Activity context,GalleryRepository repo,Photo p){View
view=context.getLayoutInflater().inflate(R.layout.dialog_photo_info,null);Glide.with(conte
xt).load(repo.source(p)).into((ImageView)view.findViewById(R.id.photo));((TextView)vie
w.findViewById(R.id.infoName)).setText(p.getFileName());((TextView)view.findViewByI
d(R.id.infoSize)).setText(PhotoHelper.formatFileSize(p.getFileSize())+" • "+p.getWidth()+"
× "+p.getHeight());((TextView)view.findViewById(R.id.infoDate)).setText(PhotoHelper.f
ormatDateTime(p.getDateTaken()));((TextView)view.findViewById(R.id.infoPath)).setTex
t(p.getFilePath()==null?"Bản sao lưu trên đám mây":"Bộ nhớ riêng của ứng dụng");((TextV
iew)view.findViewById(R.id.infoDevice)).setText(p.getDeviceName()==null?"Không có th
ông tin":p.getDeviceName());((TextView)view.findViewById(R.id.infoType)).setText(p.get
MimeType());androidx.appcompat.app.AlertDialog dialog=new androidx.appcompat.app.Al
ertDialog.Builder(context).setView(view).create();view.findViewById(R.id.close).setOnCli
ckListener(v->dialog.dismiss());view.findViewById(R.id.toolbar).setOnClickListener(v->di
alog.dismiss());dialog.show();if(dialog.getWindow()!=null)dialog.getWindow().setLayout(-
1,(int)(context.getResources().getDisplayMetrics().heightPixels*.85));}
}
```

1 1 20 RegisterActivity

Nguồn: app/src/main/java/com/example/appanh/RegisterActivity.java

Vai trò: Điểm vào màn đăng ký; kế thừa AuthActivity.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class RegisterActivity extends AuthActivity {}
```

1 1 21 SearchActivity

Nguồn: app/src/main/java/com/example/appanh/SearchActivity.java

Vai trò: Màn tìm kiếm dùng chung xử lý lọc và sắp xếp từ GalleryActivity.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class SearchActivity extends GalleryActivity {}
```

1 1 22 SettingsActivity

Nguồn: app/src/main/java/com/example/appanh/SettingsActivity.java

Vai trò: Màn Hồ sơ và Cài đặt; mở tài khoản, giao diện, dung lượng, thông báo, các mục đã ẩn, thùng rác và đăng xuất.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatDelegate;
import com.example.appanh.helpers.PhotoHelper;
import com.example.appanh.models.Photo;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;
import org.json.JSONArray;

public class SettingsActivity extends BaseActivity {
   @Override public void onCreate(Bundle state) {
      super.onCreate(state);
      screen(R.layout.activity_settings);
      click(R.id.account,()->go(AccountActivity.class));
```

```
        click(R.id.backup,()->go(BackupActivity.class));
        click(R.id.notifications,()->go(NotificationsActivity.class));
        click(R.id.trash,()->go(TrashActivity.class));
        click(R.id.hiddenMedia,()->go(HiddenMediaActivity.class));
        click(R.id.storage,this::showStorage);
        click(R.id.theme,this::chooseTheme);
        click(R.id.language,()->new AlertDialog.Builder(this).setTitle("Ngôn ngữ")
                .setSingleChoiceItems(new String[]{"Tiếng Việt"},0,(dialog,which)->dialog.dism
iss())
                .setNegativeButton("Đóng",null).show());
        click(R.id.about,()->new AlertDialog.Builder(this).setTitle("Ảnh 1.0")
                .setMessage("Lưu giữ khoảnh khắc đẹp\n\nQuản lý ảnh và video, tạo album và ch
ỉnh sửa những kỷ niệm của bạn.")
                .setPositiveButton("Đóng",null).show());
        click(R.id.help,()->new AlertDialog.Builder(this).setTitle("Trợ giúp")
                .setMessage("1. Tab Ảnh → dấu + để thêm ảnh/video.\n2. Nhấn giữ ảnh/video →
Ẩn để đưa vào mục đã ẩn.\n3. Hồ sơ → Ảnh và video đã ẩn → nhấn giữ → Bỏ ẩn.\n4. Tạo a
lbum ở tab Album.\n5. Hồ sơ → Tài khoản để sửa thông tin.\n6. Thùng rác cho phép khôi p
hục ảnh đã xóa.\n\nDữ liệu khách và từng tài khoản được lưu riêng. Sao lưu chỉ minh họa gi
ao diện. Ẩn là bộ lọc trong app, chưa có khóa mật khẩu.")
                .setPositiveButton("Đã hiểu",null).show());
        click(R.id.logout,()->{
            if(FirebaseAuth.getInstance().getCurrentUser()==null){go(LoginActivity.class);retu
rn;}
            confirm("Đăng xuất tài khoản?",()->{
                FirebaseAuth.getInstance().signOut();
                startActivity(new Intent(this,LoginActivity.class).addFlags(Intent.FLAG_ACTIV
ITY_NEW_TASK|Intent.FLAG_ACTIVITY_CLEAR_TASK));
                finish();
            });
        });
    }

    @Override protected void onResume() {
        super.onResume();
        FirebaseUser user=FirebaseAuth.getInstance().getCurrentUser();
        detail(R.id.accountDetail,user==null?"Chế độ khách · Đăng nhập":user.getEmail()==n
ull?user.getPhoneNumber():user.getEmail());
        detail(R.id.themeDetail,getSharedPreferences("settings",0).getBoolean("dark",false)?"
Tối":"Sáng");
        repo.load((photos,albums,error)->{if(!isDestroyed())summary();},()->{if(!isDestroyed(
))summary();});
    }

    private void summary() {
        long bytes=0;int active=0,trash=0,synced=0;
        for(Photo photo:repo.photos){bytes+=photo.getFileSize();if(photo.isDeleted())trash++;
else{active++;if(photo.getFirebaseUrl()!=null&&!photo.getFirebaseUrl().isEmpty())synced
++;}}
```

```
        detail(R.id.storageDetail,PhotoHelper.formatFileSize(bytes)+" · "+repo.photos.size()+"
 mục");
        detail(R.id.trashDetail,trash==0?"Không có ảnh đã xóa":trash+" mục · Có thể khôi phụ
c");
        int hidden=0;for(Photo photo:repo.photos)if(!photo.isDeleted()&&photo.isHidden())hi
dden++;
        detail(R.id.hiddenMediaDetail,hidden==0?"Quản lý các mục đã ẩn":hidden+" mục · N
hấn để xem và bỏ ẩn");
        boolean auto=getSharedPreferences("backup-"+repo.uid(),0).getBoolean("auto",false);
        detail(R.id.backupDetail,"Giao diện minh họa · Chưa kích hoạt");
        try {
            int count=new JSONArray(getSharedPreferences("notifications-"+repo.uid(),0).getS
tring("items","[]")).length();
            detail(R.id.notificationsDetail,count==0?"Chưa có hoạt động mới":count+" hoạt độn
g gần đây");
        }catch(org.json.JSONException ignored){detail(R.id.notificationsDetail,"Lịch sử hoạt
động");}
    }

    private void showStorage() {
        long bytes=0;int images=0,videos=0,trash=0;
        for(Photo photo:repo.photos){bytes+=photo.getFileSize();if(photo.isDeleted())trash++;
else if(photo.isVideo())videos++;else images++;}
        new AlertDialog.Builder(this).setTitle("Dung lượng thư viện")
                .setMessage(PhotoHelper.formatFileSize(bytes)+"\n\n"+images+" ảnh · "+videos
+" video\n"+trash+" mục trong thùng rác\n\nDung lượng tính theo các tệp trong thư viện củ
a tài khoản hiện tại.")
                .setPositiveButton("Đóng",null).setNeutralButton("Mở thùng rác",(dialog,which)-
>go(TrashActivity.class)).show();
    }

    private void chooseTheme() {
        boolean dark=getSharedPreferences("settings",0).getBoolean("dark",false);
        new AlertDialog.Builder(this).setTitle("Giao diện")
                .setSingleChoiceItems(new String[]{"Sáng","Tối"},dark?1:0,(dialog,index)->{
                    dialog.dismiss();boolean next=index==1;if(next==dark)return;
                    ThemeManager.setDark(this,next);
                }).setNegativeButton("Đóng",null).show();
    }

    private void detail(int id,String value){((TextView)findViewById(id)).setText(value);}
}
```

1 1 23 ShareActivity

Nguồn: app/src/main/java/com/example/appanh/ShareActivity.java

Vai trò: Chia sẻ file bằng FileProvider và bảng chia sẻ Android. Liên kết chỉ dùng được khi mục đã có URL cloud hợp lệ.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.content.*;
import android.net.Uri;
import android.os.Bundle;
import androidx.core.content.FileProvider;
import com.example.appanh.models.Photo;
import java.io.File;
public class ShareActivity extends BaseActivity {
 private Photo photo;
 @Override public void onCreate(Bundle state){super.onCreate(state);screen(R.layout.sheet
_share);repo.load((p,a,error)->{photo=repo.find(getIntent().getStringExtra("photoId"));resul
t(error);});click(R.id.cancel,this::finish);click(R.id.shareFile,()->{if(photo!=null)openFile(th
is,repo,photo,true);});click(R.id.shareLink,()->{if(photo==null)return;if(photo.getFirebaseU
rl()==null){toast("Sao lưu ảnh trước khi chia sẻ liên kết.");return;}new androidx.appcompat.
app.AlertDialog.Builder(this).setTitle("Chia sẻ liên kết?").setMessage("Bất kỳ ai có liên kết
 đều có thể tải ảnh này.").setNegativeButton("Hủy",null).setPositiveButton("Chia sẻ",(d,w)-
>startActivity(Intent.createChooser(new Intent(Intent.ACTION_SEND).setType("text/plain
").putExtra(Intent.EXTRA_TEXT,photo.getFirebaseUrl()),"Chia sẻ liên kết"))).show();});}
 public static void openFile(android.app.Activity context,GalleryRepository repo,Photo phot
o,boolean share){Object source=repo.source(photo);if(source==null){android.widget.Toast.
makeText(context,"Không tìm thấy file.",android.widget.Toast.LENGTH_LONG).show();r
eturn;}if(share&&!(source instanceof File)){android.widget.Toast.makeText(context,"Đang
 tải file để chia sẻ…",android.widget.Toast.LENGTH_SHORT).show();File dir=new File(c
ontext.getCacheDir(),"shared");if(!dir.exists()&&!dir.mkdirs()){android.widget.Toast.make
Text(context,"Không tạo được file chia sẻ.",android.widget.Toast.LENGTH_LONG).show(
);return;}File target=new File(dir,photo.getId());com.google.firebase.storage.FirebaseStorag
e.getInstance().getReferenceFromUrl(source.toString()).getFile(target).addOnSuccessListen
er(snapshot->{if(!context.isDestroyed())send(context,target,photo.getMimeType(),true);}).a
ddOnFailureListener(e->android.widget.Toast.makeText(context,e.getLocalizedMessage(),a
ndroid.widget.Toast.LENGTH_LONG).show());return;}send(context,source,photo.getMime
Type(),share);}
 private static void send(android.app.Activity context,Object source,String mime,boolean sh
are){Uri uri=source instanceof File?FileProvider.getUriForFile(context,context.getPackage
Name()+".files",(File)source):Uri.parse(source.toString());Intent intent=new Intent(share?In
tent.ACTION_SEND:Intent.ACTION_VIEW).addFlags(Intent.FLAG_GRANT_READ_U
RI_PERMISSION);if(share){intent.setType(mime);intent.putExtra(Intent.EXTRA_STREA
M,uri);intent.setClipData(ClipData.newRawUri("Ảnh",uri));}else intent.setDataAndType(ur
i,mime);try{context.startActivity(Intent.createChooser(intent,share?"Chia sẻ ảnh":"Phát vid
eo"));}catch(ActivityNotFoundException e){android.widget.Toast.makeText(context,"Chưa
 có ứng dụng hỗ trợ định dạng này.",android.widget.Toast.LENGTH_LONG).show();}}
}
```

1 1 24 SplashActivity

Nguồn: app/src/main/java/com/example/appanh/SplashActivity.java

Vai trò: Màn chào; đưa người dùng vào đăng nhập hoặc thư viện tùy trạng thái tài khoản. Có lựa chọn khám phá trên thiết bị.

Mã nguồn đầy đủ

```
package com.example.appanh;
import android.os.Bundle;
import com.google.firebase.auth.FirebaseAuth;
public class SplashActivity extends BaseActivity {
 @Override public void onCreate(Bundle state) {
  super.onCreate(state);
  screen(R.layout.activity_splash);
  ((com.google.android.material.appbar.MaterialToolbar)findViewById(R.id.toolbar)).setNa
vigationIcon(null);
  click(R.id.start, () -> ready(() -> {
   com.google.firebase.auth.FirebaseUser user = FirebaseAuth.getInstance().getCurrentUser(
);
   go(user == null ? LoginActivity.class : user.getEmail() != null && !user.isEmailVerified()
 ? OtpActivity.class : MainActivity.class);
   finish();
  }));
  click(R.id.guest, () -> ready(() -> {
   FirebaseAuth.getInstance().signOut();
   startActivity(new android.content.Intent(this,MainActivity.class).addFlags(android.conten
t.Intent.FLAG_ACTIVITY_NEW_TASK|android.content.Intent.FLAG_ACTIVITY_CLEA
R_TASK));
   finish();
  }));
 }
 private void ready(Runnable action) {
  findViewById(R.id.start).setEnabled(false);
  findViewById(R.id.guest).setEnabled(false);
  ((GalleryApplication)getApplication()).whenReady(() -> {
   if(isFinishing() || isDestroyed()) return;
   findViewById(R.id.start).setEnabled(true);
   findViewById(R.id.guest).setEnabled(true);
   RuntimeException error = ((GalleryApplication)getApplication()).getInitializationError();
   if(error != null) { toast("Không khởi tạo được Firebase. Kiểm tra google-services.json."); r
eturn; }
   action.run();
  });
 }
}
```

1 1 25 ThemeManager

Nguồn: app/src/main/java/com/example/appanh/ThemeManager.java

Vai trò: Lưu lựa chọn sáng tối và áp dụng AppCompatDelegate cho toàn bộ ứng dụng.

Mã nguồn đầy đủ

```
package com.example.appanh;

import android.content.Context;
import androidx.appcompat.app.AppCompatDelegate;

/** One persisted theme for every Activity, including Activities in the back stack. */
public final class ThemeManager {
   private ThemeManager() {}
   public static boolean isDark(Context context) {
      return context.getSharedPreferences("settings",0).getBoolean("dark",false);
   }
   public static void initialize(Context context) {
      AppCompatDelegate.setDefaultNightMode(isDark(context)
            ? AppCompatDelegate.MODE_NIGHT_YES : AppCompatDelegate.MODE_NIG
HT_NO);
   }
   public static void setDark(Context context,boolean dark) {
      context.getSharedPreferences("settings",0).edit().putBoolean("dark",dark).apply();
      initialize(context);
   }
}
```

1 1 26 TrashActivity

Nguồn: app/src/main/java/com/example/appanh/TrashActivity.java

Vai trò: Màn thùng rác; kế thừa xử lý khôi phục, xóa vĩnh viễn và dọn mục quá hạn từ GalleryActivity.

Mã nguồn đầy đủ

```
package com.example.appanh;
public class TrashActivity extends GalleryActivity {}
```

1 2 Thư mục editor

Đường dẫn: app/src/main/java/com/example/appanh/editor

Bộ xử lý chỉnh sửa ảnh, vùng tương tác và trạng thái giữ qua vòng đời Activity. Số file Java: 3.

Danh sách: EditorCanvasView.java, EditorDocument.java, EditorViewModel.java.

1 2 1 EditorCanvasView

Nguồn: app/src/main/java/com/example/appanh/editor/EditorCanvasView.java

Vai trò: Vẽ ảnh xem trước, khung cắt và lưới một phần ba; nhận kéo góc, di chuyển khung và nét vẽ theo tọa độ ảnh.

Mã nguồn đầy đủ

```
package com.example.appanh.editor;

import android.content.Context;
import android.graphics.*;
import android.util.AttributeSet;
import android.view.MotionEvent;
import android.view.View;
import java.util.*;

/** Image-space crop handles and vector strokes: no bitmap allocation while dragging. */
public final class EditorCanvasView extends View {
    public enum Mode { VIEW, CROP, DRAW }
    private EditorDocument document;
    private EditorDocument.State paintedState;
    private Mode mode=Mode.VIEW;
    private final RectF imageBounds=new RectF(),crop=new RectF(),dragStart=new RectF()
;
    private final Paint imagePaint=new Paint(Paint.ANTI_ALIAS_FLAG|Paint.FILTER_BIT
MAP_FLAG),paint=new Paint(Paint.ANTI_ALIAS_FLAG);
    private float scale=1,ratio,downX,downY,brushDp=6;
    private int handle,color=Color.WHITE;
    private final List<PointF> points=new ArrayList<>();
    private final Path livePath=new Path();
    private Runnable changed;
    private final float density;
    public EditorCanvasView(Context context,AttributeSet attrs){super(context,attrs);density
=getResources().getDisplayMetrics().density;setContentDescription("Vùng chỉnh sửa ảnh.
Chọn Cắt để kéo khung, hoặc Vẽ để thêm nét bút.");}
    public void bind(EditorDocument value){boolean newImage=paintedState==null||value!=
null&&paintedState.image!=value.state().image;document=value;if(newImage)resetCrop();i
nvalidate();}
    public void setOnChanged(Runnable callback){changed=callback;}
    public void setMode(Mode value){mode=value;points.clear();livePath.reset();if(value==
Mode.CROP)resetCrop();invalidate();}
    public Mode mode(){return mode;}
    public void setBrush(int value,float widthDp){color=value;brushDp=widthDp;}
    public void setAspect(float value){ratio=value;resetCrop();invalidate();}
    public Rect cropPixels(){return new Rect(Math.round(crop.left),Math.round(crop.top),M
ath.round(crop.right),Math.round(crop.bottom));}
    public void restoreCrop(Rect rect){if(document==null||rect==null)return;Bitmap image=
document.state().image;crop.set(Math.max(0,rect.left),Math.max(0,rect.top),Math.min(imag
e.getWidth(),rect.right),Math.min(image.getHeight(),rect.bottom));if(crop.width()<1||crop.h
```

```
eight()<1)resetCrop();invalidate();}
   public RectF imageBounds(){fit();return new RectF(imageBounds);}
   public void resetCrop(){if(document==null)return;Bitmap image=document.state().image
;float width=image.getWidth(),height=image.getHeight();if(ratio>0){if(width/height>ratio)
width=height*ratio;else height=width/ratio;}crop.set((image.getWidth()-width)/2,(image.ge
tHeight()-height)/2,(image.getWidth()+width)/2,(image.getHeight()+height)/2);}
   private void fit(){if(document==null)return;Bitmap image=document.state().image;float a
reaWidth=Math.max(1,getWidth()-getPaddingLeft()-getPaddingRight()),areaHeight=Math.
max(1,getHeight()-getPaddingTop()-getPaddingBottom());scale=Math.min(areaWidth/imag
e.getWidth(),areaHeight/image.getHeight());float width=image.getWidth()*scale,height=im
age.getHeight()*scale;float left=getPaddingLeft()+(areaWidth-width)/2,top=getPaddingTop
()+(areaHeight-height)/2;imageBounds.set(left,top,left+width,top+height);}
   @Override protected void onDraw(Canvas canvas){
      super.onDraw(canvas);if(document==null)return;fit();EditorDocument.State state=doc
ument.state();
      if(state!=paintedState){imagePaint.setColorFilter(EditorDocument.effect(state));painte
dState=state;}
      int saved=canvas.save();canvas.translate(imageBounds.left,imageBounds.top);canvas.s
cale(scale,scale);canvas.clipRect(0,0,state.image.getWidth(),state.image.getHeight());canvas
.drawBitmap(state.image,0,0,imagePaint);
      for(EditorDocument.Stroke stroke:state.strokes)EditorDocument.drawStroke(canvas,st
roke,paint);
      if(!points.isEmpty()){strokePaint(color,brushDp*density/scale);if(points.size()==1){pa
int.setStyle(Paint.Style.FILL);PointF point=points.get(0);canvas.drawCircle(point.x,point.y,
paint.getStrokeWidth()/2,paint);}else canvas.drawPath(livePath,paint);}
      canvas.restoreToCount(saved);
      if(mode==Mode.CROP){
         RectF box=new RectF(imageBounds.left+crop.left*scale,imageBounds.top+crop.top
*scale,imageBounds.left+crop.right*scale,imageBounds.top+crop.bottom*scale);
         paint.reset();paint.setColor(0x99000000);canvas.drawRect(imageBounds.left,image
Bounds.top,imageBounds.right,box.top,paint);canvas.drawRect(imageBounds.left,box.botto
m,imageBounds.right,imageBounds.bottom,paint);canvas.drawRect(imageBounds.left,box.t
op,box.left,box.bottom,paint);canvas.drawRect(box.right,box.top,imageBounds.right,box.bo
ttom,paint);
         paint.setAntiAlias(true);paint.setStyle(Paint.Style.STROKE);paint.setStrokeWidth(2
*density);paint.setColor(Color.WHITE);canvas.drawRect(box,paint);paint.setStrokeWidth(
density);paint.setColor(0xB3FFFFFF);
         for(int i=1;i<3;i++){float x=box.left+box.width()*i/3,y=box.top+box.height()*i/3;ca
nvas.drawLine(x,box.top,x,box.bottom,paint);canvas.drawLine(box.left,y,box.right,y,paint);
}
         paint.setStyle(Paint.Style.FILL);paint.setColor(Color.WHITE);for(float x:new float[
]{box.left,box.right})for(float y:new float[]{box.top,box.bottom})canvas.drawCircle(x,y,5*
density,paint);
      }
   }
   private void strokePaint(int color,float width){paint.reset();paint.setAntiAlias(true);paint.
setColor(color);paint.setStyle(Paint.Style.STROKE);paint.setStrokeWidth(width);paint.setS
trokeCap(Paint.Cap.ROUND);paint.setStrokeJoin(Paint.Join.ROUND);}
   @Override public boolean onTouchEvent(MotionEvent event){
```

```
        if(!isEnabled()||document==null||mode==Mode.VIEW)return super.onTouchEvent(eve
nt);
        fit();float x=clamp((event.getX()-imageBounds.left)/scale,0,document.state().image.get
Width()),y=clamp((event.getY()-imageBounds.top)/scale,0,document.state().image.getHeig
ht());
        switch(event.getActionMasked()){
            case MotionEvent.ACTION_DOWN:
                if(!imageBounds.contains(event.getX(),event.getY()))return false;
                getParent().requestDisallowInterceptTouchEvent(true);downX=x;downY=y;
                if(mode==Mode.CROP){dragStart.set(crop);float threshold=28*density/scale;han
dle=0;if(distance(x,y,crop.left,crop.top)<threshold)handle=1;else if(distance(x,y,crop.right,c
rop.top)<threshold)handle=2;else if(distance(x,y,crop.right,crop.bottom)<threshold)handle=
3;else if(distance(x,y,crop.left,crop.bottom)<threshold)handle=4;else if(crop.contains(x,y))h
andle=5;}
                else {if(document.state().strokes.size()>=100)return false;points.clear();points.add
(new PointF(x,y));livePath.reset();livePath.moveTo(x,y);}invalidate();return true;
            case MotionEvent.ACTION_MOVE:
                if(mode==Mode.CROP)moveCrop(x,y);else if(!points.isEmpty()&&points.size()<
5000){PointF previous=points.get(points.size()-1);if(distance(x,y,previous.x,previous.y)>.7
5f/scale){livePath.quadTo(previous.x,previous.y,(previous.x+x)/2,(previous.y+y)/2);points.
add(new PointF(x,y));}}invalidate();return true;
            case MotionEvent.ACTION_UP:
                if(mode==Mode.DRAW&&!points.isEmpty()){document.addStroke(new EditorD
ocument.Stroke(points,color,brushDp*density/scale));points.clear();livePath.reset();if(chang
ed!=null)changed.run();}
                handle=0;getParent().requestDisallowInterceptTouchEvent(false);performClick();
invalidate();return true;
            case MotionEvent.ACTION_CANCEL:points.clear();livePath.reset();handle=0;inval
idate();return true;
            default:return true;
        }
    }
    private void moveCrop(float x,float y){
        float imageWidth=document.state().image.getWidth(),imageHeight=document.state().i
mage.getHeight(),minimum=Math.min(32,Math.min(imageWidth,imageHeight)/4);
        if(handle==5){float left=clamp(dragStart.left+x-downX,0,imageWidth-dragStart.width
()),top=clamp(dragStart.top+y-downY,0,imageHeight-dragStart.height());crop.set(left,top,le
ft+dragStart.width(),top+dragStart.height());return;}
        if(handle==0)return;
        boolean left=handle==1||handle==4,top=handle==1||handle==2;float anchorX=left?dra
gStart.right:dragStart.left,anchorY=top?dragStart.bottom:dragStart.top;
        float width=Math.max(minimum,left?anchorX-x:x-anchorX),height=Math.max(minim
um,top?anchorY-y:y-anchorY);
        float maxWidth=left?anchorX:imageWidth-anchorX,maxHeight=top?anchorY:imageH
eight-anchorY;
        if(ratio>0){width=Math.min(Math.max(width,height*ratio),Math.min(maxWidth,max
Height*ratio));height=width/ratio;}else{width=Math.min(width,maxWidth);height=Math.m
in(height,maxHeight);}
        crop.set(left?anchorX-width:anchorX,top?anchorY-height:anchorY,left?anchorX:anch
```

```
orX+width,top?anchorY:anchorY+height);
    }
    private static float clamp(float value,float min,float max){return Math.max(min,Math.min
(max,value));}
    private static float distance(float x,float y,float a,float b){return (float)Math.hypot(x-a,y-b
);}
    @Override public boolean performClick(){super.performClick();return true;}
}
```

1 2 2 EditorDocument

Nguồn: app/src/main/java/com/example/appanh/editor/EditorDocument.java

Vai trò: Lưu trạng thái ảnh và nét vẽ, lịch sử hoàn tác, biến đổi cắt xoay, bộ lọc và kết xuất bitmap.

Mã nguồn đầy đủ

```
package com.example.appanh.editor;

import android.graphics.*;
import java.util.*;

/** Immutable edit snapshots; only transforms/export allocate a bitmap. */
public final class EditorDocument {
    public static final class Stroke {
        public final List<PointF> points;
        public final int color;
        public final float width;
        public final Path path=new Path();
        public Stroke(List<PointF> points,int color,float width){this.points=new ArrayList<>(
points);this.color=color;this.width=width;if(!points.isEmpty()){PointF previous=points.get(
0);path.moveTo(previous.x,previous.y);for(int i=1;i<points.size();i++){PointF point=points.
get(i);path.quadTo(previous.x,previous.y,(previous.x+point.x)/2,(previous.y+point.y)/2);pre
vious=point;}path.lineTo(previous.x,previous.y);}}
        public Stroke transform(Matrix matrix){float[] xy=new float[points.size()*2];for(int i=
0;i<points.size();i++){xy[i*2]=points.get(i).x;xy[i*2+1]=points.get(i).y;}matrix.mapPoints(
xy);List<PointF> transformed=new ArrayList<>();for(int i=0;i<points.size();i++)transforme
d.add(new PointF(xy[i*2],xy[i*2+1]));return new Stroke(transformed,color,width);}
    }
    public static final class State {
        public final Bitmap image;
        public final List<Stroke> strokes;
        public final int filter,brightness,contrast,version;
        State(Bitmap image,List<Stroke> strokes,int filter,int brightness,int contrast,int version
){this.image=image;this.strokes=Collections.unmodifiableList(new ArrayList<>(strokes));t
his.filter=filter;this.brightness=brightness;this.contrast=contrast;this.version=version;}
    }
    private final State original;
```

```
    private State state;
    private int version;
    private final Deque<State> undo=new ArrayDeque<>(),redo=new ArrayDeque<>();
    public EditorDocument(Bitmap image){original=new State(image,Collections.emptyList(
),0,0,0,0);state=original;}
    public State state(){return state;}
    public boolean dirty(){return state.version!=0;}
    public boolean canUndo(){return !undo.isEmpty();}
    public boolean canRedo(){return !redo.isEmpty();}
    private void remember(){undo.addLast(state);while(undo.size()>4)undo.removeFirst();re
do.clear();}
    public void addStroke(Stroke stroke){remember();List<Stroke> strokes=new ArrayList<
>(state.strokes);strokes.add(stroke);state=new State(state.image,strokes,state.filter,state.brig
htness,state.contrast,++version);}
    public void clearStrokes(){if(state.strokes.isEmpty())return;remember();state=new State(s
tate.image,Collections.emptyList(),state.filter,state.brightness,state.contrast,++version);}
    public void undo(){if(!canUndo())return;redo.addLast(state);state=undo.removeLast();}
    public void redo(){if(!canRedo())return;undo.addLast(state);state=redo.removeLast();}
    public void reset(){remember();state=original;}
    public void beginAdjustment(){remember();}
    public void adjust(int brightness,int contrast){state=new State(state.image,state.strokes,st
ate.filter,brightness,contrast,++version);}
    public void filter(int value){if(value==state.filter)return;remember();state=new State(stat
e.image,state.strokes,value,state.brightness,state.contrast,++version);}
    public void acceptTransform(State transformed){remember();state=new State(transforme
d.image,transformed.strokes,transformed.filter,transformed.brightness,transformed.contrast,
++version);}
    public static State crop(State source,Rect rect){
        int left=Math.max(0,Math.min(source.image.getWidth()-1,rect.left)),top=Math.max(0,
Math.min(source.image.getHeight()-1,rect.top));
        int width=Math.max(1,Math.min(source.image.getWidth()-left,rect.width())),height=M
ath.max(1,Math.min(source.image.getHeight()-top,rect.height()));
        Bitmap bitmap=Bitmap.createBitmap(source.image,left,top,width,height);Matrix matri
x=new Matrix();matrix.setTranslate(-left,-top);
        List<Stroke> strokes=new ArrayList<>();for(Stroke stroke:source.strokes)strokes.add(
stroke.transform(matrix));
        return new State(bitmap,strokes,source.filter,source.brightness,source.contrast,0);
    }
    public static State rotate(State source){
        Matrix rotation=new Matrix();rotation.setRotate(90);Bitmap bitmap=Bitmap.createBit
map(source.image,0,0,source.image.getWidth(),source.image.getHeight(),rotation,true);
        Matrix coordinates=new Matrix();coordinates.setValues(new float[]{0,-1,source.image
.getHeight(),1,0,0,0,0,1});
        List<Stroke> strokes=new ArrayList<>();for(Stroke stroke:source.strokes)strokes.add(
stroke.transform(coordinates));
        return new State(bitmap,strokes,source.filter,source.brightness,source.contrast,0);
    }
    public static ColorMatrixColorFilter effect(State state){
        ColorMatrix filter=new ColorMatrix();
```

```
        if(state.filter==1)filter.setSaturation(0);
        if(state.filter==2)filter.set(new float[]{.393f,.769f,.189f,0,0,.349f,.686f,.168f,0,0,.272f,.
534f,.131f,0,0,0,0,0,1,0});
        if(state.filter==3)filter.set(new float[]{1.08f,0,0,0,7,0,1.02f,0,0,2,0,0,.91f,0,-3,0,0,0,1,0
});
        if(state.filter==4)filter.set(new float[]{.92f,0,0,0,-2,0,1.02f,0,0,2,0,0,1.1f,0,8,0,0,0,1,0}
);
        float contrast=1+state.contrast/100f,offset=state.brightness+(1-contrast)*128;
        filter.postConcat(new ColorMatrix(new float[]{contrast,0,0,0,offset,0,contrast,0,0,offse
t,0,0,contrast,0,offset,0,0,0,1,0}));
        return new ColorMatrixColorFilter(filter);
    }
    public static void drawStroke(Canvas canvas,Stroke stroke,Paint paint){
        if(stroke.points.isEmpty())return;
        paint.reset();paint.setAntiAlias(true);paint.setColor(stroke.color);paint.setStrokeWidth(
stroke.width);paint.setStrokeCap(Paint.Cap.ROUND);paint.setStrokeJoin(Paint.Join.ROUN
D);paint.setStyle(Paint.Style.STROKE);
        PointF start=stroke.points.get(0);if(stroke.points.size()==1){paint.setStyle(Paint.Style.
FILL);canvas.drawCircle(start.x,start.y,stroke.width/2,paint);return;}
        canvas.drawPath(stroke.path,paint);
    }
    public static Bitmap render(State state){Bitmap output=Bitmap.createBitmap(state.image.
getWidth(),state.image.getHeight(),Bitmap.Config.ARGB_8888);Canvas canvas=new Canv
as(output);Paint paint=new Paint(Paint.ANTI_ALIAS_FLAG|Paint.FILTER_BITMAP_FL
AG);paint.setColorFilter(effect(state));canvas.drawBitmap(state.image,0,0,paint);for(Stroke
 stroke:state.strokes)drawStroke(canvas,stroke,paint);return output;}
}
```

1 2 3 EditorViewModel

Nguồn: app/src/main/java/com/example/appanh/editor/EditorViewModel.java

Vai trò: Giữ phiên chỉnh sửa khi Activity được tạo lại; giải mã, biến đổi và xuất JPEG trên luồng nền; lưu thành bản sao.

Mã nguồn đầy đủ

```
package com.example.appanh.editor;

import android.app.Application;
import android.graphics.*;
import android.os.Handler;
import android.os.Looper;
import androidx.lifecycle.AndroidViewModel;
import androidx.lifecycle.MutableLiveData;
import com.bumptech.glide.Glide;
import com.bumptech.glide.request.FutureTarget;
import com.example.appanh.GalleryRepository;
import com.example.appanh.models.Photo;
```

```
import java.io.*;
import java.util.*;
import java.util.concurrent.Executors;
import java.util.concurrent.ExecutorService;

/** Retains edits across Activity recreation; decode, crop, rotate and export run off the UI th
read. */
public final class EditorViewModel extends AndroidViewModel {
   public EditorDocument document;
   public Photo original;
   public boolean busy;
   public String tool="view";
   public int brushColor=Color.WHITE;
   public float brushWidth=6,aspect;
   public final MutableLiveData<Integer> updates=new MutableLiveData<>(0);
   public final MutableLiveData<String> message=new MutableLiveData<>();
   public final MutableLiveData<Photo> saved=new MutableLiveData<>();
   private final ExecutorService io=Executors.newSingleThreadExecutor();
   private final Handler main=new Handler(Looper.getMainLooper());
   private GalleryRepository repository;
   private int tick;
   public EditorViewModel(Application app){super(app);}
   public void changed(){updates.setValue(++tick);}
   public void load(GalleryRepository repo,Photo photo){
      if(document!=null||busy)return;repository=repo;original=photo;busy=true;changed();
      io.execute(()->{
         FutureTarget<Bitmap> request=null;
         try{
            request=Glide.with(getApplication()).asBitmap().load(repo.source(photo)).fitCent
er().submit(1600,1600);
            Bitmap bitmap=request.get().copy(Bitmap.Config.ARGB_8888,false);
            main.post(()->{document=new EditorDocument(bitmap);busy=false;message.set
Value("Chọn công cụ để chỉnh sửa ảnh");changed();});
         }catch(Exception error){failure("Không đọc được ảnh. Kiểm tra tệp hoặc kết nối mạ
ng.");}
         finally{if(request!=null)Glide.with(getApplication()).clear(request);}
      });
   }
   public void crop(Rect crop,boolean saveAfter){transform(crop,false,saveAfter);}
   public void rotate(){transform(null,true,false);}
   private void transform(Rect crop,boolean rotate,boolean saveAfter){
      if(document==null||busy)return;EditorDocument.State source=document.state();busy=t
rue;message.setValue(rotate?"Đang xoay ảnh…":"Đang cắt ảnh…");changed();
      io.execute(()->{
         try{EditorDocument.State output=rotate?EditorDocument.rotate(source):EditorDocu
ment.crop(source,crop);
            main.post(()->{document.acceptTransform(output);busy=false;tool="view";messa
ge.setValue(rotate?"Đã xoay ảnh":"Đã áp dụng vùng cắt");changed();if(saveAfter)save();});
         }catch(Exception|OutOfMemoryError error){failure("Không xử lý được ảnh. Hãy ho
```

```
àn tác và thử vùng cắt nhỏ hơn.");}
        });
    }
    public void save(){
        if(document==null||busy)return;EditorDocument.State source=document.state();busy=t
rue;message.setValue("Đang lưu bản sao…");changed();
        io.execute(()->{
            File target=null;Bitmap output=null;
            try{
                output=EditorDocument.render(source);String id=UUID.randomUUID().toString(
);File dir=new File(getApplication().getFilesDir(),"media");if(!dir.exists()&&!dir.mkdirs())t
hrow new IOException("Không tạo được thư mục ảnh");
                target=new File(dir,id+".jpg");try(OutputStream out=new FileOutputStream(targe
t)){if(!output.compress(Bitmap.CompressFormat.JPEG,94,out))throw new IOException("K
hông mã hóa được ảnh");}
                Photo photo=new Photo(repository.uid(),"EDIT_"+System.currentTimeMillis()+"
.jpg",target.getPath(),target.length(),output.getWidth(),output.getHeight(),"image/jpeg");
                photo.setId(id);photo.setCreatedAt(new Date());photo.setDateTaken(new Date());
photo.setDeviceName(android.os.Build.MODEL);photo.setAlbumIds(new ArrayList<>(ori
ginal.getAlbumIds()));photo.setHidden(original.isHidden());
                main.post(()->repository.appendLocal(photo,error->{busy=false;if(error!=null){m
essage.setValue("Chưa lưu được bản sao: "+error);changed();}else{saved.setValue(photo);c
hanged();}}));
            }catch(Exception|OutOfMemoryError error){if(target!=null)target.delete();failure("
Chưa lưu được ảnh. Kiểm tra dung lượng bộ nhớ và thử lại.");}
            finally{if(output!=null)output.recycle();}
        });
    }
    private void failure(String text){main.post(()->{busy=false;message.setValue(text);chang
ed();});}
    @Override protected void onCleared(){io.shutdown();}
}
```

1 3 Thư mục helpers

Đường dẫn: app/src/main/java/com/example/appanh/helpers

Các tiện ích và lớp hỗ trợ; có helper Firebase và quyền từ mã ban đầu. Số file Java: 3.

Danh sách: FirebaseHelper.java, PermissionHelper.java, PhotoHelper.java.

1 3 1 FirebaseHelper

Nguồn: app/src/main/java/com/example/appanh/helpers/FirebaseHelper.java

Vai trò: Lớp hỗ trợ Firebase từ phiên bản ban đầu, gồm thao tác ảnh, album và thông báo. Các Activity hiện tại chủ yếu dùng GalleryRepository, không gọi helper này.

Mã nguồn đầy đủ

```
package com.example.appanh.helpers;

import android.net.Uri;
import android.util.Log;

import androidx.annotation.NonNull;

import com.example.appanh.models.Album;
import com.example.appanh.models.Notification;
import com.example.appanh.models.Photo;
import com.example.appanh.models.User;
import com.google.android.gms.tasks.Task;
import com.google.firebase.auth.FirebaseAuth;
import com.google.firebase.auth.FirebaseUser;
import com.google.firebase.firestore.CollectionReference;
import com.google.firebase.firestore.DocumentReference;
import com.google.firebase.firestore.DocumentSnapshot;
import com.google.firebase.firestore.FieldValue;
import com.google.firebase.firestore.FirebaseFirestore;
import com.google.firebase.firestore.Query;
import com.google.firebase.firestore.QueryDocumentSnapshot;
import com.google.firebase.firestore.QuerySnapshot;
import com.google.firebase.firestore.WriteBatch;
import com.google.firebase.storage.FirebaseStorage;
import com.google.firebase.storage.StorageReference;
import com.google.firebase.storage.UploadTask;

import java.util.ArrayList;
import java.util.Collections;
import java.util.Date;
import java.util.HashMap;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.UUID;

/**
 * Singleton helper class managing all Firebase Authentication, Firestore, and Storage opera
tions.
 */
public class FirebaseHelper {

    private static final String TAG = "FirebaseHelper";

    private static FirebaseHelper instance;
    private final FirebaseAuth auth;
    private final FirebaseFirestore db;
    private final FirebaseStorage storage;
```

```
private static final String USERS_COLLECTION = "users";
private static final String PHOTOS_COLLECTION = "photos";
private static final String ALBUMS_COLLECTION = "albums";
private static final String NOTIFICATIONS_COLLECTION = "notifications";
private static final String TRASH_COLLECTION = "trash";

private FirebaseHelper() {
    auth = FirebaseAuth.getInstance();
    db = FirebaseFirestore.getInstance();
    storage = FirebaseStorage.getInstance();
}

public static synchronized FirebaseHelper getInstance() {
    if (instance == null) {
        instance = new FirebaseHelper();
    }
    return instance;
}

// ---------------------------------------------------------------------
// Auth methods
// ---------------------------------------------------------------------

public FirebaseAuth getAuth() {
    return auth;
}

public FirebaseUser getCurrentUser() {
    return auth.getCurrentUser();
}

public String getCurrentUserId() {
    FirebaseUser user = getCurrentUser();
    return user != null ? user.getUid() : null;
}

// ---------------------------------------------------------------------
// Firestore references
// ---------------------------------------------------------------------

public CollectionReference getUsersRef() {
    return db.collection(USERS_COLLECTION);
}

public CollectionReference getPhotosRef() {
    return db.collection(PHOTOS_COLLECTION);
}

public CollectionReference getAlbumsRef() {
```

```
    return db.collection(ALBUMS_COLLECTION);
}

public CollectionReference getNotificationsRef() {
    return db.collection(NOTIFICATIONS_COLLECTION);
}

public CollectionReference getTrashRef() {
    return db.collection(TRASH_COLLECTION);
}

// ---------------------------------------------------------------------
// User operations
// ---------------------------------------------------------------------

/**
 * Saves or overwrites user information in Firestore.
 */
public void saveUser(User user, OnCompleteListener listener) {
    String uid = user.getId();
    if (uid == null || uid.trim().isEmpty()) {
        uid = getCurrentUserId();
        user.setId(uid);
    }

    if (uid == null) {
        if (listener != null) listener.onFailure("Không tìm thấy User ID");
        return;
    }

    getUsersRef().document(uid)
            .set(user)
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error saving user", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Retrieves user profile from Firestore by userId.
 */
public void getUser(String userId, OnDataListener<User> listener) {
    if (userId == null || userId.trim().isEmpty()) {
        if (listener != null) listener.onFailure("User ID không hợp lệ");
        return;
    }
```

```
        getUsersRef().document(userId)
                .get()
                .addOnSuccessListener(snapshot -> {
                    if (snapshot.exists()) {
                        User user = snapshot.toObject(User.class);
                        if (user != null) {
                            user.setId(snapshot.getId());
                        }
                        if (listener != null) listener.onSuccess(user);
                    } else {
                        if (listener != null) listener.onFailure("Không tìm thấy người dùng");
                    }
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error getting user", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Updates specific fields of a user document in Firestore.
     */
    public void updateUser(String userId, Map<String, Object> updates, OnCompleteListene
r listener) {
        if (userId == null || userId.trim().isEmpty()) {
            if (listener != null) listener.onFailure("User ID không hợp lệ");
            return;
        }

        Map<String, Object> finalUpdates = new HashMap<>(updates);
        finalUpdates.put("updatedAt", FieldValue.serverTimestamp());

        getUsersRef().document(userId)
                .update(finalUpdates)
                .addOnSuccessListener(aVoid -> {
                    if (listener != null) listener.onSuccess();
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error updating user", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    // ---------------------------------------------------------------------
    // Photo operations
    // ---------------------------------------------------------------------

    /**
```

```
 * Saves a new photo or updates an existing photo record in Firestore.
 */
public void savePhoto(Photo photo, OnCompleteListener listener) {
    DocumentReference docRef;
    if (photo.getId() == null || photo.getId().trim().isEmpty()) {
        docRef = getPhotosRef().document();
        photo.setId(docRef.getId());
    } else {
        docRef = getPhotosRef().document(photo.getId());
    }

    docRef.set(photo)
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error saving photo", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Retrieves all active (non-deleted) photos for the given user, sorted descending by date.
 */
public void getPhotos(String userId, OnDataListener<List<Photo>> listener) {
    if (userId == null) {
        if (listener != null) listener.onFailure("User ID không hợp lệ");
        return;
    }

    getPhotosRef()
            .whereEqualTo("userId", userId)
            .get()
            .addOnSuccessListener(querySnapshot -> {
                List<Photo> photoList = new ArrayList<>();
                for (QueryDocumentSnapshot doc : querySnapshot) {
                    Photo p = doc.toObject(Photo.class);
                    p.setId(doc.getId());
                    if (!p.isDeleted()) {
                        photoList.add(p);
                    }
                }
                sortPhotosByDateDesc(photoList);
                if (listener != null) listener.onSuccess(photoList);
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error getting photos", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
```

```
    }

    /**
     * Retrieves all photos belonging to a specific album.
     */
    public void getPhotosByAlbum(String albumId, OnDataListener<List<Photo>> listener)
{
        if (albumId == null) {
            if (listener != null) listener.onFailure("Album ID không hợp lệ");
            return;
        }

        getPhotosRef()
                .whereArrayContains("albumIds", albumId)
                .get()
                .addOnSuccessListener(querySnapshot -> {
                    List<Photo> photoList = new ArrayList<>();
                    for (QueryDocumentSnapshot doc : querySnapshot) {
                        Photo p = doc.toObject(Photo.class);
                        p.setId(doc.getId());
                        if (!p.isDeleted()) {
                            photoList.add(p);
                        }
                    }
                    sortPhotosByDateDesc(photoList);
                    if (listener != null) listener.onSuccess(photoList);
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error getting photos by album", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Updates specific fields of a photo document.
     */
    public void updatePhoto(String photoId, Map<String, Object> updates, OnCompleteListe
ner listener) {
        if (photoId == null) {
            if (listener != null) listener.onFailure("Photo ID không hợp lệ");
            return;
        }

        Map<String, Object> finalUpdates = new HashMap<>(updates);
        finalUpdates.put("updatedAt", FieldValue.serverTimestamp());

        getPhotosRef().document(photoId)
                .update(finalUpdates)
                .addOnSuccessListener(aVoid -> {
```

```
            if (listener != null) listener.onSuccess();
        })
        .addOnFailureListener(e -> {
            Log.e(TAG, "Error updating photo", e);
            if (listener != null) listener.onFailure(e.getMessage());
        });
}

/**
 * Permanently deletes a photo document from Firestore and trash.
 */
public void deletePhoto(String photoId, OnCompleteListener listener) {
    if (photoId == null) {
        if (listener != null) listener.onFailure("Photo ID không hợp lệ");
        return;
    }

    WriteBatch batch = db.batch();
    batch.delete(getPhotosRef().document(photoId));
    batch.delete(getTrashRef().document(photoId));

    batch.commit()
        .addOnSuccessListener(aVoid -> {
            if (listener != null) listener.onSuccess();
        })
        .addOnFailureListener(e -> {
            Log.e(TAG, "Error deleting photo", e);
            if (listener != null) listener.onFailure(e.getMessage());
        });
}

/**
 * Moves a photo to trash (soft-delete).
 */
public void moveToTrash(Photo photo, OnCompleteListener listener) {
    if (photo == null || photo.getId() == null) {
        if (listener != null) listener.onFailure("Ảnh không hợp lệ");
        return;
    }

    photo.setDeleted(true);
    photo.setDeletedAt(new Date());

    WriteBatch batch = db.batch();

    DocumentReference photoRef = getPhotosRef().document(photo.getId());
    Map<String, Object> updates = new HashMap<>();
    updates.put("deleted", true);
    updates.put("deletedAt", FieldValue.serverTimestamp());
```

```
    updates.put("updatedAt", FieldValue.serverTimestamp());
    batch.update(photoRef, updates);

    DocumentReference trashRef = getTrashRef().document(photo.getId());
    batch.set(trashRef, photo);

    batch.commit()
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error moving photo to trash", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Restores a photo from trash.
 */
public void restoreFromTrash(String photoId, OnCompleteListener listener) {
    if (photoId == null) {
        if (listener != null) listener.onFailure("Photo ID không hợp lệ");
        return;
    }

    WriteBatch batch = db.batch();

    DocumentReference photoRef = getPhotosRef().document(photoId);
    Map<String, Object> updates = new HashMap<>();
    updates.put("deleted", false);
    updates.put("deletedAt", null);
    updates.put("updatedAt", FieldValue.serverTimestamp());
    batch.update(photoRef, updates);

    DocumentReference trashRef = getTrashRef().document(photoId);
    batch.delete(trashRef);

    batch.commit()
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error restoring photo from trash", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Gets all photos in the trash for a given user.
```

```
 */
public void getTrashPhotos(String userId, OnDataListener<List<Photo>> listener) {
    if (userId == null) {
        if (listener != null) listener.onFailure("User ID không hợp lệ");
        return;
    }

    getTrashRef()
            .whereEqualTo("userId", userId)
            .get()
            .addOnSuccessListener(querySnapshot -> {
                List<Photo> trashList = new ArrayList<>();
                for (QueryDocumentSnapshot doc : querySnapshot) {
                    Photo p = doc.toObject(Photo.class);
                    p.setId(doc.getId());
                    trashList.add(p);
                }
                if (listener != null) listener.onSuccess(trashList);
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error getting trash photos", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Permanently deletes all photos in the trash for a given user.
 */
public void emptyTrash(String userId, OnCompleteListener listener) {
    if (userId == null) {
        if (listener != null) listener.onFailure("User ID không hợp lệ");
        return;
    }

    getTrashRef()
            .whereEqualTo("userId", userId)
            .get()
            .addOnSuccessListener(querySnapshot -> {
                if (querySnapshot.isEmpty()) {
                    if (listener != null) listener.onSuccess();
                    return;
                }

                WriteBatch batch = db.batch();
                for (DocumentSnapshot doc : querySnapshot.getDocuments()) {
                    String id = doc.getId();
                    batch.delete(getTrashRef().document(id));
                    batch.delete(getPhotosRef().document(id));
                }
```

```
                batch.commit()
                        .addOnSuccessListener(aVoid -> {
                            if (listener != null) listener.onSuccess();
                        })
                        .addOnFailureListener(e -> {
                            Log.e(TAG, "Error committing empty trash batch", e);
                            if (listener != null) listener.onFailure(e.getMessage());
                        });
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error querying trash to empty", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
    }

    /**
     * Toggles favorite status for a photo.
     */
    public void toggleFavorite(String photoId, boolean isFavorite, OnCompleteListener listen
er) {
        if (photoId == null) {
            if (listener != null) listener.onFailure("Photo ID không hợp lệ");
            return;
        }

        Map<String, Object> updates = new HashMap<>();
        updates.put("favorite", isFavorite);
        updates.put("updatedAt", FieldValue.serverTimestamp());

        getPhotosRef().document(photoId)
                .update(updates)
                .addOnSuccessListener(aVoid -> {
                    if (listener != null) listener.onSuccess();
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error toggling favorite", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Searches photos for a user matching file name or location query.
     */
    public void searchPhotos(String userId, String query, OnDataListener<List<Photo>> liste
ner) {
        if (userId == null) {
            if (listener != null) listener.onFailure("User ID không hợp lệ");
            return;
```

```
        }

        final String cleanQuery = query != null ? query.trim().toLowerCase(Locale.getDefault(
)) : "";

        getPhotos(userId, new OnDataListener<List<Photo>>() {
            @Override
            public void onSuccess(List<Photo> allPhotos) {
                if (cleanQuery.isEmpty()) {
                    if (listener != null) listener.onSuccess(allPhotos);
                    return;
                }

                List<Photo> filtered = new ArrayList<>();
                for (Photo photo : allPhotos) {
                    boolean matchName = photo.getFileName() != null &&
                            photo.getFileName().toLowerCase(Locale.getDefault()).contains(cleanQu
ery);
                    boolean matchLocation = photo.getLocation() != null &&
                            photo.getLocation().toLowerCase(Locale.getDefault()).contains(cleanQue
ry);

                    if (matchName || matchLocation) {
                        filtered.add(photo);
                    }
                }
                if (listener != null) listener.onSuccess(filtered);
            }

            @Override
            public void onFailure(String error) {
                if (listener != null) listener.onFailure(error);
            }
        });
    }

    /**
     * Retrieves filtered photos based on filterType: "favorites", "videos", "images", "recent".
     */
    public void getFilteredPhotos(String userId, String filterType, OnDataListener<List<Phot
o>> listener) {
        getPhotos(userId, new OnDataListener<List<Photo>>() {
            @Override
            public void onSuccess(List<Photo> allPhotos) {
                if (filterType == null || filterType.isEmpty()) {
                    if (listener != null) listener.onSuccess(allPhotos);
                    return;
                }
```

```
            List<Photo> filtered = new ArrayList<>();
            String type = filterType.toLowerCase(Locale.getDefault());

            for (Photo p : allPhotos) {
                switch (type) {
                    case "favorites":
                    case "favorite":
                        if (p.isFavorite()) filtered.add(p);
                        break;
                    case "videos":
                    case "video":
                        if (p.isVideo()) filtered.add(p);
                        break;
                    case "images":
                    case "image":
                    case "photo":
                        if (!p.isVideo()) filtered.add(p);
                        break;
                    case "recent":
                        // Recent last 30 items or non-deleted
                        filtered.add(p);
                        break;
                    default:
                        filtered.add(p);
                        break;
                }
            }

            if ("recent".equals(type) && filtered.size() > 50) {
                filtered = new ArrayList<>(filtered.subList(0, 50));
            }

            if (listener != null) listener.onSuccess(filtered);
        }

        @Override
        public void onFailure(String error) {
            if (listener != null) listener.onFailure(error);
        }
    });
}

// ---------------------------------------------------------------------
// Album operations
// ---------------------------------------------------------------------

/**
 * Creates or saves an album in Firestore.
 */
```

```
public void createAlbum(Album album, OnCompleteListener listener) {
    DocumentReference docRef;
    if (album.getId() == null || album.getId().trim().isEmpty()) {
        docRef = getAlbumsRef().document();
        album.setId(docRef.getId());
    } else {
        docRef = getAlbumsRef().document(album.getId());
    }

    docRef.set(album)
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error creating album", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Retrieves all albums created by the user.
 */
public void getAlbums(String userId, OnDataListener<List<Album>> listener) {
    if (userId == null) {
        if (listener != null) listener.onFailure("User ID không hợp lệ");
        return;
    }

    getAlbumsRef()
            .whereEqualTo("userId", userId)
            .get()
            .addOnSuccessListener(querySnapshot -> {
                List<Album> albums = new ArrayList<>();
                for (QueryDocumentSnapshot doc : querySnapshot) {
                    Album album = doc.toObject(Album.class);
                    album.setId(doc.getId());
                    albums.add(album);
                }
                if (listener != null) listener.onSuccess(albums);
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error getting albums", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

/**
 * Updates specific fields of an album document.
 */
```

```
    public void updateAlbum(String albumId, Map<String, Object> updates, OnCompleteLis
tener listener) {
        if (albumId == null) {
            if (listener != null) listener.onFailure("Album ID không hợp lệ");
            return;
        }

        Map<String, Object> finalUpdates = new HashMap<>(updates);
        finalUpdates.put("updatedAt", FieldValue.serverTimestamp());

        getAlbumsRef().document(albumId)
                .update(finalUpdates)
                .addOnSuccessListener(aVoid -> {
                    if (listener != null) listener.onSuccess();
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error updating album", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Deletes an album from Firestore.
     */
    public void deleteAlbum(String albumId, OnCompleteListener listener) {
        if (albumId == null) {
            if (listener != null) listener.onFailure("Album ID không hợp lệ");
            return;
        }

        getAlbumsRef().document(albumId)
                .delete()
                .addOnSuccessListener(aVoid -> {
                    if (listener != null) listener.onSuccess();
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error deleting album", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Adds a photo to an album and updates both references.
     */
    public void addPhotoToAlbum(String albumId, String photoId, OnCompleteListener liste
ner) {
        if (albumId == null || photoId == null) {
            if (listener != null) listener.onFailure("Album ID hoặc Photo ID không hợp lệ");
            return;
```

```
        }

        WriteBatch batch = db.batch();

        DocumentReference albumRef = getAlbumsRef().document(albumId);
        batch.update(albumRef,
                "photoIds", FieldValue.arrayUnion(photoId),
                "photoCount", FieldValue.increment(1),
                "updatedAt", FieldValue.serverTimestamp());

        DocumentReference photoRef = getPhotosRef().document(photoId);
        batch.update(photoRef,
                "albumIds", FieldValue.arrayUnion(albumId),
                "updatedAt", FieldValue.serverTimestamp());

        batch.commit()
                .addOnSuccessListener(aVoid -> {
                    if (listener != null) listener.onSuccess();
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error adding photo to album", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Removes a photo from an album and updates both references.
     */
    public void removePhotoFromAlbum(String albumId, String photoId, OnCompleteListen
er listener) {
        if (albumId == null || photoId == null) {
            if (listener != null) listener.onFailure("Album ID hoặc Photo ID không hợp lệ");
            return;
        }

        WriteBatch batch = db.batch();

        DocumentReference albumRef = getAlbumsRef().document(albumId);
        batch.update(albumRef,
                "photoIds", FieldValue.arrayRemove(photoId),
                "photoCount", FieldValue.increment(-1),
                "updatedAt", FieldValue.serverTimestamp());

        DocumentReference photoRef = getPhotosRef().document(photoId);
        batch.update(photoRef,
                "albumIds", FieldValue.arrayRemove(albumId),
                "updatedAt", FieldValue.serverTimestamp());

        batch.commit()
```

```
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error removing photo from album", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
    }

    // ---------------------------------------------------------------------
    // Notification operations
    // ---------------------------------------------------------------------

    /**
     * Creates and stores a notification.
     */
    public void createNotification(Notification notification, OnCompleteListener listener) {
        DocumentReference docRef;
        if (notification.getId() == null || notification.getId().trim().isEmpty()) {
            docRef = getNotificationsRef().document();
            notification.setId(docRef.getId());
        } else {
            docRef = getNotificationsRef().document(notification.getId());
        }

        docRef.set(notification)
                .addOnSuccessListener(aVoid -> {
                    if (listener != null) listener.onSuccess();
                })
                .addOnFailureListener(e -> {
                    Log.e(TAG, "Error creating notification", e);
                    if (listener != null) listener.onFailure(e.getMessage());
                });
    }

    /**
     * Retrieves all notifications for the given user, sorted newest first.
     */
    public void getNotifications(String userId, OnDataListener<List<Notification>> listener)
{
        if (userId == null) {
            if (listener != null) listener.onFailure("User ID không hợp lệ");
            return;
        }

        getNotificationsRef()
                .whereEqualTo("userId", userId)
                .get()
                .addOnSuccessListener(querySnapshot -> {
```

```
            List<Notification> list = new ArrayList<>();
            for (QueryDocumentSnapshot doc : querySnapshot) {
                Notification n = doc.toObject(Notification.class);
                n.setId(doc.getId());
                list.add(n);
            }
            // Sort descending by createdAt
            Collections.sort(list, (n1, n2) -> {
                Date d1 = n1.getCreatedAt();
                Date d2 = n2.getCreatedAt();
                if (d1 == null && d2 == null) return 0;
                if (d1 == null) return 1;
                if (d2 == null) return -1;
                return d2.compareTo(d1);
            });
            if (listener != null) listener.onSuccess(list);
        })
        .addOnFailureListener(e -> {
            Log.e(TAG, "Error getting notifications", e);
            if (listener != null) listener.onFailure(e.getMessage());
        });
}

/**
 * Marks a notification as read.
 */
public void markNotificationRead(String notificationId, OnCompleteListener listener) {
    if (notificationId == null) {
        if (listener != null) listener.onFailure("Notification ID không hợp lệ");
        return;
    }

    getNotificationsRef().document(notificationId)
            .update("read", true)
            .addOnSuccessListener(aVoid -> {
                if (listener != null) listener.onSuccess();
            })
            .addOnFailureListener(e -> {
                Log.e(TAG, "Error marking notification as read", e);
                if (listener != null) listener.onFailure(e.getMessage());
            });
}

// ---------------------------------------------------------------------
// Storage operations
// ---------------------------------------------------------------------

/**
 * Uploads a photo to Firebase Storage with progress tracking.
```

```
     */
    public void uploadPhoto(Uri fileUri, String userId, OnUploadListener listener) {
        if (fileUri == null || userId == null) {
            if (listener != null) listener.onFailure("File URI hoặc User ID không hợp lệ");
            return;
        }

        String fileName = System.currentTimeMillis() + "_" + UUID.randomUUID().toString(
) + ".jpg";
        StorageReference storageRef = storage.getReference()
                .child("users")
                .child(userId)
                .child("photos")
                .child(fileName);

        UploadTask uploadTask = storageRef.putFile(fileUri);

        uploadTask.addOnProgressListener(snapshot -> {
            long totalBytes = snapshot.getTotalByteCount();
            if (totalBytes > 0) {
                int progress = (int) ((100.0 * snapshot.getBytesTransferred()) / totalBytes);
                if (listener != null) {
                    listener.onProgress(progress);
                }
            }
        }).continueWithTask(task -> {
            if (!task.isSuccessful()) {
                throw task.getException() != null ? task.getException() : new Exception("Upload
failed");
            }
            return storageRef.getDownloadUrl();
        }).addOnSuccessListener(uri -> {
            if (listener != null) {
                listener.onSuccess(uri.toString());
            }
        }).addOnFailureListener(e -> {
            Log.e(TAG, "Error uploading photo", e);
            if (listener != null) {
                listener.onFailure(e.getMessage());
            }
        });
    }

    // ---------------------------------------------------------------------
    // Internal Utilities
    // ---------------------------------------------------------------------

    private void sortPhotosByDateDesc(List<Photo> photos) {
        Collections.sort(photos, (p1, p2) -> {
```

```
            Date d1 = p1.getDateTaken() != null ? p1.getDateTaken() : p1.getCreatedAt();
            Date d2 = p2.getDateTaken() != null ? p2.getDateTaken() : p2.getCreatedAt();
            if (d1 == null && d2 == null) return 0;
            if (d1 == null) return 1;
            if (d2 == null) return -1;
            return d2.compareTo(d1);
        });
    }

    // ---------------------------------------------------------------------
    // Callback interfaces
    // ---------------------------------------------------------------------

    public interface OnCompleteListener {
        void onSuccess();
        void onFailure(String error);
    }

    public interface OnDataListener<T> {
        void onSuccess(T data);
        void onFailure(String error);
    }

    public interface OnUploadListener {
        void onProgress(int progress);
        void onSuccess(String downloadUrl);
        void onFailure(String error);
    }
}
```

1 3 2 PermissionHelper

Nguồn: app/src/main/java/com/example/appanh/helpers/PermissionHelper.java

Vai trò: Các hàm hỗ trợ kiểm tra và yêu cầu quyền của bản mã ban đầu. Luồng nhập hiện tại dùng trình chọn tài liệu Android, không yêu cầu đọc toàn bộ thư viện.

Mã nguồn đầy đủ

```
package com.example.appanh.helpers;

import android.Manifest;
import android.app.Activity;
import android.content.Context;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.net.Uri;
import android.os.Build;
import android.provider.Settings;
```

```
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

import com.google.android.material.dialog.MaterialAlertDialogBuilder;

import java.util.ArrayList;
import java.util.List;

/**
 * Utility helper class for runtime permission checks, requests, and rationale dialogs.
 */
public class PermissionHelper {

    public static final int REQUEST_CODE_STORAGE = 1001;
    public static final int REQUEST_CODE_CAMERA = 1002;
    public static final int REQUEST_CODE_NOTIFICATIONS = 1003;
    public static final int REQUEST_CODE_ALL = 1004;

    // ---------------------------------------------------------------------
    // Storage Permissions
    // ---------------------------------------------------------------------

    /**
     * Returns the appropriate storage permissions array depending on Android OS version.
     */
    public static String[] getStoragePermissions() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            return new String[]{
                    Manifest.permission.READ_MEDIA_IMAGES,
                    Manifest.permission.READ_MEDIA_VIDEO
            };
        } else if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.Q) {
            return new String[]{
                    Manifest.permission.READ_EXTERNAL_STORAGE
            };
        } else {
            return new String[]{
                    Manifest.permission.READ_EXTERNAL_STORAGE,
                    Manifest.permission.WRITE_EXTERNAL_STORAGE
            };
        }
    }

    /**
     * Checks if storage permissions are granted.
     */
    public static boolean hasStoragePermission(Context context) {
        if (context == null) return false;
```

```
        String[] permissions = getStoragePermissions();
        for (String permission : permissions) {
            if (ContextCompat.checkSelfPermission(context, permission) != PackageManager.P
ERMISSION_GRANTED) {
                return false;
            }
        }
        return true;
    }

    /**
     * Requests storage permissions for the activity.
     */
    public static void requestStoragePermission(Activity activity, int requestCode) {
        if (activity == null) return;
        String[] permissions = getStoragePermissions();
        ActivityCompat.requestPermissions(activity, permissions, requestCode);
    }

    // ---------------------------------------------------------------------
    // Camera Permission
    // ---------------------------------------------------------------------

    /**
     * Checks if camera permission is granted.
     */
    public static boolean hasCameraPermission(Context context) {
        if (context == null) return false;
        return ContextCompat.checkSelfPermission(context, Manifest.permission.CAMERA)
                == PackageManager.PERMISSION_GRANTED;
    }

    /**
     * Requests camera permission for the activity.
     */
    public static void requestCameraPermission(Activity activity, int requestCode) {
        if (activity == null) return;
        ActivityCompat.requestPermissions(activity, new String[]{Manifest.permission.CAM
ERA}, requestCode);
    }

    // ---------------------------------------------------------------------
    // Notification Permission
    // ---------------------------------------------------------------------

    /**
     * Checks if notification permission is granted (Android 13+).
     */
    public static boolean hasNotificationPermission(Context context) {
```

```
        if (context == null) return false;
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            return ContextCompat.checkSelfPermission(context, Manifest.permission.POST_N
OTIFICATIONS)
                    == PackageManager.PERMISSION_GRANTED;
        }
        return true;
    }

    /**
     * Requests notification permission for the activity (Android 13+).
     */
    public static void requestNotificationPermission(Activity activity, int requestCode) {
        if (activity == null) return;
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            ActivityCompat.requestPermissions(activity, new String[]{Manifest.permission.POS
T_NOTIFICATIONS}, requestCode);
        }
    }

    // ---------------------------------------------------------------------
    // Combined Permissions
    // ---------------------------------------------------------------------

    /**
     * Checks if all required gallery permissions (storage and camera) are granted.
     */
    public static boolean hasAllPermissions(Context context) {
        return hasStoragePermission(context) && hasCameraPermission(context) && hasNoti
ficationPermission(context);
    }

    /**
     * Requests all ungranted permissions in a single prompt.
     */
    public static void requestAllPermissions(Activity activity, int requestCode) {
        if (activity == null) return;

        List<String> list = new ArrayList<>();

        for (String perm : getStoragePermissions()) {
            if (ContextCompat.checkSelfPermission(activity, perm) != PackageManager.PERMI
SSION_GRANTED) {
                list.add(perm);
            }
        }

        if (!hasCameraPermission(activity)) {
            list.add(Manifest.permission.CAMERA);
```

```
        }

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU && !hasNot
ificationPermission(activity)) {
            list.add(Manifest.permission.POST_NOTIFICATIONS);
        }

        if (!list.isEmpty()) {
            ActivityCompat.requestPermissions(activity, list.toArray(new String[0]), requestCo
de);
        }
    }

    /**
     * Utility to check whether all results in an onRequestPermissionsResult callback are GR
ANTED.
     */
    public static boolean isPermissionGranted(int[] grantResults) {
        if (grantResults == null || grantResults.length == 0) return false;
        for (int result : grantResults) {
            if (result != PackageManager.PERMISSION_GRANTED) {
                return false;
            }
        }
        return true;
    }

    // ---------------------------------------------------------------------
    // Dialogs
    // ---------------------------------------------------------------------

    /**
     * Shows a rationale dialog explaining why a permission is needed before asking again.
     */
    public static void showPermissionRationaleDialog(Activity activity, String title, String m
essage, Runnable onPositive) {
        if (activity == null || activity.isFinishing()) return;

        new MaterialAlertDialogBuilder(activity)
                .setTitle(title != null ? title : "Cấp quyền truy cập")
                .setMessage(message != null ? message : "Ứng dụng cần quyền này để hiển thị và
quản lý ảnh/video trên thiết bị của bạn.")
                .setPositiveButton("Cho phép", (dialog, which) -> {
                    if (onPositive != null) {
                        onPositive.run();
                    }
                })
                .setNegativeButton("Hủy", (dialog, which) -> dialog.dismiss())
                .setCancelable(false)
```

```
            .show();
    }

    /**
     * Shows a dialog directing the user to app settings if permission was permanently denied.
     */
    public static void showSettingsDialog(Activity activity, String message) {
        if (activity == null || activity.isFinishing()) return;

        new MaterialAlertDialogBuilder(activity)
                .setTitle("Cần cấp quyền trong Cài đặt")
                .setMessage(message != null ? message : "Quyền đã bị từ chối trước đó. Vui lòng
vào Cài đặt ứng dụng để bật lại quyền này.")
                .setPositiveButton("Mở Cài đặt", (dialog, which) -> {
                    Intent intent = new Intent(Settings.ACTION_APPLICATION_DETAILS_SET
TINGS);
                    Uri uri = Uri.fromParts("package", activity.getPackageName(), null);
                    intent.setData(uri);
                    activity.startActivity(intent);
                })
                .setNegativeButton("Để sau", (dialog, which) -> dialog.dismiss())
                .setCancelable(false)
                .show();
    }
}
```

1 3 3 PhotoHelper

Nguồn: app/src/main/java/com/example/appanh/helpers/PhotoHelper.java

Vai trò: Các tiện ích cho ảnh và video, định dạng kích thước hoặc thời gian và xử lý metadata.

Mã nguồn đầy đủ

```
package com.example.appanh.helpers;

import android.content.ContentResolver;
import android.content.ContentUris;
import android.content.Context;
import android.database.Cursor;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.Matrix;
import android.media.ExifInterface;
import android.media.ThumbnailUtils;
import android.net.Uri;
import android.os.Build;
import android.os.Handler;
```

```
import android.os.Looper;
import android.provider.MediaStore;
import android.util.Log;
import android.util.Size;

import com.example.appanh.models.Photo;

import java.io.ByteArrayOutputStream;
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.text.DecimalFormat;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.Date;
import java.util.HashMap;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Locale;
import java.util.Map;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

/**
 * Utility helper class for local device media loading, EXIF extraction,
 * thumbnail generation, image formatting, and compression.
 */
public class PhotoHelper {

    private static final String TAG = "PhotoHelper";
    private static final ExecutorService EXECUTOR = Executors.newFixedThreadPool(4);
    private static final Handler MAIN_HANDLER = new Handler(Looper.getMainLooper());

    public interface OnMediaLoadedListener {
        void onMediaLoaded(List<Photo> mediaList);
        void onError(String error);
    }

    // ---------------------------------------------------------------------
    // MediaStore Loading Operations
    // ---------------------------------------------------------------------

    /**
     * Loads local photos and videos synchronously using MediaStore.
     */
    public static List<Photo> loadLocalMedia(Context context) {
        List<Photo> mediaList = new ArrayList<>();
```

```
        if (context == null) return mediaList;

        ContentResolver contentResolver = context.getContentResolver();

        // 1. Load Images
        Uri imagesUri = MediaStore.Images.Media.EXTERNAL_CONTENT_URI;
        String[] imageProjection = {
                MediaStore.Images.Media._ID,
                MediaStore.Images.Media.DISPLAY_NAME,
                MediaStore.Images.Media.DATA,
                MediaStore.Images.Media.SIZE,
                MediaStore.Images.Media.WIDTH,
                MediaStore.Images.Media.HEIGHT,
                MediaStore.Images.Media.MIME_TYPE,
                MediaStore.Images.Media.DATE_TAKEN,
                MediaStore.Images.Media.DATE_ADDED
        };

        String sortOrder = MediaStore.Images.Media.DATE_ADDED + " DESC";

        try (Cursor cursor = contentResolver.query(imagesUri, imageProjection, null, null, sort
Order)) {
            if (cursor != null) {
                int idCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media._ID);
                int nameCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media.DISP
LAY_NAME);
                int dataCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media.DATA)
;
                int sizeCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media.SIZE);
                int widthCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media.WID
TH);
                int heightCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media.HEI
GHT);
                int mimeCol = cursor.getColumnIndexOrThrow(MediaStore.Images.Media.MIM
E_TYPE);
                int dateTakenCol = cursor.getColumnIndex(MediaStore.Images.Media.DATE_T
AKEN);
                int dateAddedCol = cursor.getColumnIndex(MediaStore.Images.Media.DATE_A
DDED);

                while (cursor.moveToNext()) {
                    long id = cursor.getLong(idCol);
                    String name = cursor.getString(nameCol);
                    String path = cursor.getString(dataCol);
                    long size = cursor.getLong(sizeCol);
                    int width = cursor.getInt(widthCol);
                    int height = cursor.getInt(heightCol);
                    String mimeType = cursor.getString(mimeCol);
```

```
            if (mimeType == null || mimeType.isEmpty()) {
                mimeType = "image/jpeg";
            }

            Photo photo = new Photo();
            photo.setId(String.valueOf(id));
            photo.setFileName(name != null ? name : "IMG_" + id);
            photo.setFilePath(path);
            photo.setFileSize(size);
            photo.setWidth(width);
            photo.setHeight(height);
            photo.setMimeType(mimeType);
            photo.setVideo(false);

            long dateTakenMs = 0;
            if (dateTakenCol != -1) {
                dateTakenMs = cursor.getLong(dateTakenCol);
            }
            if (dateTakenMs <= 0 && dateAddedCol != -1) {
                dateTakenMs = cursor.getLong(dateAddedCol) * 1000L;
            }
            if (dateTakenMs > 0) {
                photo.setDateTaken(new Date(dateTakenMs));
                photo.setCreatedAt(new Date(dateTakenMs));
            } else {
                photo.setCreatedAt(new Date());
            }

            // Content URI for thumbnail or loading
            Uri contentUri = ContentUris.withAppendedId(imagesUri, id);
            photo.setThumbnailPath(contentUri.toString());

            mediaList.add(photo);
        }
    }
} catch (Exception e) {
    Log.e(TAG, "Error querying MediaStore images", e);
}

// 2. Load Videos
Uri videoUri = MediaStore.Video.Media.EXTERNAL_CONTENT_URI;
String[] videoProjection = {
        MediaStore.Video.Media._ID,
        MediaStore.Video.Media.DISPLAY_NAME,
        MediaStore.Video.Media.DATA,
        MediaStore.Video.Media.SIZE,
        MediaStore.Video.Media.WIDTH,
        MediaStore.Video.Media.HEIGHT,
        MediaStore.Video.Media.MIME_TYPE,
```

```
            MediaStore.Video.Media.DURATION,
            MediaStore.Video.Media.DATE_TAKEN,
            MediaStore.Video.Media.DATE_ADDED
        };

        String videoSortOrder = MediaStore.Video.Media.DATE_ADDED + " DESC";

        try (Cursor cursor = contentResolver.query(videoUri, videoProjection, null, null, video
SortOrder)) {
            if (cursor != null) {
                int idCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media._ID);
                int nameCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media.DISPL
AY_NAME);
                int dataCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media.DATA);
                int sizeCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media.SIZE);
                int widthCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media.WIDT
H);
                int heightCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media.HEIG
HT);
                int mimeCol = cursor.getColumnIndexOrThrow(MediaStore.Video.Media.MIME
_TYPE);
                int durationCol = cursor.getColumnIndex(MediaStore.Video.Media.DURATION)
;
                int dateTakenCol = cursor.getColumnIndex(MediaStore.Video.Media.DATE_TA
KEN);
                int dateAddedCol = cursor.getColumnIndex(MediaStore.Video.Media.DATE_AD
DED);

                while (cursor.moveToNext()) {
                    long id = cursor.getLong(idCol);
                    String name = cursor.getString(nameCol);
                    String path = cursor.getString(dataCol);
                    long size = cursor.getLong(sizeCol);
                    int width = cursor.getInt(widthCol);
                    int height = cursor.getInt(heightCol);
                    String mimeType = cursor.getString(mimeCol);
                    long duration = durationCol != -1 ? cursor.getLong(durationCol) : 0;

                    if (mimeType == null || mimeType.isEmpty()) {
                        mimeType = "video/mp4";
                    }

                    Photo video = new Photo();
                    video.setId(String.valueOf(id));
                    video.setFileName(name != null ? name : "VID_" + id);
                    video.setFilePath(path);
                    video.setFileSize(size);
                    video.setWidth(width);
                    video.setHeight(height);
```

```
                video.setMimeType(mimeType);
                video.setVideo(true);
                video.setDuration(duration);

                long dateTakenMs = 0;
                if (dateTakenCol != -1) {
                    dateTakenMs = cursor.getLong(dateTakenCol);
                }
                if (dateTakenMs <= 0 && dateAddedCol != -1) {
                    dateTakenMs = cursor.getLong(dateAddedCol) * 1000L;
                }
                if (dateTakenMs > 0) {
                    video.setDateTaken(new Date(dateTakenMs));
                    video.setCreatedAt(new Date(dateTakenMs));
                } else {
                    video.setCreatedAt(new Date());
                }

                Uri contentUri = ContentUris.withAppendedId(videoUri, id);
                video.setThumbnailPath(contentUri.toString());

                mediaList.add(video);
            }
        }
    } catch (Exception e) {
        Log.e(TAG, "Error querying MediaStore videos", e);
    }

    // Sort unified list descending by date
    mediaList.sort((p1, p2) -> {
        Date d1 = p1.getDateTaken() != null ? p1.getDateTaken() : p1.getCreatedAt();
        Date d2 = p2.getDateTaken() != null ? p2.getDateTaken() : p2.getCreatedAt();
        if (d1 == null && d2 == null) return 0;
        if (d1 == null) return 1;
        if (d2 == null) return -1;
        return d2.compareTo(d1);
    });

    return mediaList;
}

/**
 * Loads local photos and videos asynchronously.
 */
public static void loadLocalMediaAsync(Context context, OnMediaLoadedListener listen
er) {
    EXECUTOR.execute(() -> {
        try {
            List<Photo> list = loadLocalMedia(context);
```

```
            MAIN_HANDLER.post(() -> {
                if (listener != null) listener.onMediaLoaded(list);
            });
        } catch (Exception e) {
            Log.e(TAG, "Error loading media asynchronously", e);
            MAIN_HANDLER.post(() -> {
                if (listener != null) listener.onError(e.getMessage());
            });
        }
    });
}

// ---------------------------------------------------------------------
// EXIF Metadata Operations
// ---------------------------------------------------------------------

/**
 * Retrieves an ExifInterface object for the given file path.
 */
public static ExifInterface getExifData(String filePath) {
    if (filePath == null || filePath.isEmpty()) return null;
    try {
        return new ExifInterface(filePath);
    } catch (IOException e) {
        Log.e(TAG, "Failed to get EXIF data from: " + filePath, e);
        return null;
    }
}

/**
 * Extracts readable EXIF metadata details into a Map.
 */
public static Map<String, String> getExifDetails(String filePath) {
    Map<String, String> details = new LinkedHashMap<>();
    ExifInterface exif = getExifData(filePath);
    if (exif == null) return details;

    String make = exif.getAttribute(ExifInterface.TAG_MAKE);
    String model = exif.getAttribute(ExifInterface.TAG_MODEL);
    String date = exif.getAttribute(ExifInterface.TAG_DATETIME);
    String fNumber = exif.getAttribute(ExifInterface.TAG_F_NUMBER);
    String expTime = exif.getAttribute(ExifInterface.TAG_EXPOSURE_TIME);
    String iso = exif.getAttribute(ExifInterface.TAG_ISO_SPEED_RATINGS);
    String focalLength = exif.getAttribute(ExifInterface.TAG_FOCAL_LENGTH);
    String flash = exif.getAttribute(ExifInterface.TAG_FLASH);
    String width = exif.getAttribute(ExifInterface.TAG_IMAGE_WIDTH);
    String length = exif.getAttribute(ExifInterface.TAG_IMAGE_LENGTH);

    if (make != null || model != null) {
```

```
            String device = (make != null ? make : "") + " " + (model != null ? model : "");
            details.put("Thiết bị", device.trim());
        }
        if (date != null) {
            details.put("Thời gian chụp", date);
        }
        if (width != null && length != null && !width.equals("0") && !length.equals("0")) {
            details.put("Độ phân giải", width + " × " + length);
        }
        if (fNumber != null) {
            details.put("Khẩu độ", "f/" + fNumber);
        }
        if (expTime != null) {
            try {
                double sec = Double.parseDouble(expTime);
                if (sec < 1.0 && sec > 0) {
                    details.put("Tốc độ màn trập", "1/" + Math.round(1.0 / sec) + " s");
                } else {
                    details.put("Tốc độ màn trập", expTime + " s");
                }
            } catch (NumberFormatException e) {
                details.put("Tốc độ màn trập", expTime);
            }
        }
        if (iso != null) {
            details.put("ISO", "ISO-" + iso);
        }
        if (focalLength != null) {
            details.put("Tiêu cự", focalLength + " mm");
        }
        if (flash != null) {
            details.put("Flash", flash.equals("0") ? "Tắt" : "Bật");
        }

        // GPS coordinates
        float[] latLong = new float[2];
        if (exif.getLatLong(latLong)) {
            details.put("Vị trí (GPS)", String.format(Locale.getDefault(), "%.5f, %.5f", latLong[
0], latLong[1]));
        }

        return details;
    }

    /**
     * Extracts latitude and longitude from EXIF data.
     * Returns float[]{latitude, longitude} or null if not available.
     */
    public static float[] getExifGpsCoordinates(String filePath) {
```

```
        ExifInterface exif = getExifData(filePath);
        if (exif == null) return null;

        float[] latLong = new float[2];
        if (exif.getLatLong(latLong)) {
            return latLong;
        }
        return null;
    }

    // ---------------------------------------------------------------------
    // Thumbnail Generation
    // ---------------------------------------------------------------------

    /**
     * Generates a thumbnail Bitmap from a file path with specified max dimensions.
     */
    public static Bitmap generateThumbnail(String filePath, int width, int height) {
        if (filePath == null || filePath.isEmpty()) return null;

        try {
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.Q) {
                File file = new File(filePath);
                if (file.exists()) {
                    return ThumbnailUtils.createImageThumbnail(file, new Size(width, height), nul
l);
                }
            }

            // Fallback for older APIs or if file thumbnail failed
            BitmapFactory.Options options = new BitmapFactory.Options();
            options.inJustDecodeBounds = true;
            BitmapFactory.decodeFile(filePath, options);

            options.inSampleSize = calculateInSampleSize(options, width, height);
            options.inJustDecodeBounds = false;
            Bitmap bitmap = BitmapFactory.decodeFile(filePath, options);

            // Handle EXIF orientation rotation
            return rotateBitmapIfRequired(bitmap, filePath);
        } catch (Exception e) {
            Log.e(TAG, "Error generating thumbnail for: " + filePath, e);
            return null;
        }
    }

    /**
     * Generates a thumbnail Bitmap from a Content URI.
     */
```

```
    public static Bitmap generateThumbnail(Context context, Uri uri, int width, int height) {
        if (context == null || uri == null) return null;

        try {
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.Q) {
                return context.getContentResolver().loadThumbnail(uri, new Size(width, height),
null);
            } else {
                InputStream input = context.getContentResolver().openInputStream(uri);
                BitmapFactory.Options options = new BitmapFactory.Options();
                options.inJustDecodeBounds = true;
                BitmapFactory.decodeStream(input, null, options);
                if (input != null) input.close();

                options.inSampleSize = calculateInSampleSize(options, width, height);
                options.inJustDecodeBounds = false;

                InputStream input2 = context.getContentResolver().openInputStream(uri);
                Bitmap bitmap = BitmapFactory.decodeStream(input2, null, options);
                if (input2 != null) input2.close();
                return bitmap;
            }
        } catch (Exception e) {
            Log.e(TAG, "Error generating thumbnail from Uri", e);
            return null;
        }
    }

    // ---------------------------------------------------------------------
    // Formatting Utilities
    // ---------------------------------------------------------------------

    /**
     * Formats byte size into human readable string (B, KB, MB, GB).
     */
    public static String formatFileSize(long bytes) {
        if (bytes <= 0) return "0 B";
        final String[] units = new String[]{"B", "KB", "MB", "GB", "TB"};
        int digitGroups = (int) (Math.log10(bytes) / Math.log10(1024));
        if (digitGroups >= units.length) digitGroups = units.length - 1;
        return new DecimalFormat("#,##0.#").format(bytes / Math.pow(1024, digitGroups)) +
" " + units[digitGroups];
    }

    /**
     * Formats Date to dd/MM/yyyy.
     */
    public static String formatDate(Date date) {
        if (date == null) return "";
```

```
        SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy", Locale.getDefault())
;
        return sdf.format(date);
    }

    /**
     * Formats timestamp in milliseconds to dd/MM/yyyy.
     */
    public static String formatDate(long timestamp) {
        return formatDate(new Date(timestamp));
    }

    /**
     * Formats Date to dd/MM/yyyy HH:mm.
     */
    public static String formatDateTime(Date date) {
        if (date == null) return "";
        SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy HH:mm", Locale.get
Default());
        return sdf.format(date);
    }

    /**
     * Formats video duration in milliseconds to mm:ss or hh:mm:ss.
     */
    public static String formatDuration(long durationMs) {
        if (durationMs <= 0) return "00:00";
        long totalSeconds = durationMs / 1000;
        long hours = totalSeconds / 3600;
        long minutes = (totalSeconds % 3600) / 60;
        long seconds = totalSeconds % 60;

        if (hours > 0) {
            return String.format(Locale.getDefault(), "%02d:%02d:%02d", hours, minutes, seco
nds);
        } else {
            return String.format(Locale.getDefault(), "%02d:%02d", minutes, seconds);
        }
    }

    /**
     * Formats Date to a friendly relative representation in Vietnamese.
     */
    public static String formatRelativeTime(Date date) {
        if (date == null) return "";

        long now = System.currentTimeMillis();
        long diff = now - date.getTime();
```

```
        if (diff < 60 * 1000) {
            return "Vừa xong";
        } else if (diff < 60 * 60 * 1000) {
            long minutes = diff / (60 * 1000);
            return minutes + " phút trước";
        }

        Calendar nowCal = Calendar.getInstance();
        Calendar dateCal = Calendar.getInstance();
        dateCal.setTime(date);

        SimpleDateFormat timeFormat = new SimpleDateFormat("HH:mm", Locale.getDefaul
t());

        if (nowCal.get(Calendar.YEAR) == dateCal.get(Calendar.YEAR) &&
                nowCal.get(Calendar.DAY_OF_YEAR) == dateCal.get(Calendar.DAY_OF_YE
AR)) {
            return "Hôm nay, " + timeFormat.format(date);
        }

        nowCal.add(Calendar.DAY_OF_YEAR, -1);
        if (nowCal.get(Calendar.YEAR) == dateCal.get(Calendar.YEAR) &&
                nowCal.get(Calendar.DAY_OF_YEAR) == dateCal.get(Calendar.DAY_OF_YE
AR)) {
            return "Hôm qua, " + timeFormat.format(date);
        }

        return formatDateTime(date);
    }

    // ---------------------------------------------------------------------
    // Image Dimension and Compression Utilities
    // ---------------------------------------------------------------------

    /**
     * Returns dimensions of an image without loading full bitmap into memory.
     * @return int[]{width, height}
     */
    public static int[] getImageDimensions(String filePath) {
        if (filePath == null || filePath.isEmpty()) return new int[]{0, 0};

        BitmapFactory.Options options = new BitmapFactory.Options();
        options.inJustDecodeBounds = true;
        BitmapFactory.decodeFile(filePath, options);
        return new int[]{options.outWidth, options.outHeight};
    }

    /**
     * Returns dimensions of an image from a Content URI.
```

```
     */
    public static int[] getImageDimensions(Context context, Uri uri) {
        if (context == null || uri == null) return new int[]{0, 0};

        try (InputStream is = context.getContentResolver().openInputStream(uri)) {
            BitmapFactory.Options options = new BitmapFactory.Options();
            options.inJustDecodeBounds = true;
            BitmapFactory.decodeStream(is, null, options);
            return new int[]{options.outWidth, options.outHeight};
        } catch (Exception e) {
            Log.e(TAG, "Error getting image dimensions from Uri", e);
            return new int[]{0, 0};
        }
    }

    /**
     * Compresses an image file down to a JPEG byte array with maximum dimension and qu
ality.
     */
    public static byte[] compressImage(String filePath, int quality, int maxDimension) {
        if (filePath == null) return null;

        BitmapFactory.Options options = new BitmapFactory.Options();
        options.inJustDecodeBounds = true;
        BitmapFactory.decodeFile(filePath, options);

        options.inSampleSize = calculateInSampleSize(options, maxDimension, maxDimensio
n);
        options.inJustDecodeBounds = false;

        Bitmap scaledBitmap = BitmapFactory.decodeFile(filePath, options);
        if (scaledBitmap == null) return null;

        Bitmap rotatedBitmap = rotateBitmapIfRequired(scaledBitmap, filePath);

        ByteArrayOutputStream baos = new ByteArrayOutputStream();
        int safeQuality = Math.max(10, Math.min(quality, 100));
        rotatedBitmap.compress(Bitmap.CompressFormat.JPEG, safeQuality, baos);

        if (rotatedBitmap != scaledBitmap) {
            scaledBitmap.recycle();
        }

        return baos.toByteArray();
    }

    /**
     * Compresses an image file and saves the result to a cache file.
     */
```

```
    public static File compressImageToFile(Context context, String sourcePath, int quality) {
        if (context == null || sourcePath == null) return null;

        byte[] compressedData = compressImage(sourcePath, quality, 1920);
        if (compressedData == null) return null;

        try {
            File cacheDir = new File(context.getCacheDir(), "compressed_photos");
            if (!cacheDir.exists()) {
                cacheDir.mkdirs();
            }

            String fileName = "compressed_" + System.currentTimeMillis() + ".jpg";
            File destFile = new File(cacheDir, fileName);

            try (FileOutputStream fos = new FileOutputStream(destFile)) {
                fos.write(compressedData);
                fos.flush();
            }
            return destFile;
        } catch (IOException e) {
            Log.e(TAG, "Error saving compressed file", e);
            return null;
        }
    }

    // ------------------------------------------------------------------------
    // Helper calculation methods
    // ------------------------------------------------------------------------

    private static int calculateInSampleSize(BitmapFactory.Options options, int reqWidth, int
 reqHeight) {
        final int height = options.outHeight;
        final int width = options.outWidth;
        int inSampleSize = 1;

        if (height > reqHeight || width > reqWidth) {
            final int halfHeight = height / 2;
            final int halfWidth = width / 2;

            while ((halfHeight / inSampleSize) >= reqHeight && (halfWidth / inSampleSize) >
= reqWidth) {
                inSampleSize *= 2;
            }
        }
        return inSampleSize;
    }

    private static Bitmap rotateBitmapIfRequired(Bitmap img, String selectedImage) {
```

```
        if (img == null || selectedImage == null) return img;

        try {
            ExifInterface ei = new ExifInterface(selectedImage);
            int orientation = ei.getAttributeInt(ExifInterface.TAG_ORIENTATION, ExifInterfa
ce.ORIENTATION_NORMAL);

            switch (orientation) {
                case ExifInterface.ORIENTATION_ROTATE_90:
                    return rotateImage(img, 90);
                case ExifInterface.ORIENTATION_ROTATE_180:
                    return rotateImage(img, 180);
                case ExifInterface.ORIENTATION_ROTATE_270:
                    return rotateImage(img, 270);
                default:
                    return img;
            }
        } catch (IOException e) {
            Log.e(TAG, "Could not check EXIF orientation", e);
            return img;
        }
    }

    private static Bitmap rotateImage(Bitmap img, int degree) {
        Matrix matrix = new Matrix();
        matrix.postRotate(degree);
        Bitmap rotatedImg = Bitmap.createBitmap(img, 0, 0, img.getWidth(), img.getHeight(),
 matrix, true);
        if (rotatedImg != img) {
            img.recycle();
        }
        return rotatedImg;
    }
}
```

1 4 Thư mục models

Đường dẫn: app/src/main/java/com/example/appanh/models

Các lớp mô hình dữ liệu dùng cho hồ sơ, ảnh, album và thông báo. Số file Java: 4.

Danh sách: Album.java, Notification.java, Photo.java, User.java.

1 4 1 Album

Nguồn: app/src/main/java/com/example/appanh/models/Album.java

Vai trò: Mô hình album: mã, chủ sở hữu, tên, ảnh bìa và mốc thời gian.

Mã nguồn đầy đủ

```
package com.example.appanh.models;

import com.google.firebase.firestore.DocumentId;
import com.google.firebase.firestore.ServerTimestamp;
import java.io.Serializable;
import java.util.ArrayList;
import java.util.Date;
import java.util.List;
import java.util.Objects;

/**
 * Model class representing a photo album.
 */
public class Album implements Serializable {

    @DocumentId
    private String id;
    private String userId;
    private String name;
    private String description;
    private String coverPhotoUrl;
    private String coverPhotoPath;
    private int photoCount;
    private int videoCount;
    private List<String> photoIds;

    @ServerTimestamp
    private Date createdAt;

    @ServerTimestamp
    private Date updatedAt;

    // Empty constructor for Firestore
    public Album() {
        this.photoIds = new ArrayList<>();
    }

    public Album(String userId, String name, String description) {
        this.userId = userId;
        this.name = name;
        this.description = description;
        this.photoCount = 0;
        this.videoCount = 0;
        this.photoIds = new ArrayList<>();
    }

    // Getters and Setters

    public String getId() {
```

```
    return id;
}

public void setId(String id) {
    this.id = id;
}

public String getUserId() {
    return userId;
}

public void setUserId(String userId) {
    this.userId = userId;
}

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public String getDescription() {
    return description;
}

public void setDescription(String description) {
    this.description = description;
}

public String getCoverPhotoUrl() {
    return coverPhotoUrl;
}

public void setCoverPhotoUrl(String coverPhotoUrl) {
    this.coverPhotoUrl = coverPhotoUrl;
}

public String getCoverPhotoPath() {
    return coverPhotoPath;
}

public void setCoverPhotoPath(String coverPhotoPath) {
    this.coverPhotoPath = coverPhotoPath;
}

public int getPhotoCount() {
    return photoCount;
```

```
}

public void setPhotoCount(int photoCount) {
   this.photoCount = photoCount;
}

public int getVideoCount() {
   return videoCount;
}

public void setVideoCount(int videoCount) {
   this.videoCount = videoCount;
}

public List<String> getPhotoIds() {
   if (photoIds == null) {
      photoIds = new ArrayList<>();
   }
   return photoIds;
}

public void setPhotoIds(List<String> photoIds) {
   this.photoIds = photoIds != null ? photoIds : new ArrayList<>();
}

public Date getCreatedAt() {
   return createdAt;
}

public void setCreatedAt(Date createdAt) {
   this.createdAt = createdAt;
}

public Date getUpdatedAt() {
   return updatedAt;
}

public void setUpdatedAt(Date updatedAt) {
   this.updatedAt = updatedAt;
}

@Override
public boolean equals(Object o) {
   if (this == o) return true;
   if (o == null || getClass() != o.getClass()) return false;
   Album album = (Album) o;
   return Objects.equals(id, album.id);
}
```

```
    @Override
    public int hashCode() {
        return Objects.hash(id);
    }

    @Override
    public String toString() {
        return "Album{" +
                "id='" + id + '\'' +
                ", userId='" + userId + '\'' +
                ", name='" + name + '\'' +
                ", photoCount=" + photoCount +
                ", videoCount=" + videoCount +
                '}';
    }
}
```

1 4 2 Notification

Nguồn: app/src/main/java/com/example/appanh/models/Notification.java

Vai trò: Mô hình dữ liệu thông báo hoạt động.

Mã nguồn đầy đủ

```
package com.example.appanh.models;

import com.google.firebase.firestore.DocumentId;
import com.google.firebase.firestore.ServerTimestamp;
import java.io.Serializable;
import java.util.Date;
import java.util.Objects;

/**
 * Model class representing a user notification in the gallery app.
 */
public class Notification implements Serializable {

    @DocumentId
    private String id;
    private String userId;
    private String title;
    private String message;
    private String type; // "backup", "album", "share", "system"
    private boolean isRead;
    private String thumbnailUrl;

    @ServerTimestamp
    private Date createdAt;
```

```
// Empty constructor for Firestore
public Notification() {}

public Notification(String userId, String title, String message, String type) {
    this.userId = userId;
    this.title = title;
    this.message = message;
    this.type = type;
    this.isRead = false;
}

// Getters and Setters

public String getId() {
    return id;
}

public void setId(String id) {
    this.id = id;
}

public String getUserId() {
    return userId;
}

public void setUserId(String userId) {
    this.userId = userId;
}

public String getTitle() {
    return title;
}

public void setTitle(String title) {
    this.title = title;
}

public String getMessage() {
    return message;
}

public void setMessage(String message) {
    this.message = message;
}

public String getType() {
    return type;
}
```

```
public void setType(String type) {
   this.type = type;
}

public boolean isRead() {
   return isRead;
}

public void setRead(boolean read) {
   isRead = read;
}

public String getThumbnailUrl() {
   return thumbnailUrl;
}

public void setThumbnailUrl(String thumbnailUrl) {
   this.thumbnailUrl = thumbnailUrl;
}

public Date getCreatedAt() {
   return createdAt;
}

public void setCreatedAt(Date createdAt) {
   this.createdAt = createdAt;
}

@Override
public boolean equals(Object o) {
   if (this == o) return true;
   if (o == null || getClass() != o.getClass()) return false;
   Notification that = (Notification) o;
   return Objects.equals(id, that.id);
}

@Override
public int hashCode() {
   return Objects.hash(id);
}

@Override
public String toString() {
   return "Notification{" +
         "id='" + id + '\'' +
         ", userId='" + userId + '\'' +
         ", title='" + title + '\'' +
         ", type='" + type + '\'' +
```

```
            ", isRead=" + isRead +
            '}';
    }
}
```

1 4 3 Photo

Nguồn: app/src/main/java/com/example/appanh/models/Photo.java

Vai trò: Mô hình ảnh hoặc video, gồm tệp, kích thước, loại, trạng thái yêu thích, ẩn, thùng rác và danh sách album.

Mã nguồn đầy đủ

```
package com.example.appanh.models;

import com.google.firebase.firestore.DocumentId;
import com.google.firebase.firestore.ServerTimestamp;
import java.io.Serializable;
import java.util.ArrayList;
import java.util.Date;
import java.util.List;
import java.util.Objects;

/**
 * Model class representing a photo or video item in the gallery.
 */
public class Photo implements Serializable {

    @DocumentId
    private String id;
    private String userId;
    private String fileName;
    private String filePath;
    private String thumbnailPath;
    private String firebaseUrl;
    private long fileSize;
    private int width;
    private int height;
    private String mimeType; // image/jpeg, image/png, video/mp4
    private boolean isVideo;
    private long duration; // for videos, in milliseconds
    private String deviceName;
    private String location;
    private boolean isFavorite;
    private boolean isDeleted;
    private boolean hidden;
    private List<String> albumIds;
```

```
    @ServerTimestamp
    private Date createdAt;

    @ServerTimestamp
    private Date updatedAt;

    private Date deletedAt;
    private Date dateTaken;

    // Empty constructor for Firestore
    public Photo() {
        this.albumIds = new ArrayList<>();
    }

    public Photo(String userId, String fileName, String filePath, long fileSize, int width, int h
eight, String mimeType) {
        this.userId = userId;
        this.fileName = fileName;
        this.filePath = filePath;
        this.fileSize = fileSize;
        this.width = width;
        this.height = height;
        this.mimeType = mimeType;
        this.isVideo = mimeType != null && mimeType.startsWith("video");
        this.isFavorite = false;
        this.isDeleted = false;
        this.albumIds = new ArrayList<>();
    }

    // Getters and Setters

    public String getId() {
        return id;
    }

    public void setId(String id) {
        this.id = id;
    }

    public String getUserId() {
        return userId;
    }

    public void setUserId(String userId) {
        this.userId = userId;
    }

    public String getFileName() {
        return fileName;
```

```
}

public void setFileName(String fileName) {
    this.fileName = fileName;
}

public String getFilePath() {
    return filePath;
}

public void setFilePath(String filePath) {
    this.filePath = filePath;
}

public String getThumbnailPath() {
    return thumbnailPath;
}

public void setThumbnailPath(String thumbnailPath) {
    this.thumbnailPath = thumbnailPath;
}

public String getFirebaseUrl() {
    return firebaseUrl;
}

public void setFirebaseUrl(String firebaseUrl) {
    this.firebaseUrl = firebaseUrl;
}

public long getFileSize() {
    return fileSize;
}

public void setFileSize(long fileSize) {
    this.fileSize = fileSize;
}

public int getWidth() {
    return width;
}

public void setWidth(int width) {
    this.width = width;
}

public int getHeight() {
    return height;
}
```

```
public void setHeight(int height) {
   this.height = height;
}

public String getMimeType() {
   return mimeType;
}

public void setMimeType(String mimeType) {
   this.mimeType = mimeType;
   this.isVideo = mimeType != null && mimeType.startsWith("video");
}

public boolean isVideo() {
   return isVideo;
}

public void setVideo(boolean video) {
   isVideo = video;
}

public long getDuration() {
   return duration;
}

public void setDuration(long duration) {
   this.duration = duration;
}

public String getDeviceName() {
   return deviceName;
}

public void setDeviceName(String deviceName) {
   this.deviceName = deviceName;
}

public String getLocation() {
   return location;
}

public void setLocation(String location) {
   this.location = location;
}

public boolean isFavorite() {
   return isFavorite;
}
```

```
public void setFavorite(boolean favorite) {
    isFavorite = favorite;
}

public boolean isDeleted() {
    return isDeleted;
}

public boolean isHidden() { return hidden; }
public void setHidden(boolean hidden) { this.hidden = hidden; }

public void setDeleted(boolean deleted) {
    isDeleted = deleted;
}

public List<String> getAlbumIds() {
    if (albumIds == null) {
        albumIds = new ArrayList<>();
    }
    return albumIds;
}

public void setAlbumIds(List<String> albumIds) {
    this.albumIds = albumIds != null ? albumIds : new ArrayList<>();
}

public Date getCreatedAt() {
    return createdAt;
}

public void setCreatedAt(Date createdAt) {
    this.createdAt = createdAt;
}

public Date getUpdatedAt() {
    return updatedAt;
}

public void setUpdatedAt(Date updatedAt) {
    this.updatedAt = updatedAt;
}

public Date getDeletedAt() {
    return deletedAt;
}

public void setDeletedAt(Date deletedAt) {
    this.deletedAt = deletedAt;
```

```
    }

    public Date getDateTaken() {
        return dateTaken;
    }

    public void setDateTaken(Date dateTaken) {
        this.dateTaken = dateTaken;
    }

    @Override
    public boolean equals(Object o) {
        if (this == o) return true;
        if (o == null || getClass() != o.getClass()) return false;
        Photo photo = (Photo) o;
        return Objects.equals(id, photo.id) ||
                (filePath != null && Objects.equals(filePath, photo.filePath));
    }

    @Override
    public int hashCode() {
        return Objects.hash(id, filePath);
    }

    @Override
    public String toString() {
        return "Photo{" +
                "id='" + id + '\'' +
                ", fileName='" + fileName + '\'' +
                ", filePath='" + filePath + '\'' +
                ", isVideo=" + isVideo +
                ", isFavorite=" + isFavorite +
                ", isDeleted=" + isDeleted +
                '}';
    }
}
```

1 4 4 User

Nguồn: app/src/main/java/com/example/appanh/models/User.java

Vai trò: Mô hình hồ sơ người dùng và các thuộc tính tài khoản.

Mã nguồn đầy đủ

```
package com.example.appanh.models;

import com.google.firebase.firestore.DocumentId;
import com.google.firebase.firestore.ServerTimestamp;
```

```
import java.io.Serializable;
import java.util.Date;
import java.util.Objects;

/**
 * Model class representing a registered user and preferences.
 */
public class User implements Serializable {

    @DocumentId
    private String id;
    private String email;
    private String displayName;
    private String photoUrl;
    private String phoneNumber;
    private boolean isEmailVerified;
    private String provider; // "email", "google"
    private long storageUsed; // in bytes
    private boolean autoBackup;
    private boolean wifiOnlyBackup;
    private boolean darkMode;
    private String language; // "vi", "en"
    private String sortOrder; // "date_desc", "date_asc", "name", "size"

    @ServerTimestamp
    private Date createdAt;

    @ServerTimestamp
    private Date lastLoginAt;

    // Empty constructor for Firestore
    public User() {}

    public User(String id, String email, String displayName) {
        this.id = id;
        this.email = email;
        this.displayName = displayName;
        this.provider = "email";
        this.storageUsed = 0L;
        this.autoBackup = false;
        this.wifiOnlyBackup = true;
        this.darkMode = false;
        this.language = "vi";
        this.sortOrder = "date_desc";
    }

    // Getters and Setters

    public String getId() {
```

```
    return id;
}

public void setId(String id) {
    this.id = id;
}

public String getEmail() {
    return email;
}

public void setEmail(String email) {
    this.email = email;
}

public String getDisplayName() {
    return displayName;
}

public void setDisplayName(String displayName) {
    this.displayName = displayName;
}

public String getPhotoUrl() {
    return photoUrl;
}

public void setPhotoUrl(String photoUrl) {
    this.photoUrl = photoUrl;
}

public String getPhoneNumber() {
    return phoneNumber;
}

public void setPhoneNumber(String phoneNumber) {
    this.phoneNumber = phoneNumber;
}

public boolean isEmailVerified() {
    return isEmailVerified;
}

public void setEmailVerified(boolean emailVerified) {
    isEmailVerified = emailVerified;
}

public String getProvider() {
    return provider;
```

```
}

public void setProvider(String provider) {
   this.provider = provider;
}

public long getStorageUsed() {
   return storageUsed;
}

public void setStorageUsed(long storageUsed) {
   this.storageUsed = storageUsed;
}

public boolean isAutoBackup() {
   return autoBackup;
}

public void setAutoBackup(boolean autoBackup) {
   this.autoBackup = autoBackup;
}

public boolean isWifiOnlyBackup() {
   return wifiOnlyBackup;
}

public void setWifiOnlyBackup(boolean wifiOnlyBackup) {
   this.wifiOnlyBackup = wifiOnlyBackup;
}

public boolean isDarkMode() {
   return darkMode;
}

public void setDarkMode(boolean darkMode) {
   this.darkMode = darkMode;
}

public String getLanguage() {
   return language;
}

public void setLanguage(String language) {
   this.language = language;
}

public String getSortOrder() {
   return sortOrder;
}
```

```
    public void setSortOrder(String sortOrder) {
        this.sortOrder = sortOrder;
    }

    public Date getCreatedAt() {
        return createdAt;
    }

    public void setCreatedAt(Date createdAt) {
        this.createdAt = createdAt;
    }

    public Date getLastLoginAt() {
        return lastLoginAt;
    }

    public void setLastLoginAt(Date lastLoginAt) {
        this.lastLoginAt = lastLoginAt;
    }

    @Override
    public boolean equals(Object o) {
        if (this == o) return true;
        if (o == null || getClass() != o.getClass()) return false;
        User user = (User) o;
        return Objects.equals(id, user.id);
    }

    @Override
    public int hashCode() {
        return Objects.hash(id);
    }

    @Override
    public String toString() {
        return "User{" +
                "id='" + id + '\'' +
                ", email='" + email + '\'' +
                ", displayName='" + displayName + '\'' +
                ", provider='" + provider + '\'' +
                ", autoBackup=" + autoBackup +
                '}';
    }
}
```

2 Giao diện ứng dụng

Nguồn giao diện nằm trong app/src/main/res. Layout XML mô tả vị trí và cấu trúc thành phần; menu khai báo hành động; drawable chứa icon và nền; color và values cung cấp màu và style. Các thư mục night chứa tài nguyên thay thế để cùng bố cục chuyển sang chế độ tối.

HiddenMediaActivity sử dụng lại activity_main.xml và thay tiêu đề cùng trạng thái lọc; không có một layout XML riêng. Vùng ảnh của activity_edit_photo.xml sử dụng EditorCanvasView viết bằng Java để hiển thị khung cắt và nét vẽ.

2 1 Khai báo ứng dụng

2 1 1 AndroidManifest

Nguồn: app/src/main/AndroidManifest.xml

Vai trò: Khai báo Application, các Activity, quyền mạng, FileProvider và chính sách chia sẻ hoặc sao lưu dữ liệu hệ thống.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <!-- Permissions -->
    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />

    <uses-feature
        android:name="android.hardware.camera"
        android:required="false" />

    <application
        android:name=".GalleryApplication"
        android:allowBackup="false"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppẢnh"
        android:usesCleartextTraffic="false"
        tools:targetApi="31">

        <provider android:name="androidx.core.content.FileProvider" android:authorities="${
applicationId}.files" android:exported="false" android:grantUriPermissions="true">
            <meta-data android:name="android.support.FILE_PROVIDER_PATHS" android:re
source="@xml/file_paths" />
        </provider>
```

```
    <activity android:name=".AccountActivity" android:exported="false" android:window
SoftInputMode="adjustResize" />
    <activity android:name=".SearchActivity" android:exported="false" />
    <activity android:name=".HiddenMediaActivity" android:exported="false" />
    <activity android:name=".CreateAlbumActivity" android:exported="false" />
    <!-- Screen 1: Splash Screen -->
    <activity
      android:name=".SplashActivity"
      android:exported="true"
      android:theme="@style/Theme.AppẢnh.Splash"
      android:windowSoftInputMode="adjustResize">
      <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
      </intent-filter>
    </activity>

    <!-- Screen 2: Login -->
    <activity
      android:name=".LoginActivity"
      android:exported="false"
      android:windowSoftInputMode="adjustResize" />

    <!-- Screen 3: Register -->
    <activity
      android:name=".RegisterActivity"
      android:exported="false"
      android:windowSoftInputMode="adjustResize" />

    <!-- Screen 4: OTP Verification -->
    <activity
      android:name=".OtpActivity"
      android:exported="false"
      android:windowSoftInputMode="adjustResize" />

    <!-- Screen 5: Main Gallery -->
    <activity
      android:name=".MainActivity"
      android:exported="false"
      android:windowSoftInputMode="adjustPan" />

    <!-- Screen 9: Photo Detail -->
    <activity
      android:name=".PhotoDetailActivity"
      android:exported="false"
      android:theme="@style/Theme.AppẢnh.Fullscreen" />

    <!-- Screen 10: Edit Photo -->
    <activity
```

```
            android:name=".EditPhotoActivity"
            android:exported="false"
            android:windowSoftInputMode="adjustResize" />

        <!-- Screen 12: Album List -->
        <activity
            android:name=".AlbumActivity"
            android:exported="false" />

        <!-- Screen 14: Album Detail -->
        <activity
            android:name=".AlbumDetailActivity"
            android:exported="false" />

        <!-- Screen 15: Trash -->
        <activity
            android:name=".TrashActivity"
            android:exported="false" />

        <!-- Screen 16: Settings -->
        <activity
            android:name=".SettingsActivity"
            android:exported="false" />

        <!-- Screen 17: Google Sign-In -->
        <activity
            android:name=".GoogleSignInActivity"
            android:exported="false" />

        <!-- Screen 18: Backup -->
        <activity
            android:name=".BackupActivity"
            android:exported="false" />

        <!-- Screen 20: Notifications -->
        <activity
            android:name=".NotificationsActivity"
            android:exported="false" />

        <!-- Screen 21: Share (used as bottom sheet from other activities) -->
        <activity
            android:name=".ShareActivity"
            android:exported="false"
            android:theme="@style/Theme.AppẢnh.Dialog" />

    </application>

</manifest>
```

2 2 Thư mục layout

Đường dẫn: app/src/main/res/layout

Bố cục màn hình, các sheet, hộp thoại và ô dữ liệu. Số file XML: 29.

Danh sách: activity_account.xml, activity_album.xml, activity_album_detail.xml, activity_backup.xml, activity_create_album.xml, activity_dark_gallery.xml, activity_edit_photo.xml, activity_google_sign_in.xml, activity_login.xml, activity_main.xml, activity_notifications.xml, activity_otp.xml, activity_photo_detail.xml, activity_register.xml, activity_search.xml, activity_settings.xml, activity_splash.xml, activity_trash.xml, dialog_photo_info.xml, item_album.xml, item_media.xml, item_media_row.xml, item_notification.xml, item_photo_grid.xml, sheet_album_options.xml, sheet_filter.xml, sheet_options.xml, sheet_photo_context.xml, sheet_share.xml.

2 2 1 activity_account

Nguồn: app/src/main/res/layout/activity_account.xml

Vai trò: Hồ sơ tài khoản đọc và ghi Firestore.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/screenRoot" android:layout_width="match_parent"
    android:layout_height="match_parent" android:orientation="vertical" android:backgroun
d="@color/background">
    <com.google.android.material.appbar.MaterialToolbar
        android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_heig
ht="56dp"
        app:title="Tài khoản" app:titleTextAppearance="@style/GalleryToolbarTitle"
        app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_
back"
        app:navigationIconTint="@color/text_primary"/>
    <androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="0dp"
        android:layout_weight="1" android:fillViewport="true" android:clipToPadding="false
">
        <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent"
            android:orientation="vertical" android:padding="24dp">
            <FrameLayout android:layout_width="80dp" android:layout_height="80dp" android
:layout_gravity="center_horizontal">
                <TextView android:id="@+id/profileInitial" android:layout_width="match_paren
t" android:layout_height="match_parent"
                    android:background="@drawable/bg_circle_icon" android:gravity="center" an
```

droid:text="Ả"
android:textColor="@color/primary" android:textSize="32sp" android:fontFamily="sans-serif-medium"/>
<ImageView android:id="@+id/profileAvatar" android:layout_width="match_parent" android:layout_height="match_parent"
android:contentDescription="Ảnh đại diện" android:visibility="gone"/>
</FrameLayout>
<TextView android:id="@+id/profileHeading" android:layout_width="match_parent" android:layout_height="wrap_content"
android:layout_marginTop="16dp" android:text="Hồ sơ của bạn" android:textSize="24sp"
android:gravity="center" android:fontFamily="sans-serif-medium" android:textColor="@color/text_primary"/>
<TextView android:id="@+id/profileSubtitle" android:layout_width="match_parent" android:layout_height="wrap_content"
android:layout_marginTop="6dp" android:textSize="14sp" android:textColor="@color/text_secondary"
android:gravity="center" android:lineSpacingExtra="3dp"/>
<com.google.android.material.button.MaterialButton android:id="@+id/accountLogin" style="@style/PrimaryButton"
android:layout_width="match_parent" android:layout_height="52dp" android:layout_marginTop="28dp"
android:text="Đăng nhập" android:visibility="gone"/>
<LinearLayout android:id="@+id/profileForm" android:layout_width="match_parent" android:layout_height="wrap_content"
android:orientation="vertical" android:layout_marginTop="28dp">
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Thông tin cá nhân"
android:textSize="17sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif-medium"/>
<com.google.android.material.textfield.TextInputLayout style="@style/CustomEditText"
android:layout_width="match_parent" android:layout_height="wrap_content" android:layout_marginTop="16dp"
android:hint="Họ và tên" app:startIconDrawable="@drawable/ic_person" app:startIconTint="@color/text_secondary">
<com.google.android.material.textfield.TextInputEditText android:id="@+id/profileName"
android:layout_width="match_parent" android:layout_height="wrap_content" android:minHeight="56dp"
android:inputType="textPersonName|textCapWords" android:maxLength="80" android:singleLine="true"
android:textSize="16sp" android:textColor="@color/text_primary"/>
</com.google.android.material.textfield.TextInputLayout>
<com.google.android.material.textfield.TextInputLayout style="@style/CustomEditText"
android:layout_width="match_parent" android:layout_height="wrap_content" android:layout_marginTop="16dp"
android:hint="Giới thiệu về bạn" app:counterEnabled="true" app:counterMaxL

```
ength="240">
            <com.google.android.material.textfield.TextInputEditText android:id="@+id/pr
ofileBio"
                android:layout_width="match_parent" android:layout_height="wrap_content
" android:minHeight="96dp"
                android:inputType="textMultiLine|textCapSentences" android:minLines="2"
 android:maxLines="4"
                android:gravity="top|start" android:maxLength="240" android:textSize="16s
p" android:textColor="@color/text_primary"/>
            </com.google.android.material.textfield.TextInputLayout>
            <com.google.android.material.card.MaterialCardView android:layout_width="ma
tch_parent" android:layout_height="wrap_content"
                android:layout_marginTop="16dp" app:cardBackgroundColor="@color/surfac
e" app:cardCornerRadius="16dp"
                app:cardElevation="0dp" app:strokeColor="@color/field_border" app:strokeWi
dth="1dp">
                <LinearLayout android:layout_width="match_parent" android:layout_height="
wrap_content" android:orientation="vertical" android:padding="18dp">
                    <TextView android:layout_width="match_parent" android:layout_height="w
rap_content" android:text="Thông tin đăng nhập"
                        android:textColor="@color/text_primary" android:textSize="16sp" androi
d:fontFamily="sans-serif-medium"/>
                    <TextView android:layout_width="match_parent" android:layout_height="w
rap_content" android:text="Email"
                        android:textColor="@color/text_secondary" android:textSize="13sp" andr
oid:layout_marginTop="16dp"/>
                    <TextView android:id="@+id/profileEmail" android:layout_width="match_
parent" android:layout_height="wrap_content"
                        android:textColor="@color/text_primary" android:textSize="15sp" androi
d:textIsSelectable="true" android:layout_marginTop="4dp"/>
                    <TextView android:id="@+id/profileVerification" android:layout_width="m
atch_parent" android:layout_height="wrap_content"
                        android:textColor="@color/text_secondary" android:textSize="13sp" andr
oid:layout_marginTop="4dp"/>
                    <TextView android:layout_width="match_parent" android:layout_height="w
rap_content" android:text="Số điện thoại"
                        android:textColor="@color/text_secondary" android:textSize="13sp" andr
oid:layout_marginTop="16dp"/>
                    <TextView android:id="@+id/profilePhone" android:layout_width="match_
parent" android:layout_height="wrap_content"
                        android:textColor="@color/text_primary" android:textSize="15sp" androi
d:layout_marginTop="4dp"/>
                </LinearLayout>
            </com.google.android.material.card.MaterialCardView>
            <TextView android:id="@+id/status" android:layout_width="match_parent" andr
oid:layout_height="wrap_content"
                android:textColor="@color/text_secondary" android:textSize="14sp" android:li
neSpacingExtra="3dp"
                android:paddingVertical="16dp" android:accessibilityLiveRegion="polite"/>
```

```
            <ProgressBar android:id="@+id/profileProgress" android:layout_width="28dp" a
ndroid:layout_height="28dp"
                android:layout_gravity="center_horizontal" android:layout_marginBottom="12
dp" android:visibility="gone"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/saveProfi
le" style="@style/PrimaryButton"
                android:layout_width="match_parent" android:layout_height="52dp" android:t
ext="Lưu thay đổi" android:textSize="16sp"/>
        </LinearLayout>
    </LinearLayout>
  </androidx.core.widget.NestedScrollView>
</LinearLayout>
```

2 2 2 activity_album

Nguồn: app/src/main/res/layout/activity_album.xml

Vai trò: Danh sách album hai cột với nút tạo album.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
  <FrameLayout android:layout_width="match_parent" android:layout_height="64dp">
    <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" andr
oid:layout_width="match_parent" android:layout_height="56dp" app:title="Album" app:titl
eTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary"
 app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_pr
imary"/>
    <com.google.android.material.button.MaterialButton android:id="@+id/createAlbum"
style="@style/TextButton" android:layout_width="144dp" android:layout_height="40dp" a
ndroid:layout_gravity="end|center_vertical" android:layout_marginEnd="16dp" android:bac
kgroundTint="@color/primary_light" android:text="Tạo album" android:textSize="15sp" a
pp:icon="@drawable/ic_add" app:iconGravity="textStart" app:cornerRadius="22dp"/>
  </FrameLayout>
  <LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="12dp">
    <TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" an
droid:fontFamily="sans-serif" android:id="@+id/status" android:visibility="gone" android:
paddingVertical="8dp"/>
    <androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_
width="match_parent" android:layout_height="match_parent" android:clipToPadding="fals
e" android:paddingBottom="16dp"/>
  </LinearLayout>
  <com.google.android.material.bottomnavigation.BottomNavigationView android:id="@
```

+id/navigation" android:layout_width="match_parent" android:layout_height="68dp" andro
id:background="@color/surface" app:menu="@menu/bottom_navigation_menu" app:label
VisibilityMode="labeled" app:itemIconTint="@color/nav_colors" app:itemTextColor="@c
olor/nav_colors" app:itemTextAppearanceActive="@style/GalleryNavigationText" app:ite
mTextAppearanceInactive="@style/GalleryNavigationText"/>
</LinearLayout>

2 2 3 activity_album_detail

Nguồn: app/src/main/res/layout/activity_album_detail.xml

Vai trò: Nội dung một album và các hành động trên toolbar.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Album" app:title
TextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary"
app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_pri
mary"/>
<LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="12dp">
<TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" an
droid:fontFamily="sans-serif" android:id="@+id/status" android:padding="8dp"/>
<androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_
width="match_parent" android:layout_height="0dp" android:layout_weight="1" android:pa
ddingBottom="16dp" android:clipToPadding="false"/>
</LinearLayout>
</LinearLayout>

2 2 4 activity_backup

Nguồn: app/src/main/res/layout/activity_backup.xml

Vai trò: Giao diện minh họa sao lưu với hai công tắc và nút sao lưu.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">

```
    <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Sao lưu và đồng
bộ" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/te
xt_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@c
olor/text_primary"/>
    <androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="match_parent">
        <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent" android:orientation="vertical" android:padding="24dp">
            <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Giữ an toàn mọi khoảnh khắc" android:textSize="24sp" android:textCo
lor="@color/text_primary" android:fontFamily="sans-serif" android:layout_marginTop="2
0dp"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Sao lưu ảnh và video để có thể xem lại trên các thiết bị của bạn." andro
id:textSize="16sp" android:textColor="@color/text_secondary" android:fontFamily="sans-s
erif" android:layout_marginTop="12dp" android:layout_marginBottom="24dp" android:lin
eSpacingExtra="4dp"/>
            <com.google.android.material.switchmaterial.SwitchMaterial android:id="@+id/aut
oBackup" android:layout_width="match_parent" android:layout_height="72dp" android:tex
t="Sao lưu khi mở ứng dụng" android:textColor="@color/text_primary" android:textSize="
16sp"/>
            <com.google.android.material.switchmaterial.SwitchMaterial android:id="@+id/wif
iOnly" android:layout_width="match_parent" android:layout_height="72dp" android:text="
Chỉ dùng Wi-Fi" android:textColor="@color/text_primary" android:textSize="16sp" androi
d:checked="true"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Các file đã sao lưu sẽ được bỏ qua ở lần tiếp theo." android:textSize="
14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" androi
d:layout_marginTop="20dp" android:layout_marginBottom="28dp"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/backupNow
" style="@style/PrimaryButton" android:layout_width="match_parent" android:layout_heig
ht="52dp" android:text="Sao lưu ngay" android:letterSpacing="0" android:textSize="16sp"/
>
            <ProgressBar android:id="@+id/progress" style="?android:attr/progressBarStyleHor
izontal" android:layout_width="match_parent" android:layout_height="12dp" android:layo
ut_marginTop="20dp" android:max="100"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary"
 android:fontFamily="sans-serif" android:id="@+id/status" android:visibility="gone" androi
d:paddingVertical="8dp"/>
        </LinearLayout>
    </androidx.core.widget.NestedScrollView>
</LinearLayout>
```

2 2 5 activity_create_album

Nguồn: app/src/main/res/layout/activity_create_album.xml

Vai trò: Nhập tên và chọn ảnh cho album mới.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
   <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Tạo album" app:t
itleTextColor="?android:attr/textColorPrimary" app:navigationIcon="@drawable/ic_arrow_
back" app:titleTextAppearance="@style/GalleryToolbarTitle" app:navigationIconTint="@c
olor/text_primary"/>
   <LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="20dp">
      <com.google.android.material.textfield.TextInputLayout android:layout_width="match
_parent" android:layout_height="wrap_content" android:layout_marginTop="14dp" android
:hint="Tên album" style="@style/CustomEditText" app:endIconMode="none">
         <com.google.android.material.textfield.TextInputEditText android:id="@+id/album
Name" android:layout_width="match_parent" android:layout_height="wrap_content" andro
id:inputType="text" android:maxLines="1"/>
      </com.google.android.material.textfield.TextInputLayout>
      <TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="Chọn ảnh cho câu chuyện mới của bạn" android:textSize="14sp" android:
textColor="?android:attr/textColorPrimary" android:paddingVertical="8dp" android:fontFa
mily="sans-serif"/>
      <TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="" android:textSize="14sp" android:paddingVertical="8dp" android:id="@
+id/status" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" an
droid:visibility="gone"/>
      <androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_
width="match_parent" android:layout_height="0dp" android:layout_weight="1" android:cli
pToPadding="false" android:paddingBottom="16dp"/>
      <com.google.android.material.button.MaterialButton android:id="@+id/create" style="
@style/PrimaryButton" android:layout_width="match_parent" android:layout_height="56d
p" android:layout_marginTop="8dp" android:text="Tạo album" android:letterSpacing="0"/
>
   </LinearLayout>
</LinearLayout>
```

2 2 6 activity_dark_gallery

Nguồn: app/src/main/res/layout/activity_dark_gallery.xml

Vai trò: Bố cục tham chiếu giao diện tối; trang Ảnh thực tế dùng activity_main với tài nguyên theo theme.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
    <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Ảnh" app:titleTex
tAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:
navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primar
y"/>
    <LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="16dp">
        <FrameLayout android:layout_width="match_parent" android:layout_height="48dp" a
ndroid:background="@drawable/bg_search_bar">
            <ImageView android:layout_width="20dp" android:layout_height="20dp" android:l
ayout_gravity="start|center_vertical" android:layout_marginStart="14dp" android:src="@dr
awable/ic_search" app:tint="@color/text_secondary" android:importantForAccessibility="n
o"/>
            <androidx.appcompat.widget.AppCompatEditText android:id="@+id/search" androi
d:layout_width="match_parent" android:layout_height="match_parent" android:backgroun
d="@android:color/transparent" android:paddingStart="44dp" android:paddingEnd="12dp"
android:paddingTop="0dp" android:paddingBottom="0dp" android:gravity="center_vertical
" android:includeFontPadding="true" android:fontFamily="sans-serif" android:singleLine="
true" android:hint="Tìm kiếm ảnh, video…" android:textSize="15sp" android:inputType="t
ext" android:imeOptions="actionSearch" android:textColor="@color/text_primary" android
:textColorHint="@color/text_secondary"/>
        </FrameLayout>
        <com.google.android.material.chip.ChipGroup android:layout_width="match_parent"
android:layout_height="wrap_content" android:layout_marginTop="6dp" app:singleLine="f
alse" app:singleSelection="true" app:selectionRequired="true" app:chipSpacingHorizontal=
"6dp">
            <com.google.android.material.chip.Chip android:id="@+id/all" android:layout_widt
h="wrap_content" android:layout_height="40dp" android:text="Tất cả" android:textSize="1
4sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="tru
e" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/fi
eld_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerR
adius="18dp" app:ensureMinTouchTargetSize="false"/>
            <com.google.android.material.chip.Chip android:id="@+id/images" android:layout_
width="wrap_content" android:layout_height="40dp" android:text="Ảnh" android:textSize
="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked=
"false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@col
or/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCorn
erRadius="18dp" app:ensureMinTouchTargetSize="false"/>
            <com.google.android.material.chip.Chip android:id="@+id/videos" android:layout_
width="wrap_content" android:layout_height="40dp" android:text="Video" android:textSiz
e="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked
="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@c
olor/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCo
rnerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
```

```
        <com.google.android.material.chip.Chip android:id="@+id/favorites" android:layou
t_width="wrap_content" android:layout_height="40dp" android:text="♡" android:textSize=
"14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="
false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@colo
r/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCorne
rRadius="18dp" app:ensureMinTouchTargetSize="false"/>
    </com.google.android.material.chip.ChipGroup>
    <TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" an
droid:fontFamily="sans-serif" android:id="@+id/status" android:visibility="gone" android:
paddingVertical="8dp"/>
    <FrameLayout android:layout_width="match_parent" android:layout_height="0dp" an
droid:layout_weight="1">
        <androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layo
ut_width="match_parent" android:layout_height="match_parent" android:paddingBottom="
84dp" android:clipToPadding="false"/>
        <LinearLayout android:id="@+id/emptyState" android:layout_width="match_parent
" android:layout_height="match_parent" android:orientation="vertical" android:gravity="ce
nter" android:visibility="gone" android:padding="28dp">
            <ImageView android:layout_width="180dp" android:layout_height="150dp" andr
oid:src="@drawable/art_splash" android:contentDescription="Thư viện của bạn"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_
content" android:text="Lưu giữ những điều bạn yêu" android:textSize="20sp" android:textC
olor="@color/text_primary" android:fontFamily="sans-serif" android:gravity="center" andr
oid:layout_marginTop="16dp"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_
content" android:text="Thêm ảnh hoặc video đầu tiên để bắt đầu câu chuyện của bạn." andr
oid:textSize="15sp" android:textColor="@color/text_secondary" android:fontFamily="sans
-serif" android:gravity="center" android:layout_marginTop="10dp" android:lineSpacingExt
ra="4dp"/>
        </LinearLayout>
        <com.google.android.material.floatingactionbutton.FloatingActionButton android:id
="@+id/importMedia" android:layout_width="wrap_content" android:layout_height="wrap
_content" android:layout_gravity="bottom|end" android:layout_margin="16dp" android:src
="@drawable/ic_add" android:contentDescription="Thêm ảnh hoặc video" app:background
Tint="@color/button_primary" app:tint="@color/white" app:fabCustomSize="56dp"/>
    </FrameLayout>
  </LinearLayout>
  <com.google.android.material.bottomnavigation.BottomNavigationView android:id="@
+id/navigation" android:layout_width="match_parent" android:layout_height="68dp" andro
id:background="@color/surface" app:menu="@menu/bottom_navigation_menu" app:label
VisibilityMode="labeled" app:itemIconTint="@color/nav_colors" app:itemTextColor="@c
olor/nav_colors" app:itemTextAppearanceActive="@style/GalleryNavigationText" app:ite
mTextAppearanceInactive="@style/GalleryNavigationText"/>
</LinearLayout>
```

2 2 7 activity_edit_photo

Nguồn: app/src/main/res/layout/activity_edit_photo.xml

Vai trò: Bố trí vùng chỉnh sửa tùy biến, công cụ, bảng màu, tỷ lệ cắt và nút lưu bản sao.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/viewer_background">
    <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Chỉnh sửa" app:ti
tleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primar
y" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_
primary"/>
    <LinearLayout android:layout_width="match_parent" android:layout_height="44dp" and
roid:orientation="horizontal" android:paddingHorizontal="12dp">
        <com.google.android.material.button.MaterialButton android:id="@+id/undo" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
_height="44dp" android:text="Hoàn tác" android:textSize="13sp" android:letterSpacing="0
" android:paddingHorizontal="4dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/redo" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
_height="44dp" android:text="Làm lại" android:textSize="13sp" android:letterSpacing="0"
android:paddingHorizontal="4dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/resetEditor" st
yle="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="44dp" android:text="Khôi phục gốc" android:textSize="13sp" android:letter
Spacing="0" android:paddingHorizontal="4dp"/>
    </LinearLayout>
    <FrameLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:layout_marginHorizontal="16dp" android:layout_marginVe
rtical="8dp">
        <com.example.appanh.editor.EditorCanvasView android:id="@+id/photo" android:lay
out_width="match_parent" android:layout_height="match_parent" android:padding="8dp"/
>
        <ProgressBar android:id="@+id/editorBusy" android:layout_width="36dp" android:la
yout_height="36dp" android:layout_gravity="center"/>
    </FrameLayout>
    <LinearLayout android:layout_width="match_parent" android:layout_height="60dp" and
roid:orientation="horizontal">
        <com.google.android.material.button.MaterialButton android:id="@+id/crop" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
_height="60dp" android:text="Cắt" android:textSize="12sp" android:paddingHorizontal="0
dp" app:icon="@drawable/ic_crop" app:iconGravity="textTop" app:iconSize="21dp" app:i
conTint="@color/text_primary" android:textColor="@color/text_primary"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/filter" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
```

```
_height="60dp" android:text="Bộ lọc" android:textSize="12sp" android:paddingHorizontal
="0dp" app:icon="@drawable/ic_filter" app:iconGravity="textTop" app:iconSize="21dp" a
pp:iconTint="@color/text_primary" android:textColor="@color/text_primary"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/adjust" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
_height="60dp" android:text="Điều chỉnh" android:textSize="12sp" android:paddingHorizo
ntal="0dp" app:icon="@drawable/ic_settings" app:iconGravity="textTop" app:iconSize="2
1dp" app:iconTint="@color/text_primary" android:textColor="@color/text_primary"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/draw" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
_height="60dp" android:text="Vẽ" android:textSize="12sp" android:paddingHorizontal="0
dp" app:icon="@drawable/ic_edit" app:iconGravity="textTop" app:iconSize="21dp" app:ic
onTint="@color/text_primary" android:textColor="@color/text_primary"/>
    </LinearLayout>
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_cont
ent" android:orientation="vertical" android:paddingHorizontal="16dp" android:background
="@color/viewer_controls">
        <LinearLayout android:id="@+id/cropPanel" android:layout_width="match_parent" a
ndroid:layout_height="wrap_content" android:orientation="vertical" android:visibility="go
ne">
            <LinearLayout android:layout_width="match_parent" android:layout_height="44dp
" android:orientation="horizontal">
                <com.google.android.material.button.MaterialButton android:id="@+id/ratioFree
" style="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" andro
id:layout_height="44dp" android:text="Tự do" android:textSize="13sp" android:letterSpaci
ng="0" android:paddingHorizontal="4dp"/>
                <com.google.android.material.button.MaterialButton android:id="@+id/ratioSqua
re" style="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" and
roid:layout_height="44dp" android:text="1:1" android:textSize="13sp" android:letterSpacin
g="0" android:paddingHorizontal="4dp"/>
                <com.google.android.material.button.MaterialButton android:id="@+id/ratioPortr
ait" style="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" an
droid:layout_height="44dp" android:text="4:3" android:textSize="13sp" android:letterSpaci
ng="0" android:paddingHorizontal="4dp"/>
                <com.google.android.material.button.MaterialButton android:id="@+id/ratioWid
e" style="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" andr
oid:layout_height="44dp" android:text="16:9" android:textSize="13sp" android:letterSpaci
ng="0" android:paddingHorizontal="4dp"/>
            </LinearLayout>
            <LinearLayout android:layout_width="match_parent" android:layout_height="44dp
" android:orientation="horizontal">
                <com.google.android.material.button.MaterialButton android:id="@+id/rotate" st
yle="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="44dp" android:text="Xoay 90°" android:textSize="13sp" android:letterSpaci
ng="0" android:paddingHorizontal="4dp"/>
                <com.google.android.material.button.MaterialButton android:id="@+id/applyCro
p" style="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" andr
oid:layout_height="44dp" android:text="Áp dụng cắt" android:textSize="13sp" android:lett
erSpacing="0" android:paddingHorizontal="4dp"/>
```

```
        </LinearLayout>
    </LinearLayout>
    <HorizontalScrollView android:id="@+id/filterPanel" android:layout_width="match_
parent" android:layout_height="48dp" android:visibility="gone" android:scrollbars="none"
>
        <LinearLayout android:layout_width="wrap_content" android:layout_height="matc
h_parent" android:orientation="horizontal">
            <com.google.android.material.button.MaterialButton android:id="@+id/filterOrig
inal" style="@style/TextButton" android:layout_width="wrap_content" android:layout_heig
ht="44dp" android:text="Gốc" android:textSize="13sp"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/filterMon
o" style="@style/TextButton" android:layout_width="wrap_content" android:layout_height
="44dp" android:text="Trắng đen" android:textSize="13sp"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/filterSepi
a" style="@style/TextButton" android:layout_width="wrap_content" android:layout_height
="44dp" android:text="Cổ điển" android:textSize="13sp"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/filterWar
m" style="@style/TextButton" android:layout_width="wrap_content" android:layout_heigh
t="44dp" android:text="Ấm" android:textSize="13sp"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/filterCool
" style="@style/TextButton" android:layout_width="wrap_content" android:layout_height=
"44dp" android:text="Lạnh" android:textSize="13sp"/>
        </LinearLayout>
    </HorizontalScrollView>
    <LinearLayout android:id="@+id/adjustPanel" android:layout_width="match_parent"
android:layout_height="wrap_content" android:orientation="vertical" android:visibility="go
ne">
        <LinearLayout android:layout_width="match_parent" android:layout_height="44dp
" android:orientation="horizontal" android:gravity="center_vertical">
            <TextView android:layout_width="86dp" android:layout_height="wrap_content"
android:text="Độ sáng" android:textSize="13sp" android:textColor="@color/text_secondar
y"/>
            <SeekBar android:id="@+id/brightness" android:layout_width="0dp" android:lay
out_weight="1" android:layout_height="44dp" android:max="200" android:progress="100"
/>
        </LinearLayout>
        <LinearLayout android:layout_width="match_parent" android:layout_height="44dp
" android:orientation="horizontal" android:gravity="center_vertical">
            <TextView android:layout_width="86dp" android:layout_height="wrap_content"
android:text="Tương phản" android:textSize="13sp" android:textColor="@color/text_secon
dary"/>
            <SeekBar android:id="@+id/contrast" android:layout_width="0dp" android:layou
t_weight="1" android:layout_height="44dp" android:max="150" android:progress="50"/>
        </LinearLayout>
    </LinearLayout>
    <LinearLayout android:id="@+id/drawPanel" android:layout_width="match_parent" a
ndroid:layout_height="wrap_content" android:orientation="vertical" android:visibility="go
ne">
        <LinearLayout android:layout_width="match_parent" android:layout_height="44dp
```

```
" android:orientation="horizontal" android:gravity="center">
        <com.google.android.material.button.MaterialButton android:id="@+id/brushWh
ite" style="@style/PrimaryButton" android:layout_width="36dp" android:layout_height="3
6dp" android:layout_marginHorizontal="5dp" android:minWidth="0dp" android:padding="
0dp" android:backgroundTint="@color/white" android:contentDescription="Bút trắng" app:
cornerRadius="18dp" app:strokeWidth="1dp" app:strokeColor="@color/field_border"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/brushBla
ck" style="@style/PrimaryButton" android:layout_width="36dp" android:layout_height="3
6dp" android:layout_marginHorizontal="5dp" android:minWidth="0dp" android:padding="
0dp" android:backgroundTint="@color/black" android:contentDescription="Bút đen" app:c
ornerRadius="18dp" app:strokeWidth="1dp" app:strokeColor="@color/field_border"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/brushRed
" style="@style/PrimaryButton" android:layout_width="36dp" android:layout_height="36d
p" android:layout_marginHorizontal="5dp" android:minWidth="0dp" android:padding="0d
p" android:backgroundTint="#EF4444" android:contentDescription="Bút đỏ" app:cornerRa
dius="18dp" app:strokeWidth="1dp" app:strokeColor="@color/field_border"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/brushBlu
e" style="@style/PrimaryButton" android:layout_width="36dp" android:layout_height="36
dp" android:layout_marginHorizontal="5dp" android:minWidth="0dp" android:padding="0
dp" android:backgroundTint="#2563EB" android:contentDescription="Bút xanh dương" ap
p:cornerRadius="18dp" app:strokeWidth="1dp" app:strokeColor="@color/field_border"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/brushGre
en" style="@style/PrimaryButton" android:layout_width="36dp" android:layout_height="3
6dp" android:layout_marginHorizontal="5dp" android:minWidth="0dp" android:padding="
0dp" android:backgroundTint="#16A34A" android:contentDescription="Bút xanh lá" app:c
ornerRadius="18dp" app:strokeWidth="1dp" app:strokeColor="@color/field_border"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/brushYel
low" style="@style/PrimaryButton" android:layout_width="36dp" android:layout_height="
36dp" android:layout_marginHorizontal="5dp" android:minWidth="0dp" android:padding=
"0dp" android:backgroundTint="#FACC15" android:contentDescription="Bút vàng" app:co
rnerRadius="18dp" app:strokeWidth="1dp" app:strokeColor="@color/field_border"/>
      </LinearLayout>
      <LinearLayout android:layout_width="match_parent" android:layout_height="44dp
" android:orientation="horizontal" android:gravity="center_vertical">
        <TextView android:layout_width="56dp" android:layout_height="wrap_content"
android:text="Nét bút" android:textColor="@color/text_secondary" android:textSize="13sp
"/>
        <SeekBar android:id="@+id/brushWidth" android:layout_width="0dp" android:la
yout_weight="1" android:layout_height="44dp" android:max="29" android:progress="5"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/clearDra
wing" style="@style/TextButton" android:layout_width="92dp" android:layout_height="44
dp" android:text="Xóa nét" android:textSize="13sp"/>
      </LinearLayout>
    </LinearLayout>
  </LinearLayout>
  <TextView android:layout_width="match_parent" android:layout_height="wrap_content
" android:text="" android:textSize="13sp" android:textColor="@color/text_secondary" andr
oid:fontFamily="sans-serif" android:id="@+id/status" android:paddingHorizontal="20dp" a
ndroid:layout_marginTop="4dp" android:maxLines="2" android:ellipsize="end"/>
```

```
    <LinearLayout android:layout_width="match_parent" android:layout_height="64dp" and
roid:paddingHorizontal="16dp" android:paddingVertical="8dp" android:orientation="horiz
ontal">
        <com.google.android.material.button.MaterialButton android:id="@+id/cancelEdit" st
yle="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="44dp" android:text="Hủy" android:textSize="13sp" android:letterSpacing="
0" android:paddingHorizontal="4dp" android:layout_marginEnd="8dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/save" style="
@style/PrimaryButton" android:layout_width="0dp" android:layout_weight="1" android:la
yout_height="44dp" android:text="Lưu bản sao" android:textSize="13sp" android:letterSpa
cing="0" android:paddingHorizontal="4dp"/>
    </LinearLayout>
</LinearLayout>
```

2 2 8 activity_google_sign_in

Nguồn: app/src/main/res/layout/activity_google_sign_in.xml

Vai trò: Trang khởi động luồng đăng nhập Google.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
    <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Đăng nhập bằng
Google" app:titleTextColor="?android:attr/textColorPrimary" app:navigationIcon="@drawa
ble/ic_arrow_back" app:titleTextAppearance="@style/GalleryToolbarTitle" app:navigationI
conTint="@color/text_primary"/>
    <LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="20dp">
        <androidx.core.widget.NestedScrollView android:layout_width="match_parent" andro
id:layout_height="match_parent" android:fillViewport="true">
            <LinearLayout android:layout_width="match_parent" android:layout_height="wrap
_content" android:orientation="vertical" android:paddingBottom="24dp">
                <ImageView android:layout_width="88dp" android:layout_height="88dp" androi
d:layout_gravity="center" android:layout_marginTop="30dp" android:src="@drawable/ic_
app_logo" android:contentDescription="Biểu tượng Ảnh"/>
                <TextView android:layout_width="match_parent" android:layout_height="wrap_
content" android:text="Chọn tài khoản Google" android:textSize="24sp" android:textColor
="?android:attr/textColorPrimary" android:paddingVertical="8dp" android:gravity="center"
 android:fontFamily="sans-serif"/>
                <TextView android:layout_width="match_parent" android:layout_height="wrap_
content" android:text="Tài khoản được chọn qua cửa sổ bảo mật của Google." android:textS
ize="14sp" android:textColor="?android:attr/textColorPrimary" android:paddingVertical="8
dp" android:fontFamily="sans-serif"/>
```

<com.google.android.material.button.MaterialButton android:id="@+id/google" s
tyle="@style/PrimaryButton" android:layout_width="match_parent" android:layout_height
="56dp" android:layout_marginTop="8dp" android:text="Chọn tài khoản" android:letterSpa
cing="0"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_
content" android:text="" android:textSize="14sp" android:paddingVertical="8dp" android:i
d="@+id/status" android:textColor="@color/text_secondary" android:fontFamily="sans-ser
if" android:visibility="gone"/>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
</LinearLayout>
</LinearLayout>

2 2 9 activity_login

Nguồn: app/src/main/res/layout/activity_login.xml

Vai trò: Biểu mẫu đăng nhập, quên mật khẩu, đăng ký và tiếp tục với Google.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:background="@drawabl
e/bg_auth_surface">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="" app:titleTextAp
pearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:nav
igationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary" a
ndroid:visibility="gone"/>
<androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="match_parent" android:fillViewport="true" android:clipToPadding="false"
>
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent" android:orientation="vertical" android:paddingHorizontal="28dp" android:padding
Top="64dp" android:paddingBottom="20dp">
<TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Đăng nhập" android:textSize="30sp" android:fontFamily="sans-serif-
medium" android:textColor="@color/text_primary"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Chào mừng bạn quay lại!" android:textSize="16sp" android:textColor=
"@color/text_secondary" android:fontFamily="sans-serif" android:layout_marginTop="8dp
" android:layout_marginBottom="36dp"/>
<com.google.android.material.textfield.TextInputLayout style="@style/CustomEdit
Text" android:layout_width="match_parent" android:layout_height="wrap_content" androi
d:layout_marginBottom="16dp" app:hintEnabled="false" app:startIconDrawable="@drawa
ble/ic_email" app:startIconTint="@color/text_secondary" app:endIconMode="none" app:en
dIconTint="@color/text_secondary">

```
        <com.google.android.material.textfield.TextInputEditText android:id="@+id/email" android:layout_width="match_parent" android:layout_height="56dp" android:hint="Email hoặc số điện thoại" android:textColor="@color/text_primary" android:textColorHint="@color/text_secondary" android:textSize="15sp" android:fontFamily="sans-serif" android:paddingVertical="12dp" android:singleLine="true" android:inputType="textEmailAddress" android:imeOptions="actionNext"/>
    </com.google.android.material.textfield.TextInputLayout>
    <com.google.android.material.textfield.TextInputLayout style="@style/CustomEditText" android:layout_width="match_parent" android:layout_height="wrap_content" android:layout_marginBottom="16dp" app:hintEnabled="false" app:startIconDrawable="@drawable/ic_lock" app:startIconTint="@color/text_secondary" app:endIconMode="password_toggle" app:endIconTint="@color/text_secondary">
        <com.google.android.material.textfield.TextInputEditText android:id="@+id/password" android:layout_width="match_parent" android:layout_height="56dp" android:hint="Mật khẩu" android:textColor="@color/text_primary" android:textColorHint="@color/text_secondary" android:textSize="15sp" android:fontFamily="sans-serif" android:paddingVertical="12dp" android:singleLine="true" android:inputType="textPassword" android:imeOptions="actionDone"/>
    </com.google.android.material.textfield.TextInputLayout>
    <com.google.android.material.button.MaterialButton android:id="@+id/submit" style="@style/PrimaryButton" android:layout_width="match_parent" android:layout_height="52dp" android:text="Đăng nhập" android:letterSpacing="0" android:textSize="16sp" android:layout_marginTop="8dp"/>
    <com.google.android.material.button.MaterialButton android:id="@+id/forgot" style="@style/TextButton" android:layout_width="match_parent" android:layout_height="52dp" android:text="Quên mật khẩu?" android:letterSpacing="0" android:textSize="16sp" android:layout_marginTop="4dp"/>
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:gravity="center" android:orientation="horizontal" android:layout_marginTop="24dp" android:layout_marginBottom="20dp">
        <View android:layout_width="0dp" android:layout_height="1dp" android:layout_weight="1" android:background="@color/field_border"/>
        <TextView android:layout_width="wrap_content" android:layout_height="wrap_content" android:paddingHorizontal="16dp" android:text="Hoặc đăng nhập bằng" android:textSize="13sp" android:textColor="@color/text_secondary"/>
        <View android:layout_width="0dp" android:layout_height="1dp" android:layout_weight="1" android:background="@color/field_border"/>
    </LinearLayout>
    <com.google.android.material.button.MaterialButton android:id="@+id/google" style="@style/OutlinedButton" android:layout_width="match_parent" android:layout_height="52dp" android:text="Tiếp tục với Google" android:letterSpacing="0" android:textSize="16sp" app:icon="@drawable/ic_google" app:iconTint="@null" app:iconGravity="textStart" app:iconPadding="16dp"/>
    <ProgressBar android:id="@+id/authProgress" style="?android:attr/progressBarStyleHorizontal" android:layout_width="match_parent" android:layout_height="4dp" android:layout_marginTop="16dp" android:indeterminate="true" android:visibility="gone"/>
    <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/status" android:layout_marginTop="12
```

dp" android:visibility="gone" android:accessibilityLiveRegion="polite"/>
<Space android:layout_width="1dp" android:layout_height="0dp" android:layout_weight="1" android:minHeight="52dp"/>
<com.google.android.material.button.MaterialButton android:id="@+id/switchAuth" style="@style/TextButton" android:layout_width="match_parent" android:layout_height="52dp" android:text="Chưa có tài khoản? Đăng ký" android:letterSpacing="0" android:textSize="16sp" android:layout_marginTop="32dp"/>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
</FrameLayout>

2 2 10 activity_main

Nguồn: app/src/main/res/layout/activity_main.xml

Vai trò: Trang Ảnh với toolbar, tìm kiếm, chip lọc, lưới ảnh, nút nhập và thanh điều hướng.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:background="@color/background">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_height="56dp" app:title="Ảnh" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
<LinearLayout android:layout_width="match_parent" android:layout_height="0dp" android:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="16dp">
<FrameLayout android:layout_width="match_parent" android:layout_height="48dp" android:background="@drawable/bg_search_bar">
<ImageView android:layout_width="20dp" android:layout_height="20dp" android:layout_gravity="start|center_vertical" android:layout_marginStart="14dp" android:src="@drawable/ic_search" app:tint="@color/text_secondary" android:importantForAccessibility="no"/>
<androidx.appcompat.widget.AppCompatEditText android:id="@+id/search" android:layout_width="match_parent" android:layout_height="match_parent" android:background="@android:color/transparent" android:paddingStart="44dp" android:paddingEnd="12dp" android:paddingTop="0dp" android:paddingBottom="0dp" android:gravity="center_vertical" android:includeFontPadding="true" android:fontFamily="sans-serif" android:singleLine="true" android:hint="Tìm kiếm ảnh, video…" android:textSize="15sp" android:inputType="text" android:imeOptions="actionSearch" android:textColor="@color/text_primary" android:textColorHint="@color/text_secondary"/>
</FrameLayout>
<com.google.android.material.chip.ChipGroup android:layout_width="match_parent" android:layout_height="wrap_content" android:layout_marginTop="6dp" app:singleLine="false" app:singleSelection="true" app:selectionRequired="true" app:chipSpacingHorizontal=

"6dp">
<com.google.android.material.chip.Chip android:id="@+id/all" android:layout_width="wrap_content" android:layout_height="40dp" android:text="Tất cả" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="true" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
<com.google.android.material.chip.Chip android:id="@+id/images" android:layout_width="wrap_content" android:layout_height="40dp" android:text="Ảnh" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
<com.google.android.material.chip.Chip android:id="@+id/videos" android:layout_width="wrap_content" android:layout_height="40dp" android:text="Video" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
<com.google.android.material.chip.Chip android:id="@+id/favorites" android:layout_width="wrap_content" android:layout_height="40dp" android:text="♡" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
</com.google.android.material.chip.ChipGroup>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/status" android:visibility="gone" android:paddingVertical="8dp"/>
<FrameLayout android:layout_width="match_parent" android:layout_height="0dp" android:layout_weight="1">
<androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_width="match_parent" android:layout_height="match_parent" android:paddingBottom="84dp" android:clipToPadding="false"/>
<LinearLayout android:id="@+id/emptyState" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:gravity="center" android:visibility="gone" android:padding="28dp">
<ImageView android:layout_width="180dp" android:layout_height="150dp" android:src="@drawable/art_splash" android:contentDescription="Thư viện của bạn"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Lưu giữ những điều bạn yêu" android:textSize="20sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif" android:gravity="center" android:layout_marginTop="16dp"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Thêm ảnh hoặc video đầu tiên để bắt đầu câu chuyện của bạn." android:textSize="15sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:gravity="center" android:layout_marginTop="10dp" android:lineSpacingExtra="4dp"/>

```
        </LinearLayout>
        <com.google.android.material.floatingactionbutton.FloatingActionButton android:id
="@+id/importMedia" android:layout_width="wrap_content" android:layout_height="wrap
_content" android:layout_gravity="bottom|end" android:layout_margin="16dp" android:src
="@drawable/ic_add" android:contentDescription="Thêm ảnh hoặc video" app:background
Tint="@color/button_primary" app:tint="@color/white" app:fabCustomSize="56dp"/>
    </FrameLayout>
  </LinearLayout>
  <com.google.android.material.bottomnavigation.BottomNavigationView android:id="@
+id/navigation" android:layout_width="match_parent" android:layout_height="68dp" andro
id:background="@color/surface" app:menu="@menu/bottom_navigation_menu" app:label
VisibilityMode="labeled" app:itemIconTint="@color/nav_colors" app:itemTextColor="@c
olor/nav_colors" app:itemTextAppearanceActive="@style/GalleryNavigationText" app:ite
mTextAppearanceInactive="@style/GalleryNavigationText"/>
</LinearLayout>
```

2 2 11 activity_notifications

Nguồn: app/src/main/res/layout/activity_notifications.xml

Vai trò: Danh sách thông báo hoạt động và nút đánh dấu đã đọc.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
  <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Thông báo" app:t
itleTextColor="?android:attr/textColorPrimary" app:navigationIcon="@drawable/ic_arrow_
back" app:titleTextAppearance="@style/GalleryToolbarTitle" app:navigationIconTint="@c
olor/text_primary"/>
  <LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="20dp">
    <TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="Hoạt động gần đây" android:textSize="24sp" android:textColor="?androi
d:attr/textColorPrimary" android:paddingVertical="8dp" android:textStyle="bold" android:f
ontFamily="sans-serif"/>
    <TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="" android:textSize="14sp" android:paddingVertical="8dp" android:id="@
+id/status" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" an
droid:visibility="gone"/>
    <androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_
width="match_parent" android:layout_height="0dp" android:layout_weight="1" android:cli
pToPadding="false" android:paddingBottom="16dp"/>
    <com.google.android.material.button.MaterialButton android:id="@+id/markRead" sty
le="@style/OutlinedButton" android:layout_width="match_parent" android:layout_height=
```

"56dp" android:layout_marginTop="8dp" android:text="Đánh dấu tất cả đã đọc" android:let
terSpacing="0"/>
</LinearLayout>
</LinearLayout>

2 2 12 activity_otp

Nguồn: app/src/main/res/layout/activity_otp.xml

Vai trò: Màn xác thực với các ô nhập mã và thông tin gửi lại.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@drawable/bg_auth_surface">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="" app:titleTextAp
pearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:nav
igationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
<androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="match_parent">
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent" android:orientation="vertical" android:paddingHorizontal="24dp">
<TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Xác thực tài khoản" android:textSize="25sp" android:textColor="@col
or/text_primary" android:fontFamily="sans-serif" android:layout_marginTop="24dp"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="" android:textSize="16sp" android:textColor="@color/text_secondary"
android:fontFamily="sans-serif" android:id="@+id/status" android:layout_marginTop="12
dp" android:lineSpacingExtra="5dp"/>
<LinearLayout android:id="@+id/otpBoxes" android:layout_width="match_parent"
android:layout_height="wrap_content" android:orientation="horizontal" android:layout_ma
rginTop="36dp" android:layout_marginBottom="28dp">
<EditText android:id="@+id/otp1" android:layout_width="0dp" android:layout_
weight="1" android:layout_height="58dp" android:layout_marginHorizontal="3dp" android
:background="@drawable/bg_otp_box" android:textColor="@color/text_primary" android:t
extSize="22sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:
inputType="number" android:selectAllOnFocus="true" android:maxLength="6" android:im
portantForAutofill="no" android:contentDescription="Chữ số 1"/>
<EditText android:id="@+id/otp2" android:layout_width="0dp" android:layout_
weight="1" android:layout_height="58dp" android:layout_marginHorizontal="3dp" android
:background="@drawable/bg_otp_box" android:textColor="@color/text_primary" android:t
extSize="22sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:
inputType="number" android:selectAllOnFocus="true" android:maxLength="6" android:im
portantForAutofill="no" android:contentDescription="Chữ số 2"/>
<EditText android:id="@+id/otp3" android:layout_width="0dp" android:layout_

```
weight="1" android:layout_height="58dp" android:layout_marginHorizontal="3dp" android
:background="@drawable/bg_otp_box" android:textColor="@color/text_primary" android:t
extSize="22sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:
inputType="number" android:selectAllOnFocus="true" android:maxLength="6" android:im
portantForAutofill="no" android:contentDescription="Chữ số 3"/>
            <EditText android:id="@+id/otp4" android:layout_width="0dp" android:layout_
weight="1" android:layout_height="58dp" android:layout_marginHorizontal="3dp" android
:background="@drawable/bg_otp_box" android:textColor="@color/text_primary" android:t
extSize="22sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:
inputType="number" android:selectAllOnFocus="true" android:maxLength="6" android:im
portantForAutofill="no" android:contentDescription="Chữ số 4"/>
            <EditText android:id="@+id/otp5" android:layout_width="0dp" android:layout_
weight="1" android:layout_height="58dp" android:layout_marginHorizontal="3dp" android
:background="@drawable/bg_otp_box" android:textColor="@color/text_primary" android:t
extSize="22sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:
inputType="number" android:selectAllOnFocus="true" android:maxLength="6" android:im
portantForAutofill="no" android:contentDescription="Chữ số 5"/>
            <EditText android:id="@+id/otp6" android:layout_width="0dp" android:layout_
weight="1" android:layout_height="58dp" android:layout_marginHorizontal="3dp" android
:background="@drawable/bg_otp_box" android:textColor="@color/text_primary" android:t
extSize="22sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:
inputType="number" android:selectAllOnFocus="true" android:maxLength="6" android:im
portantForAutofill="no" android:contentDescription="Chữ số 6"/>
        </LinearLayout>
        <EditText android:id="@+id/otp" android:layout_width="1dp" android:layout_heig
ht="1dp" android:visibility="gone" android:inputType="number"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/verify" style
="@style/PrimaryButton" android:layout_width="match_parent" android:layout_height="5
2dp" android:text="Xác thực" android:letterSpacing="0" android:textSize="16sp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/resend" styl
e="@style/TextButton" android:layout_width="match_parent" android:layout_height="52d
p" android:text="Gửi lại xác thực" android:letterSpacing="0" android:textSize="16sp" andr
oid:layout_marginTop="12dp"/>
    </LinearLayout>
  </androidx.core.widget.NestedScrollView>
</LinearLayout>
```

2 2 13 activity_photo_detail

Nguồn: app/src/main/res/layout/activity_photo_detail.xml

Vai trò: Màn xem ảnh hoặc video, yêu thích, chỉnh sửa, chia sẻ và xóa.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
```

```
droid:background="@color/viewer_background">
    <FrameLayout android:layout_width="match_parent" android:layout_height="56dp">
        <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" andr
oid:layout_width="match_parent" android:layout_height="match_parent" app:navigationIco
n="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
        <ImageButton android:id="@+id/more" android:layout_width="48dp" android:layout_
height="48dp" android:layout_gravity="end|center_vertical" android:background="?attr/sel
ectableItemBackgroundBorderless" android:src="@drawable/ic_more_vert" app:tint="@col
or/text_primary" android:contentDescription="Thông tin và thao tác"/>
    </FrameLayout>
    <ImageView android:id="@+id/photo" android:layout_width="match_parent" android:la
yout_height="0dp" android:layout_weight="1" android:scaleType="fitCenter" android:cont
entDescription="Ảnh đang xem"/>
    <TextView android:layout_width="match_parent" android:layout_height="wrap_content
" android:text="" android:textSize="13sp" android:textColor="@color/text_secondary" andr
oid:fontFamily="sans-serif" android:id="@+id/status" android:padding="12dp" android:visi
bility="gone"/>
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_cont
ent" android:orientation="horizontal" android:background="@color/viewer_controls">
        <com.google.android.material.button.MaterialButton android:id="@+id/favorite" style
="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:lay
out_height="76dp" android:text="Yêu thích" android:textColor="@color/text_primary" and
roid:textSize="12sp" android:letterSpacing="0" android:paddingHorizontal="0dp" app:icon
="@drawable/ic_favorite_outline" app:iconGravity="textTop" app:iconTint="@color/text_
primary" app:iconSize="22dp" app:iconPadding="6dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/edit" style="@
style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout_
height="76dp" android:text="Chỉnh sửa" android:textColor="@color/text_primary" android
:textSize="12sp" android:letterSpacing="0" android:paddingHorizontal="0dp" app:icon="@
drawable/ic_edit" app:iconGravity="textTop" app:iconTint="@color/text_primary" app:ico
nSize="22dp" app:iconPadding="6dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/share" style="
@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:layout
_height="76dp" android:text="Chia sẻ" android:textColor="@color/text_primary" android:t
extSize="12sp" android:letterSpacing="0" android:paddingHorizontal="0dp" app:icon="@d
rawable/ic_share" app:iconGravity="textTop" app:iconTint="@color/text_primary" app:ico
nSize="22dp" app:iconPadding="6dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/deletePhoto" s
tyle="@style/TextButton" android:layout_width="0dp" android:layout_weight="1" android:
layout_height="76dp" android:text="Xóa" android:textColor="@color/text_primary" androi
d:textSize="12sp" app:icon="@drawable/ic_delete" app:iconGravity="textTop" app:iconTi
nt="@color/text_primary" app:iconSize="22dp"/>
    </LinearLayout>
</LinearLayout>
```

2 2 14 activity_register

Nguồn: app/src/main/res/layout/activity_register.xml

Vai trò: Biểu mẫu tạo tài khoản và chuyển về đăng nhập.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:background="@drawabl
e/bg_auth_surface">
   <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="" app:titleTextAp
pearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:nav
igationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary" a
ndroid:visibility="gone"/>
   <androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="match_parent" android:fillViewport="true" android:clipToPadding="false"
>
      <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent" android:orientation="vertical" android:paddingHorizontal="28dp" android:padding
Top="64dp" android:paddingBottom="20dp">
         <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Đăng ký" android:textSize="30sp" android:fontFamily="sans-serif-me
dium" android:textColor="@color/text_primary"/>
         <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Tạo tài khoản để bắt đầu" android:textSize="16sp" android:textColor=
"@color/text_secondary" android:fontFamily="sans-serif" android:layout_marginTop="8dp
" android:layout_marginBottom="36dp"/>
         <com.google.android.material.textfield.TextInputLayout style="@style/CustomEdit
Text" android:layout_width="match_parent" android:layout_height="wrap_content" androi
d:layout_marginBottom="16dp" app:hintEnabled="false" app:startIconDrawable="@drawa
ble/ic_person" app:startIconTint="@color/text_secondary" app:endIconMode="none" app:e
ndIconTint="@color/text_secondary">
            <com.google.android.material.textfield.TextInputEditText android:id="@+id/nam
e" android:layout_width="match_parent" android:layout_height="56dp" android:hint="Họ
và tên" android:textColor="@color/text_primary" android:textColorHint="@color/text_sec
ondary" android:textSize="15sp" android:fontFamily="sans-serif" android:paddingVertical
="12dp" android:singleLine="true" android:inputType="textPersonName" android:imeOpti
ons="actionNext"/>
         </com.google.android.material.textfield.TextInputLayout>
         <com.google.android.material.textfield.TextInputLayout style="@style/CustomEdit
Text" android:layout_width="match_parent" android:layout_height="wrap_content" androi
d:layout_marginBottom="16dp" app:hintEnabled="false" app:startIconDrawable="@drawa
ble/ic_email" app:startIconTint="@color/text_secondary" app:endIconMode="none" app:en
dIconTint="@color/text_secondary">
            <com.google.android.material.textfield.TextInputEditText android:id="@+id/ema
il" android:layout_width="match_parent" android:layout_height="56dp" android:hint="Ema
il hoặc số điện thoại" android:textColor="@color/text_primary" android:textColorHint="@
color/text_secondary" android:textSize="15sp" android:fontFamily="sans-serif" android:pa
ddingVertical="12dp" android:singleLine="true" android:inputType="textEmailAddress" an
droid:imeOptions="actionNext"/>
```

```
        </com.google.android.material.textfield.TextInputLayout>
        <com.google.android.material.textfield.TextInputLayout style="@style/CustomEdit
Text" android:layout_width="match_parent" android:layout_height="wrap_content" androi
d:layout_marginBottom="16dp" app:hintEnabled="false" app:startIconDrawable="@drawa
ble/ic_lock" app:startIconTint="@color/text_secondary" app:endIconMode="password_tog
gle" app:endIconTint="@color/text_secondary">
            <com.google.android.material.textfield.TextInputEditText android:id="@+id/pass
word" android:layout_width="match_parent" android:layout_height="56dp" android:hint="
Mật khẩu" android:textColor="@color/text_primary" android:textColorHint="@color/text_
secondary" android:textSize="15sp" android:fontFamily="sans-serif" android:paddingVerti
cal="12dp" android:singleLine="true" android:inputType="textPassword" android:imeOptio
ns="actionDone"/>
        </com.google.android.material.textfield.TextInputLayout>
        <com.google.android.material.button.MaterialButton android:id="@+id/submit" styl
e="@style/PrimaryButton" android:layout_width="match_parent" android:layout_height="
52dp" android:text="Đăng ký" android:letterSpacing="0" android:textSize="16sp" android:l
ayout_marginTop="8dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/forgot" styl
e="@style/TextButton" android:layout_width="match_parent" android:layout_height="52d
p" android:text="Quên mật khẩu?" android:letterSpacing="0" android:textSize="16sp" andr
oid:layout_marginTop="4dp" android:visibility="gone"/>
        <LinearLayout android:layout_width="match_parent" android:layout_height="wrap
_content" android:gravity="center" android:orientation="horizontal" android:layout_margin
Top="32dp" android:layout_marginBottom="20dp">
            <View android:layout_width="0dp" android:layout_height="1dp" android:layout_
weight="1" android:background="@color/field_border"/>
            <TextView android:layout_width="wrap_content" android:layout_height="wrap_
content" android:paddingHorizontal="16dp" android:text="Hoặc đăng nhập bằng" android:t
extSize="13sp" android:textColor="@color/text_secondary"/>
            <View android:layout_width="0dp" android:layout_height="1dp" android:layout_
weight="1" android:background="@color/field_border"/>
        </LinearLayout>
        <com.google.android.material.button.MaterialButton android:id="@+id/google" styl
e="@style/OutlinedButton" android:layout_width="match_parent" android:layout_height="
52dp" android:text="Tiếp tục với Google" android:letterSpacing="0" android:textSize="16s
p" app:icon="@drawable/ic_google" app:iconTint="@null" app:iconGravity="textStart" ap
p:iconPadding="16dp"/>
        <ProgressBar android:id="@+id/authProgress" style="?android:attr/progressBarStyl
eHorizontal" android:layout_width="match_parent" android:layout_height="4dp" android:l
ayout_marginTop="16dp" android:indeterminate="true" android:visibility="gone"/>
        <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary"
 android:fontFamily="sans-serif" android:id="@+id/status" android:layout_marginTop="12
dp" android:visibility="gone" android:accessibilityLiveRegion="polite"/>
        <Space android:layout_width="1dp" android:layout_height="0dp" android:layout_w
eight="1" android:minHeight="52dp"/>
        <com.google.android.material.button.MaterialButton android:id="@+id/switchAuth
" style="@style/TextButton" android:layout_width="match_parent" android:layout_height=
"52dp" android:text="Đã có tài khoản? Đăng nhập" android:letterSpacing="0" android:textS
```

ize="16sp" android:layout_marginTop="32dp"/>
    </LinearLayout>
  </androidx.core.widget.NestedScrollView>
</FrameLayout>

2 2 15 activity_search

Nguồn: app/src/main/res/layout/activity_search.xml

Vai trò: Trang tìm kiếm và lọc trong thư viện.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
  <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Tìm kiếm" app:tit
leTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary
" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_p
rimary"/>
  <LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="16dp">
    <FrameLayout android:layout_width="match_parent" android:layout_height="48dp" a
ndroid:background="@drawable/bg_search_bar">
      <ImageView android:layout_width="20dp" android:layout_height="20dp" android:l
ayout_gravity="start|center_vertical" android:layout_marginStart="14dp" android:src="@dr
awable/ic_search" app:tint="@color/text_secondary" android:importantForAccessibility="n
o"/>
      <androidx.appcompat.widget.AppCompatEditText android:id="@+id/search" androi
d:layout_width="match_parent" android:layout_height="match_parent" android:backgroun
d="@android:color/transparent" android:paddingStart="44dp" android:paddingEnd="12dp"
android:paddingTop="0dp" android:paddingBottom="0dp" android:gravity="center_vertical
" android:includeFontPadding="true" android:fontFamily="sans-serif" android:singleLine="
true" android:hint="Tìm kiếm ảnh, video…" android:textSize="15sp" android:inputType="t
ext" android:imeOptions="actionSearch" android:textColor="@color/text_primary" android
:textColorHint="@color/text_secondary"/>
    </FrameLayout>
    <com.google.android.material.chip.ChipGroup android:layout_width="match_parent"
android:layout_height="wrap_content" android:layout_marginTop="6dp" app:singleLine="f
alse" app:singleSelection="true" app:selectionRequired="true" app:chipSpacingHorizontal=
"6dp">
      <com.google.android.material.chip.Chip android:id="@+id/all" android:layout_widt
h="wrap_content" android:layout_height="40dp" android:text="Tất cả" android:textSize="1
4sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="tru
e" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/fi
eld_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerR

adius="18dp" app:ensureMinTouchTargetSize="false"/>
<com.google.android.material.chip.Chip android:id="@+id/images" android:layout_width="wrap_content" android:layout_height="40dp" android:text="Ảnh" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
<com.google.android.material.chip.Chip android:id="@+id/videos" android:layout_width="wrap_content" android:layout_height="40dp" android:text="Video" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
<com.google.android.material.chip.Chip android:id="@+id/favorites" android:layout_width="wrap_content" android:layout_height="40dp" android:text="♡" android:textSize="14sp" android:textColor="@color/chip_text" android:checkable="true" android:checked="false" app:chipBackgroundColor="@color/chip_background" app:chipStrokeColor="@color/field_border" app:chipStrokeWidth="1dp" app:checkedIconVisible="false" app:chipCornerRadius="18dp" app:ensureMinTouchTargetSize="false"/>
</com.google.android.material.chip.ChipGroup>
<com.google.android.material.button.MaterialButton android:id="@+id/sort" style="@style/TextButton" android:layout_width="match_parent" android:layout_height="52dp" android:text="Sắp xếp và bộ lọc" android:letterSpacing="0" android:textSize="16sp" app:icon="@drawable/ic_filter" app:iconGravity="textStart"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/status" android:visibility="gone" android:paddingVertical="8dp"/>
<FrameLayout android:layout_width="match_parent" android:layout_height="0dp" android:layout_weight="1">
<androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_width="match_parent" android:layout_height="match_parent" android:paddingBottom="84dp" android:clipToPadding="false"/>
<LinearLayout android:id="@+id/emptyState" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:gravity="center" android:visibility="gone" android:padding="28dp">
<ImageView android:layout_width="180dp" android:layout_height="150dp" android:src="@drawable/art_splash" android:contentDescription="Thư viện của bạn"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Lưu giữ những điều bạn yêu" android:textSize="20sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif" android:gravity="center" android:layout_marginTop="16dp"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Thêm ảnh hoặc video đầu tiên để bắt đầu câu chuyện của bạn." android:textSize="15sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:gravity="center" android:layout_marginTop="10dp" android:lineSpacingExtra="4dp"/>
</LinearLayout>
<com.google.android.material.floatingactionbutton.FloatingActionButton android:id

="@+id/importMedia" android:layout_width="wrap_content" android:layout_height="wrap_content" android:layout_gravity="bottom|end" android:layout_margin="16dp" android:src="@drawable/ic_add" android:contentDescription="Thêm ảnh hoặc video" app:backgroundTint="@color/button_primary" app:tint="@color/white" app:fabCustomSize="56dp"/>
    </FrameLayout>
  </LinearLayout>
  <com.google.android.material.bottomnavigation.BottomNavigationView android:id="@+id/navigation" android:layout_width="match_parent" android:layout_height="68dp" android:background="@color/surface" app:menu="@menu/bottom_navigation_menu" app:labelVisibilityMode="labeled" app:itemIconTint="@color/nav_colors" app:itemTextColor="@color/nav_colors" app:itemTextAppearanceActive="@style/GalleryNavigationText" app:itemTextAppearanceInactive="@style/GalleryNavigationText"/>
</LinearLayout>

2 2 16 activity_settings

Nguồn: app/src/main/res/layout/activity_settings.xml

Vai trò: Trang Hồ sơ và Cài đặt, các dòng chức năng có mô tả phụ.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:background="@color/background">
  <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_height="56dp" app:title="Cài đặt" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
  <androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:layout_height="0dp" android:layout_weight="1" android:fillViewport="true">
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="vertical" android:paddingHorizontal="12dp" android:paddingBottom="16dp">
      <LinearLayout android:id="@+id/account" android:layout_width="match_parent" android:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBottom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddingHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/bg_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" android:src="@drawable/ic_person" app:tint="@color/text_primary" android:importantForAccessibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="16dp" android:layout_marginEnd="8dp">

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            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Tài khoản" android:textSize="15sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Đang tải tài khoản…" android:textSize="13sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/accountDetail" android:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" android:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAccessibility="no"/>
    </LinearLayout>
    <LinearLayout android:id="@+id/backup" android:layout_width="match_parent" android:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBottom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddingHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/bg_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" android:src="@drawable/ic_cloud_upload" app:tint="@color/text_primary" android:importantForAccessibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Sao lưu và đồng bộ" android:textSize="15sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Chưa bật" android:textSize="13sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/backupDetail" android:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" android:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAccessibility="no"/>
    </LinearLayout>
    <LinearLayout android:id="@+id/storage" android:layout_width="match_parent" android:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBottom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddingHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/bg_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" android:src="@drawable/ic_album" app:tint="@color/text_primary" android:importantForAccessibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wra
```

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p_content" android:text="Dung lượng lưu trữ" android:textSize="15sp" android:textColor="
@color/text_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Đang tính dung lượng…" android:textSize="13sp" android:textCo
lor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/storageDe
tail" android:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
    </LinearLayout>
    <LinearLayout android:id="@+id/notifications" android:layout_width="match_pare
nt" android:layout_height="wrap_content" android:minHeight="56dp" android:layout_marg
inBottom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android
:paddingHorizontal="14dp" android:paddingVertical="12dp" android:background="@draw
able/bg_settings_row" android:foreground="?attr/selectableItemBackground" android:clicka
ble="true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_notification" app:tint="@color/text_primary" android:importantForA
ccessibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Thông báo" android:textSize="15sp" android:textColor="@color/t
ext_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Lịch sử hoạt động" android:textSize="13sp" android:textColor="
@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/notificationsDe
tail" android:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
    </LinearLayout>
    <LinearLayout android:id="@+id/theme" android:layout_width="match_parent" an
droid:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBot
tom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:padd
ingHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/b
g_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="
true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_settings" app:tint="@color/text_primary" android:importantForAcces
sibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Giao diện" android:textSize="15sp" android:textColor="@color/t
```

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ext_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Sáng" android:textSize="13sp" android:textColor="@color/text_s
econdary" android:fontFamily="sans-serif" android:id="@+id/themeDetail" android:layout_
marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
    </LinearLayout>
    <LinearLayout android:id="@+id/language" android:layout_width="match_parent"
android:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginB
ottom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:pa
ddingHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawabl
e/bg_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable
="true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_language" app:tint="@color/text_primary" android:importantForAcce
ssibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Ngôn ngữ" android:textSize="15sp" android:textColor="@color/t
ext_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Tiếng Việt" android:textSize="13sp" android:textColor="@color/t
ext_secondary" android:fontFamily="sans-serif" android:id="@+id/languageDetail" android
:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
    </LinearLayout>
    <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Thư viện" android:textSize="13sp" android:textColor="@color/text_se
condary" android:fontFamily="sans-serif" android:paddingTop="14dp" android:paddingBot
tom="8dp"/>
    <LinearLayout android:id="@+id/hiddenMedia" android:layout_width="match_pare
nt" android:layout_height="wrap_content" android:minHeight="56dp" android:layout_marg
inBottom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android
:paddingHorizontal="14dp" android:paddingVertical="12dp" android:background="@draw
able/bg_settings_row" android:foreground="?attr/selectableItemBackground" android:clicka
ble="true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_visibility_off" app:tint="@color/text_primary" android:importantFor
Accessibility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
```

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16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Ảnh và video đã ẩn" android:textSize="15sp" android:textColor="
@color/text_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Quản lý các mục đã ẩn" android:textSize="13sp" android:textColo
r="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/hiddenMedi
aDetail" android:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
    </LinearLayout>
    <LinearLayout android:id="@+id/trash" android:layout_width="match_parent" andr
oid:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBotto
m="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddin
gHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/bg
_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="tr
ue" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_delete" app:tint="@color/text_primary" android:importantForAccessi
bility="no"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Thùng rác" android:textSize="15sp" android:textColor="@color/t
ext_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Khôi phục ảnh đã xóa" android:textSize="13sp" android:textColor
="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/trashDetail"
android:layout_marginTop="3dp"/>
        </LinearLayout>
        <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
    </LinearLayout>
    <TextView android:layout_width="match_parent" android:layout_height="wrap_co
ntent" android:text="Khác" android:textSize="13sp" android:textColor="@color/text_secon
dary" android:fontFamily="sans-serif" android:paddingTop="14dp" android:paddingBottom
="8dp"/>
    <LinearLayout android:id="@+id/about" android:layout_width="match_parent" and
roid:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBott
om="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddi
ngHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/b
g_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="
true" android:focusable="true">
        <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_about" app:tint="@color/text_primary" android:importantForAccessi
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bility="no"/>
            <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
                <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Giới thiệu" android:textSize="15sp" android:textColor="@color/t
ext_primary" android:fontFamily="sans-serif"/>
            </LinearLayout>
            <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
        </LinearLayout>
        <LinearLayout android:id="@+id/help" android:layout_width="match_parent" andr
oid:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBotto
m="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddin
gHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/bg
_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="tr
ue" android:focusable="true">
            <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_help" app:tint="@color/text_primary" android:importantForAccessibi
lity="no"/>
            <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
                <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Trợ giúp" android:textSize="15sp" android:textColor="@color/te
xt_primary" android:fontFamily="sans-serif"/>
            </LinearLayout>
            <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAcc
essibility="no"/>
        </LinearLayout>
        <LinearLayout android:id="@+id/logout" android:layout_width="match_parent" an
droid:layout_height="wrap_content" android:minHeight="56dp" android:layout_marginBot
tom="3dp" android:orientation="horizontal" android:gravity="center_vertical" android:padd
ingHorizontal="14dp" android:paddingVertical="12dp" android:background="@drawable/b
g_settings_row" android:foreground="?attr/selectableItemBackground" android:clickable="
true" android:focusable="true">
            <ImageView android:layout_width="22dp" android:layout_height="22dp" androi
d:src="@drawable/ic_logout" app:tint="@color/danger" android:importantForAccessibility
="no"/>
            <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
ayout_height="wrap_content" android:orientation="vertical" android:layout_marginStart="
16dp" android:layout_marginEnd="8dp">
                <TextView android:layout_width="match_parent" android:layout_height="wra
p_content" android:text="Đăng xuất" android:textSize="15sp" android:textColor="@color/
danger" android:fontFamily="sans-serif"/>
            </LinearLayout>
            <ImageView android:layout_width="18dp" android:layout_height="18dp" androi
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d:src="@drawable/ic_chevron_right" app:tint="@color/text_hint" android:importantForAccessibility="no"/>
</LinearLayout>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/status" android:visibility="gone" android:paddingVertical="12dp"/>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
<com.google.android.material.bottomnavigation.BottomNavigationView android:id="@+id/navigation" android:layout_width="match_parent" android:layout_height="68dp" android:background="@color/surface" app:menu="@menu/bottom_navigation_menu" app:labelVisibilityMode="labeled" app:itemIconTint="@color/nav_colors" app:itemTextColor="@color/nav_colors" app:itemTextAppearanceActive="@style/GalleryNavigationText" app:itemTextAppearanceInactive="@style/GalleryNavigationText"/>
</LinearLayout>

2 2 17 activity_splash

Nguồn: app/src/main/res/layout/activity_splash.xml

Vai trò: Màn chào có logo, minh họa, nút Bắt đầu và khám phá trên thiết bị.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_width="match_parent" android:layout_height="match_parent" android:background="@drawable/bg_auth_surface">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_height="56dp" app:title="" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary" android:visibility="gone"/>
<androidx.constraintlayout.widget.ConstraintLayout android:layout_width="match_parent" android:layout_height="match_parent">
<androidx.constraintlayout.widget.Guideline android:id="@+id/heroAnchor" android:layout_width="wrap_content" android:layout_height="wrap_content" android:orientation="horizontal" app:layout_constraintGuide_percent="0.14"/>
<ImageView android:id="@+id/welcomeLogo" android:layout_width="80dp" android:layout_height="80dp" android:padding="10dp" android:background="@drawable/bg_logo_tile" android:elevation="3dp" android:src="@drawable/ic_app_logo" android:contentDescription="Ảnh" app:layout_constraintTop_toBottomOf="@id/heroAnchor" app:layout_constraintStart_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent"/>
<TextView android:id="@+id/welcomeTitle" android:layout_width="wrap_content" android:layout_height="wrap_content" android:text="Ảnh" android:textSize="46sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif-medium" android:layout_marginTop="12dp" app:layout_constraintTop_toBottomOf="@id/welcomeLogo" app:layou

t_constraintStart_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent"/>
<TextView android:id="@+id/welcomeSubtitle" android:layout_width="wrap_content
" android:layout_height="wrap_content" android:text="Lưu giữ khoảnh khắc đẹp" android:t
extColor="@color/text_secondary" android:textSize="16sp" android:layout_marginTop="4
dp" app:layout_constraintTop_toBottomOf="@id/welcomeTitle" app:layout_constraintStart
_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent"/>
<ImageView android:id="@+id/welcomeArt" android:layout_width="0dp" android:lay
out_height="0dp" android:src="@drawable/art_splash" android:scaleType="fitXY" android
:contentDescription="Những tấm ảnh giữa phong cảnh núi xanh" app:layout_constraintStart
_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent" app:layout_constraintBot
tom_toBottomOf="parent" app:layout_constraintTop_toBottomOf="@id/welcomeSubtitle"/
>
<com.google.android.material.button.MaterialButton android:id="@+id/start" style="
@style/PrimaryButton" android:layout_width="0dp" android:layout_height="52dp" android
:layout_marginHorizontal="60dp" android:text="Bắt đầu" android:textSize="16sp" android:
layout_marginBottom="8dp" app:layout_constraintBottom_toTopOf="@id/guest" app:layo
ut_constraintStart_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent"/>
<com.google.android.material.button.MaterialButton android:id="@+id/guest" style="
@style/TextButton" android:layout_width="wrap_content" android:layout_height="48dp" a
ndroid:text="Khám phá trên thiết bị" android:textSize="14sp" android:backgroundTint="@
color/primary_light" android:layout_marginBottom="36dp" app:cornerRadius="14dp" app:l
ayout_constraintBottom_toBottomOf="parent" app:layout_constraintStart_toStartOf="pare
nt" app:layout_constraintEnd_toEndOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
</FrameLayout>

2 2 18 activity_trash

Nguồn: app/src/main/res/layout/activity_trash.xml

Vai trò: Danh sách mục trong thùng rác và nút dọn.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="match_parent" android:orientation="vertical" an
droid:background="@color/background">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Thùng rác" app:ti
tleTextColor="?android:attr/textColorPrimary" app:navigationIcon="@drawable/ic_arrow_
back" app:titleTextAppearance="@style/GalleryToolbarTitle" app:navigationIconTint="@c
olor/text_primary"/>
<LinearLayout android:layout_width="match_parent" android:layout_height="0dp" andr
oid:layout_weight="1" android:orientation="vertical" android:paddingHorizontal="20dp">
<TextView android:layout_width="match_parent" android:layout_height="wrap_conte
nt" android:text="Ảnh được giữ 30 ngày. Xóa vĩnh viễn cần xác nhận." android:textSize="1
4sp" android:textColor="?android:attr/textColorPrimary" android:paddingVertical="8dp" an

droid:fontFamily="sans-serif"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:paddingVertical="8dp" android:id="@+id/status" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:visibility="gone"/>
<androidx.recyclerview.widget.RecyclerView android:id="@+id/grid" android:layout_width="match_parent" android:layout_height="0dp" android:layout_weight="1" android:clipToPadding="false" android:paddingBottom="16dp"/>
<com.google.android.material.button.MaterialButton android:id="@+id/emptyTrash" style="@style/PrimaryButton" android:layout_width="match_parent" android:layout_height="56dp" android:layout_marginTop="8dp" android:text="Dọn sạch thùng rác" android:letterSpacing="0"/>
</LinearLayout>
</LinearLayout>

2 2 19 dialog_photo_info

Nguồn: app/src/main/res/layout/dialog_photo_info.xml

Vai trò: Thông tin tên, kích thước, thời gian, vị trí lưu và loại ảnh.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:background="@color/background">
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_height="56dp" app:title="Chi tiết" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
<androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:layout_height="0dp" android:layout_weight="1">
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:paddingHorizontal="24dp" android:orientation="vertical">
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="horizontal" android:gravity="center_vertical" android:paddingVertical="16dp">
<com.google.android.material.card.MaterialCardView android:layout_width="76dp" android:layout_height="76dp" app:cardCornerRadius="10dp" app:strokeWidth="0dp">
<ImageView android:id="@+id/photo" android:layout_width="match_parent" android:layout_height="match_parent" android:scaleType="centerCrop" android:contentDescription="Ảnh xem trước"/>
</com.google.android.material.card.MaterialCardView>
<LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:layout_marginStart="16dp" android:orientation="vertical">

```
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="16sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif" android:id="@+id/infoName"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/infoSize" android:layout_marginTop="6dp"/>
        </LinearLayout>
    </LinearLayout>
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="horizontal" android:paddingVertical="16dp">
        <ImageView android:layout_width="24dp" android:layout_height="24dp" android:src="@drawable/ic_calendar" app:tint="@color/text_primary" android:contentDescription="Thời gian nhập"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:layout_marginStart="16dp" android:orientation="vertical">
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Thời gian nhập" android:textSize="15sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/infoDate" android:layout_marginTop="4dp"/>
        </LinearLayout>
    </LinearLayout>
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="horizontal" android:paddingVertical="16dp">
        <ImageView android:layout_width="24dp" android:layout_height="24dp" android:src="@drawable/ic_album" app:tint="@color/text_primary" android:contentDescription="Đường dẫn"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:layout_marginStart="16dp" android:orientation="vertical">
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Đường dẫn" android:textSize="15sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif"/>
            <TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/infoPath" android:layout_marginTop="4dp"/>
        </LinearLayout>
    </LinearLayout>
    <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="horizontal" android:paddingVertical="16dp">
        <ImageView android:layout_width="24dp" android:layout_height="24dp" android:src="@drawable/ic_device" app:tint="@color/text_primary" android:contentDescription="Thiết bị"/>
        <LinearLayout android:layout_width="0dp" android:layout_weight="1" android:l
```

ayout_height="wrap_content" android:layout_marginStart="16dp" android:orientation="vertical">
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Thiết bị" android:textSize="15sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/infoDevice" android:layout_marginTop="4dp"/>
</LinearLayout>
</LinearLayout>
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="horizontal" android:paddingVertical="16dp">
<ImageView android:layout_width="24dp" android:layout_height="24dp" android:src="@drawable/ic_file_type" app:tint="@color/text_primary" android:contentDescription="Loại file"/>
<LinearLayout android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:layout_marginStart="16dp" android:orientation="vertical">
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Loại file" android:textSize="15sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="" android:textSize="14sp" android:textColor="@color/text_secondary" android:fontFamily="sans-serif" android:id="@+id/infoType" android:layout_marginTop="4dp"/>
</LinearLayout>
</LinearLayout>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:padding="24dp" android:orientation="vertical">
<com.google.android.material.button.MaterialButton android:id="@+id/close" style="@style/PrimaryButton" android:layout_width="match_parent" android:layout_height="52dp" android:text="Đóng" android:letterSpacing="0" android:textSize="16sp"/>
</LinearLayout>
</LinearLayout>

2 2 20 item_album

Nguồn: app/src/main/res/layout/item_album.xml

Vai trò: Thẻ album có ảnh bìa, tên, số lượng và nút ba chấm.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:lay

```
out_width="match_parent" android:layout_height="wrap_content" android:layout_margin=
"5dp" app:cardCornerRadius="14dp" app:cardElevation="0dp" app:cardBackgroundColor=
"@color/surface" app:strokeWidth="0dp">
  <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_cont
ent" android:orientation="vertical">
    <FrameLayout android:layout_width="match_parent" android:layout_height="120dp"
>
      <ImageView android:id="@+id/thumbnail" android:layout_width="match_parent" a
ndroid:layout_height="match_parent" android:scaleType="centerCrop" android:contentDes
cription="Ảnh bìa album"/>
      <ImageView android:id="@+id/albumVideoBadge" android:layout_width="40dp" a
ndroid:layout_height="40dp" android:layout_gravity="center" android:background="@dra
wable/bg_video_duration" android:src="@drawable/ic_play_circle" app:tint="@color/white
" android:contentDescription="Album có video" android:visibility="gone"/>
    </FrameLayout>
    <FrameLayout android:layout_width="match_parent" android:layout_height="60dp">
      <TextView android:id="@+id/caption" android:layout_width="match_parent" andro
id:layout_height="match_parent" android:layout_marginEnd="40dp" android:paddingStart=
"10dp" android:paddingEnd="0dp" android:gravity="center_vertical" android:textColor="
@color/text_primary" android:textSize="15sp" android:lineSpacingExtra="4dp" android:ma
xLines="2" android:ellipsize="end"/>
      <ImageButton android:id="@+id/albumMore" android:layout_width="40dp" androi
d:layout_height="48dp" android:layout_gravity="end|center_vertical" android:padding="10
dp" android:background="?attr/selectableItemBackgroundBorderless" android:src="@draw
able/ic_more_vert" app:tint="@color/text_secondary" android:contentDescription="Tùy ch
ọn album"/>
    </FrameLayout>
    <ImageView android:id="@+id/check" android:layout_width="1dp" android:layout_h
eight="1dp" android:visibility="gone" android:contentDescription="Đã chọn"/>
  </LinearLayout>
</com.google.android.material.card.MaterialCardView>
```

2 2 21 item_media

Nguồn: app/src/main/res/layout/item_media.xml

Vai trò: Ô ảnh hoặc video trong lưới; huy hiệu số thể hiện thứ tự chọn.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schemas.andr
oid.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:lay
out_width="match_parent" android:layout_height="wrap_content" android:layout_margin=
"2dp" app:cardCornerRadius="8dp" app:cardElevation="0dp" app:strokeWidth="0dp">
  <androidx.constraintlayout.widget.ConstraintLayout android:layout_width="match_paren
t" android:layout_height="wrap_content">
    <ImageView android:id="@+id/thumbnail" android:layout_width="0dp" android:layo
ut_height="0dp" android:scaleType="centerCrop" android:contentDescription="Ảnh" app:l
```

ayout_constraintDimensionRatio="1:1" app:layout_constraintTop_toTopOf="parent" app:layout_constraintStart_toStartOf="parent" app:layout_constraintEnd_toEndOf="parent"/>
<TextView android:id="@+id/caption" android:layout_width="wrap_content" android:layout_height="wrap_content" android:layout_margin="6dp" android:background="#66081224" android:padding="4dp" android:textColor="@color/white" android:textSize="12sp" app:layout_constraintStart_toStartOf="parent" app:layout_constraintBottom_toBottomOf="@id/thumbnail"/>
<TextView android:id="@+id/check" android:layout_width="28dp" android:layout_height="28dp" android:layout_margin="6dp" android:visibility="gone" android:contentDescription="Đã chọn" app:layout_constraintTop_toTopOf="parent" app:layout_constraintEnd_toEndOf="parent" android:background="@drawable/bg_selection_number" android:textColor="@color/white" android:textSize="14sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:includeFontPadding="false" android:elevation="2dp"/>
</androidx.constraintlayout.widget.ConstraintLayout>
</com.google.android.material.card.MaterialCardView>

2 2 22 item_media_row

Nguồn: app/src/main/res/layout/item_media_row.xml

Vai trò: Một mục dạng danh sách, dùng trong thùng rác.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:layout_width="match_parent" android:layout_height="96dp" android:orientation="horizontal" android:gravity="center_vertical" android:paddingVertical="10dp">
<com.google.android.material.card.MaterialCardView android:layout_width="76dp" android:layout_height="76dp" app:cardCornerRadius="10dp" app:strokeWidth="0dp">
<ImageView android:id="@+id/thumbnail" android:layout_width="match_parent" android:layout_height="match_parent" android:scaleType="centerCrop" android:contentDescription="Ảnh trong thùng rác"/>
</com.google.android.material.card.MaterialCardView>
<TextView android:id="@+id/caption" android:layout_width="0dp" android:layout_weight="1" android:layout_height="wrap_content" android:layout_marginStart="16dp" android:textColor="@color/text_primary" android:textSize="15sp" android:maxLines="3" android:ellipsize="end" android:lineSpacingExtra="5dp"/>
<TextView android:id="@+id/check" android:layout_width="28dp" android:layout_height="28dp" android:visibility="gone" android:contentDescription="Đã chọn" android:background="@drawable/bg_selection_number" android:textColor="@color/white" android:textSize="14sp" android:fontFamily="sans-serif-medium" android:gravity="center" android:includeFontPadding="false" android:elevation="2dp"/>
</LinearLayout>

2 2 23 item_notification

Nguồn: app/src/main/res/layout/item_notification.xml

Vai trò: Một thông báo hoạt động gồm biểu tượng và nội dung.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:layout_width="match_parent" android:lay
out_height="wrap_content" android:minHeight="88dp" android:orientation="horizontal" an
droid:gravity="center_vertical" android:paddingVertical="16dp">
    <ImageView android:layout_width="40dp" android:layout_height="40dp" android:backg
round="@drawable/bg_circle_icon" android:padding="10dp" android:src="@drawable/ic_n
otification" app:tint="@color/primary" android:contentDescription="Thông báo"/>
    <TextView android:id="@+id/notificationMessage" android:layout_width="0dp" android
:layout_weight="1" android:layout_height="wrap_content" android:layout_marginStart="16
dp" android:textSize="15sp" android:textColor="@color/text_primary" android:lineSpacing
Extra="5dp"/>
</LinearLayout>
```

2 2 24 item_photo_grid

Nguồn: app/src/main/res/layout/item_photo_grid.xml

Vai trò: Bố cục màn hình, các sheet, hộp thoại và ô dữ liệu. Tên tài nguyên: item_photo_grid.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schemas.andr
oid.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:ns2
="http://schemas.android.com/tools" android:id="@+id/cardPhoto" android:layout_width="
match_parent" android:layout_height="wrap_content" android:layout_margin="2dp" app:ca
rdCornerRadius="4dp" app:cardElevation="0dp" app:strokeWidth="0dp" app:cardBackgrou
ndColor="@color/background_card">
    <androidx.constraintlayout.widget.ConstraintLayout android:layout_width="match_paren
t" android:layout_height="wrap_content">
        <ImageView android:id="@+id/imageViewPhoto" android:layout_width="0dp" androi
d:layout_height="0dp" android:scaleType="centerCrop" android:contentDescription="@nul
l" app:layout_constraintTop_toTopOf="parent" app:layout_constraintBottom_toBottomOf=
"parent" app:layout_constraintStart_toStartOf="parent" app:layout_constraintEnd_toEndOf
="parent" app:layout_constraintDimensionRatio="1:1" ns2:src="@drawable/ic_photos"/>
        <ImageView android:id="@+id/imageViewSelect" android:layout_width="24dp" andr
oid:layout_height="24dp" android:layout_marginTop="8dp" android:layout_marginEnd="8
dp" android:src="@drawable/ic_checkbox_unchecked" android:visibility="gone" android:c
ontentDescription="@null" app:layout_constraintTop_toTopOf="parent" app:layout_constr
aintEnd_toEndOf="parent" ns2:visibility="visible"/>
        <TextView android:id="@+id/textViewVideoDuration" android:layout_width="wrap_
```

content" android:layout_height="wrap_content" android:layout_marginStart="8dp" android:layout_marginBottom="8dp" android:background="@drawable/bg_video_duration" android:gravity="center_vertical" android:drawableStart="@drawable/ic_video" android:drawablePadding="4dp" android:textColor="@color/white" android:textSize="10sp" android:textStyle="bold" android:visibility="gone" app:layout_constraintBottom_toBottomOf="parent" app:layout_constraintStart_toStartOf="parent" ns2:text="02:45" ns2:visibility="visible" android:fontFamily="sans-serif"/>
</androidx.constraintlayout.widget.ConstraintLayout>
</com.google.android.material.card.MaterialCardView>

2 2 25 sheet_album_options

Nguồn: app/src/main/res/layout/sheet_album_options.xml

Vai trò: Menu chọn nhiều, thêm ảnh, đổi tên, bỏ thành viên hoặc xóa album.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto" android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="vertical" android:background="@drawable/bg_bottom_sheet" android:padding="12dp">
<View android:layout_width="36dp" android:layout_height="4dp" android:layout_gravity="center_horizontal" android:layout_marginVertical="8dp" android:background="@drawable/bg_handle_bar"/>
<TextView android:layout_width="match_parent" android:layout_height="wrap_content" android:text="Tùy chọn album" android:textSize="20sp" android:textColor="@color/text_primary" android:fontFamily="sans-serif" android:padding="16dp"/>
<com.google.android.material.button.MaterialButton android:id="@+id/select" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Chọn nhiều" app:icon="@drawable/ic_select_all"/>
<com.google.android.material.button.MaterialButton android:id="@+id/addPhotos" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Thêm ảnh" app:icon="@drawable/ic_album"/>
<com.google.android.material.button.MaterialButton android:id="@+id/renameAlbum" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Đổi tên" app:icon="@drawable/ic_edit"/>
<com.google.android.material.button.MaterialButton android:id="@+id/removeSelected" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Bỏ ảnh đã chọn khỏi album" app:icon="@drawable/ic_delete"/>
<com.google.android.material.button.MaterialButton android:id="@+id/deleteAlbum" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Xóa album" app:icon="@drawable/ic_delete" android:textColor="@color/danger" app:iconTint="@color/danger"/>
</LinearLayout>

2 2 26 sheet_filter

Nguồn: app/src/main/res/layout/sheet_filter.xml

Vai trò: Các cách sắp xếp thư viện.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="wrap_content" android:orientation="vertical" an
droid:background="@drawable/bg_bottom_sheet" android:padding="8dp">
    <View android:layout_width="36dp" android:layout_height="4dp" android:layout_gravit
y="center_horizontal" android:layout_marginTop="8dp" android:layout_marginBottom="8
dp" android:background="@drawable/bg_handle_bar"/>
    <com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Bộ lọc" app:titleT
extAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" a
pp:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_pri
mary"/>
    <androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="wrap_content">
        <LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent" android:orientation="vertical" android:paddingBottom="12dp">
            <com.google.android.material.button.MaterialButton android:id="@+id/newest" styl
e="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56d
p" android:text="Mới nhất" app:icon="@drawable/ic_calendar"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/oldest" style
="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56d
p" android:text="Cũ nhất" app:icon="@drawable/ic_sort"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/size" style=
"@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp"
 android:text="Kích thước" app:icon="@drawable/ic_image_size"/>
            <com.google.android.material.button.MaterialButton android:id="@+id/type" style=
"@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp"
 android:text="Tên A → Z" app:icon="@drawable/ic_file_type"/>
        </LinearLayout>
    </androidx.core.widget.NestedScrollView>
</LinearLayout>
```

2 2 27 sheet_options

Nguồn: app/src/main/res/layout/sheet_options.xml

Vai trò: Menu chọn nhiều, thêm album, ẩn, xem các mục đã ẩn và cài đặt.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
```

ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="wrap_content" android:orientation="vertical" an
droid:background="@drawable/bg_bottom_sheet" android:padding="8dp">
<View android:layout_width="36dp" android:layout_height="4dp" android:layout_gravit
y="center_horizontal" android:layout_marginTop="8dp" android:layout_marginBottom="8
dp" android:background="@drawable/bg_handle_bar"/>
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" androi
d:layout_width="match_parent" android:layout_height="56dp" app:title="Tùy chọn" app:tit
leTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary
" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_p
rimary"/>
<androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:
layout_height="wrap_content">
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_c
ontent" android:orientation="vertical" android:paddingBottom="12dp">
<com.google.android.material.button.MaterialButton android:id="@+id/select" style
="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56d
p" android:text="Chọn nhiều" app:icon="@drawable/ic_select_all"/>
<com.google.android.material.button.MaterialButton android:id="@+id/addAlbum"
style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="
56dp" android:text="Thêm vào album" app:icon="@drawable/ic_album"/>
<com.google.android.material.button.MaterialButton android:id="@+id/hide" style=
"@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp"
android:text="Ẩn các mục đã chọn" app:icon="@drawable/ic_visibility_off"/>
<com.google.android.material.button.MaterialButton android:id="@+id/hiddenMedi
a" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_heigh
t="56dp" android:text="Ảnh và video đã ẩn" app:icon="@drawable/ic_visibility_off"/>
<com.google.android.material.button.MaterialButton android:id="@+id/settings" sty
le="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56
dp" android:text="Cài đặt" app:icon="@drawable/ic_settings"/>
<com.google.android.material.button.MaterialButton android:id="@+id/help" style=
"@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp"
android:text="Trợ giúp" app:icon="@drawable/ic_help"/>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
</LinearLayout>

2 2 28 sheet_photo_context

Nguồn: app/src/main/res/layout/sheet_photo_context.xml

Vai trò: Menu khi nhấn giữ ảnh hoặc video.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h
ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_wid
th="match_parent" android:layout_height="wrap_content" android:orientation="vertical" an

droid:background="@drawable/bg_bottom_sheet" android:padding="8dp">
<View android:layout_width="36dp" android:layout_height="4dp" android:layout_gravity="center_horizontal" android:layout_marginTop="8dp" android:layout_marginBottom="8dp" android:background="@drawable/bg_handle_bar"/>
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_height="56dp" app:title="Ảnh của bạn" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
<androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:layout_height="wrap_content">
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="vertical" android:paddingBottom="12dp">
<com.google.android.material.button.MaterialButton android:id="@+id/info" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Xem chi tiết" app:icon="@drawable/ic_info"/>
<com.google.android.material.button.MaterialButton android:id="@+id/addAlbum" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Thêm vào album" app:icon="@drawable/ic_album"/>
<com.google.android.material.button.MaterialButton android:id="@+id/favorite" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Yêu thích" app:icon="@drawable/ic_favorite_outline"/>
<com.google.android.material.button.MaterialButton android:id="@+id/hide" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Ẩn / Bỏ ẩn" app:icon="@drawable/ic_visibility_off"/>
<com.google.android.material.button.MaterialButton android:id="@+id/edit" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Chỉnh sửa" app:icon="@drawable/ic_edit"/>
<com.google.android.material.button.MaterialButton android:id="@+id/share" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Chia sẻ" app:icon="@drawable/ic_share"/>
<com.google.android.material.button.MaterialButton android:id="@+id/trash" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Chuyển vào thùng rác" app:icon="@drawable/ic_delete" android:textColor="@color/danger" app:iconTint="@color/danger"/>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
</LinearLayout>

2 2 29 sheet_share

Nguồn: app/src/main/res/layout/sheet_share.xml

Vai trò: Các lựa chọn chia sẻ tệp hoặc liên kết nếu đã có URL hợp lệ.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="h

ttp://schemas.android.com/apk/res-auto" android:id="@+id/screenRoot" android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="vertical" android:background="@drawable/bg_bottom_sheet" android:padding="8dp">
<View android:layout_width="36dp" android:layout_height="4dp" android:layout_gravity="center_horizontal" android:layout_marginTop="8dp" android:layout_marginBottom="8dp" android:background="@drawable/bg_handle_bar"/>
<com.google.android.material.appbar.MaterialToolbar android:id="@+id/toolbar" android:layout_width="match_parent" android:layout_height="56dp" app:title="Chia sẻ ảnh" app:titleTextAppearance="@style/GalleryToolbarTitle" app:titleTextColor="@color/text_primary" app:navigationIcon="@drawable/ic_arrow_back" app:navigationIconTint="@color/text_primary"/>
<androidx.core.widget.NestedScrollView android:layout_width="match_parent" android:layout_height="wrap_content">
<LinearLayout android:layout_width="match_parent" android:layout_height="wrap_content" android:orientation="vertical" android:paddingBottom="12dp">
<com.google.android.material.button.MaterialButton android:id="@+id/shareFile" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Chia sẻ qua ứng dụng" app:icon="@drawable/ic_share"/>
<com.google.android.material.button.MaterialButton android:id="@+id/shareLink" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Chia sẻ liên kết sao lưu" app:icon="@drawable/ic_cloud_done"/>
<com.google.android.material.button.MaterialButton android:id="@+id/cancel" style="@style/GalleryRow" android:layout_width="match_parent" android:layout_height="56dp" android:text="Hủy" app:icon="@drawable/ic_close"/>
</LinearLayout>
</androidx.core.widget.NestedScrollView>
</LinearLayout>

2 3 Thư mục menu

Đường dẫn: app/src/main/res/menu

Khai báo menu thanh điều hướng và hành động trên toolbar. Số file XML: 6.

Danh sách: album_actions.xml, bottom_nav_menu.xml, bottom_navigation_menu.xml, context_menu.xml, gallery_actions.xml, options_menu.xml.

2 3 1 album_actions

Nguồn: app/src/main/res/menu/album_actions.xml

Vai trò: Khai báo menu thanh điều hướng và hành động trên toolbar. Tên tài nguyên: album_actions.

Mã nguồn đầy đủ

<?xml version="1.0" ?>
<menu xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://schemas.android.com/apk/res-auto">
<item android:id="@+id/addPhotos" android:title="Thêm ảnh" android:icon="@drawabl

```
e/ic_add" app:showAsAction="always"/>
   <item android:id="@+id/albumOptions" android:title="Tùy chọn album" android:icon="
@drawable/ic_more_vert" app:showAsAction="always"/>
</menu>
```

2 3 2 bottom_nav_menu

Nguồn: app/src/main/res/menu/bottom_nav_menu.xml

Vai trò: Khai báo menu thanh điều hướng và hành động trên toolbar. Tên tài nguyên: bottom_nav_menu.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
      android:id="@+id/nav_photos"
      android:icon="@drawable/ic_photos"
      android:title="@string/nav_photos" />
   <item
      android:id="@+id/nav_albums"
      android:icon="@drawable/ic_albums"
      android:title="@string/nav_albums" />
   <item
      android:id="@+id/nav_search"
      android:icon="@drawable/ic_search"
      android:title="@string/nav_search" />
   <item
      android:id="@+id/nav_profile"
      android:icon="@drawable/ic_profile"
      android:title="@string/nav_profile" />
</menu>
```

2 3 3 bottom_navigation_menu

Nguồn: app/src/main/res/menu/bottom_navigation_menu.xml

Vai trò: Khai báo menu thanh điều hướng và hành động trên toolbar. Tên tài nguyên: bottom_navigation_menu.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
      android:id="@+id/nav_photos"
      android:icon="@drawable/ic_photo"
      android:title="@string/nav_photos" />
```

```
    <item
        android:id="@+id/nav_albums"
        android:icon="@drawable/ic_album"
        android:title="@string/nav_albums" />
    <item
        android:id="@+id/nav_search"
        android:icon="@drawable/ic_search"
        android:title="@string/nav_search" />
    <item
        android:id="@+id/nav_profile"
        android:icon="@drawable/ic_person"
        android:title="@string/nav_profile" />
</menu>
```

2 3 4 context_menu

Nguồn: app/src/main/res/menu/context_menu.xml

Vai trò: Khai báo menu thanh điều hướng và hành động trên toolbar. Tên tài nguyên: context_menu.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/context_detail"
        android:icon="@drawable/ic_info"
        android:title="@string/context_view_detail" />
    <item
        android:id="@+id/context_add_album"
        android:icon="@drawable/ic_album"
        android:title="@string/context_add_to_album" />
    <item
        android:id="@+id/context_favorite"
        android:icon="@drawable/ic_favorite_outline"
        android:title="@string/context_favorite" />
    <item
        android:id="@+id/context_edit"
        android:icon="@drawable/ic_edit"
        android:title="@string/context_edit" />
    <item
        android:id="@+id/context_share"
        android:icon="@drawable/ic_share"
        android:title="@string/context_share" />
    <item
        android:id="@+id/context_delete"
        android:icon="@drawable/ic_delete"
        android:title="@string/context_delete" />
```

</menu>

2 3 5 gallery_actions

Nguồn: app/src/main/res/menu/gallery_actions.xml

Vai trò: Khai báo menu thanh điều hướng và hành động trên toolbar. Tên tài nguyên: gallery_actions.

Mã nguồn đầy đủ

```
<menu xmlns:android="http://schemas.android.com/apk/res/android" xmlns:app="http://sch
emas.android.com/apk/res-auto">
    <item android:id="@+id/galleryFilter" android:title="Bộ lọc" android:icon="@drawable/
ic_filter" app:showAsAction="always"/>
    <item android:id="@+id/galleryOptions" android:title="Tùy chọn" android:icon="@dra
wable/ic_more_vert" app:showAsAction="always"/>
</menu>
```

2 3 6 options_menu

Nguồn: app/src/main/res/menu/options_menu.xml

Vai trò: Khai báo menu thanh điều hướng và hành động trên toolbar. Tên tài nguyên: options_menu.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item
        android:id="@+id/menu_select"
        android:icon="@drawable/ic_select_all"
        android:title="@string/menu_select_multiple"
        app:showAsAction="never" />
    <item
        android:id="@+id/menu_add_album"
        android:icon="@drawable/ic_album"
        android:title="@string/menu_add_to_album"
        app:showAsAction="never" />
    <item
        android:id="@+id/menu_settings"
        android:icon="@drawable/ic_settings"
        android:title="@string/menu_settings"
        app:showAsAction="never" />
    <item
        android:id="@+id/menu_backup"
        android:icon="@drawable/ic_cloud_upload"
```

```
        android:title="@string/menu_backup"
        app:showAsAction="never" />
    <item
        android:id="@+id/menu_help"
        android:icon="@drawable/ic_help"
        android:title="@string/menu_help"
        app:showAsAction="never" />
</menu>
```

2 4 Thư mục values

Đường dẫn: app/src/main/res/values

Chuỗi, màu, theme và style mặc định của ứng dụng. Số file XML: 7.

Danh sách: colors.xml, dimens.xml, strings.xml, theme_colors.xml, theme_flags.xml, themes.xml, ui_styles.xml.

2 4 1 colors

Nguồn: app/src/main/res/values/colors.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: colors.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="primary">#2864FF</color>
    <color name="primary_dark">#1951E8</color>
    <color name="primary_light">#DCE9FF</color>
    <color name="primary_variant">#2E6DB5</color>
    <color name="accent">#FF6B35</color>
    <color name="accent_light">#FF8A5C</color>
    <color name="background">#F6F9FF</color>
    <color name="background_light">#F5F7FA</color>
    <color name="background_card">#F0F4F8</color>
    <color name="surface">#FFFFFF</color>
    <color name="text_primary">#102C63</color>
    <color name="text_secondary">#526D99</color>
    <color name="text_hint">#7084A6</color>
    <color name="text_white">#FFFFFF</color>
    <color name="success">#10B981</color>
    <color name="warning">#F59E0B</color>
    <color name="error">#EF4444</color>
    <color name="info">#3B82F6</color>
    <color name="tab_selected">#2864FF</color>
    <color name="tab_unselected">#9CA3AF</color>
```

```
    <color name="tab_indicator">#2864FF</color>
    <color name="chip_selected_bg">#E8F0FE</color>
    <color name="chip_selected_text">#2864FF</color>
    <color name="chip_unselected_bg">#F3F4F6</color>
    <color name="chip_unselected_text">#6B7280</color>
    <color name="bottom_nav_bg">#FFFFFF</color>
    <color name="bottom_nav_selected">#2864FF</color>
    <color name="bottom_nav_unselected">#9CA3AF</color>
    <color name="divider">#E5E7EB</color>
    <color name="overlay_dark">#80000000</color>
    <color name="overlay_light">#40000000</color>
    <color name="gradient_start">#2864FF</color>
    <color name="gradient_end">#6BA5E7</color>
    <color name="dark_background">#102557</color>
    <color name="dark_surface">#252540</color>
    <color name="dark_card">#2D2D4A</color>
    <color name="dark_text_primary">#FFFFFF</color>
    <color name="dark_text_secondary">#A0A0B8</color>
    <color name="trash_bg">#FEF2F2</color>
    <color name="trash_text">#DC2626</color>
    <color name="fab_bg">#2864FF</color>
    <color name="fab_icon">#FFFFFF</color>
    <color name="field_border">#CEDDF3</color>
</resources>
```

2 4 2 dimens

Nguồn: app/src/main/res/values/dimens.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: dimens.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <dimen name="spacing_xs">4dp</dimen>
    <dimen name="spacing_sm">8dp</dimen>
    <dimen name="spacing_md">16dp</dimen>
    <dimen name="spacing_lg">24dp</dimen>
    <dimen name="spacing_xl">32dp</dimen>
    <dimen name="spacing_xxl">48dp</dimen>

    <dimen name="text_xs">10sp</dimen>
    <dimen name="text_sm">12sp</dimen>
    <dimen name="text_md">14sp</dimen>
    <dimen name="text_lg">16sp</dimen>
    <dimen name="text_xl">18sp</dimen>
    <dimen name="text_xxl">24sp</dimen>
    <dimen name="text_title">28sp</dimen>
```

```
    <dimen name="text_splash">48sp</dimen>

    <dimen name="corner_radius_sm">8dp</dimen>
    <dimen name="corner_radius_md">12dp</dimen>
    <dimen name="corner_radius_lg">16dp</dimen>
    <dimen name="corner_radius_xl">24dp</dimen>
    <dimen name="corner_radius_circle">50dp</dimen>

    <dimen name="icon_sm">20dp</dimen>
    <dimen name="icon_md">24dp</dimen>
    <dimen name="icon_lg">32dp</dimen>
    <dimen name="icon_xl">48dp</dimen>
    <dimen name="icon_xxl">64dp</dimen>

    <dimen name="button_height">48dp</dimen>
    <dimen name="bottom_nav_height">56dp</dimen>
    <dimen name="toolbar_height">56dp</dimen>
    <dimen name="fab_size">56dp</dimen>

    <dimen name="photo_grid_spacing">2dp</dimen>
    <dimen name="album_card_height">120dp</dimen>
    <dimen name="album_cover_size">80dp</dimen>
</resources>
```

2 4 3 strings

Nguồn: app/src/main/res/values/strings.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: strings.

Mã nguồn đầy đủ

```
<resources>
    <string name="app_name">App Ảnh</string>

    <!-- Bottom Navigation -->
    <string name="nav_photos">Ảnh</string>
    <string name="nav_albums">Album</string>
    <string name="nav_search">Tìm kiếm</string>
    <string name="nav_profile">Hồ sơ</string>

    <!-- Options Menu -->
    <string name="menu_select_multiple">Chọn nhiều</string>
    <string name="menu_add_to_album">Thêm vào album</string>
    <string name="menu_settings">Cài đặt</string>
    <string name="menu_backup">Sao lưu</string>
    <string name="menu_help">Trợ giúp</string>

    <!-- Context Menu -->
```

```
<string name="context_view_detail">Xem chi tiết</string>
<string name="context_add_to_album">Thêm vào album</string>
<string name="context_favorite">Yêu thích</string>
<string name="context_edit">Chỉnh sửa</string>
<string name="context_share">Chia sẻ</string>
<string name="context_delete">Xóa</string>

<!-- Photo Detail View -->
<string name="action_favorite">Yêu thích</string>
<string name="action_edit">Chỉnh sửa</string>
<string name="action_share">Chia sẻ</string>
<string name="action_delete">Xóa</string>
<string name="action_more">Thêm</string>

<!-- Edit Photo -->
<string name="edit_title">Chỉnh sửa</string>
<string name="edit_crop">Cắt</string>
<string name="edit_filter">Bộ lọc</string>
<string name="edit_adjust">Điều chỉnh</string>
<string name="edit_text">Văn bản</string>
<string name="edit_rotate">Xoay</string>
<string name="btn_cancel">Hủy</string>
<string name="btn_save">Lưu</string>

<!-- Photo Info -->
<string name="info_title">Chi tiết</string>
<string name="info_filename">Tên tệp</string>
<string name="info_size">Kích thước</string>
<string name="info_date">Thời gian chụp</string>
<string name="info_location">Đường dẫn</string>
<string name="info_device">Thiết bị</string>
<string name="info_type">Loại</string>
<string name="btn_close">Đóng</string>

<!-- Album -->
<string name="album_title">Album</string>
<string name="btn_create_album">Tạo album</string>
<string name="album_default_count">0 ảnh</string>

<!-- Create Album -->
<string name="create_album_title">Tạo album</string>
<string name="hint_album_name">Tên album</string>
<string name="hint_album_desc">Mô tả</string>
<string name="btn_choose_cover">Chọn ảnh</string>
<string name="btn_create">Tạo album</string>

<!-- Album Detail -->
<string name="album_detail_title">Album chi tiết</string>
<string name="album_select_multiple">Chọn nhiều</string>
```

```
    <string name="album_add_photo">Thêm ảnh</string>
    <string name="album_rename">Đổi tên</string>
    <string name="album_delete">Xóa album</string>
    <string name="album_stats_photos">%1$d ảnh</string>
    <string name="album_stats_selected">%1$d đã chọn</string>

    <!-- Trash -->
    <string name="trash_title">Thùng rác</string>
    <string name="trash_subtitle">Các mục sẽ tự xóa vĩnh viễn sau 30 ngày</string>
    <string name="btn_empty_trash">Dọn sạch thùng rác</string>
    <string name="btn_restore">Khôi phục</string>
    <string name="btn_delete_permanent">Xóa vĩnh viễn</string>
</resources>
```

2 4 4 theme_colors

Nguồn: app/src/main/res/values/theme_colors.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: theme_colors.

Mã nguồn đầy đủ

```
<resources>
    <color name="button_primary">#2864FF</color>
    <color name="theme_on_primary">#FFFFFF</color>
    <color name="danger">#CB243C</color>
    <color name="viewer_background">#F6F9FF</color>
    <color name="viewer_controls">#FFFFFF</color>
    </resources>
```

2 4 5 theme_flags

Nguồn: app/src/main/res/values/theme_flags.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: theme_flags.

Mã nguồn đầy đủ

```
<resources><bool name="light_system_bars">true</bool></resources>
```

2 4 6 themes

Nguồn: app/src/main/res/values/themes.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: themes.

Mã nguồn đầy đủ

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme -->
    <style name="Base.Theme.AppẢnh" parent="Theme.Material3.DayNight.NoActionBar"
```

```
>
    <item name="colorPrimary">@color/primary</item>
    <item name="colorPrimaryDark">@color/primary_dark</item>
    <item name="colorAccent">@color/accent</item>
    <item name="colorOnPrimary">@color/theme_on_primary</item>
    <item name="colorSecondary">@color/primary_light</item>
    <item name="colorOnSecondary">@color/text_primary</item>
    <item name="colorSurface">@color/surface</item>
    <item name="colorSurfaceVariant">@color/background_card</item>
    <item name="colorOnSurface">@color/text_primary</item>
    <item name="colorOnSurfaceVariant">@color/text_secondary</item>
    <item name="colorOnBackground">@color/text_primary</item>
    <item name="colorOutline">@color/field_border</item>
    <item name="colorError">@color/danger</item>
    <item name="android:colorBackground">@color/background</item>
    <item name="android:textColorPrimary">@color/text_primary</item>
    <item name="android:textColorSecondary">@color/text_secondary</item>
    <item name="android:statusBarColor">@color/background</item>
    <item name="android:windowLightStatusBar">@bool/light_system_bars</item>
    <item name="android:navigationBarColor">@color/background</item>
    <item name="android:windowBackground">@color/background</item>
</style>

<style name="Theme.AppẢnh" parent="Base.Theme.AppẢnh" />

<!-- Splash Theme -->
<style name="Theme.AppẢnh.Splash" parent="Theme.AppẢnh">
    <item name="android:statusBarColor">@color/background</item>
    <item name="android:windowLightStatusBar">@bool/light_system_bars</item>
    <item name="android:windowBackground">@color/background</item>
</style>

<!-- Fullscreen Theme for photo viewing -->
<style name="Theme.AppẢnh.Fullscreen" parent="Theme.AppẢnh">
    <item name="android:statusBarColor">@android:color/transparent</item>
    <item name="android:navigationBarColor">@android:color/transparent</item>
    <item name="android:windowBackground">@color/viewer_background</item>
</style>

<!-- Dark Theme -->
<style name="Theme.AppẢnh.Dark" parent="Theme.Material3.Dark.NoActionBar">
    <item name="colorPrimary">@color/primary</item>
    <item name="android:statusBarColor">@color/dark_background</item>
    <item name="android:windowBackground">@color/dark_background</item>
</style>

<!-- Dialog Theme -->
<style name="Theme.AppẢnh.Dialog" parent="Theme.Material3.DayNight.Dialog">
    <item name="colorPrimary">@color/primary</item>
```

```
    </style>

    <!-- Custom Button Style -->
    <style name="PrimaryButton" parent="Widget.Material3.Button">
        <item name="android:textAllCaps">false</item>
        <item name="cornerRadius">14dp</item>
        <item name="backgroundTint">@color/button_primary</item>
        <item name="android:textColor">@color/white</item>
        <item name="android:textSize">16sp</item>
        <item name="android:fontFamily">sans-serif-medium</item>
        <item name="android:insetTop">0dp</item>
        <item name="android:insetBottom">0dp</item>
    </style>

    <!-- Outlined Button Style -->
    <style name="OutlinedButton" parent="Widget.Material3.Button.OutlinedButton">
        <item name="android:textAllCaps">false</item>
        <item name="cornerRadius">14dp</item>
        <item name="strokeColor">@color/field_border</item>
        <item name="android:textColor">@color/text_primary</item>
        <item name="android:textSize">16sp</item>
    </style>

    <!-- Text Button Style -->
    <style name="TextButton" parent="Widget.Material3.Button.TextButton">
        <item name="android:textAllCaps">false</item>
        <item name="android:textColor">@color/primary</item>
        <item name="android:textSize">14sp</item>
    </style>

    <!-- EditText Style -->
    <style name="CustomEditText" parent="Widget.Material3.TextInputLayout.OutlinedBo
x">
        <item name="boxCornerRadiusTopStart">12dp</item>
        <item name="boxCornerRadiusTopEnd">12dp</item>
        <item name="boxCornerRadiusBottomStart">12dp</item>
        <item name="boxCornerRadiusBottomEnd">12dp</item>
        <item name="boxStrokeColor">@color/input_stroke</item>
        <item name="boxBackgroundColor">@android:color/transparent</item>
        <item name="boxStrokeWidth">1dp</item>
        <item name="boxStrokeWidthFocused">1.5dp</item>
        <item name="hintTextColor">@color/text_hint</item>
    </style>

    <!-- Card Style -->
    <style name="PhotoCard" parent="Widget.Material3.CardView.Filled">
        <item name="cardCornerRadius">12dp</item>
        <item name="cardElevation">0dp</item>
    </style>
```

```
    <!-- Bottom Sheet Style -->
    <style name="CustomBottomSheet" parent="Widget.Material3.BottomSheet">
        <item name="shapeAppearance">@style/BottomSheetShape</item>
    </style>

    <style name="BottomSheetShape" parent="">
        <item name="cornerFamily">rounded</item>
        <item name="cornerSizeTopLeft">20dp</item>
        <item name="cornerSizeTopRight">20dp</item>
    </style>
</resources>
```

2 4 7 ui_styles

Nguồn: app/src/main/res/values/ui_styles.xml

Vai trò: Chuỗi, màu, theme và style mặc định của ứng dụng. Tên tài nguyên: ui_styles.

Mã nguồn đầy đủ

```
<resources>
    <style name="GalleryToolbarTitle" parent="TextAppearance.Material3.TitleLarge">
        <item name="android:textSize">22sp</item>
        <item name="android:fontFamily">sans-serif-medium</item>
        <item name="android:textColor">@color/text_primary</item>
    </style>
    <style name="GalleryNavigationText" parent="TextAppearance.Material3.LabelMedium
">
        <item name="android:textSize">12sp</item>
        <item name="android:fontFamily">sans-serif-medium</item>
    </style>
    <style name="GalleryRow" parent="Widget.Material3.Button.TextButton">
        <item name="android:gravity">start|center_vertical</item>
        <item name="android:textColor">@color/text_primary</item>
        <item name="android:textSize">16sp</item>
        <item name="android:textAllCaps">false</item>
        <item name="android:letterSpacing">0</item>
        <item name="android:paddingStart">16dp</item>
        <item name="iconGravity">start</item>
        <item name="iconPadding">18dp</item>
        <item name="iconSize">22dp</item>
        <item name="iconTint">@color/text_primary</item>
        <item name="cornerRadius">12dp</item>
        <item name="android:insetTop">0dp</item>
        <item name="android:insetBottom">0dp</item>
    </style>
</resources>
```

2 5 Thư mục values night

Đường dẫn: app/src/main/res/values-night

Màu, theme và giá trị thay thế trong chế độ tối. Số file XML: 3.

Danh sách: colors.xml, theme_flags.xml, themes.xml.

2 5 1 colors

Nguồn: app/src/main/res/values-night/colors.xml

Vai trò: Màu, theme và giá trị thay thế trong chế độ tối. Tên tài nguyên: colors.

Mã nguồn đầy đủ

<resources><color name="background">#10151F</color><color name="surface">#1C253
4</color><color name="text_primary">#EDF3FF</color><color name="text_secondary">
#ABBBD5</color><color name="text_hint">#9EAFCA</color><color name="field_borde
r">#52617B</color><color name="primary">#91B7FF</color><color name="primary_ligh
t">#253A61</color><color name="theme_on_primary">#102C63</color><color name="d
anger">#FF929D</color><color name="error">#FF929D</color><color name="backgroun
d_light">#192231</color><color name="background_card">#253145</color><color name
="divider">#35435A</color><color name="bottom_nav_bg">#1C2534</color><color nam
e="bottom_nav_selected">#91B7FF</color><color name="bottom_nav_unselected">#AB
BBD5</color><color name="tab_selected">#91B7FF</color><color name="tab_unselecte
d">#ABBBD5</color><color name="tab_indicator">#91B7FF</color><color name="chip
_selected_bg">#253A61</color><color name="chip_selected_text">#91B7FF</color><col
or name="chip_unselected_bg">#1C2534</color><color name="chip_unselected_text">#A
BBBD5</color><color name="trash_bg">#3A2431</color><color name="trash_text">#FF
929D</color><color name="viewer_background">#080D17</color><color name="viewer_
controls">#121C2B</color></resources>

2 5 2 theme_flags

Nguồn: app/src/main/res/values-night/theme_flags.xml

Vai trò: Màu, theme và giá trị thay thế trong chế độ tối. Tên tài nguyên: theme_flags.

Mã nguồn đầy đủ

<resources><bool name="light_system_bars">false</bool></resources>

2 5 3 themes

Nguồn: app/src/main/res/values-night/themes.xml

Vai trò: Màu, theme và giá trị thay thế trong chế độ tối. Tên tài nguyên: themes.

Mã nguồn đầy đủ

<resources/>

2 6 Thư mục values v31

Đường dẫn: app/src/main/res/values-v31

Thiết lập theme dành cho Android API 31 trở lên. Số file XML: 1.

Danh sách: themes.xml.

2 6 1 themes

Nguồn: app/src/main/res/values-v31/themes.xml

Vai trò: Thiết lập theme dành cho Android API 31 trở lên. Tên tài nguyên: themes.

Mã nguồn đầy đủ

```
<resources>
   <style name="Theme.AppẢnh.Splash" parent="Theme.AppẢnh">
      <item name="android:windowSplashScreenBackground">@color/background</item>
      <item name="android:windowSplashScreenAnimatedIcon">@drawable/ic_app_logo</
item>
      <item name="android:windowSplashScreenIconBackgroundColor">@color/backgroun
d</item>
      <item name="android:windowBackground">@color/background</item>
      <item name="android:statusBarColor">@color/background</item>
      <item name="android:windowLightStatusBar">@bool/light_system_bars</item>
   </style>
</resources>
```

2 7 Thư mục color

Đường dẫn: app/src/main/res/color

Màu có trạng thái cho chip và thanh điều hướng. Số file XML: 4.

Danh sách: chip_background.xml, chip_text.xml, input_stroke.xml, nav_colors.xml.

2 7 1 chip_background

Nguồn: app/src/main/res/color/chip_background.xml

Vai trò: Màu có trạng thái cho chip và thanh điều hướng. Tên tài nguyên: chip_background.

Mã nguồn đầy đủ

```
<selector xmlns:android="http://schemas.android.com/apk/res/android"><item android:state
_checked="true" android:color="@color/button_primary"/><item android:color="@color/s
urface"/></selector>
```

2 7 2 chip_text

Nguồn: app/src/main/res/color/chip_text.xml

Vai trò: Màu có trạng thái cho chip và thanh điều hướng. Tên tài nguyên: chip_text.

Mã nguồn đầy đủ

<selector xmlns:android="http://schemas.android.com/apk/res/android"><item android:state_checked="true" android:color="#FFFFFF"/><item android:color="@color/text_primary"/></selector>

2 7 3 input_stroke

Nguồn: app/src/main/res/color/input_stroke.xml

Vai trò: Màu có trạng thái cho chip và thanh điều hướng. Tên tài nguyên: input_stroke.

Mã nguồn đầy đủ

```
<selector xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:state_focused="true" android:color="@color/primary"/>
    <item android:color="@color/field_border"/>
</selector>
```

2 7 4 nav_colors

Nguồn: app/src/main/res/color/nav_colors.xml

Vai trò: Màu có trạng thái cho chip và thanh điều hướng. Tên tài nguyên: nav_colors.

Mã nguồn đầy đủ

```
<selector xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:state_checked="true" android:color="@color/primary"/>
    <item android:color="@color/text_secondary"/>
</selector>
```

2 8 Thư mục drawable

Đường dẫn: app/src/main/res/drawable

Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Số file XML: 98.

Danh sách: art_landscape.xml, art_splash.xml, bg_album_card.xml, bg_auth_surface.xml, bg_bottom_sheet.xml, bg_chip_selected.xml, bg_chip_unselected.xml, bg_circle_icon.xml, bg_circle_primary.xml, bg_circle_white.xml, bg_dotted_border.xml, bg_edit_text.xml, bg_fab_add.xml, bg_google_button.xml, bg_gradient_primary.xml, bg_handle_bar.xml, bg_header_rounded_bottom.xml, bg_logo_tile.xml, bg_notification_item.xml, bg_otp_box.xml, bg_photo_overlay.xml, bg_rounded_primary.xml, bg_rounded_white.xml,

bg_search_bar.xml, bg_selection_number.xml, bg_settings_item.xml, bg_settings_row.xml, bg_splash_gradient.xml, bg_trash_item.xml, bg_video_duration.xml, ic_about.xml, ic_add.xml, ic_adjust.xml, ic_album.xml, ic_albums.xml, ic_app_logo.xml, ic_arrow_back.xml, ic_back.xml, ic_calendar.xml, ic_camera.xml, ic_checkbox_checked.xml, ic_checkbox_unchecked.xml, ic_chevron_right.xml, ic_close.xml, ic_cloud_done.xml, ic_cloud_upload.xml, ic_copy.xml, ic_crop.xml, ic_dark_mode.xml, ic_delete.xml, ic_device.xml, ic_download.xml, ic_drive.xml, ic_edit.xml, ic_email.xml, ic_facebook.xml, ic_favorite.xml, ic_favorite_filled.xml, ic_favorite_outline.xml, ic_file_type.xml, ic_filter.xml, ic_gmail.xml, ic_google.xml, ic_help.xml, ic_image_size.xml, ic_info.xml, ic_instagram.xml, ic_language.xml, ic_launcher_background.xml, ic_launcher_foreground.xml, ic_location.xml, ic_lock.xml, ic_logout.xml, ic_messages.xml, ic_messenger.xml, ic_more_vert.xml, ic_notification.xml, ic_person.xml, ic_photo.xml, ic_photos.xml, ic_play_circle.xml, ic_privacy.xml, ic_profile.xml, ic_restore.xml, ic_rotate.xml, ic_search.xml, ic_select_all.xml, ic_settings.xml, ic_share.xml, ic_share_nearby.xml, ic_sort.xml, ic_sync.xml, ic_text.xml, ic_video.xml, ic_visibility.xml, ic_visibility_off.xml, ic_wifi.xml, ic_zalo.xml.

2 8 1 art_landscape

Nguồn: app/src/main/res/drawable/art_landscape.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: art_landscape.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<vector xmlns:android="http://schemas.android.com/apk/res/android" android:width="360dp" android:height="240dp" android:viewportWidth="360" android:viewportHeight="240">
    <path android:fillColor="#EAF2FF" android:pathData="M0,0h360v240h-360z"/>
    <path android:fillColor="#FFC76B" android:pathData="M264,48a28,28 0,1 0,1 0z"/>
    <path android:fillColor="#B7D4FF" android:pathData="M0,180L90,55 195,180 265,83 360,177V240H0z"/>
    <path android:fillColor="#75A9F8" android:pathData="M0,210L120,115 222,205 295,151 360,208V240H0z"/>
    <path android:fillColor="#5ECDB4" android:pathData="M0,224Q90,170 180,220Q270,172 360,222V240H0z"/>
</vector>
```

2 8 2 art_splash

Nguồn: app/src/main/res/drawable/art_splash.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: art_splash.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<vector xmlns:android="http://schemas.android.com/apk/res/android" android:width="360d
p" android:height="410dp" android:viewportWidth="360" android:viewportHeight="410">
    <path android:fillColor="#E6F0FF" android:pathData="M0,126Q30,110 55,138Q99,166
120,139Q160,104 199,142Q236,165 269,130Q312,95 360,127V410H0Z"/>
    <path android:fillColor="#C1D9FF" android:pathData="M0,211Q20,167 45,180L80,217
L124,161L172,210L218,171L262,218L304,167L360,211V410H0Z"/>
    <path android:fillColor="#96BEF8" android:pathData="M0,274L40,225L79,253L130,20
5L188,274L238,218L286,262L330,222L360,243V410H0Z"/>
    <path android:fillColor="#72A9EA" android:pathData="M0,319Q45,261 92,294Q143,32
4 184,283Q244,255 287,296Q330,266 360,297V410H0Z"/>
    <path android:fillColor="#B8DCF7" android:pathData="M106,410Q165,359 161,328Q1
56,300 204,289Q172,321 199,348Q225,379 258,410Z"/>
    <path android:fillColor="#A0D6B0" android:pathData="M0,355Q22,324 56,348Q81,326
 110,350L76,410H0Z M272,410L268,362Q300,328 324,351Q344,328 360,349V410Z"/>
    <path android:fillColor="#6BBF9E" android:pathData="M0,390Q25,356 56,372Q69,345
 92,373Q103,367 117,389L124,410H0Z M242,410Q264,369 291,383Q299,361 321,375Q3
45,358 360,381V410Z"/>
    <path android:fillColor="#D4E6FF" android:pathData="M28,57Q25,46 36,42Q49,37 54,
50Q67,47 69,59Q69,65 60,65H31Q24,64 28,57Z M286,84Q280,74 290,69Q303,62 310,76
Q322,72 325,84Q324,91 316,91H290Z"/>
    <path android:fillColor="#71B5FC" android:pathData="M61,92l5,-4l4,5l-5,4Z M281,13
9l4,-3l3,5l-4,2Z M109,64l3,-3l3,3l-3,3Z"/>
    <path android:fillColor="#FFC754" android:pathData="M165,48l5,2l-2,7l-4,-2Z M264,1
18l6,1l-1,5l-6,-1Z"/>
    <group android:translateX="88" android:translateY="92" android:rotation="-17" android:
pivotX="36" android:pivotY="36">
        <path android:fillColor="#D3E3F7" android:pathData="M2,3h77v83h-77Z"/>
        <path android:fillColor="#FFFFFF" android:pathData="M0,0h76v82h-76Z"/>
        <path android:fillColor="#B9E2FF" android:pathData="M6,6h64v60h-64Z"/>
        <path android:fillColor="#FFD168" android:pathData="M55,12a8,8 0,1 0,0 16a8,8 0,1
 0,0 -16"/>
        <path android:fillColor="#53AD98" android:pathData="M6,56L28,26L47,48L58,35L
70,54V66H6Z"/>
        <path android:fillColor="#8ED58A" android:pathData="M6,61L23,48L43,60L59,50L
70,60V66H6Z"/>
    </group>
    <group android:translateX="183" android:translateY="69" android:rotation="12" android
:pivotX="36" android:pivotY="36">
        <path android:fillColor="#D3E3F7" android:pathData="M2,3h77v83h-77Z"/>
        <path android:fillColor="#FFFFFF" android:pathData="M0,0h76v82h-76Z"/>
        <path android:fillColor="#B9E2FF" android:pathData="M6,6h64v60h-64Z"/>
        <path android:fillColor="#FFD168" android:pathData="M55,12a8,8 0,1 0,0 16a8,8 0,1
 0,0 -16"/>
        <path android:fillColor="#53AD98" android:pathData="M6,56L28,26L47,48L58,35L
70,54V66H6Z"/>
        <path android:fillColor="#8ED58A" android:pathData="M6,61L23,48L43,60L59,50L
70,60V66H6Z"/>
    </group>
```

```
    <group android:translateX="139" android:translateY="137" android:rotation="-5" androi
d:pivotX="36" android:pivotY="36">
        <path android:fillColor="#D3E3F7" android:pathData="M2,3h77v83h-77Z"/>
        <path android:fillColor="#FFFFFF" android:pathData="M0,0h76v82h-76Z"/>
        <path android:fillColor="#B9E2FF" android:pathData="M6,6h64v60h-64Z"/>
        <path android:fillColor="#FFD168" android:pathData="M55,12a8,8 0,1 0,0 16a8,8 0,1
 0,0 -16"/>
        <path android:fillColor="#53AD98" android:pathData="M6,56L28,26L47,48L58,35L
70,54V66H6Z"/>
        <path android:fillColor="#8ED58A" android:pathData="M6,61L23,48L43,60L59,50L
70,60V66H6Z"/>
    </group>
</vector>
```

2 8 3 bg_album_card

Nguồn: app/src/main/res/drawable/bg_album_card.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_album_card.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/surface" />
    <corners android:radius="16dp" />
    <stroke
        android:width="1dp"
        android:color="@color/divider" />
</shape>
```

2 8 4 bg_auth_surface

Nguồn: app/src/main/res/drawable/bg_auth_surface.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_auth_surface.

Mã nguồn đầy đủ

```
<shape xmlns:android="http://schemas.android.com/apk/res/android"><gradient android:an
gle="315" android:startColor="@color/surface" android:centerColor="@color/background"
 android:endColor="@color/background_card"/></shape>
```

2 8 5 bg_bottom_sheet

Nguồn: app/src/main/res/drawable/bg_bottom_sheet.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_bottom_sheet.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/surface" />
    <corners
        android:topLeftRadius="24dp"
        android:topRightRadius="24dp" />
</shape>
```

2 8 6 bg_chip_selected

Nguồn: app/src/main/res/drawable/bg_chip_selected.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_chip_selected.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/primary" />
    <corners android:radius="16dp" />
</shape>
```

2 8 7 bg_chip_unselected

Nguồn: app/src/main/res/drawable/bg_chip_unselected.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_chip_unselected.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/chip_unselected_bg" />
    <corners android:radius="20dp" />
</shape>
```

2 8 8 bg_circle_icon

Nguồn: app/src/main/res/drawable/bg_circle_icon.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_circle_icon.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="@color/chip_selected_bg" />
</shape>
```

2 8 9 bg_circle_primary

Nguồn: app/src/main/res/drawable/bg_circle_primary.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_circle_primary.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="@color/primary" />
</shape>
```

2 8 10 bg_circle_white

Nguồn: app/src/main/res/drawable/bg_circle_white.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_circle_white.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="@color/surface" />
</shape>
```

2 8 11 bg_dotted_border

Nguồn: app/src/main/res/drawable/bg_dotted_border.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_dotted_border.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/background_card" />
    <corners android:radius="12dp" />
    <stroke
        android:width="2dp"
        android:color="@color/primary"
        android:dashWidth="8dp"
        android:dashGap="6dp" />
</shape>
```

2 8 12 bg_edit_text

Nguồn: app/src/main/res/drawable/bg_edit_text.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_edit_text.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/background_light" />
    <corners android:radius="12dp" />
    <stroke
        android:width="1dp"
        android:color="@color/divider" />
</shape>
```

2 8 13 bg_fab_add

Nguồn: app/src/main/res/drawable/bg_fab_add.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_fab_add.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="@color/primary" />
    <size
        android:width="56dp"
        android:height="56dp" />
</shape>
```

2 8 14 bg_google_button

Nguồn: app/src/main/res/drawable/bg_google_button.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_google_button.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/surface" />
    <corners android:radius="24dp" />
    <stroke
        android:width="1dp"
        android:color="@color/divider" />
</shape>
```

2 8 15 bg_gradient_primary

Nguồn: app/src/main/res/drawable/bg_gradient_primary.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_gradient_primary.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <gradient
        android:startColor="@color/gradient_start"
        android:endColor="@color/gradient_end"
        android:angle="135" />
</shape>
```

2 8 16 bg_handle_bar

Nguồn: app/src/main/res/drawable/bg_handle_bar.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_handle_bar.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/divider" />
    <corners android:radius="4dp" />
    <size
        android:width="40dp"
        android:height="4dp" />
</shape>
```

2 8 17 bg_header_rounded_bottom

Nguồn: app/src/main/res/drawable/bg_header_rounded_bottom.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_header_rounded_bottom.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/primary_light" />
    <corners
        android:bottomLeftRadius="32dp"
        android:bottomRightRadius="32dp" />
</shape>
```

2 8 18 bg_logo_tile

Nguồn: app/src/main/res/drawable/bg_logo_tile.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_logo_tile.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#FFFFFF"/>
    <corners android:radius="24dp"/>
    <stroke android:width="1dp" android:color="#E7EEFC"/>
</shape>
```

2 8 19 bg_notification_item

Nguồn: app/src/main/res/drawable/bg_notification_item.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_notification_item.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/background_light" />
    <corners android:radius="12dp" />
</shape>
```

2 8 20 bg_otp_box

Nguồn: app/src/main/res/drawable/bg_otp_box.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_otp_box.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<selector xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:state_focused="true">
        <shape android:shape="rectangle">
            <solid android:color="@color/surface" />
            <stroke android:width="2dp" android:color="@color/primary" />
            <corners android:radius="12dp" />
        </shape>
    </item>
    <item>
        <shape android:shape="rectangle">
            <solid android:color="@color/background_light" />
            <stroke android:width="1dp" android:color="@color/field_border" />
            <corners android:radius="12dp" />
        </shape>
    </item>
</selector>
```

2 8 21 bg_photo_overlay

Nguồn: app/src/main/res/drawable/bg_photo_overlay.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_photo_overlay.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <gradient
        android:startColor="#00000000"
        android:endColor="#80000000"
        android:angle="270" />
</shape>
```

2 8 22 bg_rounded_primary

Nguồn: app/src/main/res/drawable/bg_rounded_primary.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_rounded_primary.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/primary" />
    <corners android:radius="24dp" />
</shape>
```

2 8 23 bg_rounded_white

Nguồn: app/src/main/res/drawable/bg_rounded_white.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_rounded_white.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/surface" />
    <corners android:radius="16dp" />
</shape>
```

2 8 24 bg_search_bar

Nguồn: app/src/main/res/drawable/bg_search_bar.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_search_bar.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
   <solid android:color="@color/primary_light"/>
   <corners android:radius="24dp"/>
   <stroke android:width="1dp" android:color="@color/field_border"/>
</shape>
```

2 8 25 bg_selection_number

Nguồn: app/src/main/res/drawable/bg_selection_number.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_selection_number.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android" android:shape="oval"
>
   <solid android:color="@color/button_primary"/>
   <stroke android:width="2dp" android:color="@color/white"/>
</shape>
```

2 8 26 bg_settings_item

Nguồn: app/src/main/res/drawable/bg_settings_item.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_settings_item.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<ripple xmlns:android="http://schemas.android.com/apk/res/android"
   android:color="@color/divider">
   <item>
      <shape android:shape="rectangle">
         <solid android:color="@color/surface" />
      </shape>
   </item>
</ripple>
```

2 8 27 bg_settings_row

Nguồn: app/src/main/res/drawable/bg_settings_row.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_settings_row.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="@color/surface"/>
    <corners android:radius="12dp"/>
</shape>
```

2 8 28 bg_splash_gradient

Nguồn: app/src/main/res/drawable/bg_splash_gradient.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_splash_gradient.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <gradient
        android:angle="135"
        android:startColor="#4A90D9"
        android:centerColor="#3A7BC8"
        android:endColor="#2E6DB5"
        android:type="linear" />
</shape>
```

2 8 29 bg_trash_item

Nguồn: app/src/main/res/drawable/bg_trash_item.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_trash_item.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
```

```
    <solid android:color="@color/background_light" />
    <corners android:radius="12dp" />
    <stroke
        android:width="1dp"
        android:color="@color/divider" />
</shape>
```

2 8 30 bg_video_duration

Nguồn: app/src/main/res/drawable/bg_video_duration.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: bg_video_duration.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">
    <solid android:color="@color/overlay_dark" />
    <corners android:radius="12dp" />
    <padding
        android:left="6dp"
        android:top="2dp"
        android:right="6dp"
        android:bottom="2dp" />
</shape>
```

2 8 31 ic_about

Nguồn: app/src/main/res/drawable/ic_about.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_about.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M11,17h2v-6h-2v6zm1,-15C6.48,2 2,6.48 2,12s4.48,10 10,10 10,-4.48 10,-10S17.52,2 12,2zm0,18c-4.41,0 -8,-3.59 -8,-8s3.59,-8 8,-8 8,3.59 8,8 -3.59,8 -8,8z M11,9h2V7h-2v2z"/>
</vector>
```

2 8 32 ic_add

Nguồn: app/src/main/res/drawable/ic_add.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_add.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#FFFFFF"
        android:pathData="M19,13h-6v6h-2v-6H5v-2h6V5h2v6h6v2z" />
</vector>
```

2 8 33 ic_adjust

Nguồn: app/src/main/res/drawable/ic_adjust.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_adjust.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M3,17v2h6v-2H3zm0,-10v2h10V7H3zm10,16v-2h8v-2h-8v-2h-2v
6h2zM7,9v2H3v2h4v2h2V9H7zm14,4v-2H11v2h10zm-6,-4h2V7h4V5h-4V3h-2v6z" />
</vector>
```

2 8 34 ic_album

Nguồn: app/src/main/res/drawable/ic_album.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_album.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#4A90D9"
      android:pathData="M4,6H2v14c0,1.1 0.9,2 2,2h14v-2H4V6zm16,-4H8c-1.1,0 -2,0.9 -
2,2v12c0,1.1 0.9,2 2,2h12c1.1,0 2,-0.9 2,-2V4c0,-1.1 -0.9,-2 -2,-2zm0,14H8V4h12v12zm-6
.5,-3.5l-2.5,3.01L9.5,13l-2.5,3.5h10l-3.5,-4.5z" />
</vector>
```

2 8 35 ic_albums

Nguồn: app/src/main/res/drawable/ic_albums.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_albums.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24"
   android:tint="?attr/colorControlNormal">
   <path
      android:fillColor="@android:color/white"
      android:pathData="M4,6H2v14c0,1.1 0.9,2 2,2h14v-2H4V6zm16,-4H8c-1.1,0 -2,0.9 -
2,2v12c0,1.1 0.9,2 2,2h12c1.1,0 2,-0.9 2,-2V4c0,-1.1 -0.9,-2 -2,-2zm0,14H8V4h12v12z" />
</vector>
```

2 8 36 ic_app_logo

Nguồn: app/src/main/res/drawable/ic_app_logo.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_app_logo.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<vector xmlns:android="http://schemas.android.com/apk/res/android" android:width="96dp
" android:height="96dp" android:viewportWidth="96" android:viewportHeight="96">
   <group android:pivotX="48" android:pivotY="48" android:rotation="0">
      <path android:fillColor="#FF5A31" android:fillAlpha="0.93" android:pathData="M48
,49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
```

```
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="45">
        <path android:fillColor="#FFBD25" android:fillAlpha="0.93" android:pathData="M4
8,49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="90">
        <path android:fillColor="#7DCD39" android:fillAlpha="0.93" android:pathData="M4
8,49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="135">
        <path android:fillColor="#28C68B" android:fillAlpha="0.93" android:pathData="M48
,49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="180">
        <path android:fillColor="#25BDE5" android:fillAlpha="0.93" android:pathData="M4
8,49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="225">
        <path android:fillColor="#2784F4" android:fillAlpha="0.93" android:pathData="M48,
49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="270">
        <path android:fillColor="#7057F6" android:fillAlpha="0.93" android:pathData="M48,
49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <group android:pivotX="48" android:pivotY="48" android:rotation="315">
        <path android:fillColor="#D646ED" android:fillAlpha="0.93" android:pathData="M4
8,49C34,40 32,27 38,20C43,13 53,13 58,20C65,30 60,41 48,49Z"/>
    </group>
    <path android:fillColor="#FFFFFF" android:pathData="M48,43a5,5 0,1 0,0 10a5,5 0,1 0,
0 -10"/>
</vector>
```

2 8 37 ic_arrow_back

Nguồn: app/src/main/res/drawable/ic_arrow_back.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_arrow_back.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
```

```
        android:fillColor="#1A1A2E"
        android:pathData="M20,11H7.83l5.59,-5.59L12,4l-8,8 8,8 1.41,-1.41L7.83,13H20v-2z
" />
</vector>
```

2 8 38 ic_back

Nguồn: app/src/main/res/drawable/ic_back.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_back.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M20,11H7.83l5.59,-5.59L12,4l-8,8 8,8 1.41,-1.41L7.83,13H20v-2z
" />
</vector>
```

2 8 39 ic_calendar

Nguồn: app/src/main/res/drawable/ic_calendar.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_calendar.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M19,3h-1V1h-2v2H8V1H6v2H5c-1.11,0 -1.99,0.9 -1.99,2L3,19c0
,1.1 0.89,2 2,2h14c1.1,0 2,-0.9 2,-2V5c0,-1.1 -0.9,-2 -2,-2zm0,16H5V8h14v11zM7,10h5v5
H7z" />
</vector>
```

2 8 40 ic_camera

Nguồn: app/src/main/res/drawable/ic_camera.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_camera.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#4A90D9"
        android:pathData="M9,2L7.17,4H4c-1.1,0 -2,0.9 -2,2v12c0,1.1 0.9,2 2,2h16c1.1,0 2,-0
.9 2,-2V6c0,-1.1 -0.9,-2 -2,-2h-3.17L15,2H9zm3,15c-2.76,0 -5,-2.24 -5,-5s2.24,-5 5,-5 5,2.2
4 5,5 -2.24,5 -5,5zm0,-8c-1.66,0 -3,1.34 -3,3s1.34,3 3,3 3,-1.34 3,-3 -1.34,-3 -3,-3z" />
</vector>
```

2 8 41 ic_checkbox_checked

Nguồn: app/src/main/res/drawable/ic_checkbox_checked.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_checkbox_checked.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="oval">
    <solid android:color="@color/primary" />
    <size
        android:width="24dp"
        android:height="24dp" />
</shape>
```

2 8 42 ic_checkbox_unchecked

Nguồn: app/src/main/res/drawable/ic_checkbox_unchecked.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_checkbox_unchecked.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:shape="oval">
    <solid android:color="#40FFFFFF" />
    <stroke
        android:width="2dp"
        android:color="@color/white" />
    <size
        android:width="24dp"
        android:height="24dp" />
</shape>
```

2 8 43 ic_chevron_right

Nguồn: app/src/main/res/drawable/ic_chevron_right.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_chevron_right.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<vector xmlns:android="http://schemas.android.com/apk/res/android" android:width="20dp
" android:height="20dp" android:viewportWidth="24" android:viewportHeight="24">
    <path android:fillColor="#526D99" android:pathData="M9,6l6,6l-6,6l-1.4,-1.4l4.6,-4.6l-
4.6,-4.6Z"/>
</vector>
```

2 8 44 ic_close

Nguồn: app/src/main/res/drawable/ic_close.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_close.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M19,6.41L17.59,5 12,10.59 6.41,5 5,6.41 10.59,12 5,17.59 6.41,19
 12,13.41 17.59,19 19,17.59 13.41,12z" />
</vector>
```

2 8 45 ic_cloud_done

Nguồn: app/src/main/res/drawable/ic_cloud_done.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_cloud_done.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#10B981"
        android:pathData="M19.35,10.04C18.67,6.59 15.64,4 12,4 9.11,4 6.6,5.64 5.35,8.04 2.
34,8.36 0,10.91 0,14c0,3.31 2.69,6 6,6h13c2.76,0 5,-2.24 5,-5 0,-2.64 -2.05,-4.78 -4.65,-4.9
6zM10,17l-3.5,-3.5 1.41,-1.41L10,14.17 15.59,8.58 17,10 10,17z" />
</vector>
```

2 8 46 ic_cloud_upload

Nguồn: app/src/main/res/drawable/ic_cloud_upload.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_cloud_upload.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M19.35,10.04C18.67,6.59 15.64,4 12,4 9.11,4 6.6,5.64 5.35,8.04 2.
34,8.36 0,10.91 0,14c0,3.31 2.69,6 6,6h13c2.76,0 5,-2.24 5,-5 0,-2.64 -2.05,-4.78 -4.65,-4.9
6zM14,13v4h-4v-4H7l5,-5 5,5h-3z" />
</vector>
```

2 8 47 ic_copy

Nguồn: app/src/main/res/drawable/ic_copy.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_copy.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#1A1A2E"
      android:pathData="M16,1H4c-1.1,0 -2,0.9 -2,2v14h2V3h12V1zm3,4H8c-1.1,0 -2,0.9 -
2,2v14c0,1.1 0.9,2 2,2h11c1.1,0 2,-0.9 2,-2V7c0,-1.1 -0.9,-2 -2,-2zm0,16H8V7h11v14z"/>
</vector>
```

2 8 48 ic_crop

Nguồn: app/src/main/res/drawable/ic_crop.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_crop.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#6B7280"
      android:pathData="M17,15h2V7c0,-1.1 -0.9,-2 -2,-2H9v2h8v8zM7,17V1H5v4H1v2h
4v10c0,1.1 0.9,2 2,2h10v4h2v-4h4v-2H7z" />
</vector>
```

2 8 49 ic_dark_mode

Nguồn: app/src/main/res/drawable/ic_dark_mode.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_dark_mode.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#6B7280"
```

```
      android:pathData="M12.3,2a10,10 0 0,0 -1.9,0.18 10,10 0 1,0 11.42,11.42A10,10 0 0,0
 12.3,2z" />
</vector>
```

2 8 50 ic_delete

Nguồn: app/src/main/res/drawable/ic_delete.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_delete.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#EF4444"
      android:pathData="M6,19c0,1.1 0.9,2 2,2h8c1.1,0 2,-0.9 2,-2V7H6v12zM19,4h-3.5l-1
,-1h-5l-1,1H5v2h14V4z" />
</vector>
```

2 8 51 ic_device

Nguồn: app/src/main/res/drawable/ic_device.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_device.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#6B7280"
      android:pathData="M16,1H8C6.34,1 5,2.34 5,4v16c0,1.66 1.34,3 3,3h8c1.66,0 3,-1.34
 3,-3V4c0,-1.66 -1.34,-3 -3,-3zm-2,20h-4v-1h4v1zm3.25,-3H6.75V4h10.5v14z" />
</vector>
```

2 8 52 ic_download

Nguồn: app/src/main/res/drawable/ic_download.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_download.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M19.35,10.04C18.67,6.59 15.64,4 12,4 9.11,4 6.6,5.64 5.35,8.04 2.
34,8.36 0,10.91 0,14c0,3.31 2.69,6 6,6h13c2.76,0 5,-2.24 5,-5 0,-2.64 -2.05,-4.78 -4.65,-4.9
6zM17,13l-5,5 -5,-5h3V9h4v4h3z"/>
</vector>
```

2 8 53 ic_drive

Nguồn: app/src/main/res/drawable/ic_drive.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_drive.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#FFC107"
        android:pathData="M31.5,8h-15l-9,15.5 7.5,13h15l9,-15.5L31.5,8zm-1.7,4l6.9,12h-13
.8l-6.9,-12h13.8z"/>
</vector>
```

2 8 54 ic_edit

Nguồn: app/src/main/res/drawable/ic_edit.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_edit.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M3,17.25V21h3.75L17.81,9.94l-3.75,-3.75L3,17.25zM20.71,7.04c
0.39,-0.39 0.39,-1.02 0,-1.41l-2.34,-2.34c-0.39,-0.39 -1.02,-0.39 -1.41,0l-1.83,1.83 3.75,3.7
5 1.83,-1.83z" />
</vector>
```

2 8 55 ic_email

Nguồn: app/src/main/res/drawable/ic_email.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_email.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M20,4H4C2.9,4 2.01,4.9 2.01,6L2,18C2,19.1 2.9,20 4,20H20C21.
1,20 22,19.1 22,18V6C22,4.9 21.1,4 20,4ZM20,8L12,13L4,8V6L12,11L20,6V8Z" />
</vector>
```

2 8 56 ic_facebook

Nguồn: app/src/main/res/drawable/ic_facebook.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_facebook.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
```

```
        android:fillColor="#1877F2"
        android:pathData="M24,4C12.95,4 4,12.95 4,24c0,9.99 7.34,18.26 16.88,19.76V29.38
h-5.22V24h5.22v-4.1c0,-5.15 3.07,-8 7.76,-8 2.25,0 4.6.4 4.6.4v5.06h-2.59c-2.55,0 -3.35,1.
58 -3.35,3.2V24h5.7l-0.91,5.38h-4.79v14.38C36.66,42.26 44,33.99 44,24c0,-11.05 -8.95,-2
0 -20,-20z"/>
</vector>
```

2 8 57 ic_favorite

Nguồn: app/src/main/res/drawable/ic_favorite.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_favorite.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#EF4444"
        android:pathData="M12,21.35l-1.45,-1.32C5.4,15.36 2,12.28 2,8.5 2,5.42 4.42,3 7.5,3
c1.74,0 3.41,0.81 4.5,2.09C13.09,3.81 14.76,3 16.5,3 19.58,3 22,5.42 22,8.5c0,3.78 -3.4,6.8
6 -8.55,11.54L12,21.35z" />
</vector>
```

2 8 58 ic_favorite_filled

Nguồn: app/src/main/res/drawable/ic_favorite_filled.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_favorite_filled.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#EF4444"
        android:pathData="M12,21.35l-1.45,-1.32C5.4,15.36 2,12.28 2,8.5 2,5.42 4.42,3 7.5,3
c1.74,0 3.41,0.81 4.5,2.09C13.09,3.81 14.76,3 16.5,3 19.58,3 22,5.42 22,8.5c0,3.78 -3.4,6.8
6 -8.55,11.54L12,21.35z" />
</vector>
```

2 8 59 ic_favorite_outline

Nguồn: app/src/main/res/drawable/ic_favorite_outline.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_favorite_outline.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M16.5,3c-1.74,0 -3.41,0.81 -4.5,2.09C10.91,3.81 9.24,3 7.5,3 4.42,
3 2,5.42 2,8.5c0,3.78 3.4,6.86 8.55,11.54L12,21.35l1.45,-1.32C18.6,15.36 22,12.28 22,8.5
22,5.42 19.58,3 16.5,3zm-4.4,15.55l-0.1,0.1 -0.1,-0.1C7.14,14.24 4,11.39 4,8.5 4,6.5 5.5,5 7
.5,5c1.54,0 3.04,0.99 3.57,2.36h1.87C13.46,5.99 14.96,5 16.5,5c2,0 3.5,1.5 3.5,3.5 0,2.89 -
3.14,5.74 -7.9,10.05z" />
</vector>
```

2 8 60 ic_file_type

Nguồn: app/src/main/res/drawable/ic_file_type.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_file_type.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M14,2H6c-1.1,0 -1.99,0.9 -1.99,2L4,20c0,1.1 0.89,2 1.99,2H18c1.
1,0 2,-0.9 2,-2V8l-6,-6zm2,16H8v-2h8v2zm0,-4H8v-2h8v2zm-3,-5V3.5L18.5,9H13z" />
</vector>
```

2 8 61 ic_filter

Nguồn: app/src/main/res/drawable/ic_filter.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_filter.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M4.25,5.61C6.27,8.2 10,13 10,13v6c0,0.55 0.45,1 1,1h2c0.55,0 1,-0.45 1,-1v-6c0,0 3.72,-4.8 5.74,-7.39C20.25,4.95 19.78,4 18.95,4H5.04C4.21,4 3.74,4.95 4.25,5.61z" />
</vector>
```

2 8 62 ic_gmail

Nguồn: app/src/main/res/drawable/ic_gmail.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_gmail.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#EA4335"
        android:pathData="M40,8H8c-2.21,0 -4,1.79 -4,4v24c0,2.21 1.79,4 4,4h32c2.21,0 4,-1.79 4,-4V12c0,-2.21 -1.79,-4 -4,-4zm-1.6,6L24,23.25 9.6,14h28.8zM8,36V16.83l14.78,9.5c0.75,0.48 1.69,0.48 2.44,0L40,16.83V36H8z"/>
</vector>
```

2 8 63 ic_google

Nguồn: app/src/main/res/drawable/ic_google.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_google.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
```

```
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <!-- Blue -->
    <path
        android:fillColor="#4285F4"
        android:pathData="M23.745,12.27c0,-0.7 -0.06,-1.4 -0.19,-2.07H12v4.51h6.6c-0.29,1.
52 -1.14,2.82 -2.4,3.68v3.05h3.88c2.27,-2.09 3.665,-5.17 3.665,-9.17z" />
    <!-- Green -->
    <path
        android:fillColor="#34A853"
        android:pathData="M12,24c3.24,0 5.95,-1.08 7.93,-2.91l-3.88,-3.05c-1.08,0.72 -2.45,1
.16 -4.05,1.16 -3.12,0 -5.77,-2.1 -6.72,-4.93H1.26v3.15C3.25,21.36 7.33,24 12,24z" />
    <!-- Yellow -->
    <path
        android:fillColor="#FBBC05"
        android:pathData="M5.28,14.27c-0.25,-0.72 -0.38,-1.49 -0.38,-2.27s0.14,-1.55 0.38,-2.
27V6.58H1.26C0.46,8.16 0,9.94 0,12s0.46,3.84 1.26,5.42l4.02,-3.15z" />
    <!-- Red -->
    <path
        android:fillColor="#EA4335"
        android:pathData="M12,4.75c1.77,0 3.35,0.61 4.6,1.8l3.42,-3.42C17.95,1.19 15.24,0 1
2,0 7.33,0 3.25,2.64 1.26,6.58l4.02,3.15c0.95,-2.83 3.6,-4.98 6.72,-4.98z" />
</vector>
```

2 8 64 ic_help

Nguồn: app/src/main/res/drawable/ic_help.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_help.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M12,2C6.48,2 2,6.48 2,12s4.48,10 10,10 10,-4.48 10,-10S17.52,2
12,2zm1,16h-2v-2h2v2zm1.07,-7.75l-0.9,0.92C12.45,11.9 12,12.5 12,14h-2v-0.5c0,-1.1 0.4
5,-2.1 1.17,-2.83l1.24,-1.26c0.37,-0.36 0.59,-0.86 0.59,-1.41 0,-1.1 -0.9,-2 -2,-2s-2,0.9 -2,2
H7c0,-2.76 2.24,-5 5,-5s5,2.24 5,5c0,1.04 -0.42,1.99 -1.07,2.75z" />
</vector>
```

2 8 65 ic_image_size

Nguồn: app/src/main/res/drawable/ic_image_size.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_image_size.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M19,12h-2v3h-3v2h5v-5zM7,9h3V7H5v5h2V9zm14,-6H3c-1.1,0
-2,0.9 -2,2v14c0,1.1 0.9,2 2,2h18c1.1,0 2,-0.9 2,-2V5c0,-1.1 -0.9,-2 -2,-2zm0,16.01H3V4.9
9h18v14.02z" />
</vector>
```

2 8 66 ic_info

Nguồn: app/src/main/res/drawable/ic_info.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_info.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M12,2C6.48,2 2,6.48 2,12s4.48,10 10,10 10,-4.48 10,-10S17.52,2
12,2zm1,15h-2v-6h2v6zm0,-8h-2V7h2v2z" />
</vector>
```

2 8 67 ic_instagram

Nguồn: app/src/main/res/drawable/ic_instagram.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_instagram.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#E1306C"
        android:pathData="M24,4c6.52,0 7.29,0.02 9.87,0.14 2.38,0.11 3.67,0.51 4.53,0.85 1.1
4,0.44 1.95,0.97 2.81,1.82 0.85,0.85 1.38,1.67 1.82,2.81 0.34,0.86 0.74,2.15 0.85,4.53 0.12,
2.57 0.14,3.34 0.14,9.86s-0.02,7.29 -0.14,9.86c-0.11,2.38 -0.51,3.67 -0.85,4.53 -0.44,1.14 -
0.97,1.95 -1.82,2.81 -0.85,0.85 -1.67,1.38 -2.81,1.82 -0.86,0.34 -2.15,0.74 -4.53,0.85 -2.57,
0.12 -3.34,0.14 -9.86,0.14s-7.29,-0.02 -9.86,-0.14c-2.38,-0.11 -3.67,-0.51 -4.53,-0.85 -1.14,-
0.44 -1.95,-0.97 -2.81,-1.82 -0.85,-0.85 -1.38,-1.67 -1.82,-2.81 -0.34,-0.86 -0.74,-2.15 -0.85,
-4.53C4.02,31.29 4,30.52 4,24s0.02,-7.29 0.14,-9.86c0.11,-2.38 0.51,-3.67 0.85,-4.53 0.44,-
1.14 0.97,-1.95 1.82,-2.81 0.85,-0.85 1.67,-1.38 2.81,-1.82 0.86,-0.34 2.15,-0.74 4.53,-0.85
C16.71,4.02 17.48,4 24,4zm0,6.49c-7.46,0 -13.51,6.05 -13.51,13.51S16.54,37.51 24,37.51
37.51,31.46 37.51,24 31.46,10.49 24,10.49zm0,22.22c-4.81,0 -8.71,-3.9 -8.71,-8.71s3.9,-8.7
1 8.71,-8.71 8.71,3.9 8.71,8.71 -3.9,8.71 -8.71,8.71zm14.06,-23.47c0,1.74 -1.41,3.15 -3.15,
3.15 -1.74,0 -3.15,-1.41 -3.15,-3.15 0,-1.74 1.41,-3.15 3.15,-3.15 1.74,0 3.15,1.41 3.15,3.15
z"/>
</vector>
```

2 8 68 ic_language

Nguồn: app/src/main/res/drawable/ic_language.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_language.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M11.99,2C6.47,2 2,6.48 2,12s4.47,10 9.99,10C17.52,22 22,17.52 2
2,12S17.52,2 11.99,2zm6.93,6h-2.95c-0.32,-1.25 -0.78,-2.45 -1.38,-3.56 1.84,0.63 3.37,1.91
 4.33,3.56zM12,4.04c0.83,1.2 1.48,2.53 1.91,3.96h-3.82c0.43,-1.43 1.08,-2.76 1.91,-3.96z
M4.26,14C4.1,13.36 4,12.69 4,12s0.1,-1.36 0.26,-2h3.38c-0.08,0.66 -0.14,1.32 -0.14,2 0,0.6
8 0.06,1.34 0.14,2H4.26zm0.82,2h2.95c0.32,1.25 0.78,2.45 1.38,3.56 -1.84,-0.63 -3.37,-1.9
-4.33,-3.56zm2.95,-8H5.08c0.96,-1.66 2.49,-2.93 4.33,-3.56 -0.6,1.11 -1.06,2.31 -1.38,3.56
zM12,19.96c-0.83,-1.2 -1.48,-2.53 -1.91,-3.96h3.82c-0.43,1.43 -1.08,2.76 -1.91,3.96zM14.3
4,14H9.66c-0.09,-0.66 -0.16,-1.32 -0.16,-2 0,-0.68 0.07,-1.35 0.16,-2h4.68c0.09,0.65 0.16,1
.32 0.16,2 0,0.68 -0.07,1.34 -0.16,2zm0.15,5.56c0.6,-1.11 1.06,-2.31 1.38,-3.56h2.95c-0.96,
1.65 -2.49,2.93 -4.33,3.56zM16.36,14c0.08,-0.66 0.14,-1.32 0.14,-2 0,-0.68 -0.06,-1.34 -0.1
4,-2h3.38c0.16,0.64 0.26,1.31 0.26,2s-0.1,1.36 -0.26,2h-3.38z" />
```

</vector>

2 8 69 ic_launcher_background

Nguồn: app/src/main/res/drawable/ic_launcher_background.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_launcher_background.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
   <solid android:color="#F3F7FF"/>
</shape>
```

2 8 70 ic_launcher_foreground

Nguồn: app/src/main/res/drawable/ic_launcher_foreground.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_launcher_foreground.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<inset xmlns:android="http://schemas.android.com/apk/res/android" android:drawable="@d
rawable/ic_app_logo" android:inset="18dp"/>
```

2 8 71 ic_location

Nguồn: app/src/main/res/drawable/ic_location.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_location.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
   android:width="24dp"
   android:height="24dp"
   android:viewportWidth="24"
   android:viewportHeight="24">
   <path
      android:fillColor="#6B7280"
      android:pathData="M12,2C8.13,2 5,5.13 5,9c0,5.25 7,13 7,13s7,-7.75 7,-13c0,-3.87 -3
.13,-7 -7,-7zm0,9.5c-1.38,0 -2.5,-1.12 -2.5,-2.5s1.12,-2.5 2.5,-2.5 2.5,1.12 2.5,2.5 -1.12,2.5 -
2.5,2.5z" />
```

</vector>

2 8 72 ic_lock

Nguồn: app/src/main/res/drawable/ic_lock.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_lock.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M18,8h-1V6c0,-2.76 -2.24,-5 -5,-5S7,3.24 7,6v2H6c-1.1,0 -2,0.9 -2,2v10c0,1.1 0.9,2 2,2h12c1.1,0 2,-0.9 2,-2V10c0,-1.1 -0.9,-2 -2,-2zm-6,9c-1.1,0 -2,-0.9 -2,-2s0.9,-2 2,-2 2,0.9 2,2 -0.9,2 -2,2zm3.1,-9H8.9V6c0,-1.71 1.39,-3.1 3.1,-3.1 1.71,0 3.1,1.39 3.1,3.1v2z" />
</vector>
```

2 8 73 ic_logout

Nguồn: app/src/main/res/drawable/ic_logout.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_logout.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#EF4444"
        android:pathData="M17,7l-1.41,1.41L18.17,11H8v2h10.17l-2.58,2.58L17,17l5,-5zM4,5h8V3H4c-1.1,0 -2,0.9 -2,2v14c0,1.1 0.9,2 2,2h8v-2H4V5z"/>
</vector>
```

2 8 74 ic_messages

Nguồn: app/src/main/res/drawable/ic_messages.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_messages.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#34A853"
        android:pathData="M24,4C12.95,4 4,12.95 4,24c0,3.95 1.15,7.63 3.12,10.74L4,44l9.5
3,-3.06C16.49,42.75 20.1,44 24,44c11.05,0 20,-8.95 20,-20S35.05,4 24,4zm-6,18c-1.1,0 -2,
-0.9 -2,-2s0.9,-2 2,-2 2,0.9 2,2 -0.9,2 -2,2zm6,0c-1.1,0 -2,-0.9 -2,-2s0.9,-2 2,-2 2,0.9 2,2 -0.9
,2 -2,2zm6,0c-1.1,0 -2,-0.9 -2,-2s0.9,-2 2,-2 2,0.9 2,2 -0.9,2 -2,2z"/>
</vector>
```

2 8 75 ic_messenger

Nguồn: app/src/main/res/drawable/ic_messenger.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_messenger.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#0084FF"
        android:pathData="M24,4C12.95,4 4,12.42 4,22.8c0,5.91 2.91,11.19 7.45,14.61v7.2l6.
89,-3.79c1.83,0.51 3.75,0.78 5.66,0.78 11.05,0 20,-8.42 20,-18.8S35.05,4 24,4zm2.14,25.32
l-5.46,-5.83 -10.66,5.83 11.72,-12.44 5.6,5.83 10.52,-5.83 -11.72,12.44z"/>
</vector>
```

2 8 76 ic_more_vert

Nguồn: app/src/main/res/drawable/ic_more_vert.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_more_vert.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M12,8c1.1,0 2,-0.9 2,-2s-0.9,-2 -2,-2 -2,0.9 -2,2 0.9,2 2,2zm0,2c-1.
1,0 -2,0.9 -2,2s0.9,2 2,2 2,-0.9 2,-2 -0.9,-2 -2,-2zm0,6c-1.1,0 -2,0.9 -2,2s0.9,2 2,2 2,-0.9 2,-2
 -0.9,-2 -2,-2z" />
</vector>
```

2 8 77 ic_notification

Nguồn: app/src/main/res/drawable/ic_notification.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_notification.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M12,22c1.1,0 2,-0.9 2,-2h-4c0,1.1 0.9,2 2,2zm6,-6v-5c0,-3.07 -1.6
3,-5.64 -4.5,-6.32V4c0,-0.83 -0.67,-1.5 -1.5,-1.5s-1.5,0.67 -1.5,1.5v0.68C7.64,5.36 6,7.92 6,
11v5l-2,2v1h16v-1l-2,-2zm-2,1H8v-6c0,-2.48 1.51,-4.5 4,-4.5s4,2.02 4,4.5v6z" />
</vector>
```

2 8 78 ic_person

Nguồn: app/src/main/res/drawable/ic_person.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_person.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
```

```
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M12,12c2.21,0 4,-1.79 4,-4s-1.79,-4 -4,-4 -4,1.79 -4,4 1.79,4 4,4zm
0,2c-2.67,0 -8,1.34 -8,4v2h16v-2c0,-2.66 -5.33,-4 -8,-4z" />
</vector>
```

2 8 79 ic_photo

Nguồn: app/src/main/res/drawable/ic_photo.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_photo.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#4A90D9"
        android:pathData="M21,19V5c0,-1.1 -0.9,-2 -2,-2H5c-1.1,0 -2,0.9 -2,2v14c0,1.1 0.9,2
2,2h14c1.1,0 2,-0.9 2,-2zM8.5,13.5l2.5,3.01L14.5,12l4.5,6H5l3.5,-4.5z" />
</vector>
```

2 8 80 ic_photos

Nguồn: app/src/main/res/drawable/ic_photos.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_photos.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24"
    android:tint="?attr/colorControlNormal">
    <path
        android:fillColor="@android:color/white"
        android:pathData="M21,19V5c0,-1.1 -0.9,-2 -2,-2H5c-1.1,0 -2,0.9 -2,2v14c0,1.1 0.9,2
2,2h14c1.1,0 2,-0.9 2,-2zM8.5,13.5l2.5,3.01L14.5,12l4.5,6H5l3.5,-4.5z" />
</vector>
```

2 8 81 ic_play_circle

Nguồn: app/src/main/res/drawable/ic_play_circle.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_play_circle.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#FFFFFF"
        android:pathData="M24,4C12.95,4 4,12.95 4,24s8.95,20 20,20 20,-8.95 20,-20S35.05,
4 24,4zm-4,29V15l14,9 -14,9z" />
</vector>
```

2 8 82 ic_privacy

Nguồn: app/src/main/res/drawable/ic_privacy.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_privacy.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M12,1L3,5v6c0,5.55 3.84,10.74 9,12 5.16,-1.26 9,-6.45 9,-12V5l-9
,-4zm0,10.99h7c-0.53,4.12 -3.28,7.79 -7,8.94V12H5V6.3l7,-3.11v8.8z"/>
</vector>
```

2 8 83 ic_profile

Nguồn: app/src/main/res/drawable/ic_profile.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_profile.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24"
    android:tint="?attr/colorControlNormal">
    <path
        android:fillColor="@android:color/white"
        android:pathData="M12,12c2.21,0 4,-1.79 4,-4s-1.79,-4 -4,-4 -4,1.79 -4,4 1.79,4 4,4zm
0,2c-2.67,0 -8,1.34 -8,4v2h16v-2c0,-2.66 -5.33,-4 -8,-4z" />
</vector>
```

2 8 84 ic_restore

Nguồn: app/src/main/res/drawable/ic_restore.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_restore.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#4A90D9"
        android:pathData="M12,5V1L7,6l5,5V7c3.31,0 6,2.69 6,6s-2.69,6 -6,6 -6,-2.69 -6,-6H
4c0,4.42 3.58,8 8,8s8,-3.58 8,-8 -3.58,-8 -8,-8z" />
</vector>
```

2 8 85 ic_rotate

Nguồn: app/src/main/res/drawable/ic_rotate.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_rotate.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
```

```
        android:fillColor="#6B7280"
        android:pathData="M12,4V1L8,5l4,4V6c3.31,0 6,2.69 6,6 0,1.01 -0.25,1.97 -0.7,2.8l1.46,1.46C19.54,15.03 20,13.57 20,12c0,-4.42 -3.58,-8 -8,-8zm0,14c-3.31,0 -6,-2.69 -6,-6 0,-1.01 0.25,-1.97 0.7,-2.8L5.24,7.74C4.46,8.97 4,10.43 4,12c0,4.42 3.58,8 8,8v3l4,-4 -4,-4v3z" />
</vector>
```

2 8 86 ic_search

Nguồn: app/src/main/res/drawable/ic_search.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_search.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M15.5,14h-0.79l-0.28,-0.27C15.41,12.59 16,11.11 16,9.5 16,5.91 13.09,3 9.5,3S3,5.91 3,9.5 5.91,16 9.5,16c1.61,0 3.09,-0.59 4.23,-1.57l0.27,0.28v0.79l5,4.99L20.49,19l-4.99,-5zm-6,0C7.01,14 5,11.99 5,9.5S7.01,5 9.5,5 14,7.01 14,9.5 11.99,14 9.5,14z" />
</vector>
```

2 8 87 ic_select_all

Nguồn: app/src/main/res/drawable/ic_select_all.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_select_all.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#4A90D9"
        android:pathData="M3,5h2V3c-1.1,0 -2,0.9 -2,2zm0,8h2v-2H3v2zm4,8h2v-2H7v2zM3,9h2V7H3v2zm10,-6h-2v2h2V3zm6,0v2h2c0,-1.1 -0.9,-2 -2,-2zm-6,18h2v-2h-2v2zm-8,-8h2v-2H5v2zm0,-4h2V7H5v2zm8,8v-6H7v6h6zm-2,-4h2v2h-2v-2zm8,-2h2v-2h-2v2zm-4,6
```

h2v-2h-2v2zm4,-10v2h2c0,-1.1 -0.9,-2 -2,-2zm0,6h2v-2h-2v2zm0,4c1.1,0 2,-0.9 2,-2h-2v2z" />
</vector>

2 8 88 ic_settings

Nguồn: app/src/main/res/drawable/ic_settings.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_settings.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M19.14,12.94c0.04,-0.3 0.06,-0.61 0.06,-0.94 0,-0.32 -0.02,-0.64 -0.07,-0.94l2.03,-1.58c0.18,-0.14 0.23,-0.41 0.12,-0.61l-1.92,-3.32c-0.12,-0.22 -0.37,-0.29 -0.59,-0.22l-2.39,0.96c-0.5,-0.38 -1.03,-0.7 -1.62,-0.94l-0.36,-2.54c-0.04,-0.24 -0.24,-0.41 -0.48,-0.41h-3.84c-0.24,0 -0.43,0.17 -0.47,0.41l-0.36,2.54c-0.59,0.24 -1.13,0.56 -1.62,0.94l-2.39,-0.96c-0.22,-0.08 -0.47,0 -0.59,0.22L2.74,8.87c-0.12,0.21 -0.08,0.47 0.12,0.61l2.03,1.58c-0.05,0.3 -0.09,0.63 -0.09,0.94s0.02,0.64 0.07,0.94l-2.03,1.58c-0.18,0.14 -0.23,0.41 -0.12,0.61l1.92,3.32c0.12,0.22 0.37,0.29 0.59,0.22l2.39,-0.96c0.5,0.38 1.03,0.7 1.62,0.94l0.36,2.54c0.05,0.24 0.24,0.41 0.48,0.41h3.84c0.24,0 0.44,-0.17 0.47,-0.41l0.36,-2.54c0.59,-0.24 1.13,-0.56 1.62,-0.94l2.39,0.96c0.22,0.08 0.47,0 0.59,-0.22l1.92,-3.32c0.12,-0.22 0.07,-0.47 -0.12,-0.61l-2.01,-1.58zM12,15.6c-1.98,0 -3.6,-1.62 -3.6,-3.6s1.62,-3.6 3.6,-3.6 3.6,1.62 3.6,3.6 -1.62,3.6 -3.6,3.6z" />
</vector>
```

2 8 89 ic_share

Nguồn: app/src/main/res/drawable/ic_share.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_share.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
```

```
    <path
        android:fillColor="#1A1A2E"
        android:pathData="M18,16.08c-0.76,0 -1.44,0.3 -1.96,0.77L8.91,12.7c0.05,-0.23 0.09,-0.46 0.09,-0.7s-0.04,-0.47 -0.09,-0.7l7.05,-4.11c0.54,0.5 1.25,0.81 2.04,0.81 1.66,0 3,-1.34 3,-3s-1.34,-3 -3,-3 -3,1.34 -3,3c0,0.24 0.04,0.47 0.09,0.7L8.04,9.81C7.5,9.31 6.79,9 6,9c-1.66,0 -3,1.34 -3,3s1.34,3 3,3c0.79,0 1.5,-0.31 2.04,-0.81l7.12,4.16c-0.05,0.21 -0.08,0.43 -0.08,0.65 0,1.61 1.31,2.92 2.92,2.92s2.92,-1.31 2.92,-2.92c0,-1.61 -1.31,-2.92 -2.92,-2.92z" />
</vector>
```

2 8 90 ic_share_nearby

Nguồn: app/src/main/res/drawable/ic_share_nearby.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_share_nearby.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#4285F4"
        android:pathData="M24,4C12.95,4 4,12.95 4,24s8.95,20 20,20 20,-8.95 20,-20S35.05,4 24,4zM24,34c-5.52,0 -10,-4.48 -10,-10s4.48,-10 10,-10 10,4.48 10,10 -4.48,10 -10,10zm0,-16c-3.31,0 -6,2.69 -6,6s2.69,6 6,6 6,-2.69 6,-6 -2.69,-6 -6,-6z"/>
</vector>
```

2 8 91 ic_sort

Nguồn: app/src/main/res/drawable/ic_sort.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_sort.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M16,17.01V10h-2v7.01h-3L15,21l4,-3.99h-3zM9,3L5,6.99h3V14h2V6.99h3L9,3z" />
</vector>
```

2 8 92 ic_sync

Nguồn: app/src/main/res/drawable/ic_sync.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_sync.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#4A90D9"
        android:pathData="M12,4V1L8,5l4,4V6c3.31,0 6,2.69 6,6 0,1.01 -0.25,1.97 -0.7,2.8l1.
46,1.46C19.54,15.03 20,13.57 20,12c0,-4.42 -3.58,-8 -8,-8zm0,14c-3.31,0 -6,-2.69 -6,-6 0,-
1.01 0.25,-1.97 0.7,-2.8L5.24,7.74C4.46,8.97 4,10.43 4,12c0,4.42 3.58,8 8,8v3l4,-4 -4,-4v3
z"/>
</vector>
```

2 8 93 ic_text

Nguồn: app/src/main/res/drawable/ic_text.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_text.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M5,4v3h5.5v12h3V7H19V4H5z" />
</vector>
```

2 8 94 ic_video

Nguồn: app/src/main/res/drawable/ic_video.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_video.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M17,10.5V7c0,-0.55 -0.45,-1 -1,-1H4c-0.55,0 -1,0.45 -1,1v10c0,0.
55 0.45,1 1,1h12c0.55,0 1,-0.45 1,-1v-3.5l4,4v-11l-4,4z" />
</vector>
```

2 8 95 ic_visibility

Nguồn: app/src/main/res/drawable/ic_visibility.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_visibility.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M12,4.5C7,4.5 2.73,7.61 1,12c1.73,4.39 6,7.5 11,7.5s9.27,-3.11 11
,-7.5c-1.73,-4.39 -6,-7.5 -11,-7.5zM12,17c-2.76,0 -5,-2.24 -5,-5s2.24,-5 5,-5 5,2.24 5,5 -2.2
4,5 -5,5zm0,-8c-1.66,0 -3,1.34 -3,3s1.34,3 3,3 3,-1.34 3,-3 -1.34,-3 -3,-3z" />
</vector>
```

2 8 96 ic_visibility_off

Nguồn: app/src/main/res/drawable/ic_visibility_off.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_visibility_off.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
```

```
    android:viewportHeight="24">
    <path
        android:fillColor="#6B7280"
        android:pathData="M12,7c2.76,0 5,2.24 5,5 0,0.65 -0.13,1.26 -0.36,1.82l2.92,2.92c1.5
1,-1.26 2.7,-2.89 3.44,-4.74 -1.73,-4.39 -6,-7.5 -11,-7.5 -1.4,0 -2.74,0.25 -3.98,0.7l2.16,2.16
C10.74,7.13 11.35,7 12,7zM2,4.27l2.28,2.28 0.46,0.46C3.08,8.3 1.78,10.02 1,12c1.73,4.39
6,7.5 11,7.5 1.55,0 3.03,-0.3 4.38,-0.84l0.42,0.42L19.73,22 21,20.73 3.27,3 2,4.27zM7.53,9
.8l1.55,1.55c-0.05,0.21 -0.08,0.43 -0.08,0.65 0,1.66 1.34,3 3,3 0.22,0 0.44,-0.03 0.65,-0.08l
1.55,1.55c-0.67,0.33 -1.41,0.53 -2.2,0.53 -2.76,0 -5,-2.24 -5,-5 0,-0.79 0.2,-1.53 0.53,-2.2z
m4.31,-0.78l3.15,3.15.02,-0.16c0,-1.66 -1.34,-3 -3,-3l-0.17,0.01z" />
</vector>
```

2 8 97 ic_wifi

Nguồn: app/src/main/res/drawable/ic_wifi.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_wifi.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="24dp"
    android:height="24dp"
    android:viewportWidth="24"
    android:viewportHeight="24">
    <path
        android:fillColor="#4A90D9"
        android:pathData="M12,3C7.79,3 3.97,4.64 1.1,7.34L12,21.5l10.9,-14.16C20.03,4.64
16.21,3 12,3zm0,3c3.27,0 6.27,1.21 8.57,3.22L12,19.2 3.43,9.22C5.73,7.21 8.73,6 12,6z"/>
</vector>
```

2 8 98 ic_zalo

Nguồn: app/src/main/res/drawable/ic_zalo.xml

Vai trò: Vector icon, hình minh họa và nền bo góc hoặc khung chọn. Tên tài nguyên: ic_zalo.

Mã nguồn đầy đủ

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="48dp"
    android:height="48dp"
    android:viewportWidth="48"
    android:viewportHeight="48">
    <path
        android:fillColor="#0068FF"
        android:pathData="M24,4C12.95,4 4,12.95 4,24c0,3.61 0.98,7 2.68,9.92L4,44l10.37,-
```

2.61C17.15,43.04 20.46,44 24,44c11.05,0 20,-8.95 20,-20S35.05,4 24,4zm-4,26h-6v-3l4,-5h-4v-3h6v3l-4,5h4v3zm9,0h-3V19h3v11zm5,0h-3V19h3v11z"/>
</vector>

2 9 Thư mục drawable night

Đường dẫn: app/src/main/res/drawable-night

Tài nguyên drawable thay thế khi giao diện tối được bật. Số file XML: 1.

Danh sách: bg_auth_surface.xml.

2 9 1 bg_auth_surface

Nguồn: app/src/main/res/drawable-night/bg_auth_surface.xml

Vai trò: Tài nguyên drawable thay thế khi giao diện tối được bật. Tên tài nguyên: bg_auth_surface.

Mã nguồn đầy đủ

<shape xmlns:android="http://schemas.android.com/apk/res/android"><gradient android:angle="315" android:startColor="@color/surface" android:centerColor="@color/background" android:endColor="@color/background_card"/></shape>

2 10 Thư mục mipmap anydpi v26

Đường dẫn: app/src/main/res/mipmap-anydpi-v26

Cấu hình biểu tượng ứng dụng thích ứng từ Android API 26. Số file XML: 2.

Danh sách: ic_launcher.xml, ic_launcher_round.xml.

2 10 1 ic_launcher

Nguồn: app/src/main/res/mipmap-anydpi-v26/ic_launcher.xml

Vai trò: Cấu hình biểu tượng ứng dụng thích ứng từ Android API 26. Tên tài nguyên: ic_launcher.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
    <background android:drawable="@drawable/ic_launcher_background"/>
    <foreground android:drawable="@drawable/ic_launcher_foreground"/>
</adaptive-icon>
```

2 10 2 ic_launcher_round

Nguồn: app/src/main/res/mipmap-anydpi-v26/ic_launcher_round.xml

Vai trò: Cấu hình biểu tượng ứng dụng thích ứng từ Android API 26. Tên tài nguyên: ic_launcher_round.

Mã nguồn đầy đủ

```
<?xml version="1.0" ?>
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
    <background android:drawable="@drawable/ic_launcher_background"/>
    <foreground android:drawable="@drawable/ic_launcher_foreground"/>
</adaptive-icon>
```

2 11 Thư mục xml

Đường dẫn: app/src/main/res/xml

Chính sách sao lưu hệ thống và các đường dẫn chia sẻ FileProvider. Số file XML: 3.

Danh sách: backup_rules.xml, data_extraction_rules.xml, file_paths.xml.

2 11 1 backup_rules

Nguồn: app/src/main/res/xml/backup_rules.xml

Vai trò: Chính sách sao lưu hệ thống và các đường dẫn chia sẻ FileProvider. Tên tài nguyên: backup_rules.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?><!--
   Sample backup rules file; uncomment and customize as necessary.
   See https://developer.android.com/guide/topics/data/autobackup
   for details.
   Note: This file is ignored for devices older than API 31
   See https://developer.android.com/about/versions/12/backup-restore
-->
<full-backup-content>
    <!--
   <include domain="sharedpref" path="."/>
   <exclude domain="sharedpref" path="device.xml"/>
-->
</full-backup-content>
```

2 11 2 data_extraction_rules

Nguồn: app/src/main/res/xml/data_extraction_rules.xml

Vai trò: Chính sách sao lưu hệ thống và các đường dẫn chia sẻ FileProvider. Tên tài nguyên: data_extraction_rules.

Mã nguồn đầy đủ

```
<?xml version="1.0" encoding="utf-8"?><!--
```

```
  Sample data extraction rules file; uncomment and customize as necessary.
  See https://developer.android.com/about/versions/12/backup-restore#xml-changes
  for details.
-->
<data-extraction-rules>
    <cloud-backup>
        <!-- TODO: Use <include> and <exclude> to control what is backed up.
        <include .../>
        <exclude .../>
        -->
    </cloud-backup>
    <!--
    <device-transfer>
        <include .../>
        <exclude .../>
    </device-transfer>
    -->
</data-extraction-rules>
```

2 11 3 file_paths

Nguồn: app/src/main/res/xml/file_paths.xml

Vai trò: Chính sách sao lưu hệ thống và các đường dẫn chia sẻ FileProvider. Tên tài nguyên: file_paths.

Mã nguồn đầy đủ

```
<paths xmlns:android="http://schemas.android.com/apk/res/android"><files-path name="m
edia" path="media/"/><cache-path name="shared" path="shared/"/></paths>
```

3 Kết quả sau khi chạy

Các hình dưới đây là ảnh chụp ứng dụng chạy trên máy ảo, không phải bản thiết kế tham khảo. Mỗi hình đi kèm tên bố cục, lớp xử lý, thao tác và ý nghĩa để trình bày các giao diện chính.

Số lượng ảnh hoặc album phản ánh dữ liệu demo có trên máy tại lúc chụp. Một số màn hình có dữ liệu tài khoản đang đăng nhập; tên và số mục không được gán cố định trong giao diện.

3 1 Màn hình khởi động



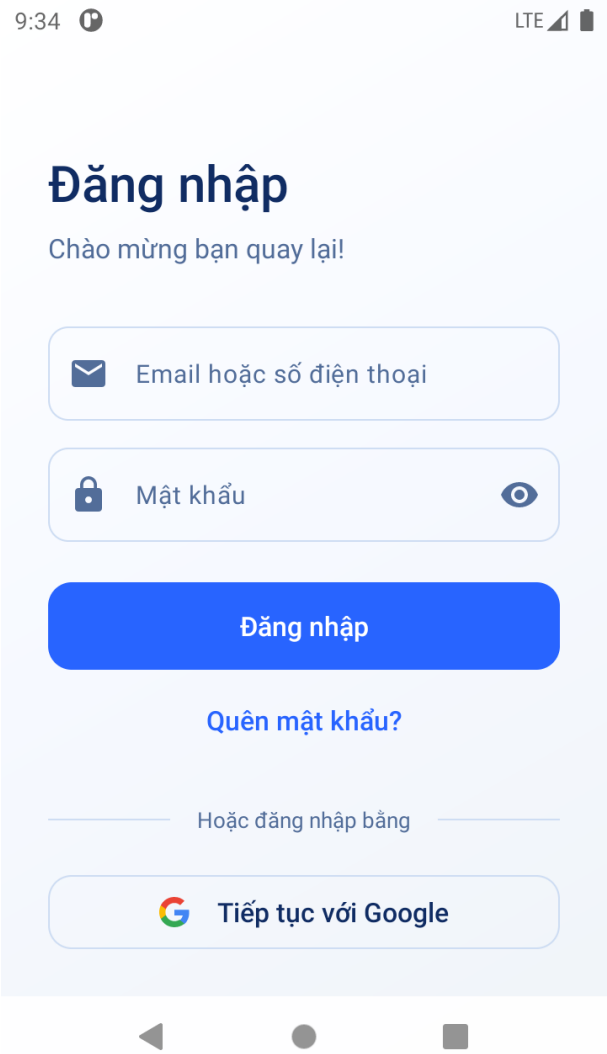
Hình 1 Màn hình khởi động

Bố cục: activity_splash.xml. Lớp xử lý: SplashActivity.

Chú thích giao diện: Hiển thị logo, tên ứng dụng, minh họa và hai lựa chọn truy cập.

Thao tác trình bày: Nhấn Bắt đầu để đi tiếp theo trạng thái đăng nhập; Khám phá trên thiết bị mở chế độ khách. Không cần thao tác đăng nhập để giới thiệu bố cục màn chào.

3 2 Màn hình đăng nhập



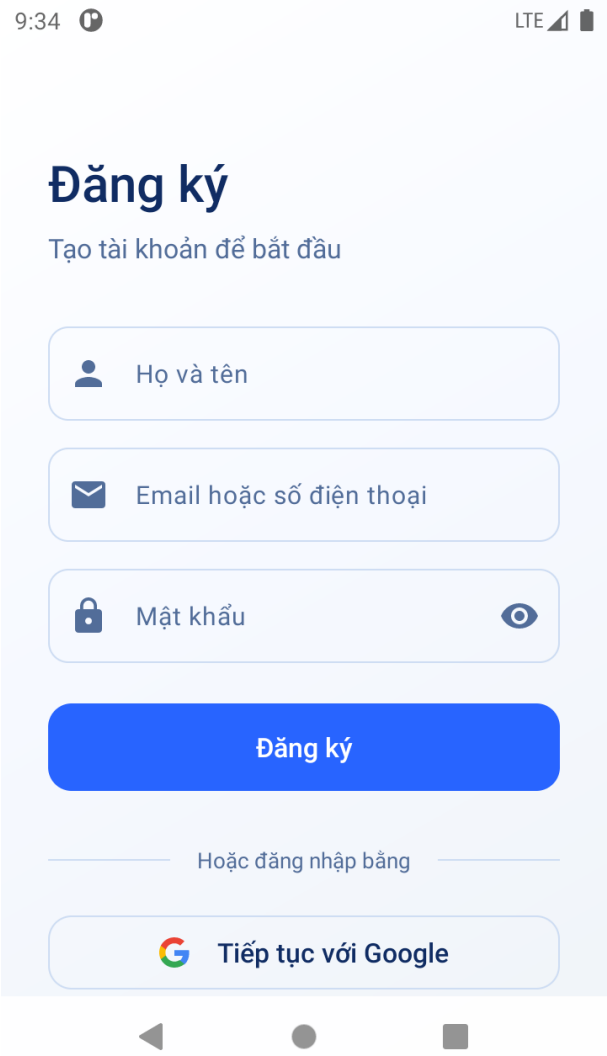
Hình 2 Màn hình đăng nhập

Bố cục: activity_login.xml. Lớp xử lý: LoginActivity và AuthActivity.

Chú thích giao diện: Biểu mẫu email hoặc số điện thoại, mật khẩu, quên mật khẩu và tiếp tục với Google.

Thao tác trình bày: Nhập tài khoản rồi nhấn Đăng nhập. Email và số điện thoại đi theo hai cách xác thực khác nhau; cửa sổ Google là luồng thật của hệ thống.

3 3 Màn hình đăng ký



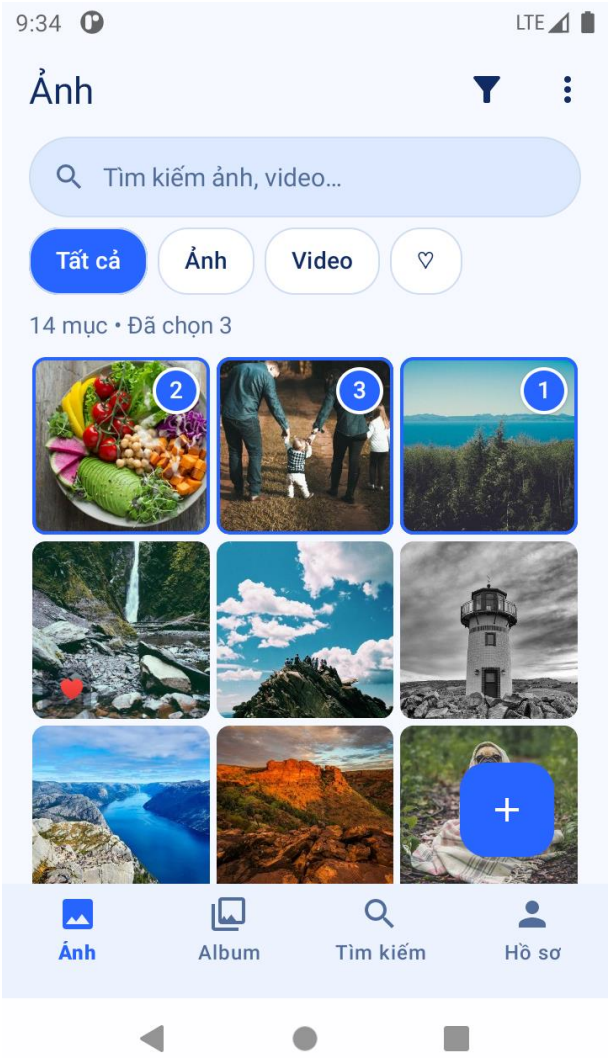
Hình 3 Màn hình đăng ký

Bố cục: activity_register.xml. Lớp xử lý: RegisterActivity và AuthActivity.

Chú thích giao diện: Biểu mẫu họ tên, email hoặc số điện thoại và mật khẩu; có đường dẫn quay về đăng nhập.

Thao tác trình bày: Các trường được kiểm tra trước khi gọi Firebase. Đăng ký email gửi liên kết xác minh; không dùng mã email giả trong bản demo.

3 4 Thư viện và thứ tự chọn nhiều



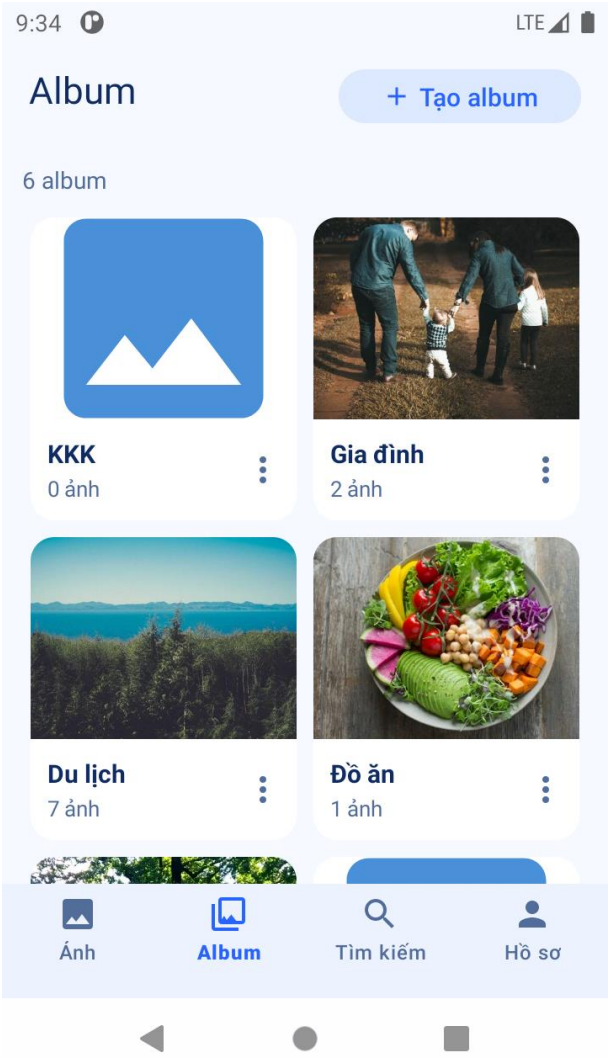
Hình 4 Thư viện và thứ tự chọn nhiều

Bố cục: activity_main.xml và item_media.xml. Lớp xử lý: MainActivity GalleryActivity và MediaAdapter.

Chú thích giao diện: Thanh tìm kiếm, bộ lọc, lưới ảnh, nút thêm tệp và bốn tab điều hướng. Huy hiệu 1, 2, 3 thể hiện thứ tự chạm chọn, không phải vị trí trong lưới.

Thao tác trình bày: Mở menu ba chấm → Chọn nhiều → chạm các mục. Bỏ một mục thì các số phía sau cập nhật; chọn lại thì mục đó được xếp cuối. Có thể thêm các mục đã chọn vào album hoặc ẩn.

3 5 Danh sách album



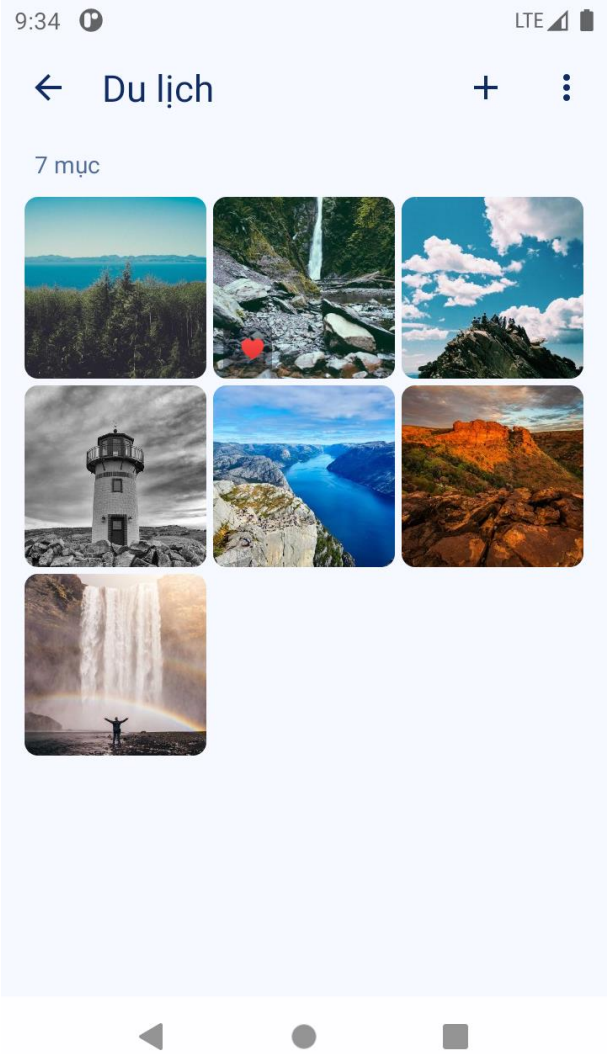
Hình 5 Danh sách album

Bố cục: activity_album.xml và item_album.xml. Lớp xử lý: AlbumActivity GalleryActivity và MediaAdapter.

Chú thích giao diện: Các thẻ album hai cột gồm ảnh bìa thật, tên và số lượng ảnh hoặc video.

Thao tác trình bày: Nhấn Tạo album để nhóm ảnh. Nhấn thẻ để xem nội dung; menu ba chấm hỗ trợ thêm ảnh, đổi tên hoặc xóa album. Xóa album không xóa tệp trong thư viện.

3 6 Nội dung album Du lịch



Hình 6 Nội dung album Du lịch

Bố cục: activity_album_detail.xml. Lớp xử lý: AlbumDetailActivity và GalleryActivity.

Chú thích giao diện: Lưới các ảnh thuộc album cùng nút thêm và menu hành động trên toolbar.

Thao tác trình bày: Nhấn + để chọn thêm ảnh từ thư viện. Menu hỗ trợ chọn nhiều và bỏ các mục khỏi album; quan hệ thành viên được lưu trong albumIds.

3 7 Tạo album mới



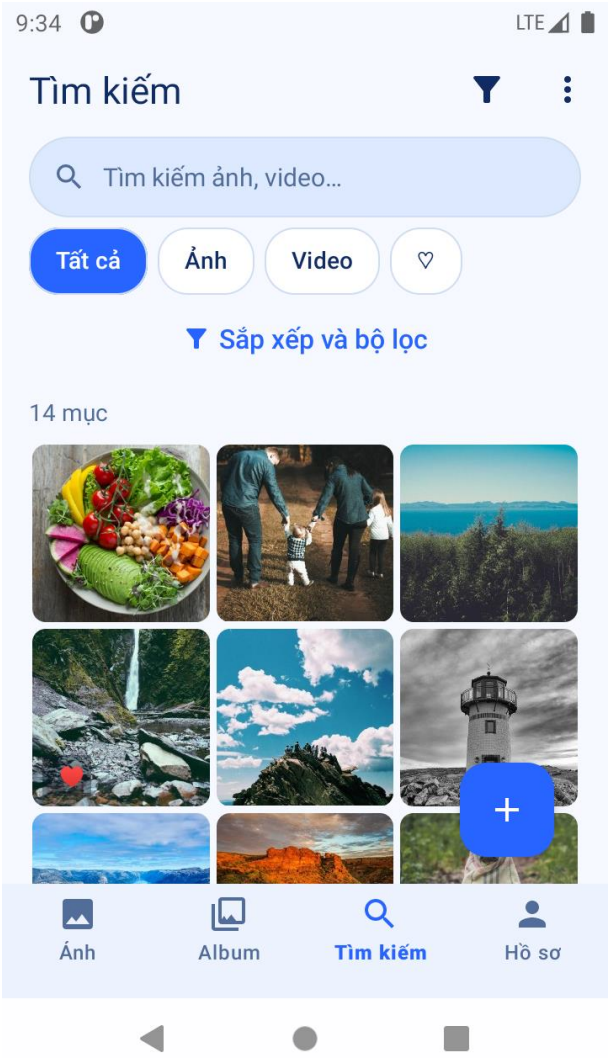
Hình 7 Tạo album mới

Bố cục: activity_create_album.xml. Lớp xử lý: CreateAlbumActivity và GalleryActivity.

Chú thích giao diện: Ô tên album và lưới chọn ảnh; các ảnh được chọn có số thứ tự.

Thao tác trình bày: Nhập tên không rỗng, chọn ảnh rồi nhấn tạo. Album mới được ghi vào thư viện của khách hoặc tài khoản đang dùng.

3 8 Tìm kiếm ảnh và video



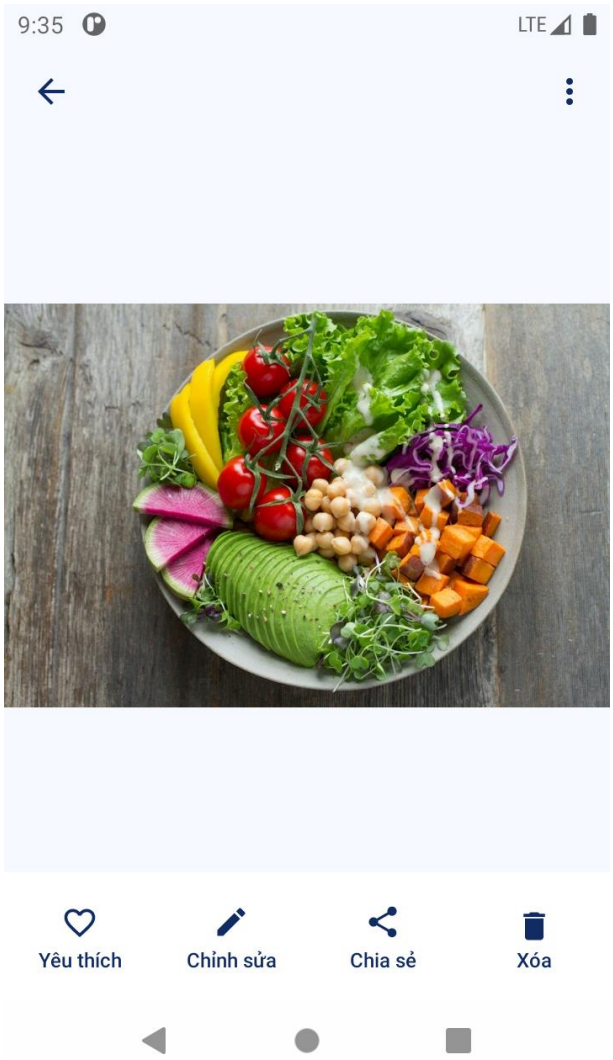
Hình 8 Tìm kiếm ảnh và video

Bố cục: activity_search.xml. Lớp xử lý: SearchActivity và GalleryActivity.

Chú thích giao diện: Ô tìm theo tên, chip loại nội dung và bộ lọc sắp xếp.

Thao tác trình bày: Gõ một phần tên tệp; chọn Ảnh hoặc Video để thu hẹp kết quả. Mục đã ẩn không xuất hiện trong kết quả tìm kiếm thường.

3 9 Xem ảnh chi tiết



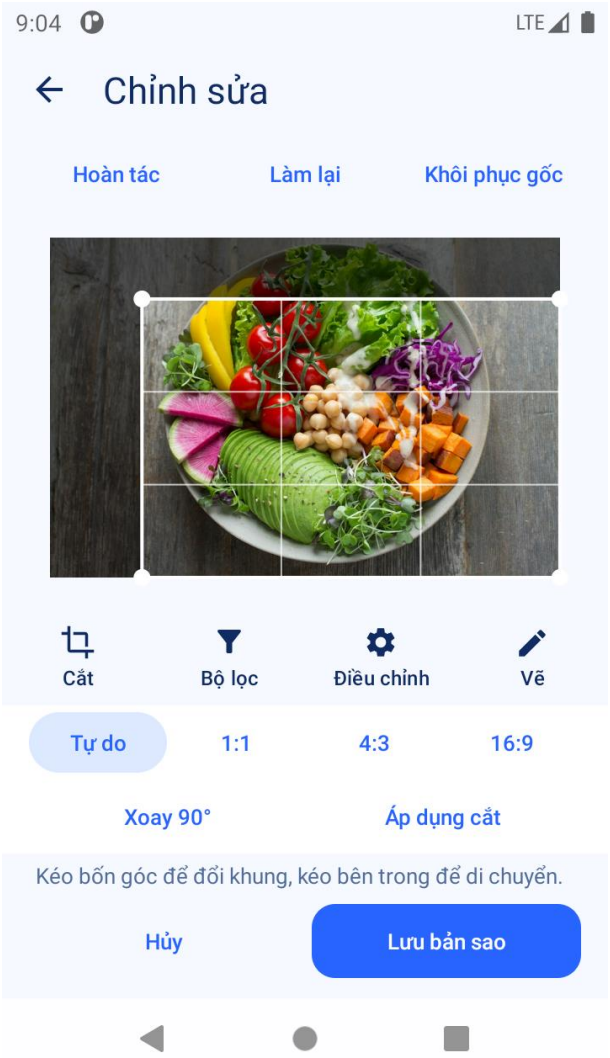
Hình 9 Xem ảnh chi tiết

Bố cục: activity_photo_detail.xml. Lớp xử lý: PhotoDetailActivity.

Chú thích giao diện: Ảnh được hiển thị lớn với hành động yêu thích, chỉnh sửa, chia sẻ và thùng rác.

Thao tác trình bày: Vuốt để chuyển ảnh; menu bổ sung mở thông tin hoặc ẩn và bỏ ẩn. Khi mở video, chạm vùng xem trước để phát bằng ứng dụng hệ thống.

3 10 Cắt ảnh bằng khung tương tác



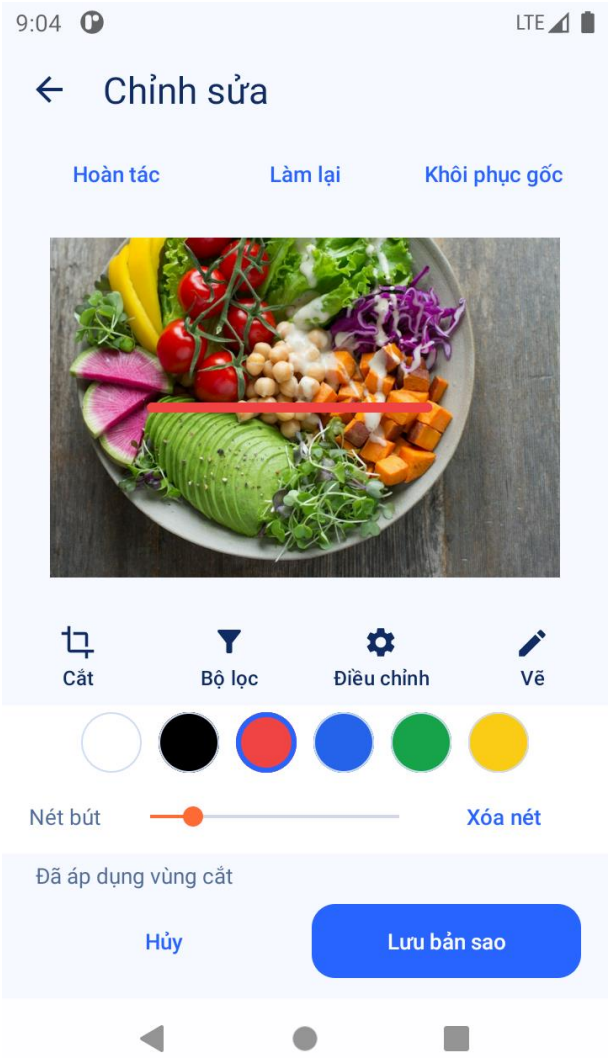
Hình 10 Cắt ảnh bằng khung tương tác

Bố cục: activity_edit_photo.xml. Lớp xử lý: EditPhotoActivity EditorCanvasView và EditorDocument.

Chú thích giao diện: Khung cắt có bốn góc kéo, lưới một phần ba và vùng ngoài được làm tối; có tỷ lệ tự do, 1:1, 4:3 và 16:9.

Thao tác trình bày: Chọn Cắt → kéo góc để thay kích thước hoặc kéo bên trong để di chuyển → Áp dụng cắt. Có thể xoay 90 độ; Lưu bản sao tạo tệp mới thay vì ghi đè ảnh gốc.

3 11 Vẽ trực tiếp lên ảnh



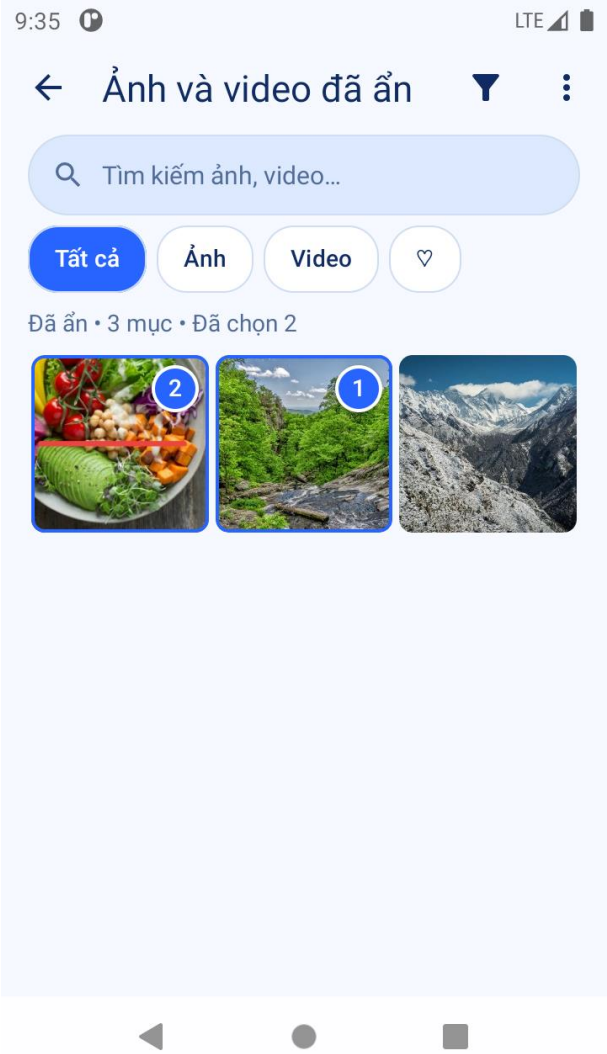
Hình 11 Vẽ trực tiếp lên ảnh

Bố cục: activity_edit_photo.xml. Lớp xử lý: EditPhotoActivity EditorCanvasView và EditorDocument.

Chú thích giao diện: Bảng sáu màu, thanh độ dày và nút xóa nét. Nét vẽ hiển thị trên ảnh theo tọa độ gốc.

Thao tác trình bày: Chọn Vẽ → chọn màu và độ dày → kéo ngón tay hoặc chuột trên ảnh. Hoàn tác và Làm lại áp dụng cho thao tác chỉnh sửa; bản xuất JPEG bao gồm cả ảnh và nét vẽ.

3 12 Ảnh và video đã ẩn



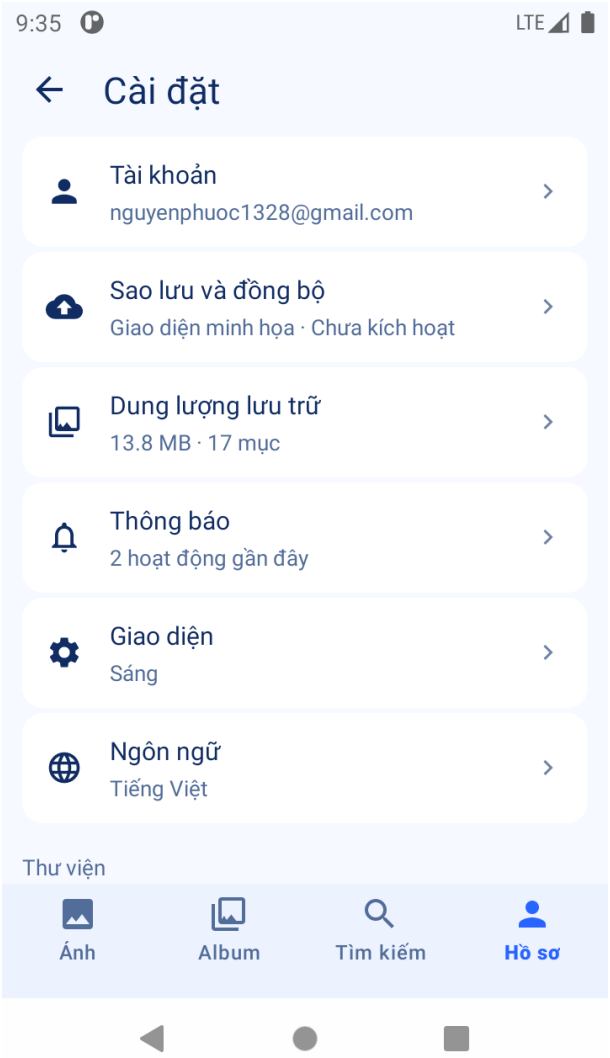
Hình 12 Ảnh và video đã ẩn

Bố cục: Dùng lại activity_main.xml và item_media.xml. Lớp xử lý: HiddenMediaActivity và GalleryActivity.

Chú thích giao diện: Màn riêng có tiêu đề, tìm kiếm, lọc ảnh hoặc video và danh sách các mục đã ẩn.

Thao tác trình bày: Mở Hồ sơ → Ảnh và video đã ẩn hoặc menu trang Ảnh. Nhấn giữ một mục → Bỏ ẩn để đưa về thư viện; có chọn nhiều. Chức năng này chưa có khóa mật khẩu.

3 13 Hồ sơ và cài đặt



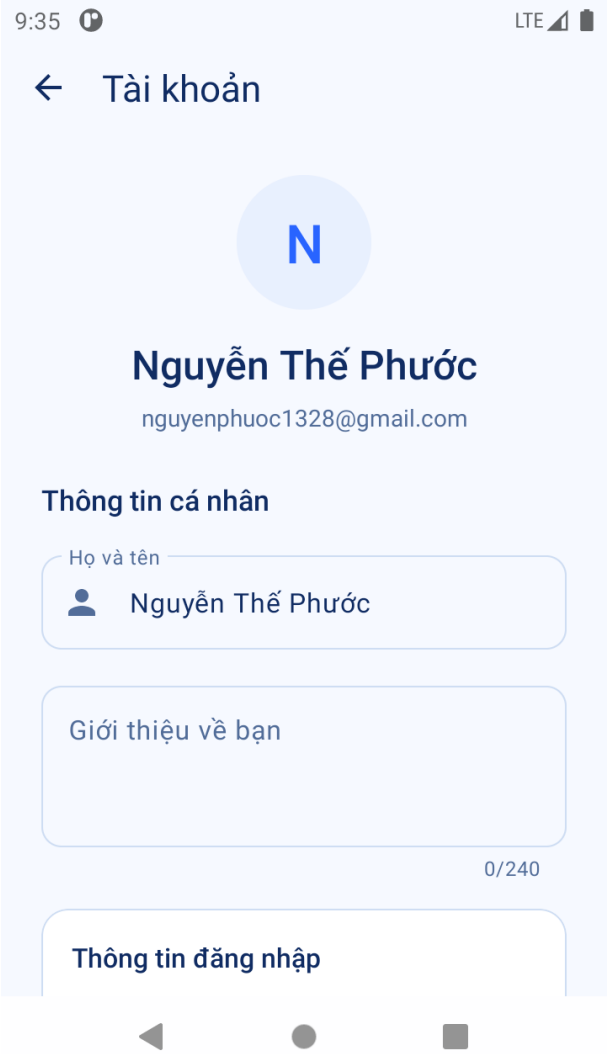
Hình 13 Hồ sơ và cài đặt

Bố cục: activity_settings.xml. Lớp xử lý: SettingsActivity.

Chú thích giao diện: Các dòng chức năng gồm tài khoản, sao lưu, dung lượng, thông báo, giao diện, ngôn ngữ và nhóm thư viện phía dưới.

Thao tác trình bày: Nhấn từng dòng để mở màn tương ứng. Chọn Giao diện để đổi sáng hoặc tối cho toàn app; lựa chọn được giữ khi mở lại ứng dụng.

3 14 Thông tin tài khoản trên Firestore



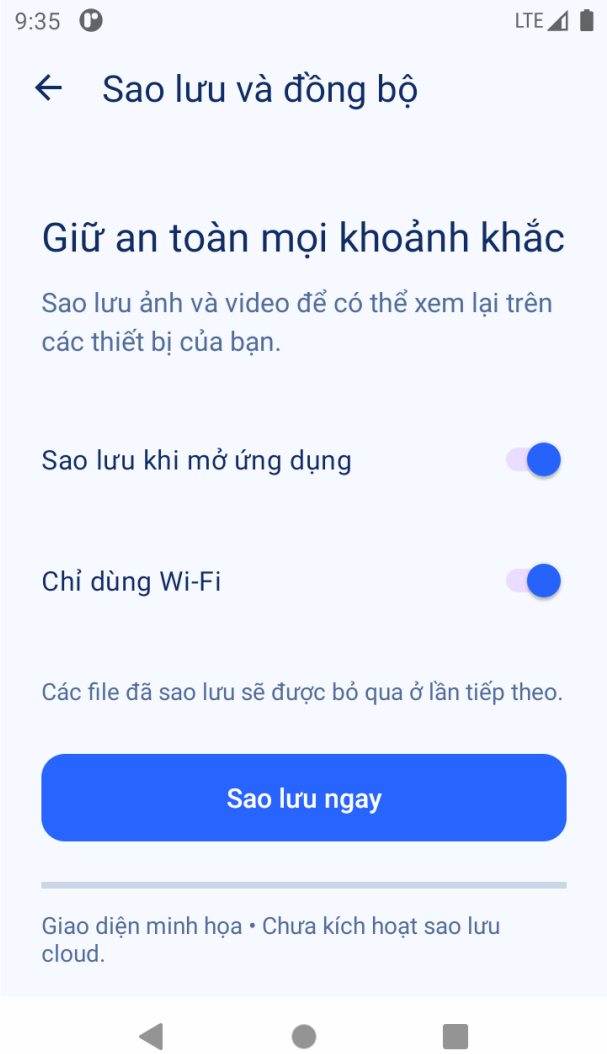
Hình 14 Thông tin tài khoản trên Firestore

Bố cục: activity_account.xml. Lớp xử lý: AccountActivity.

Chú thích giao diện: Hồ sơ riêng với họ tên, giới thiệu và thông tin đăng nhập; thay cho hộp thoại chỉ hiển thị UID.

Thao tác trình bày: Dữ liệu hồ sơ tham chiếu document users/{uid}. Chỉnh thông tin rồi lưu; chờ xác nhận trước khi đối chiếu các trường displayName và bio trên Firebase Console.

3 15 Giao diện sao lưu minh họa



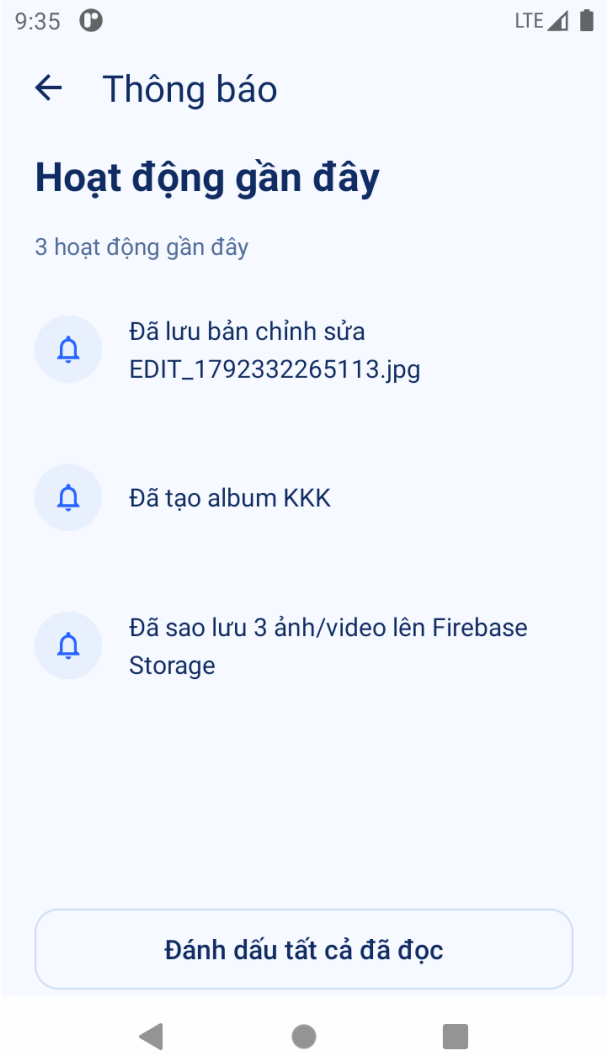
Hình 15 Giao diện sao lưu minh họa

Bố cục: activity_backup.xml. Lớp xử lý: BackupActivity.

Chú thích giao diện: Hai công tắc và nút Sao lưu ngay được giữ để trình bày bố cục. Trạng thái ghi rõ đây là màn minh họa.

Thao tác trình bày: Nhấn Sao lưu ngay chỉ mở thông báo giải thích; không tải file lên Firebase Storage, không có sao lưu tự động trong bản hiện tại.

3 16 Thông báo hoạt động



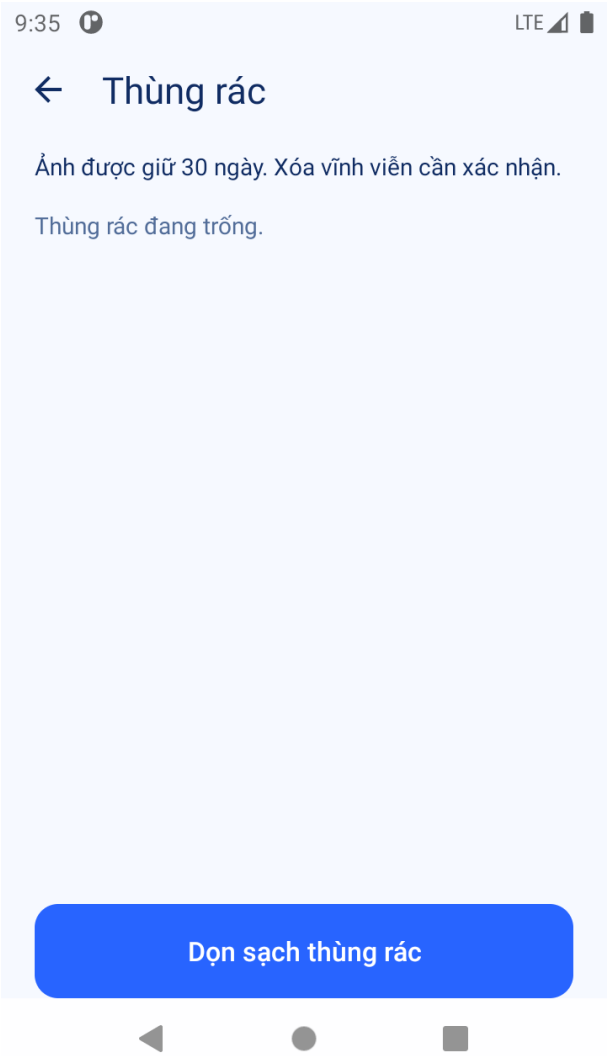
Hình 16 Thông báo hoạt động

Bố cục: activity_notifications.xml và item_notification.xml. Lớp xử lý: NotificationsActivity.

Chú thích giao diện: Danh sách các hoạt động lưu trong ứng dụng hoặc Firestore và nút đánh dấu tất cả đã đọc.

Thao tác trình bày: Đây là lịch sử hoạt động, không phải thông báo đẩy. Nếu danh sách còn câu sao lưu lên Storage từ dữ liệu cũ, câu đó không chứng minh bản hiện tại đã tải file lên cloud.

3 17 Thùng rác



Hình 17 Thùng rác

Bố cục: activity_trash.xml và item_media_row.xml. Lớp xử lý: TrashActivity và GalleryActivity.

Chú thích giao diện: Danh sách mục đã chuyển vào thùng rác cùng hành động dọn. Nếu chưa xóa mục nào, màn hình hiển thị trạng thái trống.

Thao tác trình bày: Chạm mục để khôi phục hoặc xác nhận xóa vĩnh viễn. Khi mở thùng rác, ứng dụng có xử lý các mục quá 30 ngày; không phải dịch vụ dọn trên server khi app đóng.